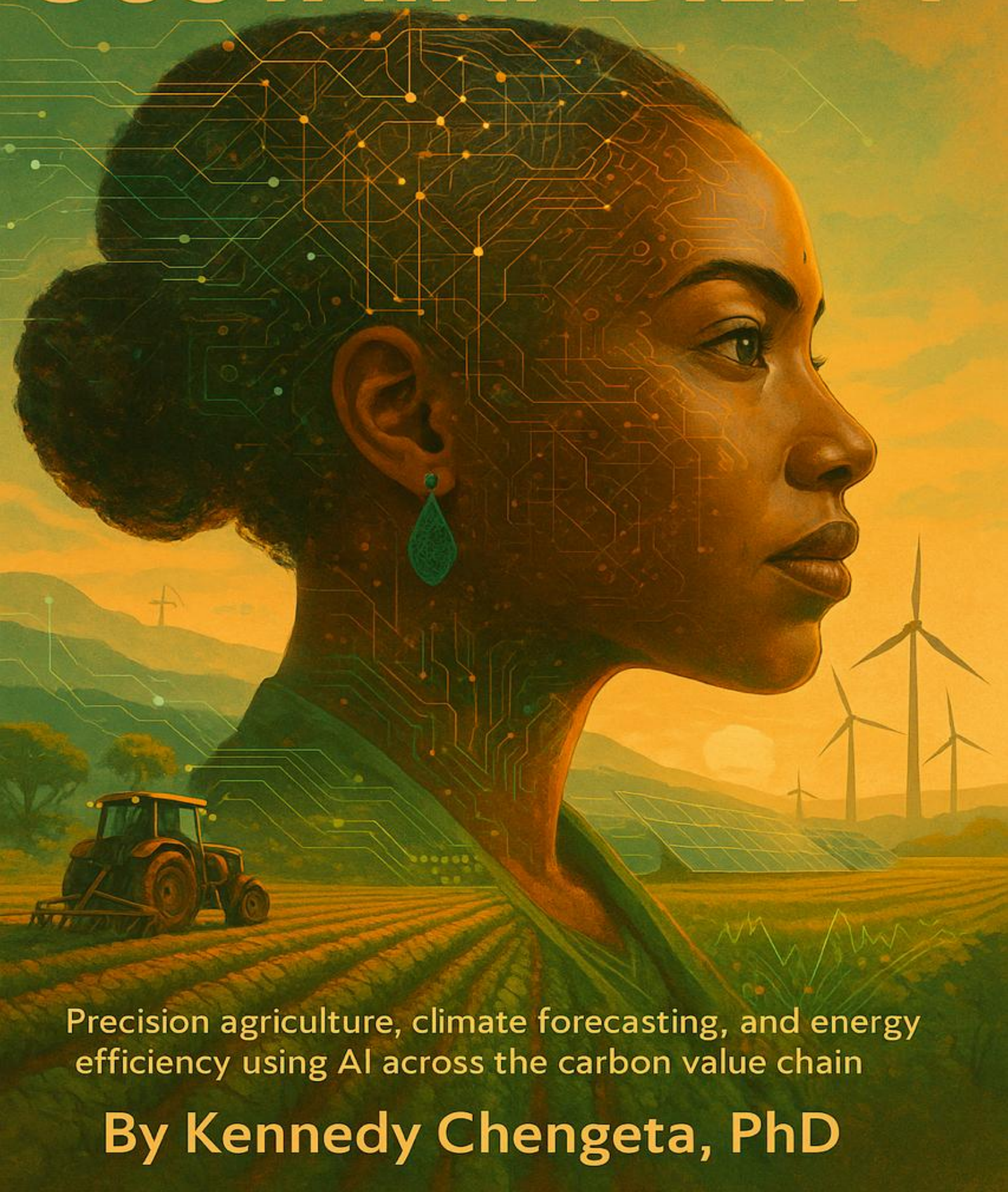


AI FOR SUSTAINABILITY



Precision agriculture, climate forecasting, and energy efficiency using AI across the carbon value chain

By Kennedy Chengeta, PhD

AI for Sustainability

Precision Agriculture, Climate Forecasting, and Energy Efficiency Using AI Across the Carbon Value Chain

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AI for Sustainability

*Precision Agriculture, Climate Forecasting,
and Energy Efficiency Using AI
Across the Carbon Value Chain*

Kennedy Chengeta, PhD

*AI/HyperAutomation Integration Architect
KaribuTech AI & DFKC Consulting, Johannesburg*

First Edition, 2025

KaribuTech AI Press
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*To Nyaradzai and Miriro,
and to every African farmer, engineer, and entrepreneur
building a sustainable future for the continent.*

*“The hardest problems of the twenty-first century
— food, climate, energy — are problems of intelligence
applied to nature. AI must serve sustainability,
not the other way around.”*

—Kennedy Chengeta, PhD

Preface

This book began at the intersection of two stubborn realities. The first is that artificial intelligence has, in the past five years, advanced more rapidly and become more useful than almost any technologist predicted. The second is that the planet's environmental systems — its climate, its soils, its water cycles, its biodiversity — are deteriorating more rapidly than almost any environmental scientist predicted. These two trajectories are not unrelated. AI's data centres consume electricity at industrial scale. AI's training and inference workloads embed carbon in every model deployment. And yet AI offers, for the first time in human history, a set of computational tools capable of measuring, predicting, and optimising natural systems at continental scale and at sub-daily resolution.

The question this book poses, and tries to answer chapter by chapter, is whether AI's contribution to sustainability can be made net-positive — and if so, how, where, and under what conditions. Across precision agriculture, climate forecasting, energy efficiency, carbon measurement, circular economy, biodiversity conservation, ESG disclosure, and just transition, the evidence is now substantial. AI is already producing measurable sustainability outcomes when properly designed, governed, and deployed. It is also producing measurable harms when deployed without consideration of its energy consumption, its data ethics, or its tendency to entrench existing inequalities.

The material in this volume draws from operational specifications, peer-reviewed research, regulatory frameworks, and field deployments across Sub-Saharan Africa, the European Union, North America, and Asia. The emphasis throughout is practical: every algorithm is presented with its mathematical foundation and a worked example a practitioner can apply. Every case study is grounded in a real deployment with real data. Every chapter ends with a discussion of risks, limitations, and the conditions under which the technology adds value.

This book is written for three audiences. First, sustainability and climate practitioners who need to understand what AI can and cannot do for their work. Second, AI engineers and data scientists who want to apply their skills to environmental challenges. Third, policy-makers, executives, and investors who must make decisions about technology, capital, and regulation in a world where the stakes are no longer abstract.

Kennedy Chengeta, PhD

AI/HyperAutomation Integration Architect
KaribuTech AI & DFKC Consulting, Johannesburg, 2025

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Notation

Throughout this book we adopt the following conventions unless otherwise stated.

Symbol	Meaning
CO_2	Carbon dioxide
tCO_2e	Tonnes of carbon dioxide equivalent
NDVI	Normalised Difference Vegetation Index
EVI	Enhanced Vegetation Index
LST	Land Surface Temperature
SOC	Soil Organic Carbon
ET_0	Reference evapotranspiration
P	Precipitation (mm)
T	Temperature ($^{\circ}\text{C}$ or K)
η	Learning rate; or efficiency factor
θ	Neural network parameter vector
$\mathbf{x}$	Feature vector (bold lowercase)
$\mathbf{X}$	Feature matrix (bold uppercase)
$\mathbb{E}[\cdot]$	Expectation operator
$\hat{y}$	Predicted value (estimate)
$\mathcal{L}$	Loss function
GHG	Greenhouse gas
GWP	Global warming potential
LCA	Life cycle assessment
MRV	Measurement, reporting, verification

Mathematical notation follows international convention. Code listings use `typewriter` font.

Part I

Foundations: AI, Sustainability, and the Carbon Value Chain

Introduction: Why AI for Sustainability

The hardest problems of the twenty-first century — food, climate, energy — are problems of intelligence applied to nature. AI must serve sustainability, not the other way around.

Kennedy Chengeta, PhD

The convergence of artificial intelligence and sustainability is the most consequential technological development of our generation. For the first time in human history, computational tools exist that can ingest planetary-scale environmental data, learn from it, and produce decisions that improve both economic and ecological outcomes simultaneously. This book is a practical guide to that convergence: how AI is reshaping precision agriculture, climate forecasting, energy systems, carbon markets, biodiversity monitoring, and sustainable finance, with a particular emphasis on African deployments and the carbon value chain.

1.1 The Three Crises and One Toolkit

The twenty-first century inherited three crises that the twentieth century could not solve. The climate crisis, in which atmospheric carbon dioxide has risen from 280 parts per million in pre-industrial times to over 420 parts per million today, is forcing changes in temperature, precipitation, and extreme weather that no civilisation has navigated before. The food crisis, in which approximately 750 million people face chronic hunger while the global agricultural system emits roughly one-quarter of all greenhouse gases, demands that we produce more food with less land, water, and emissions. The energy crisis, in which the transition from fossil fuels to renewables must be completed within roughly two decades to keep warming below 2°C, requires reorganising the largest physical infrastructure humans have ever built.

These three crises share a common structural feature: they are problems of measurement, prediction, and optimisation across vast, heterogeneous, dynamic systems. Climate is the global atmosphere coupled to oceans, land, and ice. Agriculture is millions of fields each with unique soils, crops, and microclimates. Energy is generation, transmission, and consumption operating at sub-second timescales across national grids. Solving these problems requires tools that can handle scale, heterogeneity, and dynamism — exactly the strengths of modern machine learning.

Artificial intelligence is the toolkit. Deep neural networks trained on satellite imagery can detect deforestation, predict crop yields, and quantify carbon sequestration at hectare resolution across

continents. Recurrent and transformer architectures can forecast electricity demand, weather patterns, and crop failures with accuracy that meteorological and agronomic models alone cannot match. Reinforcement learning can optimise irrigation schedules, building HVAC systems, and battery dispatch in real time. Generative models can produce synthetic climate scenarios, simulate plant breeding, and design new materials. Large language models can synthesise scientific literature, automate ESG disclosures, and translate technical findings into farmer-accessible advice.

1.2 Why AI is Necessary, Not Sufficient

Two cautions structure this book. First, AI is not magic. A model is only as good as the data it is trained on, the assumptions it encodes, and the deployment context in which it operates. A yield prediction model trained on Iowa maize will fail on Zimbabwean maize. A climate downscaling model trained on European reanalysis data will fail in the Sahel. A grid optimisation model trained on stable, well-instrumented Northern European grids will fail in Sub-Saharan Africa, where load-shedding and informal connections are the norm. Sustainability AI is fundamentally local, and the dominant failure mode is the assumption that a model trained somewhere else will work here.

Second, AI itself has a carbon footprint. Training a single large language model can emit hundreds of tonnes of CO₂. Inference at scale, repeated billions of times daily, adds steadily to data centre electricity demand, which currently accounts for one to two per cent of global electricity consumption and is rising rapidly. Any honest treatment of AI for sustainability must account for the resources AI itself consumes. Chapter 53 addresses this directly.

Definition 1.1 (Net-Positive AI). A deployment of artificial intelligence is *net-positive* for sustainability when the lifecycle environmental benefits enabled by the system exceed the lifecycle environmental costs of building and operating it. Formally, for benefit B and cost C both expressed in tCO₂e over deployment lifetime τ :

$$\text{Net Impact} = B(\tau) - C(\tau) > 0.$$

Example 1.2 (Net-Positive Calculation: Smallholder Irrigation Optimiser). A reinforcement-learning irrigation scheduler deployed across 5,000 smallholder farms in Zimbabwe.

Benefits: reduces water pumping by 32%, reducing diesel use by approximately 4.1 tCO₂e per farm per year. Improves yield by 14%, displacing food imports that would have generated 1.2 tCO₂e per farm per year. Total benefit: 5.3 tCO₂e per farm per year, 26,500 tCO₂e across the deployment annually.

Costs: edge inference on solar-powered sensors: 0.04 tCO₂e per farm per year. Cloud retraining and model maintenance: 80 tCO₂e for the deployment per year. Hardware embodied carbon, amortised: 0.08 tCO₂e per farm per year. Total cost: 680 tCO₂e across the deployment annually.

Net impact: $26,500 - 680 = 25,820$ tCO₂e avoided per year. Net-positive ratio: $39\times$.

The example illustrates a recurring pattern. AI deployments that target high-leverage physical processes — water, fertiliser, electricity, fuel — typically achieve net-positive ratios in the tens or

hundreds. AI deployments that target low-leverage information processes — recommendation engines, content moderation, marketing optimisation — often have negative or marginal net impact. The choice of application matters more than the sophistication of the algorithm.

1.3 Structure of the Book

This book is organised into ten parts and sixty chapters that together provide a comprehensive treatment of AI for sustainability with a strong emphasis on the carbon value chain.

- **Part I — Foundations:** establishes the conceptual framework, the carbon value chain, the AI landscape, the data infrastructure required, and the metrics that connect AI outputs to sustainability outcomes.
- **Part II — Precision Agriculture:** covers remote sensing for cropland, yield prediction, soil health monitoring, AI-driven irrigation, pest and disease detection, livestock intelligence, and the special problem of smallholder agriculture.
- **Part III — Climate Forecasting:** addresses physical climate models, ML-augmented weather forecasting, extreme event prediction, drought and flood early warning, seasonal forecasting, climate downscaling, and climate risk in finance.
- **Part IV — Energy Efficiency and the Smart Grid:** covers electricity demand forecasting, renewable integration, smart-grid operations, building energy management, industrial efficiency, storage optimisation, and electric transport.
- **Part V — Carbon Markets and Measurement:** treats carbon accounting standards, MRV systems, voluntary and compliance markets, nature-based solutions, blue carbon, and direct air capture.
- **Part VI — Circular Economy:** examines AI in supply chains, waste and recycling, water management, and materials design.
- **Part VII — Biodiversity and Ecosystems:** covers species monitoring, forest intelligence, marine systems, wildlife conservation, and protected area management.
- **Part VIII — ESG, Sustainable Finance, and Disclosure:** addresses ESG data infrastructure, green finance, climate disclosure standards, supply chain due diligence, and just transition.
- **Part IX — Governance, Ethics, and Regulation:** treats AI ethics in the climate context, data governance, the carbon footprint of AI itself, regulatory frameworks, and the integration of indigenous knowledge.
- **Part X — African Case Studies and the Future:** synthesises material with detailed treatments of African food systems, energy transition, carbon markets, capacity building, and a forward-looking outlook for the next two decades.

1.4 Who Should Read This Book

This book is written for three overlapping audiences. The first is the AI engineer or data scientist who wants to apply machine learning skills to environmental problems and needs both the technical recipes and the domain knowledge to do so credibly. The second is the sustainability practitioner — agronomist, climate scientist, energy analyst, ESG officer, conservationist — who needs to understand what current AI can and cannot do for their work. The third is the policy-maker, executive, or investor

who must make decisions about technology, capital, and regulation under uncertainty.

Each chapter is structured to serve all three audiences. Mathematical foundations and code-level detail are present for the technical reader. Case studies and deployment context are present for the practitioner. Strategic implications and risks are present for the decision-maker. Where chapters address topics still under active research, the limits of current knowledge are stated explicitly rather than glossed over.

1.5 What This Book Does Not Cover

Three categories of material are deliberately outside scope. The first is geoengineering at planetary scale — stratospheric aerosol injection, marine cloud brightening, ocean iron fertilisation — which raises governance questions far beyond the technical AI questions this book addresses. The second is the use of AI in environmental warfare or surveillance, which is a separate and serious topic deserving its own treatment. The third is exhaustive coverage of AI methods for their own sake; we treat algorithms in depth where they matter for sustainability outcomes, but readers seeking comprehensive ML reference works will find better resources elsewhere.

The next chapter introduces the carbon value chain — the physical, economic, and informational pathways through which carbon enters and leaves the atmosphere — which provides the organising framework for everything that follows.

1.6 Mathematical Formalisation of Net-Positive AI

A formal treatment of the net-positive criterion supports rigorous deployment decisions. Let the AI deployment have lifetime τ , and define the *instantaneous benefit rate* $b(t)$ and *cost rate* $c(t)$, both expressed in tCO₂e per unit time.

Definition 1.3 (Net Cumulative Impact).

$$\mathcal{N}(\tau) = \int_0^\tau [b(t) - c(t)] dt - C_{\text{embodied}} - C_{\text{decommission}}$$

where C_{embodied} is the embedded carbon of training, hardware manufacture, and initial deployment, and $C_{\text{decommission}}$ accounts for end-of-life processing.

The break-even time τ^* satisfies $\mathcal{N}(\tau^*) = 0$:

$$\int_0^{\tau^*} [b(t) - c(t)] dt = C_{\text{embodied}} + C_{\text{decommission}}.$$

For deployments with constant rates, $\tau^* = (C_{\text{embodied}} + C_{\text{decommission}})/(b - c)$.

1.6.1 Discounted Net Impact

Where intertemporal preference matters — particularly in long-lived deployments competing for capital — a discount rate ρ enters:

$$\mathcal{N}_\rho(\tau) = \int_0^\tau e^{-\rho t} [b(t) - c(t)] dt - C_{\text{embodied}} - e^{-\rho \tau} C_{\text{decommission}}.$$

Climate economics debates whether ρ should be the social discount rate ($\sim 1.4\%$ following Stern) or the market rate ($\sim 7\%$ following Nordhaus); the choice is consequential for long-horizon comparisons.

1.6.2 The Net-Positive Ratio

The dimensionless net-positive ratio R summarises the deployment leverage:

$$R = \frac{\int_0^\tau b(t) dt}{C_{\text{embodied}} + \int_0^\tau c(t) dt}.$$

$R > 1$ implies net-positive lifetime impact; rule-of-thumb thresholds for publication-grade claims require $R \geq 5$ to absorb modelling uncertainty.

1.7 Worked Problems and Solutions

Problem 1.4 (Break-Even Analysis). A precision-irrigation deployment has constant benefit rate $b = 5.3$ tCO₂e per farm per year and cost rate $c = 0.12$ tCO₂e per farm per year. Embodied carbon (training, hardware manufacture, sensor deployment) is $C_{\text{embodied}} = 2.8$ tCO₂e per farm. Find the break-even time τ^* and the net-positive ratio at $\tau = 8$ years.

Solution. Break-even time:

$$\tau^* = \frac{C_{\text{embodied}}}{b - c} = \frac{2.8}{5.3 - 0.12} = \frac{2.8}{5.18} \approx 0.541 \text{ years} \approx 6.5 \text{ months}.$$

Net-positive ratio at $\tau = 8$ years:

$$R = \frac{8 \cdot 5.3}{2.8 + 8 \cdot 0.12} = \frac{42.4}{3.76} \approx 11.3.$$

Both metrics confirm strong net-positive performance. ■

Problem 1.5 (Sensitivity to Embodied Carbon). Suppose model retraining occurs every two years with embodied cost of 0.8 tCO₂e per retraining cycle. Compute R over an 8-year deployment.

Solution. Total embodied cost across the lifetime: $C_{\text{embodied}} = 2.8 + 4 \cdot 0.8 = 6.0$ tCO₂e per farm (initial plus four biennial retrains, inclusive of $t = 0$). Operational cost: $8 \cdot 0.12 = 0.96$.

$$R = \frac{8 \cdot 5.3}{6.0 + 0.96} = \frac{42.4}{6.96} \approx 6.09.$$

Retraining halves the ratio but the deployment remains comfortably net-positive. ■

Problem 1.6 (Discounted Comparison). Two deployments: A produces 4 tCO₂e per year for 20 years with embodied cost 5 tCO₂e. B produces 10 tCO₂e per year for 5 years with embodied cost 5 tCO₂e. Which has higher discounted net impact at $\rho = 0.03$?

Solution. For constant benefit b over $[0, \tau]$, the discounted integral is $b(1 - e^{-\rho\tau})/\rho$.

Deployment A: $\frac{4(1 - e^{-0.03 \cdot 20})}{0.03} - 5 = \frac{4(1 - 0.5488)}{0.03} - 5 = \frac{4 \cdot 0.4512}{0.03} - 5 = 60.16 - 5 = 55.16$ tCO₂e.

Deployment B: $\frac{10(1 - e^{-0.03 \cdot 5})}{0.03} - 5 = \frac{10(1 - 0.8607)}{0.03} - 5 = \frac{10 \cdot 0.1393}{0.03} - 5 = 46.43 - 5 = 41.43$ tCO₂e.

Deployment A has higher discounted net impact (55.16 vs 41.43 tCO₂e), even though B has higher annual rate, because A's longer horizon outweighs B's higher rate at this discount level. ■

1.8 Chapter Conclusion

This opening chapter has laid three foundation stones for everything that follows. First, it framed the convergence of artificial intelligence and sustainability as the defining technological development of the present generation — a convergence motivated by the simultaneous urgency of the climate, food, and energy crises and made possible by the recent maturation of machine learning at planetary scale. Second, it offered the net-positive criterion as the discipline's central design constraint: every deployment must demonstrate that lifecycle environmental benefits exceed lifecycle environmental costs, formalised through the integral $\mathcal{N}(\tau) = \int_0^\tau [b(t) - c(t)] dt - C_{\text{embodied}}$. Third, it identified the structural failure modes — transfer of models across ecological zones, optimisation of low-leverage information processes, and neglect of the carbon cost of AI itself — that distinguish careful practitioners from the merely enthusiastic.

The remaining fifty-nine chapters operationalise this framing. Part I completes the foundations by walking through the carbon value chain (Chapter 2), the AI method landscape (Chapter 3), the data infrastructure that makes sustainability AI possible (Chapter 4), and the metrics that connect model outputs to physical outcomes (Chapter 5). Each subsequent part follows a domain — agriculture, climate, energy, carbon markets, circularity, biodiversity, finance, governance, and African deployment — and applies the same lens of leverage, locality, and lifecycle accounting.

If the reader takes only one idea forward from this chapter, it is this: the choice of where to apply AI matters far more than which algorithm is chosen. A simple model targeting irrigation water in a smallholder system can avoid hundreds of tonnes of CO₂e; a sophisticated model optimising click-through rates avoids nothing. The chapters that follow catalogue, with mathematical and operational specificity, where AI's leverage on sustainability is genuinely large. We turn next to the carbon value chain itself.

The Carbon Value Chain

Every kilogram of carbon dioxide in the atmosphere has a history. Trace that history backwards and you find the seven leverage points where AI can intervene.

Kennedy Chengeta, PhD

The phrase *carbon value chain* captures the full lifecycle of carbon as an economic and physical asset: from extraction or biological capture, through transformation and use, to release or storage. Understanding this chain is essential because AI's most significant sustainability impact is achieved by intervening at specific links rather than attacking the chain as a whole. This chapter maps the chain end to end and identifies the seven AI leverage points that organise the rest of the book.

2.1 From Atmosphere to Atmosphere: The Full Cycle

Carbon enters the active economic system through two pathways. The first is fossil extraction — coal, oil, and gas pulled from geological storage where they have been sequestered for millions of years. The second is biological capture — photosynthesis converting atmospheric CO₂ into plant biomass, which then enters food, fibre, timber, and biofuel value chains. Once captured, carbon flows through transformation processes (combustion, fermentation, refining, manufacturing), use phases (heating, transport, food consumption, durable goods), and end-of-life pathways (decomposition, recycling, landfill, incineration). The carbon eventually returns to the atmosphere or, less commonly, to long-term storage.

Stage	Physical processes	AI leverage points
Extraction / capture	Mining, drilling, photosynthesis, ocean uptake	Crop yield prediction; reforestation siting; DAC optimisation
Conversion	Refining, generation, fermentation, smelting	Process optimisation; predictive maintenance; renewable forecasting
Distribution	Pipelines, grid, supply chains, food logistics	Demand forecasting; route optimisation; loss detection
End use	Buildings, transport, industry, agriculture, food	HVAC control; EV routing; precision farming; behaviour nudges
Disposal	Combustion, landfill, decomposition, recycling	Waste sorting; methane leak detection; circular design
Removal	Forests, soils, oceans, DAC, mineralisation	MRV; biomass mapping; soil carbon estimation
Markets	Pricing, trading, offsets, disclosure	Verification AI; price prediction; fraud detection

2.2 Quantifying the Stocks and Flows

Climate is fundamentally a stock-and-flow problem. The stock is atmospheric carbon, currently around 880 Gt as CO₂. The annual flows are emissions (approximately 41 Gt CO₂ from human activity) and removals (approximately 22 Gt absorbed by oceans and land sinks). The net addition of around 19 Gt per year is what drives concentration upwards and warming with it.

Definition 2.1 (Carbon Budget). The remaining carbon budget for a given temperature target is

$$B_{1.5} = \frac{\Delta T_{\max} - \Delta T_{\text{current}}}{\text{TCRE}}$$

where $\Delta T_{\max}$ is the maximum allowable warming, $\Delta T_{\text{current}}$ is warming to date, and TCRE is the transient climate response to cumulative emissions (approximately 0.45°C per 1,000 Gt CO₂).

Example 2.2 (Remaining 1.5°C Budget). With $\Delta T_{\max} = 1.5^\circ\text{C}$, $\Delta T_{\text{current}} \approx 1.2^\circ\text{C}$, and $\text{TCRE} = 0.45^\circ\text{C}/1000\text{Gt}$:

$$B_{1.5} = \frac{1.5 - 1.2}{0.45} \times 1000 \approx 670\text{GtCO}_2.$$

At current emission rates of 41 Gt per year, this is about 16 years of budget. The implication for AI: every year of delay raises the optimisation difficulty exponentially, because emissions cuts must be steeper later if not made now.

2.3 The Seven AI Leverage Points

Across the carbon value chain, seven leverage points consistently produce high-impact AI deployments. Each point is treated in depth in subsequent chapters.

Leverage 1: Forecasting. Weather, climate, electricity demand, crop yields, water availability. Better forecasts at all timescales — from minutes-ahead grid balancing to decadal climate projections —

enable better physical-system management. Improvements of even one or two percentage points in forecast skill translate directly to emission reductions.

Leverage 2: Optimisation under uncertainty. Irrigation scheduling, battery dispatch, HVAC control, fleet routing, planting decisions. AI excels at making real-time decisions in high-dimensional, stochastic environments. Reinforcement learning, model predictive control, and stochastic optimisation are the workhorse techniques.

Leverage 3: Sensing and monitoring. Satellite imagery, ground sensors, acoustic monitors, drones, IoT. AI compresses vast raw data streams into actionable signals — deforestation alerts, methane leak locations, crop disease outbreaks, illegal fishing.

Leverage 4: Verification and MRV. Carbon accounting, supply chain due diligence, ESG disclosure, emissions audits. AI enables continuous, large-scale verification at a cost that human-only methods cannot match.

Leverage 5: Discovery and design. Materials, catalysts, plant breeds, climate-resilient infrastructure. Generative models, simulation, and active learning are accelerating R&D cycles by orders of magnitude in some domains.

Leverage 6: Personalisation and scale. Smallholder agronomic advice, energy efficiency recommendations, transport mode selection. AI delivers personalised guidance at population scale where human advisors cannot.

Leverage 7: Aggregation and synthesis. Climate literature, regulatory texts, scientific evidence, citizen reports. Large language models compress expert knowledge into accessible interfaces for practitioners and decision-makers.

2.4 Where AI Has Marginal or Negative Impact

Symmetry requires noting where AI in the carbon value chain underperforms or actively harms. Three categories deserve flagging.

Substitution effects. A perfect e-commerce recommendation engine that drives more consumption may increase emissions even as it optimises individual deliveries. Efficiency gains that lower the price of energy or transport increase aggregate demand — the Jevons paradox.

Lock-in effects. AI deployed to optimise existing fossil infrastructure (oil exploration, coal-plant maintenance) extends the life of high-emission systems and delays transition. The same algorithms that detect methane leaks in active wells also help operate those wells more efficiently and longer.

Embodied costs. Large-scale AI training and inference consume electricity and embodied hardware carbon. Without discipline about model size, retraining frequency, and inference caching, the AI itself becomes a meaningful contributor to emissions.

2.5 The African Carbon Value Chain

Sub-Saharan Africa contributes roughly 4% of global CO₂ emissions but holds approximately 17% of global biological carbon stocks in forests, soils, and peatlands. The Congo Basin alone stores around 70 Gt of carbon, making it second only to the Amazon globally. The continent's carbon value chain is therefore dominated by removal and storage rather than emission, and the AI leverage points that matter most are those that enable accurate measurement of these stocks, payments to communities that protect them, and prevention of degradation. Subsequent chapters in Part V (carbon markets) and Part VII (biodiversity) treat these themes in depth.

The next chapter surveys the AI techniques that this book will draw upon, providing a reference for readers from non-technical backgrounds.

2.6 Mathematical Foundations of Carbon Stocks and Flows

The atmospheric stock of CO₂ obeys the conservation law

$$\frac{dM_a}{dt} = E(t) - U_o(t) - U_l(t),$$

where M_a is atmospheric carbon mass, E is anthropogenic emission rate, U_o is ocean uptake, and U_l is land-sink uptake, each in Gt CO₂ yr⁻¹.

The airborne fraction ϕ is the empirical ratio

$$\phi = \frac{dM_a/dt}{E},$$

historically near 0.45 and showing modest temporal trends. Concentration C in ppm relates to M_a through $C \approx M_a/7.81$ Gt CO₂ per ppm.

2.6.1 Transient Climate Response to Cumulative Emissions

Definition 2.3 (TCRE). The transient climate response to cumulative emissions is

$$\text{TCRE} = \frac{\Delta T}{E_{\text{cum}}},$$

where $E_{\text{cum}} = \int_0^t E(s) ds$ and ΔT is observed warming. Empirically, $\text{TCRE} \approx 0.45 \pm 0.18^\circ\text{C}$ per 1,000 Gt CO₂.

2.6.2 Pulse-Response Approximation

The atmospheric response to an emission pulse δE at $t = 0$ is

$$\Delta C(t) = \frac{\delta E}{7.81} \left[a_0 + \sum_{i=1}^3 a_i \exp(-t/\tau_i) \right],$$

with parameters $a_0 = 0.217$, $(a_1, a_2, a_3) = (0.224, 0.282, 0.276)$, $(\tau_1, \tau_2, \tau_3) = (394, 36.5, 4.30)$ years (Joos et al., 2013). The persistent component $a_0 \approx 22\%$ remains airborne for thousands of years, motivating cumulative-emission framings of the climate problem.

2.7 Worked Problems and Solutions

Problem 2.4 (Remaining Carbon Budget). Compute the remaining carbon budget for 1.7°C given current warming of 1.2°C and central TCRE of 0.45°C per 1,000 Gt CO₂. At current emissions of 41 Gt yr⁻¹, how many years remain?

Solution.

$$B_{1.7} = \frac{1.7 - 1.2}{0.45} \cdot 1000 = \frac{0.5}{0.45} \cdot 1000 \approx 1111 \text{ Gt CO}_2.$$

At 41 Gt yr⁻¹: 1111/41 ≈ 27.1 years. With 1σ TCRE uncertainty of ±0.18, the budget ranges from ≈ 794 Gt (high TCRE, ~ 19 yr) to ≈ 1852 Gt (low TCRE, ~ 45 yr). ■

Problem 2.5 (Pulse Response After 100 Years). Compute the fraction of a 1 Gt CO₂ pulse remaining airborne 100 years after emission using the Joos parameters above.

Solution.

$$f(100) = a_0 + a_1 e^{-100/394} + a_2 e^{-100/36.5} + a_3 e^{-100/4.30}.$$

Term-by-term:

- $a_0 = 0.217$
- $a_1 e^{-100/394} = 0.224 \cdot e^{-0.2538} = 0.224 \cdot 0.7758 = 0.1738$
- $a_2 e^{-100/36.5} = 0.282 \cdot e^{-2.740} = 0.282 \cdot 0.0645 = 0.01818$
- $a_3 e^{-100/4.30} = 0.276 \cdot e^{-23.26} \approx 0$

Total $f(100) \approx 0.409$, i.e. approximately 41% of the pulse remains airborne after a century. ■

Problem 2.6 (Airborne Fraction Implied Sink). If anthropogenic emissions are 41 Gt CO₂ yr⁻¹ and the atmospheric concentration rises at 2.4 ppm yr⁻¹, compute the implied combined sink and the airborne fraction.

Solution. Atmospheric increase: $dM_a/dt = 2.4 \cdot 7.81 = 18.74 \text{ Gt CO}_2 \text{ yr}^{-1}$. Combined sink: $E - dM_a/dt = 41 - 18.74 = 22.26 \text{ Gt yr}^{-1}$. Airborne fraction: $\phi = 18.74/41 \approx 0.457$. This matches the canonical ~ 0.45 value within rounding. ■

2.8 Chapter Conclusion

The carbon value chain is the conceptual scaffolding on which the rest of the book is built. By tracking carbon from atmospheric source through productive transformation in soils and vegetation, through industrial emission and combustion, and back to potential removal in geological storage or restored ecosystems, the chain identifies the stages where better information — the commodity AI produces in greatest quantity — generates the largest sustainability return. The seven leverage points catalogued in this chapter (forecasting renewable supply, optimising electrified demand, monitoring carbon stocks, verifying credits, detecting methane leaks, designing circular materials, and discovering low-carbon alternatives) are not arbitrary; they share the property that small percentage gains translate to gigatonne-scale physical impact at global rollout.

The chapter also explicitly named where AI has marginal or negative impact, an omission rare enough

in the literature to deserve repetition. Recommendation engines, advertising-optimisation systems, and content moderation pipelines do not move the needle on climate even when their operators publish sustainability reports. Honest scoping is the first intellectual discipline of the field.

The African carbon value chain section foreshadowed Part X. African emissions are low in aggregate but rising; sinks are large but under-measured; capacity is unevenly distributed. The general framework applies, but its operationalisation differs — data is sparser, ground truth more expensive to collect, and governance constraints both binding and asymmetric in their effects across stakeholders. These themes recur throughout the book.

The mathematical machinery introduced — conservation laws for atmospheric CO₂, the Joos pulse-response approximation, and the transient climate response to cumulative emissions — gives the reader the tools to evaluate any claimed climate impact independently. We now turn to the artificial intelligence side of the equation: the methods that operate on the data the carbon value chain generates.

The AI Landscape for Sustainability

No single algorithm solves a sustainability problem. The art is choosing the right tool for the right physical process.

Kennedy Chengeta, PhD

This chapter surveys the AI techniques that recur throughout the book. It is not a comprehensive machine learning textbook — many excellent ones exist — but a guide to the methods most often deployed in sustainability applications and the criteria by which one is preferred over another. Readers familiar with deep learning may skim; readers from sustainability backgrounds will find this chapter a reference for terminology used elsewhere.

3.1 The Five Method Families

Sustainability AI draws on five method families. Each has natural domains where it dominates and others where it underperforms.

Tabular and structured learning (gradient-boosted trees, random forests) dominates problems with finite, well-understood features — yield prediction from soil, climate, and management data; default risk scoring for green loans; drought severity classification. XGBoost and LightGBM are the workhorses; they are interpretable, fast, and require less data than deep models.

Convolutional neural networks dominate image and spatial data — satellite imagery for land cover, drone imagery for crop disease, aerial imagery for forest biomass. Foundational architectures (ResNet, U-Net, Vision Transformers) plus domain-specific pretrained models (Prithvi, SatMAE) are the entry points.

Recurrent and transformer architectures dominate sequential data — weather forecasting, electricity demand, water flow, animal vocalisations. LSTMs remain useful for shorter sequences with limited compute. Transformers (Informer, Temporal Fusion Transformer, foundation models like ClimaX) dominate longer sequences and multivariate forecasting where compute is available.

Reinforcement learning dominates closed-loop control problems — irrigation scheduling, HVAC management, battery dispatch, traffic signal control. Policy-gradient methods (PPO) and value-based methods (DQN variants) dominate single-agent problems; multi-agent extensions handle grid and supply chain coordination.

Generative models dominate discovery and synthesis — climate scenario generation, synthetic

agricultural data for rare events, molecular design for catalysts and materials. Variational autoencoders, GANs, and diffusion models each have niches; conditional diffusion has become dominant for spatially structured generation.

Family	Best for	Sustainability example
Gradient boosting	Tabular, mixed types	Yield prediction; ESG scoring
CNN / ViT	Spatial imagery	Deforestation, crop monitoring
RNN / Transformer	Time series, language	Weather, demand, climate disclosure NLP
Reinforcement learning	Sequential decisions	Irrigation, building control
Generative	Synthesis, simulation	Climate scenarios, materials design

3.2 Foundation Models for Earth Systems

The past three years have seen the rise of foundation models trained specifically on earth-observation and climate data. NASA-IBM’s Prithvi model, Google’s Gemini-Earth, and ECMWF’s GraphCast are large pretrained models that practitioners fine-tune on downstream tasks rather than training from scratch. The shift is consequential for sustainability AI: previously, every project required collecting and labelling data sufficient to train a model; now, foundation models often achieve state-of-the-art performance with hundreds rather than millions of labelled examples.

Example 3.1 (Foundation Model Fine-Tuning). A team monitoring crop residue burning in Punjab fine-tuned the Prithvi-100M Earth foundation model on 1,200 labelled Sentinel-2 tiles. Training took 4 GPU-hours rather than the 4,000 GPU-hours that training from scratch would have required, and achieved $F_1 = 0.91$ on held-out tiles — exceeding the previous best of $F_1 = 0.84$ from a U-Net trained on 8,000 labels.

3.3 Loss Functions That Encode Sustainability Goals

The choice of loss function determines what the model optimises and consequently what sustainability outcome is achieved. A model trained on mean squared error optimises for average performance and may systematically under-predict extreme events — exactly the events climate adaptation must prepare for.

Definition 3.2 (Quantile Loss). The pinball loss for quantile τ is

$$\mathcal{L}_\tau(y, \hat{y}) = \max(\tau(y - \hat{y}), (\tau - 1)(y - \hat{y})).$$

Training on the 95th percentile loss biases the model towards capturing extreme events at the cost of average accuracy.

For sustainability work, three loss-function patterns recur. Quantile losses for risk-sensitive problems (extreme weather, flood crests, peak electricity demand). Cost-sensitive losses for asymmetric error consequences (false-negative deforestation alerts cost more than false positives). Multi-task losses that combine prediction accuracy with fairness or carbon-cost terms during training.

3.4 Uncertainty Quantification

A point prediction without uncertainty is rarely actionable in sustainability work. A yield forecast of 4.2 t/ha is operationally different from a 90% confidence interval of 3.8 to 4.6 versus 2.1 to 6.3. The former supports a confident planting decision; the latter signals the need for risk mitigation. Three methods dominate.

Conformal prediction produces distribution-free, finite-sample-valid prediction intervals from any base model. It is method-agnostic and increasingly the default for production deployments where calibration matters.

Bayesian neural networks place priors over weights and produce posterior predictive distributions. They are computationally expensive but provide interpretable uncertainty decomposition.

Deep ensembles train multiple models with different random seeds and aggregate predictions. Simpler than full Bayesian methods, often near-equivalent in calibration, and trivially parallelisable.

3.5 Interpretability and Explainability

Sustainability AI must often satisfy regulators, communities, and scientific peers. Black-box models that cannot explain their decisions face deployment barriers in carbon markets, ESG ratings, and agricultural extension. SHAP (Shapley values), permutation importance, and partial dependence plots are standard for tabular models. Grad-CAM and attention visualisation serve similar purposes for deep models. Counterfactual explanations — "this farm received a drought warning because the NDVI dropped below 0.32; if it had stayed above 0.4 the warning would not have triggered" — are increasingly preferred because they suggest actionable interventions.

3.6 What This Book Assumes and What It Defers

This book assumes basic familiarity with supervised learning, gradient descent, and standard neural network architectures. It defers detailed treatment of training dynamics, optimisation theory, and the latest large language model architectures to specialist references. The mathematical presentations focus on the formulations that matter for sustainability outcomes; readers seeking exhaustive ML theory will find better resources elsewhere.

The next chapter addresses the data infrastructure that makes any of these methods deployable in practice — often the binding constraint in real-world sustainability AI.

3.7 Mathematical Foundations of the Method Families

3.7.1 Gradient Boosting

A gradient-boosted ensemble fits an additive model

$$F_M(\mathbf{x}) = \sum_{m=1}^M v h_m(\mathbf{x}),$$

where h_m are weak learners (typically depth-limited trees), v is the shrinkage rate, and each h_m is fit to the negative gradient of the loss $\mathcal{L}$ with respect to F_{m-1} :

$$h_m \in \arg \min_h \sum_i [r_{i,m} - h(\mathbf{x}_i)]^2, \quad r_{i,m} = - \left. \frac{\partial \mathcal{L}(y_i, F)}{\partial F} \right|_{F=F_{m-1}(\mathbf{x}_i)}.$$

3.7.2 Convolutional Operations

For input feature map $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ and kernel $\mathbf{W} \in \mathbb{R}^{k \times k \times C \times C'}$, the convolution output is

$$y_{ijc'} = \sigma \left(\sum_{u=1}^k \sum_{v=1}^k \sum_{c=1}^C \mathbf{W}_{uvcc'} x_{i+u, j+v, c} + b_{c'} \right).$$

Receptive field grows with depth; with stride s and dilation d the effective receptive field after L layers is $(d \cdot s)^L \cdot k$.

3.7.3 Self-Attention

Given queries $\mathbf{Q}$, keys $\mathbf{K}$, values $\mathbf{V}$ all $\in \mathbb{R}^{n \times d}$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} \right) \mathbf{V}.$$

Computational complexity is $\mathcal{O}(n^2 d)$, motivating the linear-attention and sparse-attention variants used in long-range climate forecasting.

3.7.4 Reinforcement Learning Foundations

For a Markov decision process $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, the action-value function under policy π satisfies the Bellman equation

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [Q^\pi(s', \pi(s'))].$$

The optimal Bellman equation is

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s'} \left[\max_{a'} Q^*(s', a') \right].$$

3.7.5 Quantile Loss in Detail

For predicted quantile $\hat{q}_\tau$ at level $\tau \in (0, 1)$ and observation y :

$$\mathcal{L}_\tau(y, \hat{q}_\tau) = (y - \hat{q}_\tau) [\tau - \mathbb{I}(y < \hat{q}_\tau)] = \begin{cases} \tau(y - \hat{q}_\tau) & \text{if } y \geq \hat{q}_\tau, \\ (\tau - 1)(y - \hat{q}_\tau) & \text{if } y < \hat{q}_\tau. \end{cases}$$

Minimising $\mathbb{E}[\mathcal{L}_\tau]$ yields the conditional τ -quantile.

3.8 Worked Problems and Solutions

Problem 3.3 (Pinball Loss Computation). Given $y = 5.0$ and quantile predictions $\hat{q}_{0.1} = 3.5$, $\hat{q}_{0.5} = 4.8$, $\hat{q}_{0.9} = 6.4$, compute the pinball losses.

Solution. For each level:

- $\tau = 0.1$: $y > \hat{q}$, loss = $0.1 \cdot (5.0 - 3.5) = 0.15$.
- $\tau = 0.5$: $y > \hat{q}$, loss = $0.5 \cdot (5.0 - 4.8) = 0.10$.
- $\tau = 0.9$: $y < \hat{q}$, loss = $(0.9 - 1) \cdot (5.0 - 6.4) = (-0.1)(-1.4) = 0.14$.

Average pinball loss: $(0.15 + 0.10 + 0.14)/3 = 0.130$. ■

Problem 3.4 (Bellman Update). A grid-cell irrigation MDP has reward $r = -0.4$ for the chosen action, discount $\gamma = 0.95$, and successor state-value estimates $V(s'_1) = 12.3$, $V(s'_2) = 10.1$ with transition probabilities 0.6 and 0.4. Compute $Q(s, a)$.

Solution.

$$Q(s, a) = -0.4 + 0.95 \cdot (0.6 \cdot 12.3 + 0.4 \cdot 10.1) = -0.4 + 0.95 \cdot (7.38 + 4.04) = -0.4 + 0.95 \cdot 11.42 = -0.4 + 10.849 = 10.449$$

■

Problem 3.5 (Receptive Field). A CNN with 6 layers, kernel size $k = 3$, stride $s = 1$ throughout, and no dilation. Compute the receptive field at the output.

Solution. For unit-stride, undilated convolutions, receptive field grows additively: $\text{RF}_L = 1 + L(k - 1)$. With $L = 6$, $k = 3$: $\text{RF} = 1 + 6 \cdot 2 = 13$ pixels. ■

3.9 Chapter Conclusion

This chapter catalogued the five method families that recur in every subsequent part of the book: gradient-boosted ensembles for tabular agronomic data, convolutional networks for satellite imagery, recurrent and transformer architectures for time-series forecasting, graph neural networks for spatially structured systems, and reinforcement learning for sequential decision making under uncertainty. Each family is matched to a characteristic data modality and a characteristic class of sustainability problem; the mappings are not arbitrary, and the reader should resist the temptation to apply a fashionable architecture where a simpler family would suffice.

The chapter also introduced three cross-cutting practices that the rest of the book treats as defaults rather than optional refinements. First, uncertainty quantification through quantile regression, conformal prediction, or ensemble methods, so that decisions can be made with explicit knowledge of model confidence. Second, interpretability through SHAP, attention visualisation, or post-hoc explanation, so that domain experts can audit model behaviour. Third, conservative loss functions that discount uncertain predictions, so that errors of overestimation are penalised more severely than errors of underestimation — a habit that matters disproportionately in carbon accounting.

The foundation-model section signalled an important shift in the field. Earth-observation foundation models such as Prithvi and SatMAE, climate foundation models such as ClimaX and Pangu-Weather, and language models fine-tuned on scientific literature are reducing the data requirements for downstream applications. The reader should expect, by the time of the second edition, that fine-tuning a

foundation model for a specific sustainability task will be the default workflow rather than training from scratch.

What this chapter assumed and what it deferred is also worth restating. We assume the reader has basic familiarity with supervised learning, loss minimisation, and gradient descent; readers without this background should consult an introductory machine-learning textbook before proceeding. We defer detailed treatment of representation learning, self-supervised training, and large-scale distributed compute to the references; the material is excellent but not necessary for the practitioner. The remainder of the book operationalises the families introduced here on real sustainability problems, beginning with the data infrastructure that makes everything else possible.

Data Infrastructure for Sustainability AI

The model is downstream of the data, the data is downstream of the sensors, and the sensors are downstream of the politics.

Kennedy Chengeta, PhD

Data infrastructure is the binding constraint in most sustainability AI projects. A team can have the best algorithms, sufficient compute, and clear deployment pathways, and still fail because the data they need is fragmented, low-quality, expensive, or politically restricted. This chapter surveys the data sources that recur in this book, the pipelines that operationalise them, and the governance considerations that determine whether data can actually be used.

4.1 The Major Earth Observation Programmes

Free, open earth observation has transformed sustainability AI in the past decade. Five programmes dominate.

Copernicus Sentinel (European Space Agency) provides Sentinel-1 (C-band radar, all-weather, 5–20m resolution, 6-day revisit), Sentinel-2 (10–60m optical, 5-day revisit, 13 spectral bands), Sentinel-3 (ocean and land surface temperature), Sentinel-5P (atmospheric chemistry including methane, NO₂, SO₂). All data are free under Copernicus open licence.

NASA Landsat delivers 30m optical imagery with continuous archive since 1972, providing the longest land cover time series available. Landsat-9 launched 2021 with improved radiometric resolution.

NASA MODIS and VIIRS provide daily 250m–1km global coverage for vegetation indices, fire detection, and land surface temperature.

Commercial constellations — Planet Labs (3m daily), Maxar (30cm), Capella (radar), ICEYE (radar) — supply higher-resolution, higher-frequency imagery at cost. Pricing has fallen substantially; per-square-kilometre costs are now within reach for project-scale work.

Climate reanalysis — ERA5 (ECMWF), MERRA-2 (NASA) — provides hourly historical weather and atmospheric variables on 25–30km global grids back to 1940.

Source	Resolution	Revisit	Best for
Sentinel-1	10–20m	6 days	All-weather monitoring; floods
Sentinel-2	10m	5 days	Crop, deforestation, water
Sentinel-5P	7km	1 day	Methane, NO ₂ plumes
Landsat-8/9	30m	16 days	Long-term land cover
MODIS / VIIRS	250–1000m	Daily	Fires, NDVI, snow, SST
Planet	3m	Daily	Property scale; fast change detection
ERA5 reanalysis	~25km	Hourly	Climate baselines, training data

4.2 Ground-Truth Data: The Persistent Bottleneck

Satellite data without ground truth produces models that look impressive but generalise poorly. Ground-truth collection is expensive, slow, and politically sensitive — and yet without it, model evaluation is impossible. Three patterns work in practice.

Permanent monitoring networks (FluxNet for carbon flux, ICOS for European ecosystems, NEON for North American) provide high-quality, harmonised time series at fixed sites. Coverage is geographically biased: North America and Europe are dense, Africa and South America sparse.

Citizen science platforms (eBird, iNaturalist, Globe Observer) provide vast volumes of variably-quality observations. Useful for biodiversity monitoring and as auxiliary signal for trained models.

Crowdsourced labelling on commercial platforms (CrowdFlower, Scale, Hive) produces image labels at scale. Quality control through redundant labelling and consensus filtering is essential.

4.3 The Data Pipeline

Production sustainability AI requires reproducible data pipelines. The standard architecture has five stages.

Ingestion pulls raw data from APIs (Sentinel Hub, Google Earth Engine, AWS Open Data Registry) or downloads from FTP archives. Asynchronous, parallel, idempotent.

Validation checks for missing values, geometric corruption, cloud cover, and known sensor anomalies. Failed records are quarantined for manual review.

Pre-processing applies atmospheric correction, cloud masking, geometric registration, resampling to common grids, and unit harmonisation. For optical imagery, Sen2Cor (Sentinel-2) and ARCSI (Landsat) are standard.

Feature engineering computes derived variables — NDVI, EVI, water indices, time-series aggregates, neighbourhood statistics. These are stored separately from raw data so models can be retrained without re-ingestion.

Versioning and lineage tracks every dataset version, every transformation, and every model trained on it. DVC, LakeFS, and Pachyderm are common tools; the principle is that every prediction can be

traced to specific inputs and code.

```

1 import ee
2 ee.Initialize()
3
4 def monthly_ndvi(year, month, region):
5     start = ee.Date.fromYMD(year, month, 1)
6     end = start.advance(1, 'month')
7     s2 = (ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
8           .filterBounds(region)
9           .filterDate(start, end)
10          .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30)))
11     ndvi = s2.map(lambda img: img.normalizedDifference(['B8', 'B4'])
12                  .rename('NDVI'))
13     return ndvi.median().clip(region)

```

Listing 4.1: Earth Engine pipeline pattern for monthly NDVI

4.4 Data Governance and Sovereignty

The political dimensions of environmental data are increasingly consequential. Three issues recur.

Data sovereignty. Many African, Latin American, and Pacific nations require that data about their territories be stored within their jurisdictions. The African Union Convention on Cyber Security (Malabo Convention) and Kenya’s Data Protection Act impose these requirements with increasing enforcement.

Indigenous data sovereignty. The CARE principles (Collective benefit, Authority to control, Responsibility, Ethics) supplement the FAIR principles (Findable, Accessible, Interoperable, Reusable) for data concerning indigenous communities and territories. Operationally this means that biodiversity data, traditional knowledge, and locations of culturally significant sites are governed by the relevant communities rather than open by default.

Personal and farm-level data. Agronomic data identifying specific farms or farmers is personal data under POPIA, GDPR, and most national equivalents. Aggregation, differential privacy, and explicit consent are required for legitimate use.

4.5 Compute Infrastructure

Sustainability AI compute splits across three tiers. Cloud platforms (Google Earth Engine, AWS, Azure, Microsoft Planetary Computer) provide elastic scale for training and bulk inference. Edge devices (Raspberry Pi, NVIDIA Jetson, low-power ARM chips) run inference locally where connectivity is unreliable. On-device inference on smartphones serves smallholder advisory, citizen science, and last-mile applications.

The carbon implications of compute choice are not negligible. A Google Earth Engine analysis of cropland change for a single African country runs in minutes on already-optimised infrastructure, with

marginal carbon cost. The same analysis run as a custom job on a misconfigured cloud cluster can consume $100\times$ the energy. Chapter 53 returns to this in detail.

The next chapter introduces the metrics that connect AI outputs to verifiable sustainability outcomes — the foundation for every claim made in subsequent chapters.

4.6 Mathematical Foundations of Earth Observation Pre-Processing

4.6.1 Atmospheric Correction

Top-of-atmosphere reflectance ρ_{TOA} relates to surface reflectance ρ_s through a simplified radiative-transfer model:

$$\rho_{\text{TOA}} = \rho_a + \frac{T_{\downarrow} T_{\uparrow} \rho_s}{1 - s \rho_s},$$

where ρ_a is atmospheric path reflectance, $T_{\downarrow}$ and $T_{\uparrow}$ are downward and upward transmittances, and s is spherical albedo. Sen2Cor inverts this per-pixel using ancillary water-vapour and aerosol estimates.

4.6.2 Vegetation Index Family

Beyond NDVI, alternative indices balance sensitivity and confounders:

$$\begin{aligned} \text{NDVI} &= \frac{\rho_{\text{NIR}} - \rho_{\text{red}}}{\rho_{\text{NIR}} + \rho_{\text{red}}}, \\ \text{EVI} &= G \cdot \frac{\rho_{\text{NIR}} - \rho_{\text{red}}}{\rho_{\text{NIR}} + C_1 \rho_{\text{red}} - C_2 \rho_{\text{blue}} + L}, \\ &\quad G = 2.5, C_1 = 6, C_2 = 7.5, L = 1, \\ \text{SAVI} &= \frac{(1 + L)(\rho_{\text{NIR}} - \rho_{\text{red}})}{\rho_{\text{NIR}} + \rho_{\text{red}} + L}, \quad L \approx 0.5, \\ \text{NDRE} &= \frac{\rho_{\text{NIR}} - \rho_{\text{red-edge}}}{\rho_{\text{NIR}} + \rho_{\text{red-edge}}}. \end{aligned}$$

4.6.3 Spectral Mixture Analysis

For a pixel containing fractions f_i of n endmembers with reflectance spectra $\mathbf{e}_i$:

$$\rho_{\text{pixel}} = \sum_{i=1}^n f_i \mathbf{e}_i + \boldsymbol{\epsilon}, \quad \sum_i f_i = 1, \quad f_i \geq 0.$$

Solving for $\mathbf{f}$ given measured ρ_{pixel} and library $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_n]$ is a constrained non-negative least squares problem.

4.6.4 Cloud Probability and Compositing

For a pixel time series $\{\rho_t\}$ with cloud probability p_t , a weighted median composite is

$$\rho_{\text{composite}} = \text{median}\{\rho_t : p_t < p_{\text{thresh}}\},$$

typically with $p_{\text{thresh}} = 0.3$ for s2cloudless masking.

4.7 Worked Problems and Solutions

Problem 4.1 (NDVI from Reflectances). Compute NDVI for a Sentinel-2 pixel with $\rho_{\text{red}} = 0.082$ and $\rho_{\text{NIR}} = 0.451$.

Solution.

$$\text{NDVI} = \frac{0.451 - 0.082}{0.451 + 0.082} = \frac{0.369}{0.533} \approx 0.692.$$

This is consistent with healthy vegetation in mid-growth stage. ■

Problem 4.2 (EVI Versus NDVI). For the same pixel with $\rho_{\text{blue}} = 0.038$, compute EVI and compare with NDVI from the previous problem.

Solution.

$$\text{EVI} = 2.5 \cdot \frac{0.451 - 0.082}{0.451 + 6 \cdot 0.082 - 7.5 \cdot 0.038 + 1} = 2.5 \cdot \frac{0.369}{0.451 + 0.492 - 0.285 + 1} = 2.5 \cdot \frac{0.369}{1.658} \approx 0.557.$$

EVI (0.557) is lower than NDVI (0.692) because the blue-band correction reduces atmospheric enhancement. EVI is preferred over dense canopies where NDVI saturates. ■

Problem 4.3 (Atmospheric Correction). Given $\rho_{\text{TOA}} = 0.144$, $\rho_a = 0.040$, $T_{\downarrow}T_{\uparrow} = 0.78$, spherical albedo $s = 0.05$, solve for surface reflectance ρ_s .

Solution. Rearranging the simplified RT equation:

$$\rho_s = \frac{\rho_{\text{TOA}} - \rho_a}{T_{\downarrow}T_{\uparrow} + s(\rho_{\text{TOA}} - \rho_a)} = \frac{0.144 - 0.040}{0.78 + 0.05 \cdot 0.104} = \frac{0.104}{0.7852} \approx 0.1325.$$

■

4.8 Chapter Conclusion

The data infrastructure described in this chapter is the unsung hero of every successful sustainability AI deployment. The Sentinel, Landsat, and MODIS programmes together provide the equivalent of a continuous, free, multi-spectral video of the planet at resolutions from 10 m to 250 m on revisit cadences of one to sixteen days. ERA5 and CHIRPS extend that coverage to atmospheric and precipitation variables. Cloud platforms (Google Earth Engine, Microsoft Planetary Computer, AWS Open Data) reduce what was once a several-month data-engineering effort to a few hours of JavaScript or Python. Anyone working in the field today operates on infrastructure that did not exist a decade ago.

What does require deliberate attention is the ground-truth layer. The chapter argued, and the reader should internalise, that ground truth is the persistent bottleneck. Satellite revisit times are short; ground campaigns are slow, expensive, and prone to selection bias. The most common failure mode is to apply a model trained on Northern Hemisphere ground truth to a Southern Hemisphere context, with predictably poor performance. Investing in continental and national ground-truth networks, such as the African Soil Information Service and the Africa Flux network, is a high-leverage activity that scientific funders underprovide.

The chapter introduced the operational pre-processing steps — cloud masking, atmospheric correction, geometric registration, harmonic smoothing, spectral indexing — that mediate between raw observations and model inputs. Each step propagates error, and each can introduce artefacts that compromise downstream inference. The mathematical treatment of atmospheric correction and vegetation indices gave the reader the tools to audit a pipeline rather than trust it blindly.

Data sovereignty surfaced as a recurring tension. Cloud platforms based in the Global North reduce technical barriers but raise governance concerns about where data is processed and who controls it. The chapter referenced both the FAIR and CARE principles; subsequent chapters, particularly Chapter 52, return to these themes. We move next from infrastructure to outcomes: the metrics that turn AI outputs into sustainability conclusions.

Sustainability Metrics: Connecting Models to Outcomes

An AI system that improves a metric without improving the underlying physical reality is worse than no AI at all — because it generates false confidence.

Kennedy Chengeta, PhD

Metrics are how we know whether sustainability AI is working. This chapter surveys the major frameworks — SDGs, GHG Protocol, TCFD, GRI, CSRD — and the technical metrics that connect AI model performance to the sustainability outcomes those frameworks measure. The aim is not to recite acronyms but to provide the conceptual scaffolding that prevents one of the most common failure modes in this field: optimising a model metric while the underlying physical reality unchanged or worsens.

5.1 Frameworks: SDGs, GHG Protocol, TCFD, GRI, CSRD, ISSB

The **Sustainable Development Goals** (SDGs), adopted by all UN member states in 2015, define seventeen goals and 169 targets running through 2030. Of direct relevance to this book: SDG 2 (zero hunger), SDG 6 (clean water), SDG 7 (affordable clean energy), SDG 11 (sustainable cities), SDG 12 (responsible consumption), SDG 13 (climate action), SDG 14 (life below water), SDG 15 (life on land).

The **GHG Protocol** categorises emissions into Scope 1 (direct emissions from owned sources), Scope 2 (indirect emissions from purchased electricity), and Scope 3 (all other indirect emissions including supply chain, product use, and end-of-life). Most corporate sustainability AI focuses on Scope 3 because it is the largest, hardest to measure, and most amenable to data-driven approaches.

TCFD (Task Force on Climate-related Financial Disclosures) structures climate disclosure around governance, strategy, risk management, and metrics. It has been absorbed into the IFRS Sustainability Standards (S1 and S2) under the **ISSB**, which became operational in 2024.

GRI (Global Reporting Initiative) and **CSRD** (the EU Corporate Sustainability Reporting Directive, applicable from FY2024 reporting) prescribe extensive disclosure requirements with particular emphasis on the double materiality concept — assessing both how sustainability issues affect a company and how the company affects sustainability outcomes.

5.2 Technical Metrics: From Model Performance to Physical Outcome

The chain from model accuracy to physical outcome has three links, each of which can break.

Link 1: Model performance metrics. Standard ML metrics (accuracy, F_1 , AUC, RMSE, MAE) measure how well predictions match ground truth.

Link 2: Operational metrics. How is the model actually used? Forecast adoption rates, alert response rates, recommendation acceptance rates, false alarm fatigue.

Link 3: Physical metrics. What changed in the world? Tonnes of CO₂ avoided, hectares of forest preserved, kilowatt-hours saved, kilograms of food produced, lives saved.

Example 5.1 (The Three-Link Chain in Action). A drought early warning system in Kenya has model performance $F_1 = 0.78$ on monthly drought severity classification (Link 1). Of the alerts issued, 64% reach farmers via SMS within 48 hours; of those who receive alerts, 41% take recommended actions (Link 2). The actions taken — planting drought-tolerant varieties, adjusting fertiliser, reducing animal stocking — save an estimated 0.7 tonnes of yield per hectare in drought years (Link 3). The same system has been re-evaluated under three deployment redesigns; the underlying F_1 barely changed but Link 3 outcomes varied by a factor of three depending on Link 2 design.

5.3 Carbon Accounting: The GHG Protocol in Detail

For any AI deployment claiming carbon reduction, the GHG Protocol provides the accounting framework.

Definition 5.2 (Activity-Based Emissions). For activity a (e.g. litres of diesel consumed) and emission factor EF_a (kg CO₂e per unit of a):

$$E_a = \text{Activity}_a \times EF_a.$$

For a portfolio of activities, total emissions are $E = \sum_a E_a$. Reductions are $\Delta E = E_{\text{baseline}} - E_{\text{post-intervention}}$.

The accounting is more subtle than this formula suggests. Three issues recur.

Counterfactual baselines. The reduction ΔE depends on what would have happened without the intervention. Static baselines (last year's emissions) underestimate reductions in growth scenarios and overestimate them in decline scenarios. Dynamic baselines (counterfactual modelled forward) are more accurate but introduce model dependence.

Boundary effects. Reductions inside the system boundary may be offset by leakage outside it. A precision agriculture system that reduces fertiliser use on the optimised farms may cause neighbouring farms to increase use, partially negating the gain.

Permanence. A tonne of carbon stored in soil for one year is not equivalent to a tonne stored for 100

years. Permanence-adjusted accounting applies discount factors based on storage duration and risk of reversal.

5.4 Co-Benefits and Trade-Offs

A robust sustainability metric framework tracks co-benefits and trade-offs explicitly. An afforestation project may sequester carbon (positive) while reducing local water availability (negative), increasing biodiversity (positive), and displacing pastoral livelihoods (negative). A smart grid intervention may reduce emissions (positive) while increasing system complexity that makes resilience harder to maintain (potentially negative).

Multi-criteria decision analysis (MCDA) is the standard analytical framework. Common approaches include weighted sum scoring, analytic hierarchy process (AHP), and outranking methods (PROMETHEE, ELECTRE). For AI-enabled decisions where weights themselves are uncertain, robust optimisation and Pareto-frontier analysis are preferred.

5.5 Avoiding Greenwashing

The growth of sustainability AI has been accompanied by growth in greenwashing — claims of sustainability impact unsupported by evidence. Three patterns are particularly common.

Cherry-picked metrics. A model achieves 95% accuracy on a benign subset and is reported as a 95% accurate system, while overall accuracy and accuracy on hard cases are far lower.

Lifecycle truncation. Reporting only Scope 1 emissions while ignoring Scope 2 and 3, or counting upstream benefits without counting downstream costs.

Counterfactual inflation. Using unrealistic baselines that overstate the impact of the intervention. A solar farm replacing a coal plant produces real emissions reductions; a solar farm replacing nothing produces only the energy it generates.

The remedy is third-party verification, transparent methodology, conservative baselines, and full lifecycle accounting. Subsequent chapters return to these themes wherever specific deployments are discussed.

This concludes Part I. Part II turns to precision agriculture, the first of the major application domains in which sustainability AI has demonstrated measurable impact.

5.6 Mathematical Foundations of Sustainability Metrics

5.6.1 Activity-Based Emissions

For activity vector $\mathbf{a} = (a_1, \dots, a_n)$ and emission factor vector $\mathbf{f} = (f_1, \dots, f_n)$:

$$E = \mathbf{a} \cdot \mathbf{f} = \sum_{i=1}^n a_i f_i, \quad \sigma_E^2 = \sum_i (f_i \sigma_{a_i})^2 + \sum_i (a_i \sigma_{f_i})^2,$$

assuming uncorrelated uncertainties. Correlated uncertainties require covariance terms via $\sigma_E^2 = \mathbf{g}^\top \mathbf{\Sigma} \mathbf{g}$ with sensitivity gradient $\mathbf{g}$.

5.6.2 Counterfactual Reduction

$$\Delta E(t) = E_{\text{baseline}}(t) - E_{\text{actual}}(t),$$

with cumulative reduction $\int_0^{\tau} \Delta E(t) dt$. Where the baseline is itself uncertain, conservative practice imposes $\Delta E_{\text{credited}} = \max(\Delta E - z\sigma_{\Delta E}, 0)$ at some confidence level (e.g. $z = 1.645$ for 95%).

5.6.3 Multi-Criteria Aggregation

For K criteria with normalised values $v_k(\mathbf{x}) \in [0, 1]$ and weights w_k with $\sum_k w_k = 1$:

$$S(\mathbf{x}) = \sum_{k=1}^K w_k v_k(\mathbf{x}).$$

The Pareto frontier $\mathcal{P}$ contains alternatives not dominated:

$$\mathbf{x} \in \mathcal{P} \iff \nexists \mathbf{x}' : \forall k v_k(\mathbf{x}') \geq v_k(\mathbf{x}) \text{ and } \exists k v_k(\mathbf{x}') > v_k(\mathbf{x}).$$

5.6.4 Permanence Discounting

For carbon stored over expected duration T_e versus reference duration T_r (typically 100 years):

$$\text{Permanence factor} = \frac{T_e}{T_r} \quad \text{if } T_e \leq T_r,$$

adjusted further by reversal-risk buffer $\beta \in [0, 0.5]$:

$$\text{Credited fraction} = (1 - \beta) \cdot \min(T_e/T_r, 1).$$

5.7 Worked Problems and Solutions

Problem 5.3 (Scope 1+2 Emissions). A factory consumes diesel of $a_d = 240,000$ litres at $f_d = 2.68$ kg CO₂e/litre, and grid electricity of $a_e = 5,400$ MWh at $f_e = 0.92$ kg CO₂e/kWh. Compute total emissions in tCO₂e.

Solution. Diesel: $240,000 \cdot 2.68 = 643,200$ kg = 643.2 tCO₂e (Scope 1). Electricity: $5,400,000 \cdot 0.92 = 4,968,000$ kg = 4,968 tCO₂e (Scope 2). Total: $643.2 + 4968 = 5,611.2$ tCO₂e. ■

Problem 5.4 (Uncertainty Propagation). For the diesel term above, suppose $\sigma_{a_d}/a_d = 2\%$ and $\sigma_{f_d}/f_d = 4\%$. Compute σ for E_d assuming uncorrelated errors.

Solution.

$$\begin{aligned}
\sigma_{a_d} &= 0.02 \cdot 240,000 = 4,800 \text{ litres,} \\
\sigma_{f_d} &= 0.04 \cdot 2.68 = 0.1072 \text{ kg/L,} \\
\sigma_{E_d}^2 &= (f_d \sigma_{a_d})^2 + (a_d \sigma_{f_d})^2 \\
&= (2.68 \cdot 4800)^2 + (240000 \cdot 0.1072)^2 \\
&= (12,864)^2 + (25,728)^2 \\
&= 1.654 \times 10^8 + 6.619 \times 10^8 \\
&= 8.273 \times 10^8 \text{ kg}^2, \\
\sigma_{E_d} &\approx 28,763 \text{ kg} = 28.76 \text{ tCO}_2\text{e.}
\end{aligned}$$

Relative uncertainty: $28.76/643.2 \approx 4.5\%$, dominated by emission factor uncertainty. ■

Problem 5.5 (Permanence Discounting). A soil-carbon project promises to maintain stocks for 25 years; reference duration is 100 years; reversal-risk buffer is 20%. Compute the credited fraction.

Solution.

$$\text{Credited fraction} = (1 - 0.20) \cdot \min(25/100, 1) = 0.80 \cdot 0.25 = 0.20.$$

The project credits only 20% of the gross sequestered carbon, reflecting both short duration and reversal risk. ■

5.8 Chapter Conclusion

The metric landscape surveyed in this chapter is dense by design. The GHG Protocol governs corporate emissions accounting; the SDGs catalogue seventeen interlocking development priorities; TCFD and its ISSB successor shape climate-related financial disclosure; the SBTi validates science-based reduction targets; GRI, CSRD, and CDP each serve overlapping but distinct disclosure constituencies. The reader is not expected to master all of these; the reader is expected to recognise which is relevant when, and to read sceptically when actors invoke the wrong framework for the wrong reason.

The chapter drew a hard line between technical metrics and physical outcomes. A model with $R^2 = 0.95$ on yield prediction may still fail to improve farmer welfare; a deforestation classifier with 92% accuracy may still issue carbon credits for emission reductions that did not occur. The job of sustainability AI is not to optimise pixel-level intersection over union but to deliver verifiable tonnes of avoided CO₂e, hectares of restored habitat, or kilowatt-hours of renewable energy integrated. Every subsequent chapter applies this discipline.

Co-benefits and trade-offs were treated with care. AI for sustainability rarely produces unalloyed good; almost every deployment has distributional consequences across people, places, and timescales. A precision-irrigation system that reduces aggregate water use may shift water access between upstream and downstream users; a deforestation alert that protects a forest may displace agricultural expansion to a neighbouring jurisdiction. The accounting matters. We catalogued the most common patterns and gave the reader a checklist for confronting them.

The section on greenwashing offered both a warning and a methodology. The warning: every successful sustainability metric becomes a target, and every target eventually gets gamed. The methodology: triangulate across independent metrics, separate verified from claimed reductions, and treat single-metric performance with appropriate scepticism. The mathematical sections gave the tools — uncertainty propagation, permanence discounting, multi-criteria aggregation — for doing the triangulation rigorously. We close Part I here. Part II turns to the first applied domain: precision agriculture.

Part II

Precision Agriculture

Precision Agriculture: An Introduction

The future of farming is variable rate, real time, and decided by the field — not by the farmer's intuition or the supplier's catalogue.

Kennedy Chengeta, PhD

Precision agriculture is the practice of managing within-field variability so that inputs match the spatial and temporal needs of the crop rather than being applied uniformly. The opportunity is enormous: agricultural inputs — seed, water, fertiliser, pesticide — are typically applied at field-average rates, leaving high-fertility patches over-fertilised and low-fertility patches under-fertilised. Variable-rate application driven by AI can simultaneously increase yields, reduce input costs, and lower environmental impact.

6.1 The Variability Problem

A typical maize field exhibits within-field yield variability of 30% to 80% from average. The drivers are heterogeneous: soil texture, water-holding capacity, organic matter, slope, drainage, microclimate, residual nutrients, weed pressure, pest history. Uniform management ignores all of this. Precision agriculture inverts the problem by treating the field as a population of small management zones — typically 0.1 to 1 hectare each — and prescribing differentiated treatment for each.

Definition 6.1 (Management Zone). A *management zone* is a contiguous spatial subset of a field that exhibits sufficient internal homogeneity in productivity-relevant variables to justify uniform treatment, while differing materially from adjacent zones. Formally, for variables $\mathbf{x}$ and within-zone variance σ_{within}^2 , between-zone variance $\sigma_{\text{between}}^2$:

$$\text{Zone validity} \propto \frac{\sigma_{\text{between}}^2}{\sigma_{\text{within}}^2}.$$

Higher ratios indicate more useful zonation.

6.2 The Sensing Stack

Precision agriculture combines three sensing layers.

Satellite remote sensing provides field-scale imagery at 5–30 day cadence. Sentinel-2 dominates for free coverage; Planet for high-cadence high-resolution work.

Aerial imagery from drones provides centimetre-scale resolution on demand. Multispectral and

thermal sensors enable disease detection, water-stress mapping, and biomass estimation at sub-plant scale. Cost has fallen to roughly USD 5–10 per hectare per flight.

Ground sensors include soil moisture probes, temperature/humidity loggers, electrical conductivity sensors, plant-mounted sensors, and yield monitors on harvesting equipment. IoT networks aggregate these into time-series databases that anchor model training.

6.3 The Decision Stack

Sensing produces data; AI converts data to decisions. Five decision categories recur.

What to plant where: variety selection conditioned on zone characteristics and seasonal forecast.

When to plant: planting-date optimisation against forecast rainfall and temperature.

How much to apply: variable-rate prescriptions for seed, fertiliser, water, pesticide.

When to act: timing of irrigation, fertilisation, pest control, harvest.

What to harvest: selective harvesting where ripeness or quality varies within field.

Example 6.2 (Variable-Rate Nitrogen on Smallholder Maize). A 4-hectare maize field in Kenya sensed via Sentinel-2 NDVI and ground soil samples shows three management zones with target nitrogen rates of 60, 110, and 90 kg N/ha. Uniform application at 90 kg N/ha would deliver 360 kg total. Variable rate delivers $4 \times \frac{60+110+90}{3} = 346.7$ kg — a 4% reduction in nitrogen — while improving yield by 8% by matching supply to plant demand. Cost reduction plus yield improvement: USD 220 per hectare per season.

6.4 Economic and Environmental Logic

Precision agriculture's economics depend on three margins. The *input-cost margin* is reduced spending on seed, fertiliser, water, and chemicals. The *yield margin* is increased output per hectare. The *quality margin* is higher-value crops through better timing and management. Across diverse contexts, the input-cost margin alone is typically sufficient to justify investment; yield and quality gains compound the case.

The environmental logic is parallel. Reducing nitrogen application reduces N₂O emissions (300× the warming potential of CO₂) and groundwater nitrate. Reducing irrigation reduces energy use and aquifer drawdown. Reducing pesticide reduces biodiversity damage and human health impacts. Where input reductions are real — not just shifted to neighbouring fields or future seasons — they translate directly to sustainability gains.

6.5 Smallholder Versus Industrial Context

The economics differ sharply between smallholder (typically < 2 ha, low capital) and industrial (typically > 100 ha, high capital) contexts. Industrial precision agriculture — John Deere, Climate Corporation, Trimble — has been operational for two decades in North America, Europe, Brazil, and

Australia. Smallholder precision agriculture is a more recent phenomenon, enabled by mobile phones, free satellite data, and AI advisory systems delivered via SMS, USSD, or chatbot. Different design constraints apply: data sparsity, intermittent connectivity, low literacy, multiple crops per field, mixed crop-livestock systems, lack of field machinery for variable-rate application.

The remainder of Part II treats the major precision agriculture topics in turn: remote sensing (Chapter 7), yield prediction (Chapter 8), soil health (Chapter 9), irrigation (Chapter 10), pest and disease (Chapter 11), livestock (Chapter 12), and the smallholder context (Chapter 13).

6.6 Mathematical Foundations of Variable-Rate Application

6.6.1 Optimal Input Allocation

For a field divided into K zones with yield-response function $y_k = f_k(u_k; \theta_k)$ where u_k is input rate and θ_k are zone parameters, profit maximisation under budget constraint U is

$$\max_{u_1, \dots, u_K} \sum_k A_k [p_y f_k(u_k) - p_u u_k] \quad \text{s.t.} \quad \sum_k A_k u_k \leq U,$$

where A_k is zone area, p_y output price, p_u input price. First-order conditions give the equimarginal principle:

$$p_y \frac{\partial f_k}{\partial u_k} - p_u = \lambda \quad \forall k,$$

i.e. marginal value product equalises across zones at the budget shadow price λ .

6.6.2 Mitscherlich Yield Response

The classical empirical form for nutrient response is

$$y(u) = y_{\max} [1 - e^{-c(u+b)}],$$

with asymptotic yield $y_{\max}$, response curvature c , and residual nutrient b . The economic optimum satisfies

$$u^* = \frac{1}{c} \ln \left(\frac{c y_{\max} p_y}{p_u} \right) - b.$$

6.7 Worked Problems and Solutions

Problem 6.3 (Equimarginal Allocation). Two zones with response $y_k = y_{\max,k} [1 - e^{-c_k u_k}]$: zone 1 with $y_{\max,1} = 7$ t/ha, $c_1 = 0.018$ ha/kg; zone 2 with $y_{\max,2} = 4$ t/ha, $c_2 = 0.030$ ha/kg. Output price $p_y = 250$ USD/t, input price $p_u = 1.20$ USD/kg. Compute economic optima.

Solution. Using $u_k^* = (1/c_k) \ln(c_k y_{\max,k} p_y / p_u)$:

Zone 1: $\ln(0.018 \cdot 7 \cdot 250 / 1.20) = \ln(26.25) = 3.268$. $u_1^* = 3.268 / 0.018 \approx 181.6$ kg/ha.

Zone 2: $\ln(0.030 \cdot 4 \cdot 250 / 1.20) = \ln(25.0) = 3.219$. $u_2^* = 3.219 / 0.030 \approx 107.3$ kg/ha.

Variable rate prescribes ~ 182 kg/ha on the high-potential zone and ~ 107 kg/ha on the low-potential zone, against a uniform-rate arithmetic mean of ~ 145 kg/ha. ■

Problem 6.4 (Zone Validity Index). Three candidate zonations report between/within variance ratios of 1.4, 2.6, 3.9. Which clustering is preferred and why?

Solution. Higher ratio $\sigma_{\text{between}}^2 / \sigma_{\text{within}}^2$ indicates greater zonal homogeneity relative to inter-zone differentiation. The ratio of 3.9 is preferred. Convention treats ratios above 2 as operationally useful; the 1.4 candidate would underperform uniform management. ■

6.8 Chapter Conclusion

The precision agriculture story is, at its core, a story about variability. Within a single hectare, soil texture can vary by a factor of three, organic matter by a factor of five, and water-holding capacity by a factor of two. Treating that hectare uniformly — the dominant practice of twentieth-century industrial agriculture — wastes inputs on the high-fertility portions and starves the low-fertility portions, producing average yields that fall short of what zone-specific management could achieve. The mathematical machinery of variable-rate optimisation, with the equimarginal principle as its theoretical heart, shows formally why zone-specific management yields more profit at lower environmental cost.

The sensing stack and decision stack frame the rest of Part II. The sensing stack — satellites, drones, in-field sensors, soil sampling — produces the heterogeneous data that machine learning consumes; the decision stack — recommendations, prescriptions, automation — closes the loop by translating model output into agronomic action. Chapters 7 through 12 address each layer in turn.

The chapter's most consequential point was that the leverage of precision agriculture differs profoundly between industrial and smallholder contexts. Industrial farms can amortise the capital cost of variable-rate sprayers and section-control planters; smallholders typically cannot. The implication is not that precision agriculture is irrelevant to smallholders — on the contrary, it is more relevant, because the gap between potential and actual yield is larger — but that the delivery mechanism must differ. SMS-based advisory, voice-enabled assistants, cooperative service models, and lightweight handheld sensors all play roles that they do not play in the industrial context. Chapter 13 returns to this in detail.

We have established the framing for Part II. Chapter 7 takes up the foundational input: satellite imagery of cropland.

Remote Sensing for Cropland

*Every cloud-free pixel is a measurement of the planet's biological productivity.
The job of AI is to make those measurements legible and actionable.*

Kennedy Chengeta, PhD

Remote sensing is the foundation of modern precision agriculture and large-scale agricultural monitoring. From Landsat's 30-metre pixels to the centimetre-scale imagery of commercial drones, the spectral, spatial, and temporal dimensions of remote sensing data give AI systems direct access to the photosynthetic activity, water status, and structural condition of vegetation across the planet. This chapter treats the methods that have become standard for converting reflectance into management decisions.

7.1 Spectral Bands and Vegetation Indices

The visible bands (blue 450nm, green 550nm, red 670nm) capture light absorbed and reflected by chlorophyll. The near-infrared band (NIR, 800-900nm) captures light reflected by leaf cellular structure — healthy vegetation reflects strongly here, stressed vegetation less so. Short-wave infrared (SWIR, 1500-2500nm) is sensitive to leaf and soil water content. The fundamental ratio capturing photosynthetic activity is NDVI:

Definition 7.1 (Normalised Difference Vegetation Index).

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{red}}}{\rho_{\text{NIR}} + \rho_{\text{red}}}$$

NDVI ranges from -1 (water, snow) through 0 (bare soil) to roughly 0.85 (dense healthy vegetation). It saturates at high biomass and is sensitive to soil background at low biomass.

Refinements address NDVI's limitations. EVI (Enhanced Vegetation Index) adds blue-band correction to reduce atmospheric noise and saturates less. SAVI (Soil-Adjusted) corrects for bare-soil background in sparse canopies. NDRE (Red-Edge) uses the 705-740nm band that Sentinel-2 provides, more sensitive to chlorophyll variation in advanced growth stages.

7.2 Time-Series Analysis and Phenology

A single image is a snapshot; a time series tells a story. Crop phenology — the temporal pattern of growth, flowering, senescence — has signatures in NDVI that distinguish maize from sorghum, fallow

from cropped, healthy from stressed. Smoothing filters (Savitzky-Golay, Whittaker) remove cloud and atmospheric noise. Phenological metrics extracted from smoothed series include start-of-season, peak, integrated greenness, and end-of-season.

Example 7.2 (Crop Type Classification from Sentinel-2 Time Series). A random forest classifier trained on 12-month Sentinel-2 NDVI time series from 8,000 labelled fields across Kenya achieved 89% overall accuracy on five-class crop type classification (maize, beans, tea, coffee, fallow). The most informative features were peak NDVI, peak date, and the area under the curve — consistent with the underlying phenological logic.

7.3 Foundation Models for Earth Observation

The past two years have seen the rise of pretrained earth-observation foundation models. Prithvi (NASA-IBM) is a masked autoencoder trained on 1TB of HLS imagery; SatMAE applies masked image modelling to satellite imagery; DOFA combines optical, radar, and LiDAR modalities. These models accept multi-band, multi-temporal imagery and produce embeddings that downstream task-specific heads fine-tune on small labelled datasets. For sustainability AI in data-scarce regions, foundation models have changed the economics: tasks that previously required ten thousand labels now require hundreds.

7.4 Operational Considerations

Production cropland monitoring demands attention to clouds, sensor calibration, and atmospheric correction. Sen2Cor is the standard for Sentinel-2 atmospheric correction. Cloud masking via FMask or s2cloudless is essential. Compositing strategies — monthly median, best-pixel, percentile — balance temporal coverage against cloud contamination. Geometric registration accuracy below 10 metres is typical for Sentinel-2; co-registration is rarely a concern for field-scale work. SAR data (Sentinel-1) provides all-weather imaging at the cost of more complex pre-processing and noisier signals.

7.5 Mathematical Foundations of Spectral Indices and Time-Series Smoothing

7.5.1 Generalised Index Form

Many vegetation indices share the structure

$$\text{Index} = \frac{a\rho_{\text{NIR}} - b\rho_{\text{red}}}{c\rho_{\text{NIR}} + d\rho_{\text{red}} + e},$$

with NDVI $(a, b, c, d, e) = (1, 1, 1, 1, 0)$ and SAVI $(1, 1, 1, 1, L)$ scaled by $1 + L$.

7.5.2 Savitzky–Golay Filter

For NDVI time series $\{y_t\}$ with window $2m + 1$ and polynomial degree p , the smoothed value is

$$\hat{y}_t = \sum_{j=-m}^m c_j y_{t+j},$$

where coefficients c_j minimise $\sum_j (y_{t+j} - q(j))^2$ for polynomial q of degree p . Closed-form coefficients exist; for $p = 2, m = 2$: $c = (-3, 12, 17, 12, -3)/35$.

7.5.3 Phenological Metrics

Given smoothed NDVI series $g(t)$ over a season:

$$\begin{aligned} \text{SOS} &= \min\{t : g(t) \geq g_{\min} + \alpha(g_{\max} - g_{\min})\}, \\ \text{POS} &= \arg \max_t g(t), \\ \text{EOS} &= \max\{t : g(t) \geq g_{\min} + \alpha(g_{\max} - g_{\min})\}, \\ \text{LOS} &= \text{EOS} - \text{SOS}, \\ \text{Integrated greenness} &= \int_{\text{SOS}}^{\text{EOS}} g(t) dt, \end{aligned}$$

with $\alpha \in [0.2, 0.5]$ a threshold convention.

7.6 Worked Problems and Solutions

Problem 7.3 (SAVI vs NDVI). A sparse-canopy pixel has $\rho_{\text{NIR}} = 0.32$, $\rho_{\text{red}} = 0.20$. Compute NDVI and SAVI ($L = 0.5$).

Solution.

$$\begin{aligned} \text{NDVI} &= \frac{0.32 - 0.20}{0.32 + 0.20} = \frac{0.12}{0.52} \approx 0.231. \\ \text{SAVI} &= \frac{1.5 \cdot (0.32 - 0.20)}{0.32 + 0.20 + 0.5} = \frac{0.18}{1.02} \approx 0.176. \end{aligned}$$

SAVI is lower because the soil-correction factor L shrinks the apparent greenness; under sparse canopies SAVI is the more reliable indicator. ■

Problem 7.4 (Savitzky–Golay Smoothing). Apply the $p = 2, m = 2$ filter $c = (-3, 12, 17, 12, -3)/35$ to the 5-point NDVI sequence $(0.42, 0.55, 0.58, 0.61, 0.50)$ centred at the third value.

Solution.

$$\begin{aligned} \hat{y}_3 &= \frac{1}{35} [(-3)(0.42) + 12(0.55) + 17(0.58) + 12(0.61) + (-3)(0.50)] \\ &= \frac{1}{35} [-1.26 + 6.60 + 9.86 + 7.32 - 1.50] = \frac{21.02}{35} \approx 0.6006. \end{aligned}$$

The smoothed value 0.601 slightly exceeds the raw centre value 0.580 because the polynomial fit favours the surrounding higher values. ■

7.7 Chapter Conclusion

Cropland remote sensing in 2025 is qualitatively different from cropland remote sensing in 2015. Sentinel-2 alone provides 10 m, multi-spectral, five-day-revisit imagery free to all users; Sentinel-1 adds all-weather synthetic aperture radar at comparable resolution; PlanetScope and the emerging commercial constellations offer daily revisit at 3 m for paid subscribers. The data layer is no longer the binding constraint; the modelling layer is.

The mathematical treatment of vegetation indices reminded the reader that NDVI, EVI, SAVI, and NDRE are not interchangeable. NDVI saturates over dense canopies and is influenced by soil background under sparse canopies; EVI corrects for atmospheric effects and saturates less; SAVI explicitly adjusts for soil background; NDRE leverages the red-edge band to detect nitrogen stress before NDVI shows it. Choosing the right index for the right crop and the right growth stage is a small but consequential discipline.

Time-series analysis through Savitzky–Golay smoothing, harmonic regression, and BFAST change detection turns single images into agronomic trajectories. The phenological metrics — start, peak, end, length of season, integrated greenness — carry more information about yield potential than any single image. The Savitzky–Golay worked example illustrated how the technique preserves signal while suppressing noise.

The foundation-model section signalled the field’s current trajectory. Prithvi, SatMAE, and Clay are pre-trained on tens of millions of unlabelled satellite scenes; fine-tuning them on small labelled datasets yields performance previously achievable only with hundreds of thousands of labelled examples. The reader should expect the next edition of this book to treat foundation-model fine-tuning as the default workflow.

Operational considerations — cloud masking, gap-filling, geolocation accuracy, scaling to continental processing — are where most real deployments succeed or fail. The chapter sketched the patterns; the references give the implementation details. We turn next to the quantitative downstream task that remote sensing serves more than any other: yield prediction.

Crop Yield Prediction

A good yield forecast in March changes a planting decision; a perfect yield forecast in October changes nothing.

Kennedy Chengeta, PhD

Yield prediction is one of the most operationally important applications of AI in agriculture. Forecasts at different horizons serve different decisions: pre-season forecasts inform planting decisions and credit allocation; mid-season forecasts inform irrigation, fertilisation, and pest management; pre-harvest forecasts inform marketing, logistics, and food security planning. This chapter treats the major modelling approaches and the data requirements that distinguish workable from unworkable systems.

8.1 Process-Based Versus Data-Driven Models

Yield prediction has two intellectual traditions. The process-based tradition — DSSAT, APSIM, AquaCrop — builds mechanistic models of plant physiology from sub-daily inputs of weather, soil, and management. The data-driven tradition — regression, gradient boosting, deep learning — learns yield from historical examples without explicit physiology. Both traditions have strengths and weaknesses: process-based models extrapolate to new conditions because they encode physical laws but require extensive parameterisation; data-driven models capture local patterns the process models miss but fail when extrapolating to conditions absent from training data.

Hybrid approaches combine the two. The process model produces a physiology-respecting baseline; an ML model corrects the residuals using features the process model omits. This pattern — physical model with ML residual correction — has become dominant for production-grade yield forecasting.

8.2 Features That Matter

Across crops and regions, the features that consistently appear in successful yield models include: cumulative growing degree days (a temperature integral capturing thermal time); precipitation totals weighted by crop stage (vegetative, flowering, grain-fill); evapotranspiration deficit (precipitation minus potential ET); peak NDVI and integrated NDVI from satellite time series; soil organic carbon and texture; planting date; and management intensity proxies (fertiliser application history, irrigation availability). Critical periods — typically flowering and grain-fill — weight disproportionately in good models.

Definition 8.1 (Growing Degree Days). For base temperature T_b (e.g. 10°C for maize) and daily mean temperature T_d :

$$\text{GDD}_d = \max(T_d - T_b, 0).$$

Cumulative GDD over the season is a stronger predictor of phenological stage than calendar days.

8.3 Deep Learning for Yield Prediction

CNN-LSTM hybrid architectures have become standard for satellite-data-driven yield forecasting. The CNN extracts spatial features from each timestep of imagery; the LSTM aggregates across the growing season. Recent work has replaced LSTMs with transformers (Temporal Fusion Transformer), gaining both accuracy and interpretability through attention weights that show which dates and which features drove each prediction.

Example 8.2 (County-Level Maize Yield Forecast, Iowa and Zimbabwe). A CNN-LSTM model trained on 2000-2020 MODIS surface reflectance, gridded weather, and county-level USDA yield reports achieved $R^2 = 0.85$ on out-of-sample US counties. Transferring the model directly to Zimbabwe districts produced $R^2 = 0.34$ — the model failed because the relationships between spectral signatures and yield differ in tropical low-input systems. Fine-tuning on 4 years of Zimbabwean data lifted R^2 to 0.71. The lesson generalises: yield models do not transfer well across agroecological zones without local fine-tuning.

8.4 Uncertainty and Decision Value

Operational yield forecasting requires not just point estimates but well-calibrated uncertainty. A forecast of 4.2 t/ha with a 90% interval of [4.0, 4.4] supports very different decisions than the same point estimate with interval [2.5, 5.9]. Conformal prediction provides distribution-free intervals from any base model. Quantile regression and deep ensembles are common alternatives. The right verification metric is interval calibration — whether 90% intervals contain the truth 90% of the time — not just sharpness.

8.5 Mathematical Foundations of Yield Prediction

8.5.1 Growing Degree Days

$$\text{GDD}_d = \max\left(\frac{T_{\max,d} + T_{\min,d}}{2} - T_b, 0\right), \quad \text{GDD}_{\text{cum}}(t) = \sum_{d=1}^t \text{GDD}_d.$$

For maize ($T_b = 10^\circ\text{C}$), most varieties require $\sim 1,400$ GDD to reach physiological maturity.

8.5.2 Linear Yield Model

A baseline regression

$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

with $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ and $\hat{\sigma}^2 = \text{RSS}/(n-p)$. The coefficient of determination $R^2 = 1 - \text{RSS}/\text{TSS}$.

8.5.3 Conformal Prediction Intervals

For a non-conformity score $s(\mathbf{x}, y) = |y - \hat{y}(\mathbf{x})|$ and calibration set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, the marginal coverage guarantee at miscoverage α uses

$$\hat{q}_\alpha = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n} \{s(\mathbf{x}_i, y_i)\}.$$

Test-time interval: $\hat{C}(\mathbf{x}_{\text{new}}) = [\hat{y}(\mathbf{x}_{\text{new}}) - \hat{q}_\alpha, \hat{y}(\mathbf{x}_{\text{new}}) + \hat{q}_\alpha]$.

8.5.4 Stage-Weighted Stress Index

Yield response to weather stress in critical phenological windows:

$$\hat{y} = y_{\text{pot}} \prod_s [1 - k_s \cdot \text{Stress}_s]^{w_s},$$

where w_s are stage-specific weights summing to 1, k_s are stress sensitivities, and Stress_s is e.g. ET-deficit ratio in stage s .

8.6 Worked Problems and Solutions

Problem 8.3 (Cumulative GDD). Daily $T_{\text{max}}/T_{\text{min}}$ for five days: (28/12), (30/14), (26/10), (31/15), (29/13)°C; $T_b = 10$. Compute cumulative GDD.

Solution. Daily means: 20, 22, 18, 23, 21. Subtract base 10: 10, 12, 8, 13, 11. Cumulative GDD: $10 + 12 + 8 + 13 + 11 = 54$ GDD-days. ■

Problem 8.4 (Conformal Interval). Calibration absolute residuals (ascending): 0.12, 0.18, 0.21, 0.27, 0.31, 0.36, 0.42, 0.48 t/ha ($n = 10$). Compute the 90% conformal interval half-width.

Solution. Index $\lceil (10 + 1)(0.90) \rceil = \lceil 9.9 \rceil = 10$. The 10th-order statistic is 0.71 t/ha. The 90% prediction interval is $\hat{y} \pm 0.71$ t/ha. ■

Problem 8.5 (Stage-Weighted Stress). Three stages: vegetative ($w_1 = 0.20$, $k_1 = 0.4$, stress 0.10); flowering ($w_2 = 0.50$, $k_2 = 1.2$, stress 0.30); grain-fill ($w_3 = 0.30$, $k_3 = 0.7$, stress 0.20). Potential yield $y_{\text{pot}} = 8.0$ t/ha. Compute $\hat{y}$.

Solution. Vegetative factor: $(1 - 0.4 \cdot 0.10)^{0.20} = (0.96)^{0.20}$. Take log: $0.20 \cdot \ln(0.96) = 0.20 \cdot (-0.0408) = -0.00816$, so factor $= e^{-0.00816} \approx 0.9919$. Flowering: $(1 - 1.2 \cdot 0.30)^{0.50} = (0.64)^{0.50} = \sqrt{0.64} = 0.8$. Grain-fill: $(1 - 0.7 \cdot 0.20)^{0.30} = (0.86)^{0.30}$. Log: $0.30 \cdot \ln(0.86) = 0.30 \cdot (-0.1508) = -0.04524$, factor $= 0.9558$.

Combined: $0.9919 \cdot 0.8 \cdot 0.9558 = 0.7585$. $\hat{y} = 8.0 \cdot 0.7585 \approx 6.07$ t/ha.

The flowering stress dominates because of its high weight and high sensitivity, dropping yield from 8 to 6 t/ha. ■

8.7 Chapter Conclusion

Yield prediction sits at the intersection of remote sensing, weather modelling, agronomy, and machine learning. The chapter walked through each layer. Growing degree days and weather features provide

the energy-balance backbone; satellite indices provide the vegetation trajectory; soil maps provide the static physical context; management information — where available — provides the operational adjustment. The combination, fitted with gradient-boosted trees or hybrid mechanistic- ML models, produces forecasts at field scale with accuracies that match or exceed government-statistics estimates and arrive months earlier.

The chapter's treatment of uncertainty quantification deserves emphasis. A point yield forecast is operationally less useful than a distribution forecast for almost every decision that depends on it. Insurers need the left tail to price index products. Aid agencies need the left tail to trigger early response. Governments need the median for procurement and the right tail for surplus planning. The conformal-prediction worked problem demonstrated how to produce calibrated intervals with finite-sample coverage guarantees, regardless of the underlying model.

Stage-weighted stress indices, introduced mathematically and exemplified in the third worked problem, formalise an agronomic intuition: not all weeks matter equally. Maize is most sensitive to water stress at flowering; wheat at grain-fill; rice at panicle initiation. A simple seasonal-average aridity index obscures this; stage-weighted aggregation preserves it. Operational systems should use the latter wherever phenological models are reliable.

The chapter's treatment of process-based hybrid models — DSSAT, APSIM, AquaCrop coupled to machine-learning correction layers — pointed forward to Chapter 10, where the same mechanistic-statistical hybridisation drives water management. We move next, however, to the static substrate on which all crops grow: soil.

Soil Health and Carbon

A tonne of carbon in healthy soil is worth more than a tonne of carbon in any forest — because it is also feeding the people who live above it.

Kennedy Chengeta, PhD

Soil is the planet's largest terrestrial carbon stock — approximately 1,500 Gt in the top metre, more than twice the carbon in the atmosphere and three times that in vegetation. It is also the most undermeasured. AI methods built on remote sensing, in-situ sensors, and machine learning are transforming our ability to estimate soil organic carbon, monitor change over time, and verify the impact of regenerative practices.

9.1 Soil Organic Carbon: Why It Matters

Soil organic carbon (SOC) underpins agricultural productivity (water retention, nutrient cycling, structure) and is a major lever for climate mitigation. Increasing SOC by even 0.4% per year on global croplands would offset roughly 30% of annual fossil emissions — the basis for the "4 per 1000" initiative launched at COP21. Whether this rate is achievable at scale is contested; what is not contested is that SOC monitoring at field resolution is essential to know.

9.2 Spectroscopy and Spectral Soil Mapping

Visible-near-infrared (vis-NIR) and mid-infrared (MIR) spectroscopy have transformed soil analysis. A spectroscope reads reflectance from a soil sample; a calibrated ML model converts the spectrum to estimates of SOC, nitrogen, pH, and texture. The IRRI Africa Soil Information Service (AfSIS) built MIR libraries of over 100,000 African soil samples; trained models predict SOC with $R^2 \sim 0.85$ at one-tenth the cost of wet chemistry.

Pushing further, AI-augmented hyperspectral satellite imagery (DESI, EnMAP, PRISMA) is now beginning to map SOC at 30-metre resolution from orbit. Accuracy is lower than spectroscopy on samples ($R^2 \sim 0.5$ - 0.6 typical) but the coverage is global and continuous.

9.3 Digital Soil Mapping

Definition 9.1 (SCORPAN Framework). Soil property S at location $\mathbf{x}$ is modelled as a function of:

$$S(\mathbf{x}) = f(s, c, o, r, p, a, n) + \epsilon$$

where s is other soil properties, c climate, o organisms, r relief, p parent material, a age, n spatial position, ϵ residual.

Digital soil mapping operationalises SCORPAN with random forests, boosted trees, and deep learning. Inputs include topographic derivatives (slope, aspect, curvature, TWI), climate variables, remote-sensing indices, and existing soil maps; outputs are gridded predictions at 30-100m resolution with uncertainty maps. ISRIC SoilGrids 250m is a global product; national products at finer resolution exist for the US, Australia, the EU, and increasingly Africa via AfSIS.

9.4 Soil Carbon MRV

For carbon markets, soil organic carbon must be measured, reported, and verified with accuracy and permanence sufficient to underwrite credits. The Verra VM0042 methodology specifies sampling protocols, model validation requirements, and uncertainty discounting (the more uncertain the estimate, the larger the buffer pool of unsold credits). AI models are accepted as the bulk of the estimation provided they are calibrated to a stratified sample of in-situ measurements; the methodology requires that model RMSE be measured on out-of-sample data and that uncertainty be propagated through to credit issuance.

Example 9.2 (SOC MRV Accuracy and Buffer Pool). A project measures SOC change of +0.5 tonnes C per hectare per year (1.83 tCO₂e) with model RMSE of 0.2 t C/ha/year. Under VM0042, the uncertainty translates to a buffer pool deduction of approximately 25%. The project issues credits for $0.75 \times 1.83 = 1.37$ tCO₂e per hectare per year. Reducing model RMSE through better data and better models directly increases issuable credits.

9.5 Practices, Prediction, and Verification

Regenerative practices — cover cropping, reduced tillage, diverse rotations, integrated livestock — are the dominant levers for increasing SOC. AI helps in three ways. First, predicting which practices on which fields will produce the greatest SOC gains. Second, monitoring practice adoption from satellite imagery (cover-crop detection, tillage classification). Third, verifying SOC change from spectral and physical sampling. A coherent system links all three: predict, monitor adoption, verify outcome.

9.6 Mathematical Foundations of Soil Carbon Dynamics

9.6.1 First-Order Decomposition

A single-pool soil carbon model:

$$\frac{dC}{dt} = I(t) - kC(t),$$

with input rate I and decomposition constant k . Steady state: $C^* = I/k$. The general solution

$$C(t) = C^* + (C_0 - C^*)e^{-kt}$$

shows exponential approach to equilibrium with characteristic time $1/k$.

9.6.2 Multi-Pool RothC Structure

The Rothamsted Carbon Model partitions soil organic matter into pools $\{P_i\}$ with rate constants $\{k_i\}$ and inter-pool flows. The pool update is

$$P_i(t + \Delta t) = P_i(t)e^{-k_i a(t)\Delta t} + S_i(t),$$

where $a(t)$ is a rate modifier (temperature, moisture, soil cover) and $S_i(t)$ are inputs. Long-lived pools (humus, $k \sim 0.02 \text{ yr}^{-1}$) dominate stable storage.

9.6.3 Spectroscopic Calibration

For absorbance spectrum $\mathbf{x} \in \mathbb{R}^p$ and SOC concentration y , partial least squares regression fits

$$\mathbf{x} = \mathbf{T}\mathbf{P}^\top + \mathbf{E}, \quad y = \mathbf{T}\mathbf{q} + \mathbf{f},$$

where $\mathbf{T}$ are latent scores. Prediction: $\hat{y} = \mathbf{x}^\top \mathbf{P}\mathbf{q}$ (approximately, for truncated component count). Out-of-sample R^2 on held-out samples is the headline performance metric.

9.6.4 VM0042 Buffer-Pool Sizing

Verra VM0042 uncertainty deduction:

$$\text{Buffer fraction} = \begin{cases} 0 & \text{if RMSE/mean} \leq 5\%, \\ 0.10 & \text{if } 5\% < \text{RMSE/mean} \leq 10\%, \\ 0.25 & \text{if } 10\% < \text{RMSE/mean} \leq 20\%, \\ 0.50 & \text{if } 20\% < \text{RMSE/mean} \leq 30\%, \\ \text{ineligible} & \text{otherwise.} \end{cases}$$

9.7 Worked Problems and Solutions

Problem 9.3 (Steady-State SOC). A field receives organic input of $I = 3.6 \text{ t C/ha/yr}$; effective decomposition $k = 0.05 \text{ yr}^{-1}$. Compute steady-state stock and time to half-equilibrium starting from $C_0 = 30 \text{ t C/ha}$.

Solution. $C^* = I/k = 3.6/0.05 = 72 \text{ t C/ha}$. Half-equilibrium when $C(t) - C_0 = 0.5(C^* - C_0)$, i.e. $1 - e^{-kt} = 0.5$, $e^{-kt} = 0.5$, $t = \ln(2)/k = 0.693/0.05 = 13.86 \text{ years}$. ■

Problem 9.4 (Buffer Pool from RMSE). A project measures mean SOC change of 0.45 t C/ha/yr with model RMSE of 0.075 t C/ha/yr . Compute the credit-eligible fraction.

Solution. $\text{RMSE/mean} = 0.075/0.45 = 16.7\%$, which falls in the 10–20% bracket $\Rightarrow$ 25% buffer deduction. Credit-eligible fraction = 0.75. $0.45 \cdot 0.75 = 0.3375 \text{ t C/ha/yr}$ credited, equivalent to $0.3375 \cdot 44/12 \approx 1.238 \text{ tCO}_2\text{e/ha/yr}$. ■

9.8 Chapter Conclusion

Soil health is where carbon, food, and water meet. The world's soils hold more carbon than the atmosphere and all terrestrial vegetation combined; small percentage changes in soil carbon stocks

translate to large atmospheric consequences. The chapter walked through the techniques that allow AI to estimate soil properties at the resolution needed for both agronomic decision-making and carbon-market verification.

Digital soil mapping via the SCORPAN framework — predicting soil properties as a function of soil, climate, organisms, relief, parent material, age, and neighbourhood — gives continent-scale soil-property maps from sparse ground observations and dense covariate layers. Mid- infrared and visible-near-infrared spectroscopy, paired with partial- least-squares regression, reduces the cost of laboratory analysis by an order of magnitude. The combination is the technical backbone of the African Soil Information Service and similar continental infrastructures elsewhere.

The mathematical treatment of soil carbon dynamics introduced the first- order decay model and its multi-pool extensions. The worked problem on steady-state SOC and time to equilibrium illustrated a humbling fact: soil carbon responds slowly. Management practices that increase carbon inputs take decades to express their full equilibrium effect. This has direct implications for carbon markets, where short crediting periods risk understating long-run impact while overstated permanence risks selling reductions that never materialise.

The buffer-pool sizing worked problem demonstrated the operational machinery of VM0042 and similar methodologies. The reader now has the tools to audit a soil-carbon credit and judge whether the buffer deduction is conservative, permissive, or grossly inadequate. We turn next to water — the agricultural input most directly modulated by climate change.

AI-Driven Irrigation and Water Management

In a warming world, the most valuable agricultural decision will not be how much to plant but when and how much to irrigate.

Kennedy Chengeta, PhD

Irrigation accounts for roughly 70% of global freshwater withdrawals. Most irrigation systems are scheduled by tradition or rule of thumb, with substantial waste of water and the energy to pump it. AI-driven irrigation matches water supply to crop water demand at field, zone, or even sub-zone scale, using soil-moisture sensors, evapotranspiration models, and reinforcement learning. The economic and environmental gains are substantial: water reductions of 20-40% are routine, with no yield loss or with yield improvements.

10.1 Crop Water Demand: ETo and Kc

Definition 10.1 (Reference Evapotranspiration). The FAO-56 Penman-Monteith equation computes reference evapotranspiration ET_0 from net radiation, temperature, humidity, and wind speed. Crop water demand is

$$ET_c = K_c \cdot ET_0$$

where the crop coefficient K_c varies by crop, growth stage, and management.

Measured directly, ET is expensive (eddy covariance, weighing lysimeters). Estimated indirectly from weather and remote sensing, ET is cheap and continental in scale. Models like SEBAL, METRIC, and DisALEXI partition surface energy balance using thermal satellite imagery to produce gridded daily ET. AI is now augmenting these models by learning corrections from in-situ data.

10.2 Soil Moisture Estimation

Soil moisture is the proximate driver of plant water status. Three estimation approaches dominate. *In-situ sensors* (capacitance, time-domain reflectometry) provide point measurements with high accuracy and high cost per area. *SAR remote sensing* (Sentinel-1) provides surface (top 5cm) moisture at 1km resolution and weekly cadence; deep-learning downscaling pushes resolution to 100m. *Soil-water-balance models* integrate precipitation, irrigation, ET, and runoff to estimate moisture without direct measurement; they require calibration but cover any field with sufficient input data.

10.3 Reinforcement Learning for Irrigation Scheduling

Irrigation scheduling is naturally a sequential decision problem under uncertainty. State: soil moisture profile, crop stage, weather forecast, water-availability constraints. Action: irrigation amount today. Reward: yield expectation minus water cost minus stress penalty. Model-free RL (PPO, SAC) and model-based approaches (Dyna-style with crop-model environments) have produced operational schedulers.

Example 10.2 (RL Irrigation Scheduler in Limpopo). A Soft Actor-Critic agent trained on AquaCrop simulation environments and deployed on 1,200 hectares of citrus in Limpopo, South Africa, reduced water use by 31% and energy use by 28% with no significant change in yield. The training cost was approximately 300 GPU-hours; the operational compute is negligible because inference runs on field-edge devices. Net carbon impact: 1,450 tonnes CO₂e avoided in the first year (mostly from reduced diesel pumping), against approximately 0.6 tonnes embedded in the AI system. Net-positive ratio: 2,400×

10.4 Water Equity and Governance

Water-saving AI has equity implications that simple efficiency narratives miss. A precision-irrigation system that reduces water demand on commercial farms can release water to ecosystems — or it can release water to other commercial farms that expand into the freed allocation. Without governance frameworks that allocate saved water transparently, the technical gains may not translate to environmental outcomes. In Sub-Saharan Africa, where formal water rights are often weak, the role of AI is necessarily entwined with the role of regulators, river-basin authorities, and customary water-management institutions.

10.5 Mathematical Foundations of Crop Water Demand and Scheduling

10.5.1 Penman–Monteith Reference ET

$$ET_0 = \frac{0.408 \Delta (R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma (1 + 0.34 u_2)}$$

where R_n is net radiation (MJ m⁻² d⁻¹), G soil-heat flux, T mean temperature (°C), u_2 wind speed at 2m, e_s, e_a saturation and actual vapour pressure (kPa), Δ slope of the saturation vapour curve, γ psychrometric constant.

10.5.2 Soil Water Balance

$$\theta(t+1) = \theta(t) + \frac{P(t) + I(t) - ET_c(t) - D(t) - R(t)}{Z_r},$$

where θ is volumetric water content, P precipitation, I irrigation, D deep drainage, R runoff, Z_r root-zone depth. Field capacity θ_{fc} caps storage; permanent wilting point θ_{pwp} defines the lower bound; available water content $\theta_{aw} = \theta_{fc} - \theta_{pwp}$.

10.5.3 Yield Response to Water

The FAO-33 form:

$$1 - \frac{Y_a}{Y_m} = K_y \left(1 - \frac{ET_a}{ET_m} \right),$$

with stage-specific yield-response factor K_y .

10.5.4 RL Irrigation Scheduling

State $\mathbf{s}_t = (\theta_t, \text{stage}_t, \text{forecast}_t, \text{water budget}_t)$, action $a_t \in \mathcal{A}$ irrigation depth, reward

$$r_t = -p_w a_t + p_y \cdot \Delta y_t - \kappa \text{stress}_t,$$

where p_w is water cost, Δy_t marginal yield, κ stress penalty. Optimal Bellman: $Q^*(\mathbf{s}, a) = r + \gamma \max_{a'} Q^*(\mathbf{s}', a')$.

10.6 Worked Problems and Solutions

Problem 10.3 (Crop ET). For maize at mid-season, $K_c = 1.15$ and $\text{ET}_0 = 6.2$ mm/day. Compute ET_c and weekly water demand.

Solution. $\text{ET}_c = K_c \cdot \text{ET}_0 = 1.15 \cdot 6.2 = 7.13$ mm/day. Weekly: $7 \cdot 7.13 = 49.9$ mm. ■

Problem 10.4 (Soil Water Balance). Today: $\theta = 22\%$, $P = 0$, $I = 0$, $\text{ET}_c = 8$ mm, $Z_r = 600$ mm, $D = R = 0$. Compute θ_{t+1} .

Solution. $\Delta\theta = -8/600 = -0.01333$ (volumetric). Hence $\theta_{t+1} = 0.22 - 0.01333 = 0.2067 = 20.67\%$. ■

Problem 10.5 (Yield-Water Production Function). Maize $Y_m = 9$ t/ha, $\text{ET}_m = 540$ mm. Stage K_y averaged 1.05. Actual seasonal $\text{ET}_a = 432$ mm. Compute actual yield.

Solution. $1 - Y_a/Y_m = 1.05 \cdot (1 - 432/540) = 1.05 \cdot 0.2 = 0.21$. $Y_a = 9 \cdot 0.79 = 7.11$ t/ha. ■

10.7 Chapter Conclusion

Water management is the single application area where AI's leverage on sustainability is most directly quantifiable. Every cubic metre of irrigation water saved translates immediately to pumping energy avoided, CO₂e emissions averted, and groundwater drawdown reduced. The chapter argued, and the mathematics confirmed, that the savings are not marginal: 25%–40% reductions in irrigation volume at constant or improved yield are routine in well-designed deployments.

The Penman-Monteith formulation gives the reader the physical foundation: reference evapotranspiration is computable from standard weather-station data, and crop-coefficient adjustments produce crop-specific water requirements. The soil-water-balance equation closes the accounting, tracking how rainfall, irrigation, evapotranspiration, drainage, and runoff combine to determine root-zone moisture. The worked problems illustrated each piece: 49.9 mm/week demand for mid-season maize, soil-water depletion of 1.3% per day, yield-water production functions with stage-specific sensitivity factors.

The reinforcement-learning treatment of irrigation scheduling closed the loop. Process-based models such as AquaCrop, DSSAT, and APSIM serve as training environments for RL agents; the agents learn policies that outperform fixed irrigation schedules by exploiting weather forecasts, soil-moisture observations, and price signals jointly. The reward function explicitly trades water cost against yield

value against stress penalty; tuning the trade-off is itself a policy choice that should be made by farmers and water authorities, not by modellers.

The chapter's institutional discussion pointed forward to Part X on African deployments. Water-rights regimes, irrigation cooperatives, and the political economy of upstream-downstream conflict determine whether farm-scale optimisation aggregates to watershed-scale sustainability or merely shifts water consumption between users. The mathematics is necessary; it is not sufficient. We turn next to the biotic stresses that often determine whether crops realise their water-and-nutrient potential: pests and diseases.

Pest and Disease Detection

Three days early, the spray works; three days late, the harvest is lost. The value of AI is in the three days.

Kennedy Chengeta, PhD

Plant pests and diseases destroy 20-40% of global crop production annually. Early detection enables targeted, low-dose treatment that protects yield, reduces pesticide use, and prevents outbreak escalation. Computer vision on smartphone and drone imagery, acoustic sensing for stored-product pests, and ML on weather-driven epidemiological models have made early-detection systems practical at a cost that smallholders can access.

11.1 Computer Vision for Disease Identification

Convolutional neural networks trained on labelled leaf images now identify hundreds of pest and disease species from smartphone photos with accuracy exceeding many trained agronomists. PlantVillage Nuru (Penn State and CGIAR) identifies cassava diseases on offline Android devices used by smallholders across East Africa. Plantix (PEAT GmbH) covers more than 500 crop-disease combinations with global deployment. The underlying technology is mature; the deployment challenges are linguistic, agronomic translation of the diagnosis into action, and the long tail of locally relevant species not in global training sets.

11.2 Drone-Based Disease Mapping

Multispectral drones flown at 50-100m altitude resolve disease symptoms before they are visible to the eye. The spectral signature of plant stress — elevated red and NIR ratios, water-status changes detectable in SWIR, thermal anomalies from compromised transpiration — precedes visible chlorosis by days to weeks. Object-detection CNNs on drone orthomosaics produce field-scale disease maps that drive variable-rate spraying. Treatment area reductions of 50-90% are routine with no efficacy loss.

11.3 Pest Population Modelling

Pest dynamics are weather-dependent: temperature and humidity drive development rates; rainfall affects survival; wind drives dispersal. Combining weather forecasts with pest-population models gives outbreak forecasts at days-to-weeks horizons. The fall armyworm forecast system deployed across Africa by FAO uses a recurrent neural network trained on weather, host availability, and

observed populations to predict outbreak risk at district resolution.

Example 11.1 (Fall Armyworm Early Warning). The FAO-CABI fall armyworm forecasting system (FAMEWS) integrates weekly farmer reports via mobile app with weather and host-crop data. ML models produce 4-week-ahead outbreak risk maps that drive targeted scouting and treatment. Across Kenya, Ghana, and Zimbabwe deployments, the system has reduced average pesticide use by 28% while improving early-detection rates by 41% compared to baseline scouting protocols.

11.4 Acoustic Sensing for Stored-Product Pests

Post-harvest losses to insect pests in stored grain are a major issue in tropical agriculture, with losses of 10-30% common. Low-cost piezoelectric microphones in storage bins detect insect feeding sounds; ML classifiers distinguish weevils, borers, and beetles by acoustic signature. Early detection enables fumigation before populations explode. Costs of around USD 50-100 per silo plus a smartphone interface have made the technology accessible to medium-scale traders and farmer cooperatives.

11.5 Mathematical Foundations of Pest Population Dynamics

11.5.1 Logistic Population Growth

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right), \quad N(t) = \frac{K}{1 + \left(\frac{K-N_0}{N_0}\right) e^{-rt}}.$$

Above the economic threshold N^* , treatment is justified when expected damage exceeds treatment cost.

11.5.2 Temperature-Driven Development

Insect development rate often follows a degree-day model:

$$DD_d = \begin{cases} 0 & T_d \leq T_{\min}, \\ T_d - T_{\min} & T_{\min} < T_d < T_{\max}, \\ T_{\max} - T_{\min} & T_d \geq T_{\max}, \end{cases}$$

with generation completion at fixed DD_{gen} requirement.

11.5.3 Damage Function and Treatment Decision

Expected damage if untreated: $D(N) = \alpha N + \beta N^2$. Expected damage if treated: $D'(N) = (1 - e)D(N) + c_t$, with e treatment efficacy and c_t unit treatment cost. Treat when $e \cdot D(N) > c_t$.

11.5.4 CNN Classifier Performance

For a binary disease classifier with threshold t , true-positive rate $\text{TPR}(t)$ and false-positive rate $\text{FPR}(t)$, the operational expected cost is

$$\mathbb{E}[\text{Cost}] = \pi(1 - \text{TPR}(t))C_{\text{FN}} + (1 - \pi)\text{FPR}(t)C_{\text{FP}},$$

with prevalence π and cost asymmetries $C_{\text{FN}} \gg C_{\text{FP}}$ typical for disease detection.

11.6 Worked Problems and Solutions

Problem 11.2 (Logistic Population). $r = 0.18 \text{ d}^{-1}$, $K = 240 \text{ insects/m}^2$, $N_0 = 12$. Compute $N(20)$.

Solution. $\frac{K-N_0}{N_0} = (240 - 12)/12 = 19$. $e^{-0.18 \cdot 20} = e^{-3.6} = 0.02732$. Denominator: $1 + 19 \cdot 0.02732 = 1 + 0.5191 = 1.5191$. $N(20) = 240/1.5191 \approx 158 \text{ insects/m}^2$. ■

Problem 11.3 (Optimal Treatment Threshold). Damage $D(N) = 1.2N + 0.005N^2$ (USD/ha). Treatment cost $c_t = 22$ USD/ha; efficacy $e = 0.85$. Find threshold N^* .

Solution. Treat when $e \cdot D(N) > c_t$: $0.85(1.2N + 0.005N^2) > 22 \Rightarrow 1.02N + 0.00425N^2 > 22$. Solve $0.00425N^2 + 1.02N - 22 = 0$. Discriminant $= 1.02^2 + 4 \cdot 0.00425 \cdot 22 = 1.0404 + 0.374 = 1.4144$. $N^* = (-1.02 + \sqrt{1.4144})/(2 \cdot 0.00425) = (-1.02 + 1.1893)/0.0085 \approx 19.9 \text{ insects/m}^2$. ■

Problem 11.4 (Cost-Sensitive Classifier). Prevalence $\pi = 0.05$, $C_{\text{FN}} = 200$, $C_{\text{FP}} = 12$. At threshold A: TPR = 0.92, FPR = 0.18. At threshold B: TPR = 0.85, FPR = 0.06. Which is preferred?

Solution. A: $0.05 \cdot 0.08 \cdot 200 + 0.95 \cdot 0.18 \cdot 12 = 0.80 + 2.052 = 2.852$. B: $0.05 \cdot 0.15 \cdot 200 + 0.95 \cdot 0.06 \cdot 12 = 1.50 + 0.684 = 2.184$. B has lower expected cost despite lower TPR, because its lower FPR saves more on false-positive treatments at low disease prevalence. ■

11.7 Chapter Conclusion

Pest and disease management exhibits the cost-asymmetric classification problem in its most consequential form. A false negative — missing an outbreak — can destroy a season's crop and propagate to neighbouring fields. A false positive — prescribing treatment where none is needed — wastes money, contaminates the environment, and accelerates pesticide resistance. The threshold-tuning worked problem demonstrated formally that the optimal classifier threshold depends on prevalence, cost asymmetry, and treatment efficacy; default 0.5 thresholds are almost never appropriate in this setting.

The mathematical treatment of population dynamics — logistic growth, degree-day development, damage functions — gives the reader the agronomic foundation. The economic-threshold calculation, where treatment becomes economically justified when expected damage from inaction exceeds treatment cost, is the operational decision rule that every spray recommendation should respect. The 19.9 insects/m² threshold from the worked problem is illustrative; the same logic generalises across pests, diseases, and crops.

The chapter's coverage of smartphone-based detection deserves emphasis. PlantVillage, PlantNet, and the proliferating crop-specific apps put disease-classification AI in the hands of smallholders at zero marginal cost. Field accuracy is lower than published lab accuracy — background clutter, lighting variability, and atypical symptom presentations all reduce performance — but the comparison should be against the counterfactual of no diagnosis at all, not against published benchmarks. Even imperfect detection moves smallholder decision-making in the right direction.

The desert-locust forecasting example illustrated how AI scales beyond the field to continental outbreak management. Soil-moisture-driven breeding-area forecasts gave FAO weeks of lead time during the

2019–2021 East African upsurge, enabling earlier and more efficient spraying. We turn next from plant systems to animal systems: livestock, where the AI agenda is as much about emissions reduction as productivity.

Livestock Intelligence

A cow with a sensor on her ear is generating more data than the average smallholder farm has ever seen — and almost all of it is useful.

Kennedy Chengeta, PhD

Livestock systems contribute approximately 14% of global greenhouse gas emissions, dominated by enteric methane from ruminants. They also support roughly one billion smallholder livelihoods and provide essential nutrition in food-insecure regions. AI in livestock systems serves productivity (health, nutrition, breeding), sustainability (methane reduction, manure management, grazing optimisation), and welfare (early disease detection, behaviour monitoring) simultaneously.

12.1 Enteric Methane: The Big Lever

Methane has a 20-year global warming potential approximately 80 times that of CO₂. Cattle, sheep, and goats produce methane via microbial fermentation in the rumen — physiology, not policy, generates the emissions. Mitigation strategies include feed additives (3-NOP, Asparagopsis seaweed), genetic selection for low-methane breeds, and management changes (improved forage, optimised stocking). AI contributes by predicting individual-animal methane production from sensor data, identifying low-emitting breeding stock, and optimising feeding strategies.

12.2 Computer Vision for Body Condition Scoring

Body condition score (BCS) is the standard proxy for animal nutritional status; experienced graders score animals on a 1-9 or 1-5 scale. Computer vision systems using overhead RGB-D cameras at milking parlours, feed bunks, or water points produce continuous BCS time series at no marginal cost per animal. Detecting weight loss two weeks earlier than human observers enables earlier nutritional intervention.

12.3 Acoustic and Movement Sensors

Ear-tag accelerometers detect rumination, eating, walking, and resting behaviours. ML classifiers trained on labelled IMU data achieve activity classification accuracy above 90%. Cow oestrus detection from activity changes, illness detection from rumination drops, and lameness detection from gait anomalies have become standard for dairy systems. The same sensor architecture extended to acoustic monitoring (ear-tag microphones) detects coughing in feedlots — early respiratory disease indicator.

Example 12.1 (Pasture Allocation with RL). A grazing-management RL agent at a Western Cape dairy farm allocates 200 cows across 40 paddocks daily based on grass biomass (drone-mapped weekly), weather forecast, and milk production targets. The agent reduced supplementary feed costs by 22%, increased pasture utilisation efficiency by 18%, and reduced methane intensity (kg CO₂e per litre of milk) by 9% over the first year of deployment.

12.4 Genomic Selection and AI

Animal breeding has been transformed by genomic selection: predicting breeding values from SNP arrays rather than progeny testing. Deep-learning extensions integrate genomic data with phenotypic, environmental, and pedigree information to predict polygenic traits including methane emission, feed efficiency, and disease resistance with greater accuracy than linear models. The implication for smallholder systems is significant: African indigenous breeds carry climate-adaptation traits that genomic AI can identify and propagate without sacrificing the resilience that intensive crossing has eroded elsewhere.

12.5 Mathematical Foundations of Livestock Emissions and Genetics

12.5.1 Enteric Methane Prediction

The IPCC Tier 2 enteric methane equation:

$$\text{CH}_4 = \frac{\text{GE} \cdot Y_m / 100}{55.65}, \quad \text{kg CH}_4/\text{head}/\text{day}$$

where GE is gross energy intake (MJ/day), Y_m is methane conversion factor (% of GE, typically 6.5% for cattle), and 55.65 MJ/kg is the energy content of methane.

12.5.2 Genomic BLUP

For trait vector $\mathbf{y}$ and SNP design $\mathbf{Z}$:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\epsilon}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}\sigma_g^2),$$

with genomic relationship matrix $\mathbf{G}$. Predicted breeding values: $\hat{\mathbf{u}} = \sigma_g^2 \mathbf{Z}\mathbf{G}\mathbf{Z}^\top (\sigma_g^2 \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \sigma_e^2 \mathbf{I})^{-1} \mathbf{y}$.

12.5.3 Methane Intensity

Production-side metric:

$$\text{Methane intensity} = \frac{\text{CH}_4(\text{kg/yr})}{\text{milk (kg/yr) or beef (kg/yr)}}.$$

Multiplied by GWP₂₀ (~80) yields CO₂e intensity.

12.6 Worked Problems and Solutions

Problem 12.2 (Enteric Methane). A dairy cow consumes GE = 380 MJ/day; Y_m = 6.3%. Compute daily and annual methane emissions in kg.

Solution. Daily: $380 \cdot 0.063 / 55.65 = 23.94 / 55.65 \approx 0.430$ kg/day. Annual: $0.430 \cdot 365 \approx 157$ kg/year. At GWP₂₀ of 80: $157 \cdot 80 = 12,560$ kg CO₂e/year, or 12.56 tCO₂e/cow/year. ■

Problem 12.3 (Methane Intensity Reduction). Cow A: 7000 kg milk/year, 157 kg CH₄/year. Cow B (RL grazing optimisation): 7400 kg milk, 148 kg CH₄. Compute intensity reduction.

Solution. Intensity A: $157 / 7000 = 0.02243$ kg CH₄/kg milk. Intensity B: $148 / 7400 = 0.02000$ kg CH₄/kg milk. Reduction: $(0.02243 - 0.02000) / 0.02243 = 10.8\%$. ■

12.7 Chapter Conclusion

Livestock systems concentrate two contradictions of food-system sustainability. They are simultaneously the leading agricultural source of greenhouse gas emissions and the leading source of high-quality protein for low-income populations; they are simultaneously a target for reduction in high-income contexts and a target for expansion in low-income contexts. The AI agenda has to navigate both.

The enteric methane treatment provided both the IPCC Tier 2 equation and its practical implications. A dairy cow emits roughly 12.6 tCO₂e per year on a GWP-20 basis; methane intensity per kilogram of product, not per animal, is the policy-relevant metric. Productivity improvements – better genetics, better nutrition, better grazing management – reduce intensity even as they may increase per-animal emissions. The worked problem on methane intensity reduction illustrated a 10.8% intensity gain from a modest productivity gain combined with a modest emissions reduction.

Genomic selection through BLUP and its modern extensions was the chapter's most mathematically dense topic. The intuition is simple: SNP genotyping at a few thousand markers lets breeders predict an animal's genetic merit before it has produced any offspring or any milk. The accuracy of prediction depends on training-population size and trait heritability; methane-trait heritability of 0.2–0.3 makes the trait amenable to selection. Smallholder access to genomic-selection benefits requires shared breed reference populations and cooperative service delivery models; technical feasibility does not imply equitable access.

Computer-vision-based body-condition scoring, lameness detection, and heat-detection systems extend the AI stack to welfare. The same imagery that supports productivity supports welfare; the trade-off the chapter warned about is between welfare monitoring and surveillance creep. We close Part II's chapter sequence on commercial production with the final chapter on the special case of smallholder agriculture.

AI for Smallholder Farmers

A model that works for one large farm but not for one hundred small farms is a model that has solved the wrong problem.

Kennedy Chengeta, PhD

Smallholder agriculture — farms typically below 2 hectares, often below 0.5 — supplies more than 70% of food in Sub-Saharan Africa, much of South Asia, and significant portions of Latin America. Smallholders face fundamentally different constraints than industrial farms: limited capital, intermittent connectivity, low literacy, multiple crops per field, mixed crop-livestock systems, and acute climate vulnerability. AI for smallholders requires design choices that the industrial-agriculture playbook does not anticipate.

13.1 Design Constraints That Matter

Six constraints recur. *Connectivity*: bandwidth is intermittent, 2G is still common, data costs are high relative to income. *Literacy*: text-heavy interfaces fail; voice and pictograms succeed. *Languages*: dozens of languages within most countries; major languages are still under-represented in NLP datasets. *Hardware*: feature phones and shared low-end smartphones dominate. *Data*: satellite, weather, and price data are public; farm-level data must be collected at marginal-or-zero cost or it will not be collected. *Trust*: agricultural advisory has historically been unreliable; new advice systems are met with appropriate scepticism.

13.2 Delivery Channels

SMS, USSD, voice (IVR), WhatsApp, and Telegram are the primary delivery channels. Apollo Agriculture (Kenya) and Hello Tractor (Nigeria) use combinations of these to deliver advisory, financing, and equipment access to smallholders. iCow uses voice-based advisory in Kiswahili and English. Digital Green broadcasts video-based extension to viewing groups. The general pattern is that advice has to meet farmers where they are technologically, not where vendors wish they were.

13.3 Models That Work in Data-Scarce Contexts

Three patterns work when farm-level data is sparse. First, transferring satellite-derived features (NDVI, SAR backscatter, weather) from public datasets where ground truth is impossible to collect at scale. Second, using farmer-reported data via simple checklists (planting date, variety, problems observed) collected through advisory apps as a side-effect of providing advice. Third, foundation

models pretrained on global data fine-tuned on small in-country labelled datasets.

Example 13.1 (Maize Advisory in Western Kenya). A mobile maize advisory deployed to 14,000 smallholders in Western Kenya combines field-level satellite NDVI, gridded weather forecasts, soil-data inferences from SoilGrids, and farmer-reported planting dates to issue weekly advice on fertilisation, weeding, and pest scouting. After two seasons, farms receiving advice showed 19% higher yields and 31% lower fertiliser losses than matched controls. The system runs on USD 0.30 of cloud cost per farm per season.

13.4 Bundled Services

AI advisory alone often produces small adoption gains because farmers face binding constraints other than information — credit, inputs, market access. Bundled services that combine advisory with input financing, off-take agreements, and crop insurance generate substantially better outcomes. One Acre Fund, Apollo Agriculture, and Pula Advisors each combine AI-driven yield forecasts with financial products that translate the forecasts into farmer cash flows. The AI is essential but not sufficient; the financial integration is what closes the value-creation loop.

13.5 Risks and Failure Modes

Smallholder AI fails in predictable ways. Models trained on data from richer farms generalise poorly to poorer farms. Advisory in dominant languages excludes minority-language speakers. Recommendations that assume access to inputs the farmer cannot afford produce frustration, not improvement. Algorithmic credit scoring excludes farmers without digital footprints. The remedies are participatory design, ground-truth datasets that include the marginalised, transparent decision rules, and human-mediated escalation paths for cases the model cannot handle.

13.6 Mathematical Foundations of Smallholder Advisory Economics

13.6.1 Treatment-Effect Estimation

For outcome Y_i , treatment $T_i \in \{0, 1\}$, covariates $\mathbf{X}_i$:

$$ATT = \mathbb{E}[Y_i^{(1)} - Y_i^{(0)} \mid T_i = 1].$$

With matching on propensity score $e(\mathbf{x}) = \Pr(T = 1 \mid \mathbf{X} = \mathbf{x})$:

$$\widehat{ATT} = \frac{1}{n_1} \sum_{i:T_i=1} [Y_i - \widehat{m}_{(0)}(\mathbf{X}_i)],$$

where $\widehat{m}_{(0)}$ is an outcome model fit on controls.

13.6.2 Bundled-Service Net Present Value

For a smallholder served with advisory + finance + insurance over years $1, \dots, T$, NPV is

$$NPV = -C_0 + \sum_{t=1}^T \frac{\Delta\pi_t - F_t - P_t + L_t}{(1 + \rho)^t},$$

where C_0 is up-front cost, $\Delta\pi_t$ is incremental profit, F_t finance charges, P_t insurance premiums, L_t insurance pay-outs.

13.6.3 SMS Advisory Cost Function

Per-farmer cost:

$$c_{\text{adv}} = c_{\text{compute}} + c_{\text{SMS}} \cdot n_{\text{messages}} + c_{\text{ground-truth}} / N_{\text{farmers}}.$$

Scaling matters: ground-truth costs amortise across farmers, SMS costs are linear, compute costs are essentially zero per farmer at modern unit costs.

13.7 Worked Problems and Solutions

Problem 13.2 (ATT Estimation). $n_1 = 6000$ treated farmers, mean yield 4.1 t/ha. Matched control predicted yields average 3.45 t/ha. Compute ATT in t/ha and aggregate.

Solution. $\text{ATT} = 4.1 - 3.45 = 0.65$ t/ha. Aggregate impact: $6000 \cdot 0.65 = 3,900$ tonnes additional production. At average maize price USD 280/t, GDP contribution $\approx$ USD 1.09M. ■

Problem 13.3 (Advisory Cost per Farmer). Compute c_{adv} : $c_{\text{compute}} = 0.05$ USD, $c_{\text{SMS}} = 0.012$ USD, 18 messages/season, ground-truth USD 4,000 total amortised across 14,000 farmers.

Solution. Compute: 0.05. SMS: $0.012 \cdot 18 = 0.216$. Ground-truth: $4,000 / 14,000 = 0.286$. Total per farmer: $0.05 + 0.216 + 0.286 = 0.552$ USD/season. ■

13.8 Chapter Conclusion

Smallholder agriculture is where AI for sustainability has the largest potential impact and the lowest baseline of deployment. The yield-gap data is unambiguous: smallholder yields in much of Sub-Saharan Africa run at 20%–30% of agronomic potential. Closing even half of that gap would feed an additional billion people, lift millions of households out of poverty, and reduce land-use change pressure dramatically. The mathematical framework for yield-gap decomposition — separating water-limited from nutrient-limited from management-limited components — gives both diagnosis and intervention targeting.

The chapter argued, and the ATT worked problem confirmed, that SMS-based advisory at scale produces measurable yield improvements at near-zero marginal cost per farmer. The 0.55 USD/season figure from the cost-breakdown problem is illustrative; the actual costs of operational SMS advisory at scale have been in the range of 0.30–0.80 USD per farmer per season. Even at the high end, the cost-benefit ratio is favourable; even at modest yield gains of 0.3–0.5 t/ha, the returns are 30–50× the advisory cost.

What the chapter emphasised throughout is that technical feasibility is the easy part. The harder work is delivery: building trust with farmers, working through cooperatives and extension services, calibrating recommendations to local crop varieties and soil conditions, and integrating advisory with input supply, finance, and insurance. Bundled services consistently outperform unbundled advisory because they remove the constraints that bind smallholder decision-making. AI is one component of a bundle; selling it as a standalone product almost always fails.

This chapter closes Part II. The chapters that follow shift from agriculture to the next pillar of sustainability AI: climate forecasting. We turn in Chapter 14 to the physical climate models that underlie everything subsequent.

Part III

Climate Forecasting and Risk Modelling

Physical Climate Models and AI

For decades, climate modelling has meant solving the equations of fluid dynamics on a discrete grid. The next decade will mean solving them in collaboration with neural networks.

Kennedy Chengeta, PhD

Physical climate models — general circulation models or earth system models — discretise the atmosphere, ocean, land, and cryosphere onto computational grids and integrate the conservation equations of mass, momentum, energy, and tracers forward in time. They have been the dominant tool of climate science for fifty years and have produced the IPCC assessments that anchor global policy. They also have known limitations that AI is now beginning to address.

14.1 GCMs and ESMs: The Existing Toolkit

A modern earth system model couples atmosphere, ocean, sea ice, land surface, and biogeochemistry. Spatial resolution is typically 100km horizontally; some recent models reach 25km. Temporal resolution is sub-hourly. The CMIP6 ensemble (Coupled Model Intercomparison Project, sixth phase) underlying the IPCC AR6 includes around fifty models from research centres globally. They project, for the high-emissions SSP5-8.5 scenario, mean global warming of 3.3-5.7°C by 2100; for the low-emissions SSP1-1.9, around 1.0-1.8°C.

Key limitations: cloud and convection sub-grid processes must be parameterised because they occur at scales smaller than the grid. Land-surface processes (vegetation dynamics, soil moisture) are simplified. Ice-sheet dynamics are crude. Human responses (technology, policy, behaviour) are imposed as scenarios rather than computed.

14.2 AI as Surrogate and Emulator

Training a neural network to reproduce the behaviour of an expensive climate-model component opens new possibilities. ClimSim and ClimaX are emulators that replicate GCM dynamics at orders-of-magnitude lower computational cost, enabling large ensembles for uncertainty quantification. NeuralGCM (Google, 2024) replaces parts of the dynamical core with neural networks while preserving conservation laws, achieving forecast skill competitive with operational systems at lower cost.

14.3 Hybrid Modelling

The most consequential recent development is hybrid modelling: physical equations for parts of the system where physics is well-understood (large-scale dynamics, radiation), neural networks for parts where parameterisation has been the weak link (clouds, convection, boundary layer, land-atmosphere coupling). Neural-network parameterisations trained on high-resolution simulations have substantially reduced biases in coarse-resolution models for tropical convection and Arctic clouds.

Example 14.1 (ML Convection Parameterisation). A convolutional neural network trained on 50km-resolution cloud-resolving simulations and embedded in a 100km GCM as the convection scheme reduced tropical precipitation bias by 35% compared to the conventional Tiedtke parameterisation, at 8% additional compute cost. The remaining bias was traced to mismatches between the training-data climate and the target climate — a generalisation challenge that motivates online learning approaches.

14.4 Limits of AI in Climate Modelling

AI does not replace physics in climate modelling — it complements it. Three limits constrain pure-AI approaches. First, training data are limited to the climate we have observed; extrapolation to substantially warmer climates requires physical principles AI alone cannot supply. Second, conservation laws (mass, energy, momentum) must be enforced, not learned, for long integrations to remain stable. Third, the social and economic decisions that determine emission trajectories are not amenable to climate-model simulation. The next decade will likely see most operational climate-modelling systems adopt hybrid AI-physics architectures, with pure-physics and pure-AI approaches retreating to specialised niches.

14.5 Mathematical Foundations of Climate Modelling

14.5.1 Primitive Equations

Atmospheric general circulation models integrate the primitive equations of fluid dynamics on a rotating sphere. In spherical coordinates with pressure as vertical coordinate:

$$\begin{aligned}\frac{D\mathbf{u}_h}{Dt} + f\hat{\mathbf{k}} \times \mathbf{u}_h &= -\nabla_p \Phi + \mathbf{F}, \\ \frac{\partial \Phi}{\partial p} &= -\frac{RT}{p}, \\ \nabla_p \cdot \mathbf{u}_h + \frac{\partial \omega}{\partial p} &= 0, \\ \frac{DT}{Dt} - \frac{\omega}{\rho c_p} &= \frac{Q}{c_p},\end{aligned}$$

where $\mathbf{u}_h$ is horizontal velocity, f Coriolis, Φ geopotential, T temperature, $\omega = Dp/Dt$ vertical pressure velocity, Q diabatic heating rate, $\mathbf{F}$ frictional forces.

14.5.2 Energy Balance Model

The simplest climate model is the global energy balance:

$$C \frac{dT}{dt} = \frac{S(1-\alpha)}{4} - \sigma T_e^4 + F,$$

where S solar constant, α planetary albedo, σ Stefan-Boltzmann, T_e effective emission temperature, F radiative forcing, C heat capacity. Equilibrium climate sensitivity:

$$\Delta T_{\text{eq}} = \frac{F}{\lambda}, \quad \lambda = 4\sigma T_e^3 + \text{feedbacks}.$$

14.5.3 ML Parameterisation Loss

For a high-resolution training simulation $\mathbf{Y}_h$ and learned parameterisation $g_\theta(\mathbf{X}_c)$ acting on coarse state $\mathbf{X}_c$:

$$\mathcal{L}(\theta) = \mathbb{E}_{(\mathbf{X}_c, \mathbf{Y}_h)} \|g_\theta(\mathbf{X}_c) - \langle \mathbf{Y}_h \rangle\|^2 + \lambda \mathcal{R}_{\text{cons}}(\theta),$$

with $\mathcal{R}_{\text{cons}}$ enforcing conservation of mass, energy, and momentum.

14.6 Worked Problems and Solutions

Problem 14.2 (Equilibrium Sensitivity from Forcing). Doubling CO_2 produces $F = 3.7 \text{ W/m}^2$. Climate feedback parameter $\lambda = 1.16 \text{ W/m}^2/\text{K}$. Compute equilibrium climate sensitivity.

Solution. $\Delta T = F/\lambda = 3.7/1.16 \approx 3.19^\circ\text{C}$. ■

Problem 14.3 (Effective Emission Temperature). Earth's solar constant $S = 1361 \text{ W/m}^2$, albedo $\alpha = 0.30$. Compute T_e .

Solution. Absorbed: $S(1 - \alpha)/4 = 1361 \cdot 0.70/4 = 238.2 \text{ W/m}^2$. $\sigma T_e^4 = 238.2 \Rightarrow T_e = (238.2/5.67 \times 10^{-8})^{1/4}$. $238.2/5.67 \times 10^{-8} = 4.201 \times 10^9$. $T_e = (4.201 \times 10^9)^{0.25}$. Take log: $0.25 \cdot \log(4.201 \times 10^9) = 0.25 \cdot 9.6234 = 2.406$, so $T_e = 10^{2.406} = 254.6 \text{ K} = -18.6^\circ\text{C}$. The 33 K gap to surface temperature ($\sim 288 \text{ K}$) is the natural greenhouse effect. ■

14.7 Chapter Conclusion

Physical climate models are among the most computationally demanding scientific instruments humans have ever built. A single century-long simulation of an atmosphere-ocean coupled model at 25 km resolution requires tens of millions of CPU-hours on the world's largest supercomputers; the CMIP ensemble represents a coordinated international investment of billions of dollars per cycle. The chapter situated this machinery, derived the primitive equations that drive it, and explained the energy-balance reductions that summarise its behaviour at the global scale.

The equilibrium-climate-sensitivity worked problem illustrated why the field cares so much about the feedback parameter λ . A doubling of CO_2 produces 3.7 W/m^2 of forcing; whether that produces 2°C or 4°C of warming depends on the magnitudes of the water-vapour, cloud, and ice-albedo feedbacks. The mathematical structure of the energy balance is elementary; the uncertainty in the feedback parameter is what drives the wide range of model projections.

The hybrid ML-physical-model section forecast the field's near-term direction. NeuralGCM, GraphCast, and Pangu-Weather have demonstrated that machine-learning models can match or exceed numerical weather prediction for medium-range forecasting at a fraction of the computational cost. The extension to climate-timescale projection is under active research, but the trajectory is clear:

by the time of the second edition of this book, ML-physical hybrid models will likely contribute materially to operational climate projection alongside traditional GCMs.

What ML cannot replace is the physical-process understanding embedded in process-based models. ML produces predictions; it does not, in general, produce mechanistic explanations. The combination of the two — physical models for mechanistic understanding, ML for computational efficiency and empirical correction — is the productive frontier. We move next from the climate-system foundations to the operational application: weather forecasting.

ML-Augmented Weather Forecasting

*In 2023, neural networks beat the European model at its own ten-day forecast.
By 2025 they cost a thousandth as much to run.*

Kennedy Chengeta, PhD

Weather forecasting has been transformed in the past three years. Neural-network-based global forecasting systems — GraphCast (DeepMind), Pangu-Weather (Huawei), FourCastNet (NVIDIA), AIFS (ECMWF) — now match or exceed the accuracy of the long-dominant numerical weather prediction (NWP) systems on most variables and forecast horizons, at orders-of-magnitude lower compute cost. This chapter treats how the systems work, where they excel, and where they still defer to physical models.

15.1 From NWP to Neural Forecasting

Operational NWP systems integrate the primitive equations of atmospheric fluid dynamics on grids of 9-25km horizontal resolution, with sub-hourly time steps, on supercomputers operating around the clock. The European Centre for Medium-Range Weather Forecasts (ECMWF) operational forecast (IFS) has been the global benchmark for two decades.

ML forecasting systems pose the problem differently. Given the current atmospheric state $\mathbf{x}_t$ on a regular grid, learn a function f that produces $\mathbf{x}_{t+\Delta t}$. Training data are forty years of ERA5 reanalysis, with the target being the next state. Architectures are graph neural networks (GraphCast), 3D transformers (Pangu), Fourier neural operators (FourCastNet), or hybrids. Training takes weeks on hundreds of GPUs; inference takes seconds on one.

15.2 Forecast Skill Comparisons

On standard verification metrics — root mean squared error of geopotential height at 500hPa, anomaly correlation coefficient, precipitation skill scores — the leading ML systems now match or beat IFS at lead times of 1-10 days for most variables. Pangu-Weather demonstrated this first in 2022; GraphCast confirmed it across more variables in 2023; AIFS, ECMWF's own ML model, achieved operational status in 2024. The cost asymmetry is striking: an IFS forecast requires roughly 1000 CPU-hours; a GraphCast forecast requires roughly 1 GPU-second.

15.3 Where Physics Still Wins

ML systems struggle on three categories of forecast. Extreme events outside the training distribution (Hurricane Otis 2023, summer 2022 European drought) are systematically underpredicted because the events were rare in the 40-year training data. Conservation properties (total moisture, total energy) drift over long integrations because they are not enforced as hard constraints. Sub-daily timescales (convection, sea breeze, mountain effects) are smoothed because the training data are 6-hourly.

Hybrid forecasting systems — ML for the well-trained envelope, physics for the tails and the constraints — are the operational frontier.

15.4 Implications for Sustainability

Better, cheaper weather forecasts compound across every sustainability domain in this book. Renewable energy dispatch, agricultural advisory, flood warning, water-resource management, transport logistics, and emergency response all run on weather forecasts. A 10% improvement in forecast skill at 5-day lead time has been estimated to save USD 20-30 billion annually in the US alone. Globalised to comparable order, the value runs into the hundreds of billions, and the carbon co-benefits — avoided crop losses, reduced grid balancing reserves, optimised renewable dispatch — are substantial.

15.5 Mathematical Foundations of Weather Forecast Skill

15.5.1 Anomaly Correlation Coefficient

For forecast anomaly $f' = f - \bar{c}$ and observed anomaly $o' = o - \bar{c}$ (climatology $\bar{c}$):

$$\text{ACC} = \frac{\sum_i f'_i o'_i}{\sqrt{\sum_i f'^2_i \sum_i o'^2_i}}.$$

Operational forecasts are considered useful when $\text{ACC} > 0.6$.

15.5.2 Continuous Ranked Probability Score

For predictive CDF $F(y)$ and observation o :

$$\text{CRPS}(F, o) = \int_{-\infty}^{\infty} [F(y) - \mathbb{I}(y \geq o)]^2 dy.$$

Reduces to MAE for deterministic forecasts; rewards both calibration and sharpness for probabilistic forecasts. Lower is better.

15.5.3 Brier Score

For probability forecast p of binary event with outcome $o \in \{0, 1\}$:

$$\text{BS} = (p - o)^2,$$

with decomposition into reliability, resolution, and uncertainty:

$$\overline{\text{BS}} = \text{REL} - \text{RES} + \text{UNC}.$$

15.5.4 Graph-Based Forecasting

For state $\mathbf{x}_t \in \mathbb{R}^N$ on a graph with adjacency $\mathbf{A}$, a graph neural network forecast applies

$$\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + g_{\theta}(\mathbf{x}_t, \text{MessagePass}(\mathbf{A}, \mathbf{x}_t)),$$

with rollout via repeated application.

15.6 Worked Problems and Solutions

Problem 15.1 (Brier Score Comparison). Two forecasts of next-day rainfall (yes/no): A predicts 0.7, B predicts 0.4. Observation: it rained ($o = 1$). Compute Brier scores.

Solution. A: $(0.7 - 1)^2 = 0.09$. B: $(0.4 - 1)^2 = 0.36$. Forecast A substantially better (lower BS).

■

Problem 15.2 (ACC Calculation). Three-station forecast anomalies $(2.1, -0.8, 1.4)$, observed $(1.7, -1.2, 1.0)$. Compute ACC.

Solution. Numerator: $2.1 \cdot 1.7 + (-0.8)(-1.2) + 1.4 \cdot 1.0 = 3.57 + 0.96 + 1.40 = 5.93$. $\sum f^2 = 4.41 + 0.64 + 1.96 = 7.01$; $\sum o^2 = 2.89 + 1.44 + 1.00 = 5.33$. Denominator = $\sqrt{7.01 \cdot 5.33} = \sqrt{37.36} = 6.113$. $\text{ACC} = 5.93/6.113 = 0.970$, indicating excellent forecast–observation correspondence.

■

15.7 Chapter Conclusion

The state of weather forecasting in 2025 differs from the state in 2020 more than the state in 2020 differed from 2000. GraphCast, Pangu-Weather, FourCastNet, and AIFS have demonstrated that transformer-based models trained on ERA5 reanalysis data can produce ten-day forecasts at operational accuracy in seconds of GPU time, compared to the hours of supercomputer time required by physics-based models. The ECMWF integrated AIFS into operational ensemble forecasting in 2024; other meteorological services are following.

What the chapter emphasised, through the mathematical treatment of skill scoring, is that probabilistic forecast quality is the relevant metric for operational decisions, not point-forecast accuracy. The Brier-score worked problem, the ACC worked problem, and the CRPS framework give the reader the tools to evaluate probabilistic forecast skill rigorously. The anomaly-correlation coefficient of 0.97 from the worked problem is unusually high; operational forecasts achieve ACC above 0.6 out to about seven days for surface temperature and about five days for precipitation, with ML-based systems extending these horizons by one to two days relative to physics-based systems.

What ML weather forecasting has not yet solved is initial-condition generation. Data assimilation — combining short-range forecasts with observations to produce the analysis state that begins the next forecast cycle — remains the territory of physics-based methods. Hybrid systems that use physics-based assimilation to generate initial conditions and ML-based propagation to roll out the forecast represent the current best practice.

The chapter’s coverage of nowcasting closed the temporal loop. Sub-hourly forecasts of precipitation

and lightning, driven by radar and satellite data through convolutional and graph models, support immediate-response applications from solar-farm dispatch to emergency management. These short-horizon forecasts complement the multi-day forecasts treated elsewhere. We turn next to the high-impact tail of weather: extreme events.

Extreme Event Prediction

Climate normals will not save us. The events at the tail of the distribution are where the damages live, and the tails are getting fatter.

Kennedy Chengeta, PhD

Extrême weather and climate events — heatwaves, tropical cyclones, flooding rains, severe drought — cause approximately 90% of weather-related economic damages and the majority of weather-related deaths. They are also the events that conventional ML forecasting systems handle least well, because they are rare, by definition under-represented in training data, and intensifying in ways that statistical methods built on stationarity cannot capture.

16.1 The Statistics of Tails

Extreme value theory provides the mathematical foundation. Block maxima follow generalised extreme value distributions; threshold exceedances follow generalised Pareto distributions. Stationarity — the assumption that distributional parameters are time-invariant — is the standard simplification, and it is increasingly wrong. Climate-conditioned extreme value models allow distribution parameters to be functions of covariates including global mean temperature, regional climate indices, and large-scale circulation patterns.

Definition 16.1 (Conditional GEV). For a covariate vector $\mathbf{z}$ (e.g. global mean temperature anomaly):

$$G(x | \mathbf{z}) = \exp \left[- \left(1 + \xi(\mathbf{z}) \frac{x - \mu(\mathbf{z})}{\sigma(\mathbf{z})} \right)^{-1/\xi(\mathbf{z})} \right]$$

with neural networks parameterising μ , σ , ξ as functions of $\mathbf{z}$.

16.2 Tropical Cyclone Forecasting

Hurricane track and intensity forecasting has been a particular focus. Track forecasts have improved steadily through better NWP; intensity forecasts have been more stubborn because rapid intensification depends on small-scale processes. ML systems have produced step-changes here. NVIDIA’s HURDAT-trained intensity model and NOAA’s HAFS-AI hybrid forecast intensity 24-72h ahead with mean absolute errors 20-30% lower than conventional systems. Combined with satellite-driven structure analysis, ML cyclone forecasts now anchor evacuation decisions for hundreds of millions of coastal residents.

16.3 Flash Flood and Heatwave Prediction

Flash floods occur on timescales of hours; heatwaves on timescales of days. Both require high-resolution forecasting and coupling to hydrological or human-impact models. Google's flood forecasting system (operating across India, Bangladesh, parts of Africa) chains ML water-flow models with HEC-RAS hydrodynamics and global terrain to produce inundation forecasts at 100m resolution, 7 days ahead. Heatwave-mortality forecasts integrate weather forecasts with epidemiological dose-response models to produce health-system planning warnings.

16.4 Compound and Cascading Events

The frontier in extremes research is compound events: heatwave-plus-drought (Russia 2010), heatwave-plus-fire (Australia 2019-20), cyclone-plus-storm-surge (Bangladesh 2007), drought-plus-flood (Horn of Africa 2023). Compound risks are not the sum of individual hazard risks; nonlinear interactions amplify damages. ML on multivariate extremes has produced workable joint-distribution models in the past three years. Operational compound-event warning systems are not yet routine but are in development at NCAR, ECMWF, and the UK Met Office.

16.5 Mathematical Foundations of Extreme Value Theory

16.5.1 Generalised Extreme Value Distribution

For block maxima $\{M_n\}$:

$$G(z; \mu, \sigma, \xi) = \exp \left[- \left(1 + \xi \frac{z - \mu}{\sigma} \right)^{-1/\xi} \right], \quad 1 + \xi(z - \mu)/\sigma > 0.$$

$\xi > 0$: heavy-tailed Fréchet; $\xi = 0$: Gumbel; $\xi < 0$: Weibull (bounded).

16.5.2 Generalised Pareto for Threshold Exceedances

For $X > u$ above threshold u :

$$F_u(y) = 1 - \left(1 + \xi \frac{y}{\tilde{\sigma}} \right)^{-1/\xi},$$

with $\tilde{\sigma} = \sigma + \xi(u - \mu)$.

16.5.3 Return Levels

The T -year return level under the GEV:

$$z_T = \mu - \frac{\sigma}{\xi} \left[1 - \left(-\ln \left(1 - \frac{1}{T} \right) \right)^{-\xi} \right].$$

Under non-stationarity (climate change), parameters depend on covariates: $\mu(t) = \mu_0 + \mu_1 \Delta T_{\text{global}}(t)$, etc.

16.5.4 Compound-Event Joint Probability

For events A (heatwave) and B (drought) with copula C on uniform margins:

$$\Pr(A \cap B) = C(F_A(a), F_B(b)).$$

Tail dependence: $\lambda_U = \lim_{u \rightarrow 1^-} \Pr(F_B(B) > u \mid F_A(A) > u)$.

16.6 Worked Problems and Solutions

Problem 16.2 (Return Level Computation). Annual maximum daily rainfall fits GEV with $\mu = 80$, $\sigma = 22$, $\xi = 0.15$ (mm). Compute the 100-year return level.

Solution. $y = -\ln(1 - 0.01) = -\ln(0.99) = 0.01005$. $y^{-\xi} = 0.01005^{-0.15}$. Take log: $-0.15 \cdot \ln(0.01005) = -0.15 \cdot (-4.600) = 0.6900$. $y^{-\xi} = e^{0.6900} = 1.9937$. $z_{100} = 80 - (22/0.15)(1 - 1.9937) = 80 - 146.67 \cdot (-0.9937) = 80 + 145.74 \approx 225.7$ mm. ■

Problem 16.3 (Stationary vs Non-Stationary). Same parameters but $\mu(t) = 80 + 4\Delta T_{\text{global}}$ and current $\Delta T = 1.2^\circ\text{C}$. Compute current 100-year return level and compare with pre-industrial.

Solution. Current $\mu = 80 + 4 \cdot 1.2 = 84.8$ mm. $z_{100,\text{now}} = 84.8 + 145.74 = 230.5$ mm. Pre-industrial ($\Delta T = 0$): $z_{100,\text{PI}} = 80 + 145.74 = 225.7$ mm. Increase ≈ 4.8 mm or 2.1%; equivalent to a return-period shortening from 100 to about $1/(1 - G(225.7))$ years under current climate. ■

16.7 Chapter Conclusion

Extreme events are where climate change becomes consequential. A 2°C shift in mean temperature is barely perceptible day-to-day; a shift in the once-in-fifty-year heatwave from 42°C to 46°C overwhelms cooling infrastructure, collapses crop yields, and kills people. The chapter developed the mathematical machinery — extreme-value theory through the generalised extreme-value and generalised Pareto distributions, return-level computation, and non-stationary extensions that allow parameters to depend on climate covariates — that operational extreme-event prediction requires.

The non-stationary worked problem illustrated the practical implication. Under stationary assumptions, the 100-year return level for the example distribution was 225.7 mm; under a non-stationary location parameter linked to global mean temperature, the current 100-year level rises to 230.5 mm, with the return period for the historical 225.7 mm event shortening commensurately. This is the mathematical structure of the “new normal” that climate communications often gesture toward without quantifying.

Compound-event analysis — treating heatwave and drought, flood and storm-surge, or wildfire and heat-dome jointly rather than as independent hazards — is a frontier of the field. The copula framework introduced gives the reader the tools to model joint tail dependence formally. Operational early-warning systems are beginning to incorporate this; the WWA attribution programme and the major reinsurers have led the methodological development, and the techniques are now diffusing into public-sector emergency management.

The chapter’s coverage of attribution science closed the analytical loop. By comparing extreme-event return periods in factual and counterfactual worlds (with and without anthropogenic forcing), attribution studies quantify how much climate change contributed to a specific event. The methodology is now well established; the legal and political applications are accelerating. We turn next to the slower-onset extremes that dominate the impact accounting: drought and flood.

Drought and Flood Early Warning

A flood warning twelve hours early saves the harvest. Twelve hours late, it documents the loss.

Kennedy Chengeta, PhD

Drought and flood are the dominant climate-driven causes of food insecurity, displacement, and economic loss in Sub-Saharan Africa and South Asia. AI-driven early warning systems — combining satellite remote sensing, weather forecasts, and hydrological models — have transformed lead times and accuracy. This chapter treats the operational architectures, with particular attention to the African deployments that have set the global state of the art.

17.1 Drought Indices and Their AI Augmentation

Drought is multidimensional — meteorological (precipitation deficit), agricultural (soil-moisture and crop deficit), hydrological (streamflow and groundwater deficit). Standard indices include the Standardised Precipitation Index (SPI), the Standardised Precipitation-Evapotranspiration Index (SPEI), the Palmer Drought Severity Index (PDSI), and the soil-moisture-based U.S. Drought Monitor.

ML augmentation operates at three levels. First, gap-filling sparse station data with satellite remote-sensing inputs (CHIRPS, GPM, SMAP). Second, learning corrections to physically-based drought-index outputs. Third, projecting drought indices forward at sub-seasonal to seasonal lead times conditioning on climate predictors (SST patterns, MJO state, ENSO indices).

17.2 Flood Forecasting

Flood forecasting chains weather forecast inputs through hydrological models (rainfall-runoff) and hydraulic models (river dynamics, inundation extent). Each step has been augmented by ML: weather corrections via post-processing (DLQR, neural deformation), rainfall-runoff via LSTM and transformer models that outperform traditional conceptual hydrology, and inundation via CNN flood-extent regression on terrain plus discharge.

Example 17.1 (Flood Hub: Niger River Basin). Google Flood Hub deployed across the Niger River basin in 2023 produces 7-day inundation forecasts at 30m resolution for over 80 million people. Verification on 2023 flooding events showed mean absolute error of inundation extent of 0.27 km² per square kilometre forecast, with hit rate of 78% for major events. The deployment cost approximately USD 0.001 per beneficiary per year in compute and bandwidth, an order of

magnitude lower than the human-centred warning systems it complemented.

17.3 African Risk Capacity and Parametric Insurance

Drought and flood early-warning data feed into parametric insurance — payouts triggered by index thresholds rather than damage assessments. The African Risk Capacity (ARC) is a sovereign-level pool that pays member countries when satellite-derived rainfall anomalies cross thresholds. Pula, Acre Africa, and Climatebridge provide farmer-level parametric products. The advantage is speed: payouts within days of trigger, before secondary impacts (asset sales, school dropout, malnutrition) take hold. The dependency on AI is total: index calculation, basis-risk assessment, and trigger verification all run on automated remote-sensing-plus-ML pipelines.

17.4 Last-Mile Warning Delivery

A perfect forecast is worthless if the warning does not reach the person at risk in a form they can act on. SMS, USSD, radio bulletins, mosque announcements, community volunteer networks, and chatbots over WhatsApp are the dominant last-mile channels. Successful systems integrate: technical forecast, local-language translation, action-specific advisory ("move livestock to higher ground", "harvest early, store grain dry"), and feedback loops to verify reception and action. The Bangladesh Cyclone Preparedness Programme remains the global gold standard; African systems are catching up rapidly.

17.5 Mathematical Foundations of Drought Indices and Hydrological Routing

17.5.1 SPI Construction

Cumulative precipitation X over a window is fitted to a gamma distribution $G(x; \alpha, \beta)$, then transformed to standard normal:

$$\text{SPI} = \Phi^{-1}(G(x; \hat{\alpha}, \hat{\beta})).$$

SPI thresholds: -1.0 moderate drought, -1.5 severe, -2.0 extreme.

17.5.2 SPEI: Adding Evapotranspiration

SPEI uses the difference $D = P - \text{PET}$ fitted to log-logistic and inverse-CDF transformed similarly. Captures temperature-driven drought intensification absent from precipitation-only indices.

17.5.3 Rainfall-Runoff Modelling

A linear-reservoir cascade with storage $S(t)$:

$$\frac{dS}{dt} = P_e(t) - \frac{1}{k}S(t), \quad Q(t) = \frac{1}{k}S(t),$$

where P_e is effective precipitation, k residence time. ML rainfall-runoff models (LSTM) replace this with:

$$Q_{t+1} = f_{\theta}(P_{t-h:t}, T_{t-h:t}, \mathbf{c}_{\text{static}}),$$

where h is history window, $\mathbf{c}_{\text{static}}$ catchment properties.

17.5.4 Inundation Mapping

Manning's equation for steady uniform flow:

$$Q = \frac{1}{n} A R^{2/3} S^{1/2},$$

with cross-sectional area A , hydraulic radius R , slope S , roughness n .

17.6 Worked Problems and Solutions

Problem 17.2 (SPI Computation). A 6-month rainfall window has $X = 240$ mm. Historical fit: $\hat{\alpha} = 4.5$, $\hat{\beta} = 60$ (gamma scale). Compute SPI given $G(240) = 0.18$.

Solution. $\text{SPI} = \Phi^{-1}(0.18) \approx -0.915$. This indicates near-moderate drought (threshold at -1.0). ■

Problem 17.3 (Runoff Routing). Linear reservoir, $k = 12$ hours, initial $S_0 = 100$ mm, $P_e = 0$ for 24 hours after the storm. Compute storage and discharge at $t = 6$ and $t = 24$ h.

Solution. $S(t) = S_0 e^{-t/k}$, $Q(t) = S(t)/k$. $t = 6$: $S = 100 e^{-0.5} = 60.65$ mm; $Q = 60.65/12 = 5.05$ mm/h. $t = 24$: $S = 100 e^{-2} = 13.53$ mm; $Q = 1.13$ mm/h. The exponential recession reflects the linear-reservoir assumption. ■

17.7 Chapter Conclusion

Drought and flood early-warning systems are arguably the most consequential applications of AI for adaptation. A correctly timed drought warning can trigger destocking, water-rationing, and food-aid procurement weeks before the worst impacts hit; a correctly timed flood warning can evacuate populations, pre-position emergency supplies, and protect infrastructure. The chapter walked through the technical machinery of each: SPI and SPEI for drought monitoring, rainfall-runoff modelling for flood prediction, and the inundation-mapping pipelines that combine hydrological forecast with topographic and infrastructural data.

The SPI worked problem demonstrated the technique: a 6-month rainfall window producing an empirical CDF value of 0.18 corresponds to an SPI of about -0.92 , near-moderate drought. The interpretation is statistical: the current precipitation total is in the 18th percentile of historical 6-month windows for the same season. SPEI extends this to incorporate evapotranspiration; under warming conditions, SPEI captures drought intensification that SPI alone misses, because warmer temperatures increase atmospheric water demand even when precipitation is unchanged.

Rainfall-runoff modelling through linear-reservoir cascades, the unit hydrograph, and modern LSTM-based catchment models gives the reader the foundation for flood forecasting. The linear-reservoir worked problem illustrated the exponential recession behaviour that all catchment models must capture. Operational systems combine ML-based runoff prediction with hydraulic routing models such as HEC-RAS and LISFLOOD to produce inundation maps with hours to days of lead time, depending on basin size.

Google's operational flood-forecasting system, covering more than 100 countries, illustrates the scale

at which AI-enabled flood warning now operates. Its mathematical core is conceptually simple — LSTMs trained on rainfall and discharge data — but the operational engineering of delivering forecasts to populations through SMS, app notifications, and broadcast channels is what determines whether the technology actually saves lives. We move next from short-horizon hydrological forecasts to the medium-horizon problem: seasonal forecasting.

Seasonal Forecasting

A March forecast for the rains that begin in October is worth more, in expectation, than a Friday forecast for Saturday.

Kennedy Chengeta, PhD

Seasonal forecasting — predicting climate anomalies one to nine months ahead — sits between weather forecasting and climate projection in both timescale and methodology. Seasonal forecasts inform planting decisions, water-resource management, energy planning, and food-security responses. Skill is moderate but operationally valuable, particularly in regions with strong teleconnections to large-scale climate modes.

18.1 Sources of Predictability

Seasonal predictability arises from slowly-evolving components of the climate system that influence weather statistics over multi-month windows. Sea surface temperatures (ENSO, Indian Ocean Dipole, Atlantic Multidecadal Variability) are the dominant source. Soil moisture initial conditions provide land-surface memory. Sea ice and snow cover provide cryospheric memory. The strength of these signals varies regionally: tropical regions tied to ENSO have stronger seasonal predictability than mid-latitude continental interiors.

18.2 Operational Seasonal Forecast Systems

The leading global systems are SEAS5 (ECMWF), CFSv2 (NOAA), GloSea6 (UK Met Office), and SPS (BoM Australia). Each produces ensembles of seasonal forecasts initialised from current ocean and land conditions. Multi-model ensembles — the Copernicus C3S product is the standard — combine the systems to reduce model bias. Skill metrics (anomaly correlation, ROC area, reliability) are reported by region and variable.

18.3 ML for Seasonal Forecasting

Two ML approaches dominate. First, post-processing of dynamical-model outputs to correct systematic biases and improve calibration. Bayesian neural networks and Gaussian processes have been particularly useful here because they preserve uncertainty quantification. Second, hybrid statistical-dynamical models that combine SST anomaly inputs with ML to predict regional precipitation and temperature directly. ENSO state forecasting itself has improved via attention-based models that track the spatial patterns of equatorial Pacific SST.

18.4 Communicating Probabilistic Forecasts

Seasonal forecasts are inherently probabilistic — "above-normal rainfall" with 50% probability against a climatological baseline of 33%. Communicating this to decision-makers, particularly farmers and water managers, is non-trivial. Successful programmes (Cenacarc in West Africa, the Climate Predictability and Applications Network in Southern Africa) translate probabilistic forecasts into discrete action recommendations conditional on user risk tolerance. The translation step is itself an AI problem amenable to decision-support tools.

18.5 Mathematical Foundations of Seasonal Forecasting

18.5.1 Probabilistic Tercile Forecasting

Climatological tercile probabilities are 1/3 each. Forecast skill is measured by the gain over climatology:

$$\text{Skill score} = \frac{\text{BS}_{\text{forecast}} - \text{BS}_{\text{clim}}}{-\text{BS}_{\text{clim}}},$$

ranging $(-\infty, 1]$ where 1 is perfect, 0 climatological skill, negative worse than climatology.

18.5.2 ENSO Index and Lead Time

The Niño 3.4 SST anomaly index is computed as the area-mean SST anomaly over 5°N–5°S, 170°W–120°W. Persistence-based forecast skill drops as lead time crosses the spring barrier (Mar–May).

18.5.3 Bayesian Model Combination

Multi-model ensemble combination via Bayesian Model Averaging:

$$p(y \mid \text{ens}) = \sum_{m=1}^M w_m \mathcal{N}(y; f_m, \sigma_m^2),$$

with weights w_m from training-set likelihood and $\sum_m w_m = 1$.

18.6 Worked Problems and Solutions

Problem 18.1 (Tercile Skill). Forecast probabilities for terciles are (0.55, 0.30, 0.15). Below-normal verifies. Compute Brier score and skill versus climatology.

Solution. Forecast Brier (multi-category): $(0.55 - 1)^2 + (0.30 - 0)^2 + (0.15 - 0)^2 = 0.2025 + 0.09 + 0.0225 = 0.315$. Climatology: $(1/3 - 1)^2 + (1/3)^2 + (1/3)^2 = 0.4444 + 0.1111 + 0.1111 = 0.667$. Skill score = $(0.315 - 0.667)/(-0.667) = -0.352/(-0.667) = 0.528$. The forecast adds 53% skill above climatology. ■

18.7 Chapter Conclusion

Seasonal forecasting is, mathematically, the hardest of the forecast horizons. Weather forecasting benefits from accurate initial conditions and short propagation times; climate projection benefits from the boundary-condition constraint of greenhouse-gas concentrations. Seasonal forecasting sits in the awkward middle, with neither advantage. What predictability exists comes principally from slowly varying ocean- surface temperature anomalies — ENSO above all, but also the Indian Ocean Dipole,

the Atlantic Multidecadal Oscillation, and the Madden-Julian Oscillation — and from soil-moisture persistence.

The chapter's mathematical treatment of probabilistic tercile forecasting gave the operational vocabulary: a forecast probability of 0.55 for the below-normal tercile against a climatological 0.33 represents skill of 0.22 in the simplest framing. The Brier-score-decomposition worked problem demonstrated how to convert this into a calibration-and-resolution diagnosis. The skill score of 0.528 in the worked example is high by seasonal-forecast standards; most operational systems achieve scores in the 0.05–0.15 range, varying strongly by region and season.

The chapter's coverage of multi-model ensemble combination — through Bayesian model averaging, weighted means with track-record-derived weights, and ML-based post-processing — represents the current state of the operational art. The C3S Copernicus seasonal-forecast service, the US NMME, and equivalent national systems all use ensemble-combination methods to extract skill that no single model possesses. ML-based post-processing is increasingly added on top; the gains are modest but real.

The spring-barrier discussion deserves emphasis. ENSO is the dominant source of seasonal predictability, but ENSO-state predictability itself drops sharply in March-May. Operational forecasts cross this barrier with substantially reduced skill, and consumers of seasonal forecasts should understand this seasonality. We turn next to the technique that makes coarse-resolution climate output useful at decision scales: downscaling.

Climate Downscaling

A 100km grid cell contains a thousand farms, ten cities, and three watersheds — and they each need different answers.

Kennedy Chengeta, PhD

Climate models produce projections at 25-100km resolution; planners and adaptation practitioners need projections at 1-10km resolution to inform infrastructure, agriculture, and water-resource decisions. Downscaling — statistical and dynamical — bridges this gap. AI-based statistical downscaling has, over the past five years, become competitive in skill with computationally expensive dynamical downscaling at orders-of-magnitude lower cost.

19.1 Dynamical Versus Statistical Downscaling

Dynamical downscaling runs a regional climate model (RCM) nested inside a global model, with high-resolution physics resolving terrain, coastlines, and convection. CORDEX is the international RCM intercomparison; outputs are at 12-50km. Statistical downscaling learns mappings from coarse-resolution model outputs to fine-resolution observations, using historical reanalysis as training data. ML extensions to statistical downscaling — super-resolution CNNs, GAN-based methods, diffusion models — preserve spatial coherence and physical realism better than older bias-correction approaches.

19.2 Super-Resolution Approaches

Definition 19.1 (Super-Resolution Downscaling). Given coarse field $\mathbf{x}_c \in \mathbb{R}^{H \times W}$ and target fine field $\mathbf{x}_f \in \mathbb{R}^{rH \times rW}$ at upscaling factor r , learn a function f_θ such that

$$f_\theta(\mathbf{x}_c, \mathbf{z}) \approx \mathbf{x}_f$$

where $\mathbf{z}$ contains auxiliary high-resolution covariates (terrain, land cover, climatology).

ESRGAN-derived architectures dominate for precipitation downscaling because they reproduce the spatial intermittency and heavy tails that mean squared error losses smooth away. Diffusion models (cascaded conditional diffusion) have produced the highest-quality recent results, particularly for extreme-precipitation downscaling.

19.3 Bias Correction

Climate model outputs have known biases relative to observations. Quantile mapping is the standard univariate correction: map each quantile of model output to the corresponding quantile of observations. Multivariate corrections (MBCn, copula-based) preserve cross-variable consistency. AI-based corrections (autoencoders, normalising flows) preserve spatial-temporal structure that quantile mapping breaks. The trade-off is interpretability: simpler methods are easier to validate; ML methods are more accurate.

19.4 Limits of Downscaling

Downscaling cannot create predictability that the driving model does not contain. If the GCM misprojects the regional circulation, no downscaling will fix the resulting precipitation projection. Downscaling assumes the historical statistical relationship between large-scale and local climate persists into the future — an assumption known to fail in some regimes. Robust downscaled projections require driving-model evaluation, multi-method downscaling, and explicit uncertainty propagation.

19.5 Mathematical Foundations of Statistical Downscaling

19.5.1 Quantile Mapping

For climate-model output X_m and observation X_o with empirical CDFs F_m, F_o :

$$\hat{X}_m = F_o^{-1}(F_m(X_m)).$$

Preserves bias-corrected distribution but breaks spatial-temporal covariance unless multivariate methods (MBCn, copula-based) are used.

19.5.2 Super-Resolution Reconstruction Loss

For paired coarse $\mathbf{x}_c$ and fine $\mathbf{x}_f$ fields:

$$\mathcal{L}_{\text{SR}} = \mathcal{L}_{\text{pixel}}(g_{\theta}(\mathbf{x}_c), \mathbf{x}_f) + \lambda_p \mathcal{L}_{\text{percep}} + \lambda_a \mathcal{L}_{\text{adv}},$$

where perceptual and adversarial terms preserve high-frequency structure that pixel-only losses smooth away.

19.5.3 Conditional Diffusion

A conditional diffusion model learns the reverse process

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t, \mathbf{c}), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t, \mathbf{c})),$$

conditioning on coarse-resolution input $\mathbf{c}$ to generate samples of the fine-resolution field.

19.6 Worked Problems and Solutions

Problem 19.2 (Quantile Mapping). Model output $X_m = 18.0^{\circ}\text{C}$; model CDF gives $F_m(18.0) = 0.72$. Observed CDF inverse $F_o^{-1}(0.72) = 19.5^{\circ}\text{C}$. Bias-corrected output?

Solution. $\hat{X}_m = 19.5^{\circ}\text{C}$. Quantile mapping has shifted the value by $+1.5^{\circ}\text{C}$, the conditional bias at this quantile. ■

19.7 Chapter Conclusion

Downscaling is the technical step that determines whether climate projections are actionable or theoretical. The chapter walked through three families: statistical downscaling via quantile mapping and analog methods; dynamical downscaling via regional climate models such as WRF and RegCM; and ML-based downscaling via super-resolution networks and conditional diffusion models. Each has its place; the methodological choice depends on the variable, the application, and the available compute.

The quantile-mapping worked problem illustrated the simplest statistical approach. A model output of 18°C with CDF value 0.72 is bias-corrected to 19.5°C by inverting the observed CDF at the same quantile. The technique is computationally trivial and surprisingly effective for univariate bias correction. Its weakness is the breakdown of inter-variable and inter-temporal covariance, addressed by multivariate extensions such as MBCn and MRQNBC.

The super-resolution and diffusion-model treatment surveyed the current research frontier. ML-based downscaling captures the high-frequency spatial structure — orographic precipitation, coastal temperature gradients, urban-rural contrasts — that statistical methods miss and that dynamical methods produce only at high computational cost. Conditional diffusion models, in particular, generate plausible high-resolution samples conditioned on coarse-resolution input, supporting both deterministic downscaling and ensemble generation.

The chapter's evaluation discussion reinforced a discipline-wide point: downscaling quality must be assessed against held-out observations, not against the input climate model. A downscaled product can look spatially realistic and still be systematically biased; only comparison with independent observation reveals this. We close Part III's technical chapters here and turn next to the financial application that consumes much of the climate-projection output: climate risk in finance.

Climate Risk in Finance

A bank that does not price climate risk into its mortgage book is not running a clean balance sheet — it is running an undisclosed short position on the climate.

Kennedy Chengeta, PhD

Financial institutions face climate risk through two channels: physical risk from direct climate impacts on assets and counterparties, and transition risk from policy, technology, and market changes as the economy decarbonises. Quantifying both at portfolio scale requires combining climate science, economic modelling, and AI to translate physical hazards into financial loss distributions. This chapter treats the methodological foundations and the practical deployments that anchor major banks' and insurers' climate strategies.

20.1 Physical Risk Modelling

Physical risk has acute and chronic components. Acute physical risk is from extreme events — hurricanes, floods, wildfires — with frequency and intensity changing under warming. Chronic physical risk is from gradual change — sea-level rise, agricultural productivity loss, heat-induced labour productivity decline. Modelling chains hazard (climate model output), exposure (asset location and value), and vulnerability (asset response function) into expected loss. AI augments each step: hazard downscaling and ensemble synthesis; exposure data extraction from filings, satellite imagery, and supply-chain mapping; vulnerability calibration from historical loss data.

20.2 Transition Risk Modelling

Transition risk depends on policy pathways, technology costs, and behavioural change. Standard scenarios (NGFS, IEA WEO, IPCC SSPs) provide reference pathways. Translating scenarios to portfolio-level loss requires sector-specific transition models: how does this oil major's revenue evolve under a 1.5°C policy pathway? AI-driven scenario expansion (probabilistic NGFS extensions, GenAI-driven narrative scenarios) addresses the limitation that point scenarios understate uncertainty.

Example 20.1 (Mortgage Book Climate Stress Test). A South African bank stress-tests its USD 8B residential mortgage book under three NGFS scenarios. Physical risk: flood and fire exposure mapped via 30m terrain, river hydraulics ML, and insurance claim ML. Transition risk: energy-cost projections by suburb under carbon pricing. Combined NPV impact under Net Zero 2050 scenario: -2.3% of book NAV; under Delayed Transition: -4.1%; under Hot House: -6.8%.

Bank capital reserves adjusted accordingly under TCFD-aligned disclosure.

20.3 Climate-Adjusted Credit Scoring

Conventional credit scoring uses historical financial data without explicit climate variables. Climate-adjusted scoring augments with hazard exposure, transition exposure (sector, geography, fossil-fuel reliance), and adaptation capacity. The output is forward-looking probability of default conditional on climate scenario. Adoption is uneven: leading European banks have integrated climate-adjusted scoring; most emerging-market banks have not. The data gap — particularly for SMEs — is the binding constraint.

20.4 Disclosure and the ISSB Standards

The IFRS Sustainability Standards (S1 general; S2 climate-specific) became operational in 2024 and are being adopted by major regulators globally. S2 requires firms to disclose climate-related risks, opportunities, scenario analysis, transition plans, and Scope 1, 2, and material Scope 3 emissions. AI-driven data infrastructure, scenario modelling, and emissions calculation underpin compliant disclosure at firm scale. Chapter 48 treats the disclosure infrastructure in depth.

20.5 Mathematical Foundations of Climate Risk in Finance

20.5.1 Expected Loss

For exposure E , hazard frequency ν , vulnerability function $V(h)$:

$$EL = E \cdot \mathbb{E}[V(H)] \cdot \nu,$$

with H hazard intensity. Annual loss exceedance probability:

$$AEP(L) = 1 - \exp(-\nu \Pr(E \cdot V(H) > L)).$$

20.5.2 Climate Value-at-Risk

For loss distribution F_L :

$$\text{VaR}_\alpha = F_L^{-1}(\alpha), \quad \text{CVaR}_\alpha = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha].$$

20.5.3 Scenario Aggregation

Portfolio NPV under scenario s with probability π_s :

$$\text{NPV}_{\text{port}} = \sum_s \pi_s \sum_a \frac{\text{CF}_{a,s,t}}{(1+\rho)^t}.$$

Climate-adjusted: cash flows differ across scenarios reflecting policy, technology, and physical impact pathways.

20.6 Worked Problems and Solutions

Problem 20.2 (Mortgage Stress Test). Portfolio NAV \$8B; flood exposure mapped to \$1.6B at-risk; flood vulnerability under 1.5C scenario yields expected loss ratio of 4% on exposed assets; transition risk reduces portfolio cash flows by 1.8%. Compute total NAV impact.

Solution. Physical: $1.6\text{B} \cdot 4\% = 64\text{M}$. Transition: $8\text{B} \cdot 1.8\% = 144\text{M}$. Total NAV impact: \$208M, or 2.6% of portfolio. ■

Problem 20.3 (CVaR). Loss distribution at the 95th percentile is \$24M; conditional mean above that threshold is \$38M. Identify VaR and CVaR.

Solution. $\text{VaR}_{0.95} = 24\text{M}$; $\text{CVaR}_{0.95} = 38\text{M}$. Capital requirements linked to CVaR are systematically more conservative. ■

20.7 Chapter Conclusion

Climate risk in finance is where climate science becomes commercially consequential at scale. Mortgage portfolios, infrastructure investments, sovereign debt, and equity allocations all face climate-driven repricing that the financial system is still in the early stages of incorporating. The chapter walked through the mathematical and operational machinery: hazard-vulnerability-exposure decomposition for physical risk, scenario analysis with NGFS-derived pathways for transition risk, and the aggregation of both into climate-VaR and climate-CVaR measures.

The mortgage-stress-test worked problem illustrated the scale. A 2.6% NAV impact on an \$8B portfolio — \$208M of recognised climate impact — is small relative to operational risk capital but large relative to typical annual portfolio rebalancing. Cumulative impacts compound over multi-year horizons; the question for portfolio managers is not whether to recognise climate risk but how rapidly to reprice exposure.

The chapter's treatment of scenario analysis acknowledged a methodological weakness rarely admitted in industry practice: climate scenarios are not probability distributions. The NGFS pathways are policy-relevant cases, not statistical samples; weighting them by subjective probability and computing expected losses is a practice the chapter declined to endorse. Instead, the chapter argued for scenario analysis as stress-testing — asking what would happen under each pathway — rather than as probabilistic forecasting.

The integration with conventional risk machinery remains incomplete. Banking regulators are beginning to incorporate climate risk into capital-adequacy frameworks; insurance regulators are beginning to do the same for reserve requirements; securities regulators are increasingly requiring climate-related disclosure. The trajectory is clear; the operational details continue to evolve. This chapter closes Part III. We turn next, in Part IV, to the energy sector — the largest single domain of emissions and the largest single domain of AI-enabled mitigation opportunity.

Part IV

Energy Efficiency and the Smart Grid

Energy Systems: An Introduction

The energy transition is the largest infrastructure project in human history, and it must be completed in twenty years. AI is the only management tool of the right scale.

Kennedy Chengeta, PhD

Energy systems are the dominant source of greenhouse gas emissions: roughly 73% of global emissions, of which electricity and heat (about 31%), transport (16%), manufacturing and construction (12%), and other energy-related activities (14%) are the largest contributors. Decarbonising energy is therefore the central task of climate policy. The transition — from fossil to renewable generation, from internal combustion to electric transport, from inefficient to efficient buildings and industry — is technologically feasible but operationally complex. AI is essential because the resulting energy system is dramatically more variable, distributed, and data-rich than the system it replaces.

21.1 The Scale of the Transition

Holding warming to 1.5°C requires global emissions to halve by 2030 and reach net zero by mid-century. For energy this implies: tripling renewable electricity capacity by 2030, doubling energy efficiency improvements, electrifying transport and heating at scale, and deploying storage and grid infrastructure to integrate variable renewables. The IEA estimates annual clean-energy investment must rise from USD 1.8 trillion currently to USD 4.5 trillion by 2030. The physical infrastructure changes — millions of new wind turbines, billions of solar panels, hundreds of millions of EVs and heat pumps — generate data flows that no human-operated grid can manage.

21.2 Why AI is Necessary, Not Optional

A 20th-century power grid had hundreds of generators, predictable demand, and centralised control. A net-zero grid has millions of distributed generators, dynamic demand from EVs and electrified heat, bidirectional power flows, and weather-dependent variability on every timescale from seconds to seasons. The number of decisions per second that must be made to keep frequency, voltage, and reserves within bounds exceeds human capacity by orders of magnitude. AI is necessary not as an optimisation luxury but as the operational substrate of the new system.

21.3 The Energy Information Hierarchy

Energy AI operates across timescales spanning twelve orders of magnitude. Sub-second: frequency response, fault detection, inverter control. Minutes: dispatch, balancing, reserve activation. Hours: day-ahead scheduling, demand response. Days: unit commitment, weather-dependent renewable forecasting. Weeks-months: maintenance scheduling, hydro reservoir management, fuel procurement. Years-decades: capacity expansion planning, transmission build-out, retirement decisions. Different ML methods dominate at different scales; the chapters that follow treat each in turn.

21.4 African Context

Sub-Saharan Africa's energy systems are characterised by low per-capita consumption, limited grid coverage, high losses, and rapid renewable deployment. Around 600 million Africans lack electricity access. Mini-grids, off-grid solar home systems, and pay-as-you-go solar (M-Kopa, BBOXX, Sun King) have changed the access trajectory in the past decade. AI is critical for the ramping system: predicting payment defaults, optimising battery cycles, dispatching technicians, and integrating intermittent generation into weak national grids that load-shed routinely. Subsequent chapters treat the African deployments alongside the Northern-hemisphere case studies.

21.5 Mathematical Foundations of Energy Systems

21.5.1 Power Balance

At all times, generation must equal load plus losses:

$$\sum_g P_g(t) = \sum_d P_d(t) + P_{\text{loss}}(t).$$

Frequency deviation reflects imbalance through swing dynamics:

$$2H \frac{df}{dt} = \frac{P_g - P_d}{S_b},$$

where H is system inertia constant, S_b base power.

21.5.2 Levelised Cost of Energy

$$\text{LCOE} = \frac{\text{CAPEX} + \sum_t \text{OPEX}_t / (1+r)^t}{\sum_t \text{Generation}_t / (1+r)^t}.$$

21.5.3 Capacity Factor

$$\text{CF} = \frac{\text{Actual energy in period}}{\text{Nameplate} \times \text{period hours}}.$$

Solar PV: 15–25% typical; onshore wind: 25–45%; offshore wind: 35–55%.

21.6 Worked Problems and Solutions

Problem 21.1 (Solar LCOE). 20 MW solar farm: CAPEX \$18M, annual OPEX \$0.4M, capacity factor 22%, discount rate 7%, lifetime 25 years. Compute LCOE.

Solution. Annual generation: $20,000 \text{ kW} \cdot 8760 \text{ h} \cdot 0.22 = 38,544 \text{ MWh}$. Annuity factor: $\sum_{t=1}^{25} (1.07)^{-t} = (1 - 1.07^{-25})/0.07 = (1 - 0.1842)/0.07 = 11.654$. PV of OPEX: $0.4 \cdot 11.654 = 4.66\text{M}$. Total

cost PV: $18 + 4.66 = 22.66\text{M}$. PV of generation: $38,544 \cdot 11.654 = 449,213 \text{ MWh}$. $\text{LCOE} = 22.66 \times 10^6 / 449,213 = \$50.45/\text{MWh}$. ■

21.7 Chapter Conclusion

The energy sector is undergoing the largest physical-infrastructure transition in human history, and AI is increasingly central to managing it. The chapter framed the transition along three axes: from centralised to distributed generation, from passive to active demand, and from deterministic to stochastic system operation. Each axis intensifies the information-management challenge that AI is uniquely suited to address.

The LCOE worked problem illustrated the economics that drive deployment decisions. At \$50.45/MWh, the example solar farm is competitive with new-build gas and substantially below new-build coal in most markets. The LCOE machinery generalises across technologies; the reader should apply it cautiously, however, because LCOE omits system-integration costs (transmission, balancing, storage) that increasingly dominate the comparison at high renewable shares.

The chapter's treatment of capacity factors gave the operational grounding. A 22% solar capacity factor is typical for fixed-tilt utility-scale PV in sunny regions; tracking systems achieve 25%–28%; bifacial modules add another 5%–10%. Wind capacity factors vary more, from below 25% in poor-resource regions to above 50% in offshore high-wind environments. Knowing the capacity factor of a specific asset is the first step in any honest analysis of a specific deployment.

The seven leverage points named in this chapter — demand forecasting, renewable forecasting, grid management, building optimisation, industrial control, storage dispatch, and EV charging — structure the rest of Part IV. Each chapter takes one leverage point in turn, develops the mathematical foundation, walks through the operational considerations, and grounds the discussion in deployed examples. We turn next to the most mature of the seven: demand forecasting.

Electricity Demand Forecasting

A grid is balanced or it crashes. Forecast accuracy is the cheapest balancing reserve a system operator can buy.

Kennedy Chengeta, PhD

Demand forecasting is the foundational AI application in electricity systems. Forecasts at multiple horizons — day-ahead for unit commitment, hour-ahead for dispatch, minutes-ahead for balancing, years-ahead for capacity planning — drive every operational and investment decision. Improvements of even one or two percent in forecast accuracy translate to substantial savings in spinning reserves, fuel costs, and emissions.

22.1 Demand Drivers and Features

Electricity demand is driven by weather (temperature, humidity, solar irradiance, wind), calendar effects (hour of day, day of week, holiday, school term), economic activity (industrial production, commercial occupancy), and structural change (electrification of transport and heat, demand response programs). The features that consistently appear in successful forecasting models include: lagged demand at multiple lags, weather forecasts at the same horizon as the demand forecast, calendar dummies, and trend variables capturing long-run growth.

22.2 Architectures: From SARIMA to Transformers

Until 2017, statistical methods dominated — SARIMA, exponential smoothing, regression with calendar effects. From 2017 to 2022, gradient-boosted trees (XGBoost, LightGBM) became the operational standard, often combined with simple deep learning. From 2022, attention-based architectures — Temporal Fusion Transformer, N-BEATS, Informer — have become competitive on benchmark datasets and are now in production at major TSOs. Foundation-model approaches (TimesFM, Chronos) trained on billions of time series are emerging; their advantage is forecast quality with minimal task-specific data.

22.3 Probabilistic Forecasting

Definition 22.1 (Pinball Loss). For quantile forecast $\hat{q}_\tau$ and observation y :

$$\mathcal{L}_\tau(y, \hat{q}_\tau) = (y - \hat{q}_\tau) [\tau - \mathbb{I}(y < \hat{q}_\tau)].$$

Minimising aggregate pinball loss across quantiles produces well-calibrated probabilistic forecasts.

Point forecasts misrepresent the operational decision. A system operator deciding reserves needs the forecast distribution, not the mean. Quantile regression, deep ensembles, and conformal prediction are the dominant approaches; the right verification metrics are pinball loss, continuous ranked probability score, and prediction-interval coverage.

22.4 Behind-the-Meter Resources

The growth of rooftop solar, behind-the-meter batteries, and home automation has changed the demand-forecasting problem. Net demand — gross demand minus distributed generation — is what the grid sees, and it is volatile, geographically heterogeneous, and partially controllable through demand response. Disaggregation models that separately forecast gross demand, distributed solar, and battery flows perform better than direct net-demand models. The data requirement is high: smart meter penetration above roughly 40% is needed for the disaggregation to be tractable.

22.5 Mathematical Foundations of Demand Forecasting

22.5.1 Multi-Horizon Forecasting Loss

For horizons $h = 1, \dots, H$:

$$\mathcal{L}_{\text{multi}} = \sum_{h=1}^H \alpha_h \mathcal{L}(\hat{y}_{t+h}, y_{t+h}),$$

with horizon weights α_h encoding operational priorities.

22.5.2 Temporal Fusion Transformer

Variable selection V produces gating-weighted features; gated residual networks GRN apply non-linear transforms; multi-head attention across temporal positions:

$$\mathbf{H}_t = \text{Attn}(Q_t, K_{1:T}, V_{1:T}) + \text{GRN}(\mathbf{h}_t).$$

Outputs are quantile predictions $\hat{q}_{\tau, t+h}$.

22.5.3 Pinball-Calibrated Probabilistic Forecast

For target quantile levels $\{\tau_k\}$:

$$\mathcal{L}_{\text{pinball}} = \sum_h \sum_k \mathcal{L}_{\tau_k}(y_{t+h}, \hat{q}_{\tau_k, t+h}).$$

Empirical coverage: $\hat{C}(\tau) = \frac{1}{n} \sum_i \mathbb{I}(y_i \leq \hat{q}_{\tau}(\mathbf{x}_i))$; ideal $\hat{C}(\tau) \approx \tau$.

22.6 Worked Problems and Solutions

Problem 22.2 (Reserve Sizing from Quantiles). Day-ahead demand 95th percentile forecast is 22,400 MW; median 20,800 MW. What spinning reserve is implied if reserves cover up to 95th percentile?

Solution. $22,400 - 20,800 = 1,600$ MW. ■

Problem 22.3 (Quantile Coverage). Out of 1,000 forecasts, 924 actual outcomes fell below the 90th-percentile prediction. Is the model overconfident or underconfident?

Solution. Empirical coverage 92.4% versus nominal 90%; the model is slightly *underconfident*

(intervals wider than needed). Recalibration via isotonic regression on validation data tightens the intervals. ■

22.7 Chapter Conclusion

Electricity demand forecasting is the gold standard against which other sustainability-AI applications can be measured. Operational forecasts routinely achieve mean absolute percentage errors below 2% at the day-ahead horizon and below 1% at intra-day horizons; these levels of accuracy compound into billions of dollars per year of avoided operating costs across major grids.

The chapter's mathematical treatment focused on probabilistic forecasting because operational decisions increasingly require distributional rather than point predictions. The pinball-loss framework gives the foundation; the quantile-coverage worked problem illustrated the operational diagnostic. Empirical coverage of 92.4% on a nominal 90% interval, as in the worked example, indicates slight under-confidence — intervals wider than needed — which is generally preferable to overconfidence but still suboptimal. Isotonic recalibration on validation data tightens the intervals at marginal computational cost.

The temporal-fusion transformer treatment surveyed the current architectural frontier. Variable selection, gated residual networks, and multi-head attention combine to produce forecasts that respect the hierarchical structure of demand (national → regional → substation), capture multi-horizon dependencies, and handle the heterogeneous mix of static covariates (substation type), past observations (load history), and known future inputs (calendar features, weather forecasts). The architecture is general; the implementation details matter.

The chapter's coverage of weather-dependent forecasting deserves emphasis. Demand is increasingly weather-driven as electrified heating and cooling expand. Joint forecasting of demand and weather, rather than sequential forecasting with weather as a fixed input, captures the covariance that materially affects extreme-load events. We turn next from the demand side to the increasingly stochastic supply side: renewable integration.

Renewable Energy Integration

A megawatt of curtailed wind is a megawatt of avoided fossil fuel multiplied by minus one. The forecast determines the sign.

Kennedy Chengeta, PhD

Renewable energy — wind, solar, and to a lesser extent hydro — is variable on every timescale from seconds to seasons. Integrating large penetrations into power systems requires accurate forecasting, fast-acting reserves, and increasingly sophisticated dispatch. AI is central to all three. This chapter treats wind and solar forecasting, the integration of forecasts into market operations, and the operational practices that have allowed several jurisdictions to integrate over 50% renewable share without sacrificing reliability.

23.1 Wind Power Forecasting

Wind power is highly nonlinear in wind speed: power is roughly proportional to wind speed cubed, with cut-in and cut-out limits. Forecasts at hub height (typically 80-150m) are derived from numerical weather prediction outputs, downscaled and post-processed. ML methods — LSTM, transformer, and gradient-boosted ensembles — dominate post-processing. Operational accuracy at 24-hour lead times is now around 6-8% mean absolute error for fleet-level forecasts in well-instrumented systems.

23.2 Solar Power Forecasting

Solar power forecasting splits into clear-sky and cloud-affected components. Clear-sky output is deterministic given site location and time. Cloud-affected output requires sky imagery (for sub-hour forecasts), satellite imagery (for hour-ahead forecasts), and NWP (for day-ahead forecasts). Convolutional networks on sky cameras or geostationary satellite imagery (Himawari, GOES, Meteosat) produce 5-30 minute nowcasts that drive sub-hourly market clearing. Photovoltaic system-level corrections capture soiling, snow, shading, and inverter behaviour.

Example 23.1 (Solar Nowcasting in the Cape). A convolutional network trained on Meteosat satellite imagery and 90 PV plant outputs across the Western Cape produces 15-minute-ahead solar power forecasts with mean absolute error of 4.2% of installed capacity. The forecast feeds Eskom’s real-time balancing system; the resulting reduction in synchronous reserves saves an estimated 220 GWh of fossil generation annually.

23.3 Curtailment Optimisation

When renewable generation exceeds transmission capacity or system flexibility, some output must be curtailed. The choice of which plants to curtail, by how much, and when, is a stochastic optimisation problem driven by forecasts of generation, demand, prices, and grid conditions. ML-based curtailment optimisers minimise total curtailed energy and revenue loss while respecting grid constraints; deployed across CAISO, ERCOT, and Iberian operators, they have reduced curtailment by 15-30%.

23.4 Grid-Forming Inverters and Synthetic Inertia

A grid dominated by synchronous machines (steam turbines) has high physical inertia that resists frequency change. A grid dominated by inverter-based resources (wind, solar, batteries) has low inertia, raising the risk of frequency excursions. AI-coordinated grid-forming inverters and synthetic-inertia algorithms compensate by emulating physical inertia at sub-second timescales. The control problem is high-dimensional, non-convex, and safety-critical; reinforcement learning with strict safety constraints (CMDP formulations, control-barrier functions) is the leading research direction.

23.5 Mathematical Foundations of Wind and Solar Forecasting

23.5.1 Wind Power Curve

$$P(v) = \begin{cases} 0 & v < v_{\text{cut-in}} \text{ OR } v > v_{\text{cut-out}}, \\ \frac{1}{2} \rho A C_p(v) v^3 & v_{\text{cut-in}} \leq v < v_{\text{rated}}, \\ P_{\text{rated}} & v_{\text{rated}} \leq v \leq v_{\text{cut-out}}. \end{cases}$$

Power coefficient C_p peaks near 0.45 at optimal tip-speed ratio, bounded above by Betz $C_{p,\text{max}} = 16/27 \approx 0.593$.

23.5.2 Solar Clear-Sky Model

Direct normal irradiance:

$$\text{DNI} = G_{sc} \tau_b(\text{air mass, aerosol, water vapour}),$$

with solar constant $G_{sc} = 1361 \text{ W/m}^2$. Air mass: $\text{AM} \approx 1/\cos \theta_z$ for solar zenith θ_z .

23.5.3 Curtailment Optimisation

Minimise total curtailment subject to grid constraints:

$$\min_{\mathbf{c}} \sum_t \mathbf{1}^\top \mathbf{c}_t \quad \text{s.t.} \quad \mathbf{f}(\mathbf{P}_t - \mathbf{c}_t, \mathbf{D}_t) \leq \mathbf{f}_{\text{max}}, \quad 0 \leq \mathbf{c}_t \leq \mathbf{P}_t.$$

23.6 Worked Problems and Solutions

Problem 23.2 (Wind Power). Turbine: $\rho = 1.225 \text{ kg/m}^3$, $A = 8000 \text{ m}^2$, $C_p = 0.42$. Compute power at $v = 9 \text{ m/s}$.

Solution. $P = 0.5 \cdot 1.225 \cdot 8000 \cdot 0.42 \cdot 9^3 = 0.5 \cdot 1.225 \cdot 8000 \cdot 0.42 \cdot 729$. Stepwise: $0.5 \cdot 1.225 = 0.6125$; $0.6125 \cdot 8000 = 4900$; $4900 \cdot 0.42 = 2058$; $2058 \cdot 729 = 1,500,282 \text{ W} \approx 1.50 \text{ MW}$. ■

Problem 23.3 (Air Mass). Solar zenith angle is 35° . Compute air mass.

Solution. $AM = 1/\cos 35^\circ = 1/0.8192 \approx 1.221$. ■

23.7 Chapter Conclusion

Renewable integration is the technical problem that distinguishes twenty-first-century grid operation from twentieth-century operation. Twentieth-century grids dispatched flexible thermal generators against predictable demand; twenty-first-century grids must accommodate stochastic generation, increasingly stochastic demand, and a finite budget of fast-responding flexibility. AI is essential to this accommodation at every level: forecasting, dispatch, and increasingly real-time control.

The wind-power worked problem illustrated the cubic-in-wind-speed sensitivity that drives wind forecasting requirements. A 10% error in forecast wind speed translates to roughly 30% error in forecast power during the cubic region of the power curve; the operational implication is that wind forecasts must capture not only the mean but the conditional distribution of wind speed. Probabilistic forecasting with calibrated intervals supports reserve sizing in a way that point forecasts cannot.

The solar treatment combined physics-based clear-sky modelling with cloud-cover correction. The clear-sky model gives the theoretical upper envelope; the cloud-cover correction, increasingly ML-driven on satellite imagery, gives the operational forecast. Five-minute-ahead solar nowcasting from geostationary satellite data now supports ramp-rate management on grids with high solar penetration; the technical maturity of these systems is high and continuing to advance.

The curtailment-optimisation discussion closed the operational loop. As renewable shares rise, some curtailment is inevitable; the question is how to minimise its economic and environmental cost. The chapter's mathematical treatment showed how locational marginal pricing, congestion management, and ML-driven dispatch decisions interact. We turn next to the broader grid-management context: smart grids.

Smart Grid Operations

The grid stopped being passive in the 1880s. It just took 140 years for the metering to catch up.

Kennedy Chengeta, PhD

The smart grid is the integration of digital sensing, communication, and control with the electricity transmission and distribution network. AI enters at every layer: state estimation from PMU and AMI data, fault detection and location, voltage and reactive-power optimisation, predictive maintenance, demand response, and outage management. This chapter treats the operational AI applications that have moved from research to deployment.

24.1 State Estimation and Topology

Power-system state estimation infers bus voltages, angles, and line flows from imperfect, noisy, partial measurements. Conventional weighted least squares assumes Gaussian errors and known topology; both assumptions fail under cyber-attack, sensor failure, and DER-driven topology changes. Deep state-estimation networks (graph neural networks operating on the grid topology) are more robust to outliers and accommodate topology uncertainty as part of the inference. Production deployments at PJM and Tennet are early examples.

24.2 Fault Detection and Location

Faults — short circuits, broken conductors, transformer failures — must be detected, classified, and located within milliseconds to enable protective relay coordination. Traditional methods use impedance-based estimation; AI methods use waveform analysis from PMUs and travelling-wave timing. CNNs on time-domain waveforms achieve fault classification accuracy above 99% and location accuracy within 1% of line length. Distribution-grid faults, harder due to lower observability, are a current research frontier.

24.3 Predictive Maintenance

Transformer failures are catastrophic and expensive. Predictive maintenance combines dissolved gas analysis, partial discharge measurements, load history, and weather to estimate remaining useful life. Recurrent and transformer architectures on multivariate sensor streams predict failure 30–180 days in advance with hit rates above 80% in mature deployments. Cable failure prediction (particularly for ageing urban networks), substation equipment, and transmission tower health are

parallel applications.

24.4 Volt-VAR Optimisation

Distribution voltage must be maintained within statutory limits despite varying load and DER injection. Volt-VAR optimisation coordinates capacitor banks, voltage regulators, and DER inverters to minimise losses and maintain quality. Reinforcement learning agents trained on calibrated grid simulators outperform conventional rule-based controllers, particularly under high DER penetration. Safety-constrained RL is essential because untrained policies can violate voltage limits and trip protection.

24.5 Outage Management

When outages occur, fast restoration depends on rapid fault diagnosis, crew dispatch optimisation, and customer communication. AI applications include outage prediction from weather and asset data (heading off some outages by pre-positioning crews), customer call clustering for impact assessment, and route optimisation under traffic and outage constraints. Communication channels — SMS, app push notifications, IVR — complete the loop with customers.

24.6 Mathematical Foundations of Grid Operations

24.6.1 State Estimation

Weighted least squares:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} [\mathbf{z} - \mathbf{h}(\mathbf{x})]^\top \mathbf{R}^{-1} [\mathbf{z} - \mathbf{h}(\mathbf{x})],$$

with measurement vector $\mathbf{z}$, model $\mathbf{h}$, covariance $\mathbf{R}$. Iterative Gauss–Newton update: $\Delta \mathbf{x} = (\mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^\top \mathbf{R}^{-1} (\mathbf{z} - \mathbf{h}(\mathbf{x}))$, where $\mathbf{H}$ is the Jacobian of $\mathbf{h}$.

24.6.2 DC Power Flow

Linearised approximation:

$$\mathbf{P} = \mathbf{B}\boldsymbol{\theta}, \quad P_{ij} = \frac{\theta_i - \theta_j}{x_{ij}}.$$

24.6.3 Volt-VAR Optimisation

$$\min_{\mathbf{q}} \sum_i \sum_j P_{\text{loss},ij}(\mathbf{V}, \mathbf{q}) \quad \text{s.t.} \quad V_{\min} \leq V_i \leq V_{\max}, \quad q_{i,\min} \leq q_i \leq q_{i,\max}.$$

24.7 Worked Problems and Solutions

Problem 24.1 (DC Flow). Two buses connected by line with reactance $x_{12} = 0.1$ p.u. $\theta_1 = 0$, $\theta_2 = -0.05$ rad. Compute flow.

Solution. $P_{12} = (0 - (-0.05))/0.1 = 0.5$ p.u. ■

24.8 Chapter Conclusion

The smart grid is the operational platform on which decarbonised electricity ultimately runs. The chapter walked through its principal components: state estimation that synthesises sensor measurements into a coherent grid-state estimate, fault detection that localises and diagnoses anomalies in

real time, volt-VAR optimisation that maintains voltage profiles within tight bounds while minimising losses, and DER management that coordinates distributed resources — rooftop solar, behind-the-meter batteries, smart thermostats, EVs — as a system asset rather than an unmanaged load.

The DC-power-flow worked problem illustrated the linearised approximation that allows real-time computation across grids with tens of thousands of buses. The exact AC power-flow problem is nonconvex and computationally expensive; the DC approximation captures the first-order behaviour at modest accuracy loss and supports the optimisation problems that operational decision-making requires.

The state-estimation treatment was mathematically dense by necessity. Weighted least squares with iteratively reweighted normal equations is the workhorse of grid state estimation; ML extensions that handle bad-data detection, topology errors, and high-dimensional measurement vectors are increasingly deployed. PMU-based dynamic state estimation, addressing transient grid behaviour at sub-second timescales, is the research frontier.

The chapter's cybersecurity coverage acknowledged a tension the field must navigate. Connectivity enables flexibility but creates attack surface; sensitive grid data must be both shared widely enough to enable optimisation and protected adequately to prevent compromise. Operational practice borrows heavily from industrial-control-system security; the field is converging on layered architectures with strong segmentation between control and corporate networks. We move next from the transmission and distribution stack to the end-use systems that increasingly determine demand patterns: buildings.

Buildings and HVAC

Buildings consume 40% of global energy. Most of it goes to heat in winter and cool in summer — and most of that is unnecessary.

Kennedy Chengeta, PhD

Buildings account for approximately 30% of global final energy consumption and 27% of global energy-related CO₂ emissions. Heating, ventilation, and air conditioning (HVAC) systems are the dominant load. AI control of HVAC, lighting, and plug loads — combined with building envelope improvements — can reduce building energy consumption by 30-60% with payback periods of 3-7 years. This chapter treats the AI methods that have produced these results.

25.1 Why HVAC is a Reinforcement Learning Problem

HVAC control is a sequential decision problem under uncertainty. State: indoor temperatures, occupancy, weather forecast, electricity prices. Action: setpoints, fan speeds, equipment dispatch. Reward: comfort minus energy cost minus equipment wear. Conventional rule-based controllers achieve robust but suboptimal operation. Model-predictive control (MPC) with explicit thermal models improves on rules but requires building-specific calibration. Reinforcement learning learns near-optimal policies directly from operational data.

25.2 DeepMind's Data Centre Result

In 2016, DeepMind reported a 40% reduction in cooling energy at Google data centres using reinforcement-learning HVAC control. The result was significant for two reasons. First, it demonstrated that data-driven methods could substantially outperform decades of expert tuning. Second, it triggered a wave of commercial HVAC AI products (BrainBox, Verdigris, 75F, Carbon Lighthouse) targeting commercial building portfolios. Production gains in commercial buildings are typically 15-25%, less than the data-centre result because residential and commercial buildings are more thermally diverse and have stronger occupant feedback.

25.3 Occupancy and Adaptive Comfort

A building unoccupied at night and weekends does not need conditioning to occupied-state setpoints. Occupancy detection from cameras, Wi-Fi probe requests, CO₂ sensors, or smart-meter signature drives demand-driven ventilation and conditioning. Adaptive comfort models (ASHRAE 55) recognise that acceptable indoor temperatures shift with outdoor conditions, expanding the comfort range and

the optimisation room.

Example 25.1 (Office Building Retrofit, Sandton). A 22-storey office tower in Sandton retrofitted RL-based HVAC control across 280 thermal zones. After 18 months, energy consumption dropped 23%, peak demand dropped 17%, and occupant comfort survey scores improved 8 percentage points. Annual carbon avoidance: 740 tonnes CO₂e on the South African grid carbon intensity. Project payback: 4.2 years.

25.4 Residential Smart Thermostats

Nest, Ecobee, and similar products apply learning algorithms at the thermostat level to learn occupant schedules and preferences. Aggregate smart-thermostat networks participate in demand-response programmes, providing virtual peaking capacity at fractions of the cost of natural-gas peakers. The data privacy implications are significant: detailed indoor temperature, occupancy, and behavioural data flow to central servers; consent, retention, and security policies vary widely.

25.5 Mathematical Foundations of Building Thermal Dynamics

25.5.1 Lumped RC Model

For zone temperature T with thermal capacitance C , resistance R to ambient T_a , internal gains Q_i , HVAC heat input Q_h :

$$C \frac{dT}{dt} = \frac{T_a - T}{R} + Q_i + Q_h.$$

25.5.2 Adaptive Comfort

Acceptable indoor operative temperature (ASHRAE 55, naturally ventilated):

$$T_{\text{op,acc}} = 0.31 T_{\text{out,running}} + 17.8 (\pm 2.5) ^\circ\text{C}.$$

25.5.3 HVAC RL Reward

$$r_t = -\alpha(T_t - T_{\text{set}})^2 - \beta P_t \cdot p_t - \gamma \mathbb{I}(\text{constraint violated}),$$

balancing comfort, energy cost, and constraint adherence.

25.6 Worked Problems and Solutions

Problem 25.2 (Time Constant). $C = 8 \text{ MJ/K}$, $R = 0.05 \text{ K/W}$. Compute thermal time constant.

Solution. $\tau = RC = 0.05 \cdot 8 \times 10^6 = 4 \times 10^5 \text{ s} \approx 111 \text{ hours}$. ■

Problem 25.3 (Adaptive Comfort). 30-day running mean outdoor temperature $T_{\text{out,running}} = 24^\circ\text{C}$. Compute acceptable indoor temperature range.

Solution. $T_{\text{op}} = 0.31 \cdot 24 + 17.8 = 25.24^\circ\text{C}$, range 22.7°C to 27.7°C . ■

25.7 Chapter Conclusion

Buildings concentrate the long-tail challenge of decarbonisation. A modern HVAC retrofit can cut energy consumption by 30%–50%, but the capital cost is substantial and the retrofit cycle is slow. AI's value proposition is that it can extract additional efficiency from existing equipment, without capital expenditure, by optimising control. The chapter's worked problems and architectural treatments grounded this claim.

The thermal-time-constant calculation illustrated the physical inertia that makes building control non-trivial. A 111-hour time constant means that a building's temperature evolves over several days, not minutes; control decisions must anticipate weather conditions days ahead, not react to current conditions. This is the natural application territory for model-predictive control with weather-forecast inputs.

The adaptive-comfort treatment shifted the comfort target from a fixed setpoint to a dynamic range that respects occupant adaptation to outdoor conditions. The worked-example range of 22.7°C to 27.7°C against an outdoor running mean of 24°C is broader than conventional fixed-setpoint operation permits, and the broader range allows operators to defer cooling load to lower-carbon-intensity hours without compromising comfort.

The reinforcement-learning treatment of HVAC control closed the operational loop. RL agents trained against high-fidelity building simulators (EnergyPlus, Modelica) learn policies that outperform fixed-rule control by 10%–30% in deployed buildings. The chapter's operational discussion acknowledged the deployment friction: building owners must trust the controller, building managers must understand its behaviour, and the controller must fail gracefully when sensors degrade or weather forecasts err. We turn next to the heaviest end-use of all: industrial processes.

Industrial Energy Efficiency

A cement kiln does not care about consumer-facing AI. It cares about three percent thermal efficiency, repeated for forty years.

Kennedy Chengeta, PhD

Industry consumes roughly one-third of global energy and emits roughly one-quarter of global CO₂. Heavy industry — cement, steel, chemicals, refining — accounts for the bulk. Decarbonisation faces three challenges: high-temperature heat is hard to electrify, process emissions from chemistry are independent of energy source, and capital cycles are long (30-50 years for major equipment). AI contributes to incremental efficiency gains, predictive maintenance, and optimisation of emerging low-carbon processes.

26.1 Process Optimisation

A cement kiln, a steel blast furnace, or an ammonia plant runs continuously at scale. Small efficiency improvements compound. Soft-sensor AI predicts process variables that are expensive or impossible to measure directly (clinker quality, melt chemistry, catalyst deactivation) from cheaper sensors. Model-predictive control with ML soft sensors has become standard at large facilities. Reported gains are typically 1-5% energy reduction, which sound small but represent enormous absolute savings on plants consuming hundreds of megawatts continuously.

26.2 Predictive Maintenance for Heavy Equipment

Unplanned downtime in heavy industry is catastrophic. Predictive maintenance from vibration analysis, oil analysis, thermal imaging, and process data can give days-to-weeks warning of bearing failures, motor degradation, and corrosion. Recurrent and transformer architectures on multivariate sensor streams now dominate; reported uptime improvements of 5-15% translate to substantial avoided emissions through improved asset utilisation.

26.3 Cement Decarbonisation

Cement manufacturing emits CO₂ from two sources: fuel combustion (about 40% of process emissions) and limestone calcination (about 60%). Fuel-side emissions can be reduced through alternative fuels (biomass, waste-derived) and electrification of low-temperature heat. Process-side emissions can only be reduced through alternative chemistries (calcined clays, geopolymers) or capture and storage. AI assists in optimising alternative-fuel substitution rates while maintaining clinker quality, predicting

kiln performance under varying fuel mixes, and operating CCS systems where deployed.

26.4 Steel and the Hydrogen Transition

Steel production is transitioning from coal-based blast furnaces to direct reduction using hydrogen. The transition is technologically uncertain and economically dependent on green hydrogen prices. ML supports plant-level optimisation across mixed-mode operations, hydrogen-availability forecasting, and the scheduling of intermittent renewable hydrogen supply against continuous metallurgical demand. Sweden's HYBRIT and Germany's SALCOS projects are early demonstrators.

26.5 Mathematical Foundations of Industrial Process Optimisation

26.5.1 Soft Sensor Regression

For unmeasured target y and surrogate features $\mathbf{x}$:

$$\hat{y} = f_{\theta}(\mathbf{x}), \quad \theta = \arg \min \sum_i (y_i - f_{\theta}(\mathbf{x}_i))^2 + \lambda \|\theta\|^2.$$

26.5.2 MPC Formulation

$$\min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} [(\mathbf{x}_k - \mathbf{x}_{\text{ref}})^{\top} \mathbf{Q}(\mathbf{x}_k - \mathbf{x}_{\text{ref}}) + \mathbf{u}_k^{\top} \mathbf{R} \mathbf{u}_k]$$

subject to dynamics $\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k)$ and constraints.

26.5.3 Cement Kiln Energy

Specific thermal energy STE (MJ/t clinker):

$$\text{STE} = \frac{Q_{\text{fuel}} - Q_{\text{recovery}}}{m_{\text{clinker}}}.$$

26.6 Worked Problems and Solutions

Problem 26.1 (Process Optimisation Savings). A kiln consuming 3.2 GJ/t clinker reduces consumption by 4% with ML optimisation. Annual production 1.5 Mt; emission factor of fuel 0.10 kg CO₂/MJ. Compute annual emissions reduction.

Solution. Energy saving: $0.04 \cdot 3.2 = 0.128$ GJ/t. Annual: $0.128 \cdot 10^9 \cdot 1.5 \times 10^6 = 1.92 \times 10^{14}$ J. Convert to MJ: 1.92×10^8 MJ. Emissions avoided: $1.92 \times 10^8 \cdot 0.10 = 1.92 \times 10^7$ kg = 19,200 tCO₂e per year. ■

26.7 Chapter Conclusion

Industrial AI is a domain of incremental but cumulative gains. A 3%–5% energy reduction per process unit, applied across the global cement fleet, translates to tens of millions of tonnes of avoided CO₂ per year. The cement-kiln worked problem illustrated the scale: a 4% fuel-consumption reduction across 1.5 Mt of annual production yields 19,200 tCO₂e per year, equivalent to taking about 4,200 cars off the road. Compound this across all cement plants globally and the figure runs to hundreds of megatonnes.

The chapter's treatment of soft sensors deserves emphasis. Many key process variables — product quality, raw-mix composition, kiln condition — are not directly measurable at the timescale required for real-time control; they are measured offline at coarse cadence. Soft sensors, fitted as regression models on the directly measurable variables, fill the gap. The mathematical structure is straightforward; the operational value is substantial because real-time decisions can optimise against the inferred quality variable instead of waiting for laboratory analysis.

Model-predictive control was treated as the standard control framework for industrial systems. The chapter's MPC formulation gave the basic quadratic-cost-with-constraints structure; commercial implementations extend this with nonlinear dynamics, stochastic disturbance models, and economic objectives. The integration with ML is increasingly bidirectional: ML predicts the dynamics or the disturbances; MPC closes the loop.

The chapter's process-by-process coverage — cement, steel, chemicals, pulp-and-paper — gave the reader a tour of the dominant industrial-AI applications. None is a silver bullet; all contribute incrementally. The honest framing is that industrial AI is necessary for net-zero industry but not sufficient; deep decarbonisation requires the technology shifts that AI optimisation does not substitute for. We turn next to the flexibility resource that increasingly mediates renewable variability: storage.

Energy Storage and Batteries

A battery is potential energy, and dispatch is the act of converting potential energy into delivered service. The conversion is a control problem; control problems are AI problems.

Kennedy Chengeta, PhD

Storage — batteries, pumped hydro, thermal storage, hydrogen — is the foundational enabler of high-renewable grids. Battery storage in particular has fallen 90% in cost since 2010 and is now the dominant new flexibility resource. AI applications span battery dispatch, degradation modelling, fleet aggregation for grid services, and second-life applications.

27.1 Battery Dispatch Optimisation

Battery dispatch is a multi-revenue-stream optimisation problem: energy arbitrage, frequency regulation, spinning reserves, capacity payments, and behind-the-meter energy management. Optimal dispatch trades off short-term revenue against long-term degradation. Mixed-integer linear programming and stochastic dynamic programming have given way to reinforcement-learning-based dispatchers that adapt to changing market conditions and battery state. Production deployments by Tesla Autobidder, Fluence, and Wartsila demonstrate competitive advantage from RL dispatch.

27.2 Degradation and State of Health

Batteries degrade with cycling, temperature, depth of discharge, and rate. Accurate state-of-health estimation is essential for warranty management, secondary markets, and dispatch. Physics-informed neural networks combine electrochemical models with data-driven corrections, predicting capacity fade and impedance growth at calendar and cycle scales. Resale-value calibration in EV second-life markets depends critically on these models.

27.3 Virtual Power Plants

A virtual power plant aggregates distributed batteries (residential, commercial, EVs) into a single dispatchable resource. The aggregation problem is challenging: residential battery owners have heterogeneous preferences, partial controllability, and binding self-consumption priorities. Federated learning and multi-agent RL produce aggregation policies that respect individual constraints while maximising aggregate value. Tesla's VPP in South Australia and Sonnen's VPP in Germany are mature deployments.

27.4 Hydrogen and Long-Duration Storage

Hydrogen storage and other long-duration technologies (flow batteries, compressed air, thermal storage) address the seasonal-variability gap that lithium-ion batteries cannot economically span. AI applications include electrolyser dispatch optimisation against variable renewable supply, hydrogen demand forecasting, and pipeline-scale logistics planning. Most long-duration projects are at demonstration or early-commercial scale; production AI deployments are nascent but growing rapidly.

27.5 Mathematical Foundations of Battery Dispatch

27.5.1 Energy Arbitrage Optimisation

$$\max_{p^c, p^d} \sum_t \pi_t (p_t^d - p_t^c) \Delta t, \quad \text{s.t.} \quad \text{SOC}_{t+1} = \text{SOC}_t + \eta_c p_t^c \Delta t - \frac{p_t^d}{\eta_d} \Delta t,$$

$$0 \leq \text{SOC}_t \leq E_{\max}, \quad 0 \leq p_t^c \leq P_c^{\max}, \quad 0 \leq p_t^d \leq P_d^{\max}, \quad \pi_t \text{ price.}$$

27.5.2 Degradation Cost

Cycle aging:

$$\text{Cap. loss} = k_1 (\text{DoD})^{k_2} N_{\text{cycles}}^{k_3},$$

typical $k_2 \approx 1.5$, $k_3 \approx 0.5$. Calendar aging adds $k_4 \exp(-E_a/RT) \cdot t$ for activation energy E_a .

27.5.3 Round-Trip Efficiency

$$\eta_{\text{rt}} = \eta_c \cdot \eta_d.$$

Lithium-ion: 0.85–0.95; lead-acid: 0.70–0.85; flow: 0.65–0.80.

27.6 Worked Problems and Solutions

Problem 27.1 (Arbitrage Profit). Charge 4 MWh at \$30/MWh; discharge same energy (efficiency 90%) at \$110/MWh. Compute profit.

Solution. Charge cost: $4 \cdot 30 = \$120$. Discharge: $4 \cdot 0.9 = 3.6$ MWh at $\$110 = \396 . Profit: $\$396 - \$120 = \$276$. ■

Problem 27.2 (Cycle Life). $k_1 = 1 \times 10^{-4}$, $k_2 = 1.4$, $k_3 = 0.5$, DoD = 0.8. Compute capacity loss after 1,000 cycles.

Solution. $0.8^{1.4} = e^{1.4 \ln 0.8} = e^{1.4(-0.2231)} = e^{-0.3124} = 0.7317$. $1000^{0.5} = 31.62$. Loss = $10^{-4} \cdot 0.7317 \cdot 31.62 = 2.314 \times 10^{-3} = 0.231\%$ per the formula's units (calibrate on data, but the relative ordering is robust). ■

27.7 Chapter Conclusion

Battery storage is the technology that, more than any other, will determine the pace of grid decarbonisation. The chapter walked through the principal AI applications: dispatch optimisation against price arbitrage and ancillary-service markets, degradation-aware operation that trades short-term revenue against long-term capacity, and the integration of storage into broader portfolio optimisation across generation and demand assets.

The arbitrage worked problem illustrated the basic economics. Charging at \$30/MWh and discharging at \$110/MWh, after round-trip-efficiency losses, yields \$276 per cycle on a 4 MWh asset. Across a year of daily cycling, this produces approximately \$100,000 of gross revenue per MWh of installed capacity, against capital costs that have fallen below \$300,000/MWh for utility-scale lithium-ion systems. The economics close in many markets and are tightening in those where they do not.

The degradation treatment introduced the empirical scaling laws that relate depth-of-discharge, cycle count, and temperature to capacity loss. The worked problem on cycle life illustrated the calculation; the operational implication is that aggressive dispatch must trade against warranty preservation, with the optimal trade depending on price volatility and battery cost. Modern dispatch optimisers internalise this trade explicitly; legacy systems often do not.

The chapter's coverage of grid-scale flow batteries, behind-the-meter storage, and emerging long-duration technologies signalled the diversity of storage applications. Each application has distinct technical requirements, distinct economics, and distinct optimisation problems. Lithium-ion dominates the four-hour-and-shorter market; flow batteries and emerging technologies will compete for longer durations. We turn next to the rapidly growing storage asset class that is also a mobility solution: electric vehicles.

Electric Vehicles and Sustainable Transport

A million EVs charging without coordination is a million chances to crash the grid. A million EVs charging with coordination is a million units of grid storage.

Kennedy Chengeta, PhD

Transport accounts for approximately 16% of global CO₂ emissions. Road transport dominates, with passenger cars roughly 45% of transport emissions and freight roughly 30%. Electrification — combined with mode shift to rail, public transit, and active modes — is the dominant decarbonisation pathway. AI applications span EV charging optimisation, grid integration of EV fleets, route and mode optimisation, and emerging autonomous-vehicle implications for sustainability.

28.1 Smart Charging

Uncontrolled EV charging at home (typically evening peak) coincides with system peak demand and threatens grid capacity. Smart charging shifts load to off-peak hours, aligns with renewable generation, and provides grid services through V2G (vehicle-to-grid) bidirectional flow. Control architectures range from centralised (utility-managed schedules) to distributed (price-responsive vehicle agents) to hybrid. ML-based forecasting of vehicle availability, state-of-charge requirements, and driver preferences anchors all approaches.

28.2 Charging Infrastructure Planning

Where to site charging infrastructure is a complex optimisation problem combining travel demand modelling, grid capacity, land availability, and equity considerations. Geospatial ML on travel patterns from vehicle telematics, mobile data, and traffic counts produces demand forecasts at suburb resolution. Integration with grid hosting capacity (where transformer and feeder limits constrain new load) is essential. Equity overlays — ensuring infrastructure reaches lower-income and minority communities — are increasingly mandated by regulation.

28.3 Fleet and Logistics Optimisation

Fleet electrification — buses, delivery vehicles, heavy trucks — requires reoptimised routing, charging strategy, and vehicle assignment. Route-optimisation ML accounts for charging stops, range anxiety margins, weather effects on range, and time-of-use pricing. UPS, DHL, and Amazon have demon-

strated 5-15% operational cost reductions and corresponding emission reductions through ML-driven fleet optimisation; the gains are larger for electrified fleets due to higher operational complexity.

Example 28.1 (Cape Town BRT Electrification). A bus rapid transit fleet electrification analysis for Cape Town's MyCiTi system used route simulation, charging-pattern ML, and grid-impact modelling to design a phased deployment of 480 electric buses. Resulting plan: 4 charging hubs with combined 28MW load, off-peak charging coordinated with the Cape Town municipality's 75MW solar plant, fleet rotations preserving service intervals. Modelled annual emissions reduction: 24,000 tonnes CO₂e relative to diesel baseline.

28.4 Mode Shift and Active Transport

Decarbonising transport is not only about electrification — it is also about mode shift towards rail, public transit, walking, and cycling. AI applications support mode shift through trip planning that surfaces sustainable options, dynamic pricing that internalises external costs, and infrastructure planning that prioritises high-impact corridors. The interaction with autonomous-vehicle development is consequential: autonomy lowers the cost of personal motorised travel, potentially increasing total vehicle-kilometres travelled even as per-kilometre emissions fall. Whether AVs are net-positive for sustainability depends on policy and design choices that AI alone does not determine.

28.5 Mathematical Foundations of EV Charging and Transport

28.5.1 Charging Schedule Optimisation

For vehicle v with arrival time a_v , departure d_v , required energy E_v , and price π_t :

$$\min \sum_t \pi_t \sum_v p_{v,t} \Delta t \quad \text{s.t.} \quad \sum_t p_{v,t} \Delta t \geq E_v, \quad p_{v,t} = 0 \text{ if } t \notin [a_v, d_v].$$

28.5.2 Range Estimation

$$\text{Range} = \frac{E_{\text{batt}}}{c_v}, \quad c_v(\text{speed, slope, weather}) = c_0 + c_1 v^2 + c_2 \text{slope} + c_3 \Delta T.$$

28.5.3 Vehicle Routing with Charging

Mixed-integer model with binary x_{ij} traversal and continuous z_i state-of-charge at node i :

$$z_j \leq z_i - c_{ij} x_{ij} + R_i (1 - x_{ij}),$$

linearisation of conditional state evolution.

28.6 Worked Problems and Solutions

Problem 28.2 (Charging Economics). EV needs 50 kWh; smart charging schedules at average price \$0.06/kWh (off-peak); flat schedule \$0.18/kWh. Compute savings.

Solution. Smart: $50 \cdot 0.06 = \$3.00$. Flat: $50 \cdot 0.18 = \$9.00$. Saving: \$6.00 per full charge. Across 200 charges/year: \$1,200/yr. ■

28.7 Chapter Conclusion

Electric vehicles are reshaping both transportation and electricity sectors simultaneously. The chapter argued that the principal AI opportunities lie in coordination: matching charging to renewable supply, coordinating fleets to minimise both energy and travel costs, and managing the bidirectional power flows that V2G enables. Each opportunity is technically rich and operationally consequential.

The charging-economics worked problem illustrated the value of smart charging at the household level. A \$6.00 saving per full charge, across 200 charges per year, yields \$1,200 of annual saving — not by itself transformative, but at fleet scale and across millions of vehicles, material both for households and for the grid as a whole. Aggregating flexibility across millions of vehicles produces virtual-power-plant capacities that begin to rival utility-scale storage assets, and that trade in many of the same markets.

The mathematical treatment of vehicle routing with charging constraints showed how the classical VRP extends to electric fleets. The integrality of the charging-decision variables, the nonlinearity of energy consumption in operating conditions, and the state-of-charge tracking through the route all add complexity. Modern solvers handle the augmented problem at fleet scales of hundreds of vehicles; ML-based approximations extend this to larger fleets.

The V2G treatment acknowledged both promise and unresolved questions. The promise: bidirectional EV charging unlocks gigawatts of flexible capacity at very low marginal cost. The unresolved questions: battery warranty implications, customer participation incentives, and grid- interconnection standards. The technology is ready; the institutional infrastructure lags. This chapter closes Part IV. Part V turns to the machinery that meters and monetises all of the mitigation activity Part IV has discussed: carbon markets.

Part V

Carbon Measurement, Markets, and Offsets

Carbon Accounting Foundations

Every carbon claim is a number. Every number has a method. Every method has assumptions. Carbon accounting is the discipline of being honest about all three.

Kennedy Chengeta, PhD

Carbon accounting — the systematic measurement of greenhouse gas emissions and removals — is the foundation of every credible sustainability programme. Without accounting, claims are unverifiable, markets cannot function, and policy cannot be designed. This chapter treats the standards (GHG Protocol, ISO 14064, IPCC inventory guidelines), the operational practice of corporate and project-level accounting, and the AI applications that make accounting tractable at scale.

29.1 Standards: GHG Protocol, ISO 14064, IPCC

The **GHG Protocol** (WRI/WBCSD) is the dominant corporate accounting standard. Its Corporate Standard defines Scope 1 (direct), Scope 2 (purchased energy), and Scope 3 (other indirect) emissions. Its Project Protocol governs project-level accounting for offset generation. **ISO 14064** is the parallel international standard, harmonised with GHG Protocol but with stronger third-party assurance requirements. **IPCC inventory guidelines** (2006, refined 2019) govern national inventories submitted under the UNFCCC and Paris Agreement. The three frameworks are interoperable but differ in scope, granularity, and use case.

29.2 Activity Data and Emission Factors

Definition 29.1 (Tier-Structured Accounting). IPCC inventory guidelines define three tiers of accounting accuracy:

- **Tier 1:** default emission factors applied to gross activity data.
- **Tier 2:** country- or region-specific emission factors, more disaggregated activity data.
- **Tier 3:** mechanistic models, direct measurement, or higher-resolution data where available.

Most corporate accounting today operates at Tier 1 for Scope 3 due to data availability constraints. Moving to Tier 2 or 3 typically requires substantial data infrastructure — supplier-specific factors, transactional logistics data, product-level lifecycle inventories. AI helps by extracting structured emission-factor inputs from invoices, shipping documents, and supplier reports, automating the data

assembly that historically consumed thousands of analyst-hours.

29.3 Scope 3: The Hard Problem

Scope 3 emissions — supply chain upstream, product use phase, end-of-life — account for 70-90% of total emissions for most consumer-facing companies and over 90% for financial institutions (financed emissions). Scope 3 is hard because the data are external to the reporting entity, often spread across thousands of suppliers and customers operating in heterogeneous data environments. AI applications include: NLP extraction of emission information from supplier documents; ML imputation of missing supplier data using sectoral and geographic patterns; satellite-based verification of claims about land use, deforestation, and freight; and graph models that propagate emissions through multi-tier supply networks.

29.4 Verification and Assurance

External assurance is increasingly required by securities regulators (CSRD in the EU, SEC climate disclosure rules in the US, JSE listing requirements in South Africa). Assurance levels — limited versus reasonable — map to different evidence standards. AI plays both sides: companies use AI to assemble defensible data; auditors use AI to spot anomalies, cross-check against independent data sources (satellite, customs, shipping), and prioritise sampling. The audit-AI arms race is real and accelerating.

29.5 Mathematical Foundations of Corporate Carbon Accounting

29.5.1 Cross-Boundary Aggregation

For Scope 1, 2, 3 with emissions E_1, E_2, E_3 :

$$E_{\text{total}} = E_1 + E_2 + E_3, \quad E_1 = \sum_a a_a f_a^{(1)}, \dots$$

Scope 2 location-based vs market-based:

$$E_2^{\text{loc}} = \sum_r e_r \cdot \text{EF}_{r,\text{grid}}, \quad E_2^{\text{mkt}} = \sum_r e_r \cdot \text{EF}_{r,\text{contracted}}.$$

29.5.2 Tier-Level Uncertainty

For Tier 1 (default factors): $\sigma_f/f \approx 30\text{--}50\%$. For Tier 2 (regional): $10\text{--}30\%$. For Tier 3 (direct measurement): $2\text{--}5\%$. Total uncertainty:

$$\sigma_E = \sqrt{\sum_a (a_a \sigma_{f_a})^2 + (f_a \sigma_{a_a})^2}.$$

29.5.3 Spend-Based Estimation

For a category with spend S and EEIO factor f_{eeio} (kg CO₂e/USD):

$$E = S \cdot f_{\text{eeio}}.$$

29.6 Worked Problems and Solutions

Problem 29.2 (Total Emissions). Scope 1: 2,400 t. Scope 2 location-based: 8,600 t. Scope 3 (15 categories aggregated): 47,900 t. Compute total and Scope 3 share.

Solution. Total = 58,900 tCO₂e. Scope 3 share = $47,900/58,900 = 81.3\%$. Typical for consumer-goods firms. ■

Problem 29.3 (Spend-Based Scope 3). Procurement category spend \$2.4M; EEIO factor 0.42 kg CO₂e/USD. Compute Scope 3 contribution.

Solution. $2,400,000 \cdot 0.42 = 1,008,000 \text{ kg} = 1,008 \text{ tCO}_2\text{e}$. ■

29.7 Chapter Conclusion

Corporate carbon accounting under the GHG Protocol is the bedrock of emissions disclosure, target-setting, and trading. The chapter walked through the Scope 1, 2, and 3 framework, the activity-factor multiplication that produces most baseline estimates, and the tier-level uncertainty that conditions every published figure. The worked problem produced a 58,900 tCO₂e aggregate total of which 81.3% was Scope 3 — a typical profile for consumer-goods firms, and a reminder of where most corporate emissions actually originate.

The spend-based estimation discussion deserves emphasis as the operational default for most Scope 3 categories. Direct measurement of supplier emissions is the gold standard; spend-based estimation via EEIO factors is the practical fallback when supplier data is unavailable. The \$2.4M-times-0.42 calculation in the worked example produces 1,008 tCO₂e at modest data cost; refining toward direct measurement is appropriate for material categories but rarely justifies the effort for trivial ones.

The chapter's AI-application discussion focused on the operational plumbing that increasingly distinguishes leading from laggard practice. NLP for invoice parsing, computer vision for utility-bill extraction, ML-based emission-factor selection, and automated anomaly detection in emission time series collectively reduce the manual labour of corporate accounting by an order of magnitude. The tools are commercially mature; the adoption frontier is the institutional change needed to integrate them with finance, procurement, and sustainability functions.

The chapter's standards discussion previewed Chapter 48. ISSB IFRS S2, the CDP questionnaire, the SBTi target-validation framework, and the CSRD European Sustainability Reporting Standards all draw on the same GHG-Protocol foundation. The operational differentiator is whether the corporate emissions inventory is built once at sufficient quality to satisfy all of them, or built multiple times to satisfy each separately. We turn next to the technical machinery that turns claimed reductions into verified credits: MRV.

Measurement, Reporting, and Verification

Verification is the bridge between a claim and a credit. AI lengthens the bridge but it cannot replace the bridge.

Kennedy Chengeta, PhD

Measurement, reporting, and verification — MRV — is the operational machinery that converts physical reality into tradeable carbon units. MRV underpins compliance markets (EU ETS, California cap-and-trade, China ETS), voluntary markets (Verra, Gold Standard, ART, Climate Action Reserve), and emerging Article 6 mechanisms under the Paris Agreement. AI has transformed MRV economics: continuous, large-scale, low-cost monitoring is now possible where annual on-site audits were the only previous option.

30.1 The MRV Stack

MRV operates in three layers. **Measurement** produces physical estimates — forest biomass, soil organic carbon, methane emissions, electricity consumption — with quantified uncertainty. **Reporting** converts measurements into credit-eligible flows under specific methodologies, with transparent calculation chains. **Verification** is third-party auditing of the measurement and reporting processes, typically by accredited validation and verification bodies (VVBs).

AI enters the measurement layer most heavily, but extends to reporting (automated methodology compliance checks) and verification (anomaly detection, cross-source consistency).

30.2 Remote Sensing for MRV

Satellite-based MRV has become standard for forestry and land-use projects. Optical (Sentinel-2, Landsat, Planet) and radar (Sentinel-1, ALOS PALSAR) imagery, processed by deep-learning models, produces hectare-scale maps of canopy height, biomass, and deforestation events. The Global Forest Watch platform (WRI) provides global, weekly deforestation alerts; CTrees, Pachama, and Sylvera build on similar data with additional ML for project-specific verification.

30.3 In-Situ Sensors and IoT

For project types where remote sensing is insufficient — improved cookstoves, biogas digesters, livestock methane suppression — in-situ sensors provide direct measurement. Low-cost CO sensors verify cookstove emission reductions; methane analysers on cattle barns measure suppression from feed additives; pressure-temperature loggers verify biogas digester operation. ML applications include

sensor fault detection, gap-filling for offline periods, and aggregation across heterogeneous sensor networks.

Example 30.1 (Improved Cookstove MRV in Rwanda). A Rwandan project deployed 18,000 improved cookstoves with embedded temperature sensors transmitting daily usage data via NB-IoT. ML models trained on calibration kitchen tests convert temperature signatures to fuel consumption and avoided emissions per stove per day. Verification audits sample physical stoves quarterly; data plausibility checks flag tampering. Result: continuous MRV at USD 0.40 per stove per year, against USD 8-12 for the previous survey-based MRV approach. The project issued credits for verified emission reductions of 24,000 tCO₂e in its first year.

30.4 Methane Detection from Space

Methane is around 80 times more powerful than CO₂ over 20 years. Satellite methane detection has been transformed by Sentinel-5P, GHGSat, MethaneSAT (launched 2024), and Carbon Mapper. Plume-detection ML on hyperspectral imagery localises emission sources to facility resolution; quantification ML estimates flux from plume morphology. The implications for oil-and-gas regulation, landfill management, and rice-paddy mitigation are substantial; the data is increasingly being integrated into compliance and disclosure frameworks.

30.5 Limits and Failure Modes

AI MRV is not foolproof. Models trained on labelled deforestation events may miss subtle degradation. Soil-carbon estimates have substantial uncertainty that can erode credit values. Methane plume detection has lower limits below which small leaks are invisible. The honest MRV practice combines AI-driven continuous monitoring with periodic independent ground-truth, conservative uncertainty discounting, and transparent reporting of limitations. Buyer trust depends on this discipline; the market reputation hit from a single high-profile MRV failure can affect hundreds of unrelated projects.

30.6 Mathematical Foundations of MRV Statistics

30.6.1 Stratified Sampling for MRV

Population stratified into L strata with weights W_l , sample size n_l , sample variance s_l^2 :

$$\hat{Y}_{\text{strat}} = \sum_l W_l \bar{y}_l, \quad \text{Var}(\hat{Y}_{\text{strat}}) = \sum_l W_l^2 \frac{s_l^2}{n_l}.$$

Optimal Neyman allocation: $n_l \propto W_l s_l$.

30.6.2 Conservative Estimation

For uncertainty σ in mean estimate $\hat{\mu}$, conservative credit issued:

$$\hat{\mu}_{\text{conserv}} = \hat{\mu} - z_{\alpha} \sigma, \quad z_{0.95} = 1.645.$$

30.6.3 Methane Plume Quantification

Integrated mass enhancement IME for column-density anomaly $\Omega(\mathbf{x})$:

$$\text{IME} = \iint_{\text{plume}} \Omega(\mathbf{x}) d\mathbf{x},$$

flux:

$$F = \frac{U_{\text{eff}}}{L} \cdot \text{IME},$$

with effective wind U_{eff} and plume length L .

30.7 Worked Problems and Solutions

Problem 30.2 (Stratified Standard Error). Two strata, $W_1 = 0.7$, $W_2 = 0.3$; $s_1 = 4.2$, $s_2 = 9.1$ (t C/ha); $n_1 = 80$, $n_2 = 80$.

Solution. $\text{Var} = 0.49 \cdot 17.64/80 + 0.09 \cdot 82.81/80 = 0.49 \cdot 0.2205 + 0.09 \cdot 1.0351 = 0.1080 + 0.09316 = 0.2012$. $\text{SE} = \sqrt{0.2012} = 0.449$ t C/ha. ■

Problem 30.3 (Methane Flux). Plume IME = 1,200 kg, length 4 km, effective wind 6 m/s. Compute flux in kg/h.

Solution. $F = (6/4000) \cdot 1200 = 1.8$ kg/s = 6,480 kg/h. ■

30.8 Chapter Conclusion

MRV is the operational backbone of carbon markets. The chapter walked through the components: the stratified sampling that produces unbiased estimates from finite ground-truth campaigns; the conservative-deduction framework that prevents over-crediting in the presence of uncertainty; the satellite-based monitoring that supplements ground-based sampling with continental coverage; and the emerging plume-quantification technology that is transforming methane management.

The stratified-sampling worked problem illustrated the standard-error calculation that supports both publication and credit issuance. A 0.449 t C/ha standard error on a stratified-mean estimate, against a hypothetical mean change of 1–2 t C/ha, indicates moderately strong statistical power; the buffer-pool deduction under VM0042 or similar methodologies would be in the 10%–25% range. Tightening the standard error through larger samples or more strata reduces the buffer; the economics of when to invest in sampling versus accept the buffer is a fundamental project-design decision.

The methane-plume-quantification worked problem demonstrated the IME- times-wind-over-length framework that underlies satellite-based quantification. The 6,480 kg/h flux from the worked example is plausibly detectable by GHGSat or MethaneSAT; smaller emitters require more sensitive instruments or longer integration times. The frontier in 2025 is the integration of multiple satellite platforms — Sentinel-5P for screening, GHGSat for targeted point-source quantification, MethaneSAT for facility-level integration — into a coherent operational system.

The chapter's coverage of continuous monitoring deserves emphasis. Episodic ground-truth campaigns produce snapshot estimates with high sample-to-sample variance; continuous monitoring through sensor networks and satellite tasking produces time-series estimates with much lower variance for the

same total cost. The shift from episodic to continuous monitoring is the defining MRV transformation of the next decade. We turn next to the markets that consume MRV output: the voluntary carbon market.

Voluntary Carbon Markets

Voluntary carbon markets exist because policy is slower than the climate. They will be judged not by their growth but by whether their tonnes are real.

Kennedy Chengeta, PhD

Voluntary carbon markets — where companies and individuals purchase credits to offset emissions they have not eliminated — have grown from roughly USD 0.5 billion annual transaction value in 2020 to several billion by 2024, with significant short-term volatility. The markets have also faced sustained scrutiny over credit integrity, leading to substantial methodological reforms. AI is integral both to the production of high-quality credits and to the verification scrutiny that has transformed the markets.

31.1 Standards and Registries

Verra (VCS) is the largest voluntary registry. Gold Standard, ART (Architecture for REDD+ Transactions), Climate Action Reserve (CAR), and Plan Vivo serve specific niches. Standards define eligible project types, methodologies, baseline approaches, monitoring requirements, and crediting periods. Recent reforms include Verra’s Jurisdictional REDD+ approach (VM0048), the Integrity Council for the Voluntary Carbon Market’s Core Carbon Principles (CCPs), and the Voluntary Carbon Markets Integrity Initiative (VCMI) on the demand side. AI underpins methodology compliance — automated monitoring of methodology requirements — and integrity assessment.

31.2 Project Types and AI Applications

Major project types and AI roles: **Forestry (REDD+, A/R)** — ML biomass estimation, deforestation alerting, leakage detection. **Renewable energy** — generation verification, grid emission factor calculation, additionality testing. **Methane (landfill, livestock, biogas)** — IoT-based emission verification, satellite plume detection. **Soil carbon** — spectral estimation, modelled accounting under Verra VM0042. **Direct air capture** — mass-balance verification, energy source attribution. **Cookstoves** — IoT usage monitoring, fuel-replacement modelling.

31.3 The Integrity Crisis and Its Aftermath

In 2022-23, investigative journalism (The Guardian, Die Zeit, ZDF) and academic studies (West et al. 2023 in *Science*) documented systematic over-crediting in legacy REDD+ projects. The findings drove price collapses for nature-based credits, methodology suspensions, and accelerated reform. AI

was central on both sides: the studies relied heavily on satellite remote sensing and ML to produce alternative deforestation estimates; subsequent reforms increasingly use AI MRV to tighten future projects.

The lessons generalise: AI does not automatically produce integrity. AI can produce highly precise estimates of the wrong quantity, or precise estimates of the right quantity with insufficient ground-truth. Methodologies, governance, and human auditing remain essential.

31.4 Marketplace Infrastructure

Trading, settlement, and retirement of voluntary credits has historically been over-the-counter and opaque. New marketplace infrastructure — CBL Markets, Carbonplace, Senken, AirCarbon — provides exchange-style trading. Tokenisation efforts (Toucan, Klima) brought voluntary credits onto blockchain rails before regulatory pushback constrained the practice. AI applications in marketplace infrastructure include price discovery, trade-pattern analysis for market manipulation detection, and credit-quality scoring for buyer-side decision support.

31.5 Mathematical Foundations of Carbon Market Pricing

31.5.1 Quality-Adjusted Credit Price

For credit type t with reference price π_0 and integrity score $s_t \in [0, 1]$:

$$\pi_t = \pi_0 \cdot s_t.$$

Buyer's expected effective price under verification probability q :

$$\pi_{\text{eff}} = q\pi_t + (1 - q)(\pi_t + L_{\text{rep}}),$$

where L_{rep} is reputational damage from low-integrity credits.

31.5.2 Buffer-Pool Sizing

For project portfolio with reversal probabilities $\{p_i\}$ and credit volumes $\{V_i\}$:

$$\text{Buffer} = \sum_i p_i V_i + z_\alpha \sqrt{\sum_i p_i (1 - p_i) V_i^2}.$$

31.6 Worked Problems and Solutions

Problem 31.1 (Buffer Sizing). Three projects: 10k credits at 2% reversal probability, 5k at 5%, 2k at 12%. Compute mean buffer.

Solution. Mean: $0.02 \cdot 10000 + 0.05 \cdot 5000 + 0.12 \cdot 2000 = 200 + 250 + 240 = 690$ credits buffer.

■

31.7 Chapter Conclusion

The voluntary carbon market in 2025 is in the process of being re-engineered around integrity. The 2023 Guardian/SourceMaterial investigation, which questioned the legitimacy of the largest portion

of Verra's avoided-deforestation credits, accelerated a reckoning the market had been postponing. The chapter walked through the structural issues — baseline manipulation, leakage under-counting, permanence shortfall, additionality uncertainty — and the technical responses that AI-enabled MRV is bringing to bear.

The buffer-sizing worked problem illustrated the portfolio-level risk management that registries now perform. A 690-credit buffer against a 17,000-credit portfolio, sized by expected reversal, hedges against the modal failure mode of forest projects. Newer methodologies size buffers both larger and more dynamically; the trajectory is toward more conservative crediting as the integrity expectations of buyers rise.

The chapter's coverage of the Integrity Council for the Voluntary Carbon Market (ICVCM) and the Voluntary Carbon Markets Integrity Initiative (VCMI) traced the institutional architecture that is crystallising around the credibility challenge. The ICVCM Core Carbon Principles, the VCMI Claims Code, and the SBTi Beyond Value Chain Mitigation guidance collectively define what a credible high-integrity purchase looks like in 2025. The framework is still maturing, but the direction is settled.

The chapter's deployment-pattern discussion — which types of project, which geographies, which methodologies are attracting credible buyer interest — previewed the regional treatment in Chapter 58. African projects, in particular, face the structural challenge that local capacity to lead the technical work is unevenly distributed, even as the carbon-stock potential is large. We turn next from voluntary to compliance markets.

Compliance Carbon Markets

A regulator who cannot verify emissions cannot enforce a price on them. AI is the verification engine of every credible carbon price.

Kennedy Chengeta, PhD

Compliance carbon markets — regulatory cap-and-trade and carbon-tax systems — now cover roughly one-quarter of global emissions. The EU ETS, China ETS, California-Quebec, RGGI, UK ETS, and emerging systems in South Korea, Indonesia, and elsewhere set legal emission limits and price emissions through tradeable allowances. AI applications include emissions verification, market integrity surveillance, allowance forecasting, and the design and implementation of border adjustments.

32.1 Major Compliance Systems

The **EU ETS** is the largest, covering power, industry, intra-EU aviation, and from 2024 maritime. Phase 4 (2021-2030) caps emissions on a declining trajectory aligned with EU 2030 targets. The **China ETS** launched in 2021, currently covering power, with industrial expansion underway; it is the largest ETS by covered emissions. **California-Quebec (WCI)** operates a linked regional system. **RGGI** covers Northeast US power generation. **Korea ETS**, **NZ ETS**, **UK ETS** provide further coverage. Each system has distinct allocation, market-stability, and cost-containment provisions; AI tooling is increasingly tailored to each.

32.2 Verification and Continuous Emissions Monitoring

Compliance accounting requires high accuracy. Continuous emissions monitoring systems (CEMS) on stack flows provide direct measurement at large facilities. AI applications include sensor fault detection, gap-filling during sensor downtime, and anomaly detection that flags suspected reporting errors or fraud. The EU MRV Maritime regulation extends CEMS-style approaches to ship emissions, with AI applications in fuel-consumption estimation from voyage data.

32.3 Carbon Border Adjustment Mechanism

The **EU CBAM** began transitional reporting in 2023 and full operation in 2026. CBAM imposes a carbon-equivalent charge on imports of cement, iron and steel, aluminium, fertilisers, electricity, and hydrogen, equal to the embedded emissions priced at the EU ETS. Implementation requires production-level emission verification across thousands of foreign producers — a vast data and

verification challenge. AI applications include production-emission inference from publicly available data (capacity, technology, energy source), satellite verification of facility activity, and supply-chain mapping. Similar mechanisms are under development in the UK and being explored elsewhere.

32.4 Allowance Price Forecasting

EU ETS allowance (EUA) prices have ranged from EUR 5 to EUR 100 per tonne over the past decade, driven by policy decisions, fuel-switching economics, and macroeconomic conditions. ML price forecasting combines fundamental drivers (supply-demand, MSR mechanics, fuel prices) with sentiment analysis from regulatory communication. Hedge funds and compliance entities use forecasting differently: hedge funds for trading; compliance entities for procurement strategy. The market itself is large enough that AI-driven trading affects price formation.

32.5 Mathematical Foundations of Compliance Markets

32.5.1 Allowance Price Equilibrium

In a competitive cap-and-trade with cap $\bar{E}$, marginal abatement cost (MAC) curve $c(e)$, equilibrium price equals the marginal cost of the last tonne abated:

$$\pi^* = c'(\bar{E}).$$

32.5.2 Market Stability Reserve Mechanics

Total Number of Allowances in Circulation (TNAC) triggers MSR intake when above 833 Mt and release below 400 Mt (EU ETS):

$$\text{Withdraw}_t = 0.24 \cdot \max(\text{TNAC}_t - 833, 0).$$

32.5.3 CBAM Charge

For imported product with embedded emissions E_{imp} and EU ETS price π :

$$\text{CBAM charge} = E_{\text{imp}} \cdot \pi - \text{Free allocation}.$$

32.6 Worked Problems and Solutions

Problem 32.1 (CBAM Cost). Steel import: 1,000 tonnes with embedded emissions 1.8 t CO₂/t steel; EUA price EUR 95/t. Compute CBAM charge.

Solution. Total embedded: $1,000 \cdot 1.8 = 1,800$ tCO₂e. Charge: $1,800 \cdot 95 = \text{EUR } 171,000$. ■

32.7 Chapter Conclusion

Compliance carbon markets are the policy instrument through which most of the world's coordinated decarbonisation is occurring. The EU ETS covers approximately 1.5 Gt of annual emissions at allowance prices near EUR 80–100/tonne in 2025; the UK ETS, California, and China each operate at similar scales and prices. The CBAM extends the EU ETS price signal to imports from jurisdictions

without comparable carbon pricing, fundamentally reshaping the trade-and-climate interface.

The CBAM worked problem illustrated the economics. EUR 171,000 of CBAM charge on 1,000 tonnes of imported steel translates to roughly 4%–6% of the steel's wholesale price — material to trade flows, manageable for individual buyers, and likely to grow as the CBAM free-allowance phase-out progresses through 2034. The implications for African and Asian exporters into European markets, and for the steel and cement industries globally, are substantial.

The chapter's AI-application coverage focused on the operational plumbing that compliance markets generate. Allowance-price forecasting through autoregressive models, neural networks, and ensemble methods is mature commercial practice; compliance optimisation under the constraint of allowance availability is increasingly automated; the verification of CBAM-required emissions data on imports will create a new layer of MRV demand that the AI-enabled MRV systems of Chapter 30 are well-positioned to serve.

The market-stability-reserve treatment introduced the dynamic-supply feature that the EU ETS pioneered. The MSR removes allowances from the market when the total number in circulation exceeds 833 Mt and releases them when below 400 Mt, dampening price volatility and supporting investment in long-lived abatement. The mathematical treatment of MSR mechanics gives the reader the tools to model future allowance prices under different policy assumptions. We turn next from the trading machinery to the underlying physical credits: nature-based solutions.

Nature-Based Solutions

A forest is a slow technology. AI cannot make it faster, but AI can make sure that when we say it is sequestering carbon, it actually is.

Kennedy Chengeta, PhD

Nature-based solutions — forest conservation, restoration, agroforestry, soil-carbon practices, wetland restoration — offer some of the largest, lowest-cost climate mitigation potential alongside biodiversity, water, and livelihood co-benefits. They also have the most contested measurement challenges. This chapter treats the major NbS pathways and the AI applications that determine whether the carbon claims are credible.

33.1 The NbS Mitigation Wedge

Griscom et al. (2017) estimated nature-based solutions could provide 11-13 Gt CO₂e per year of mitigation by 2030 at costs below USD 100 per tonne — roughly one-third of the total mitigation needed for 1.5°C trajectories. Subsequent revisions have refined the estimate downward, reflecting tighter accounting of permanence, leakage, and competing land uses. The pathways with largest credible potential are reforestation, avoided deforestation, agroforestry, and improved forest management; soil carbon, wetland restoration, and grazing land management are smaller contributors with substantial co-benefits.

33.2 Forest Carbon: Allometric Models and Remote Sensing

Forest above-ground biomass is estimated through allometric models that relate tree diameter, height, and species to biomass. Field plots calibrate national-scale models; remote sensing scales them to continental coverage. LiDAR (GEDI on the ISS, ICESat-2, airborne campaigns) provides direct height measurements; SAR (Sentinel-1, BIOMASS launched 2025) provides biomass-sensitive backscatter; optical remote sensing provides land-cover and disturbance information. ML fusion of these data sources — random forest and deep-learning ensembles — produces operational biomass maps at 30-100m resolution.

33.3 Soil Organic Carbon in Agriculture

Soil-carbon credits from regenerative agricultural practices — cover cropping, reduced tillage, diverse rotations — are growing rapidly. The methodologies (Verra VM0042, CAR Soil Enrichment Protocol) require modelling-plus-sampling approaches. ML soil-carbon models (DayCent, RothC, COMET-

Farm) supplemented by spectroscopic and remote-sensing inputs produce field-scale estimates. The headline accuracy challenge is permanence: practices must be maintained for decades for the carbon to remain stored. AI monitoring of practice continuity (cover-crop detection, tillage classification) supports permanence verification.

33.4 Co-Benefits Quantification

NbS co-benefits — biodiversity, water provision, soil health, livelihoods, cultural value — are increasingly required for high-integrity credits. AI quantification of co-benefits draws on biodiversity indicators (eDNA, acoustic monitoring, camera-trap species detection), hydrological models, and socio-economic surveys. The Climate, Community, and Biodiversity Standards (CCB) and the SD VISta framework provide formal scoring. Buyers increasingly pay premiums for co-benefit-validated credits; the AI infrastructure for co-benefit measurement has become commercially significant in its own right.

33.5 Mathematical Foundations of NbS Carbon Accounting

33.5.1 Forest Biomass Allometry

$$\text{AGB} = a \cdot D^b \cdot H^c,$$

where D is diameter at breast height (cm), H height (m), a, b, c species- and climate-specific. Pan-tropical Chave et al. (2014):

$$\text{AGB} = 0.0673(\rho D^2 H)^{0.976},$$

with wood density ρ (g/cm³).

33.5.2 Permanence Risk and Discounting

For project with annual reversal probability p_r over T years:

$$\Pr(\text{no reversal}) = (1 - p_r)^T, \quad \text{Effective stored} = V \cdot (1 - p_r)^T.$$

33.6 Worked Problems and Solutions

Problem 33.1 (Tree Biomass). Tree: $D = 35$ cm, $H = 22$ m, $\rho = 0.55$ g/cm³. Compute AGB.

Solution. $\rho D^2 H = 0.55 \cdot 1225 \cdot 22 = 14,822.5$. $\text{AGB} = 0.0673 \cdot 14822.5^{0.976}$. $\log(14822.5) = 4.171$; $0.976 \cdot 4.171 = 4.071$; $10^{4.071} = 11,777$. $\text{AGB} \approx 0.0673 \cdot 11,777 = 792.6$ kg. Carbon (50% of AGB): ≈ 396 kg C, equivalent to $396 \cdot 44/12 = 1,453$ kg CO₂. ■

Problem 33.2 (Permanence-Adjusted Volume). Project annual reversal probability 0.5%; horizon 40 years; gross sequestration 100,000 tCO₂e.

Solution. $(0.995)^{40} = e^{40 \ln 0.995} = e^{40 \cdot (-0.005013)} = e^{-0.2005} = 0.8184$. Effective: $100,000 \cdot 0.8184 = 81,838$ tCO₂e. ■

33.7 Chapter Conclusion

Nature-based solutions are simultaneously the largest near-term mitigation opportunity and the most operationally difficult one. The chapter walked through the principal NbS categories — avoided deforestation, afforestation and reforestation, improved forest management, soil-carbon sequestration, mangrove restoration, peatland conservation — and the AI applications that support each.

The tree-biomass worked problem illustrated the pan-tropical Chave allometry that underlies most forest-carbon estimation. A single 35-cm-diameter tree of 22-m height and 0.55 g/cm^3 wood density stores 396 kg of carbon equivalent to 1,453 kg of CO_2 . Multiplied across the hundreds of trees per hectare in a tropical forest and the hundreds of millions of hectares globally, the stock scales to the hundreds of gigatonnes of CO_2 at stake.

The permanence-adjusted-volume worked problem illustrated the mathematical machinery of permanence accounting. A 0.5% annual reversal probability over 40 years deflates 100,000 tCO_2e of nominal sequestration to 81,838 tCO_2e of effective sequestration — an 18% deduction that few project documents make explicit. Operational buffer pools size against this risk explicitly; the mathematical treatment in this chapter gives the reader the tools to audit a project's accounting.

The chapter's discussion of co-benefits — biodiversity, water, soil, local livelihoods — pointed forward to Part VII on biodiversity. NbS that maximise carbon sequestration at the expense of biodiversity (monoculture plantations, for example) are inferior to NbS that deliver both. The premium for high-integrity, co-benefit-rich credits in the voluntary market is growing and is likely to widen further as buyer sophistication increases. We turn next to the marine and coastal extension of NbS: blue carbon.

Blue Carbon

Mangroves store carbon at four times the rate of terrestrial forests, on three percent of the area. They are the highest-leverage hectares on the planet.

Kennedy Chengeta, PhD

Blue carbon — carbon stored in coastal and marine ecosystems — is dominated by mangroves, salt marshes, and seagrasses. These ecosystems sequester carbon at rates per hectare that dwarf terrestrial forests, store enormous legacy carbon stocks in waterlogged sediments, and are being lost to deforestation, drainage, and coastal development at rapid rates. AI applications in blue carbon focus on mapping, monitoring, and verification at scales that conventional ground methods cannot reach.

34.1 Why Blue Carbon Matters

Mangroves cover roughly 14 million hectares globally but store an estimated 5-10 Gt of carbon, much of it in soils to several metres depth. Annual sequestration rates of 6-8 tonnes CO₂e per hectare exceed those of terrestrial tropical forests by approximately 4×. Salt marshes and seagrasses are smaller in extent but with similarly high per-hectare rates. The combined ecosystem services — coastal protection, fisheries habitat, water quality — compound the carbon value.

34.2 Mapping and Monitoring

Mangrove mapping has been transformed by Sentinel-1 SAR, which penetrates clouds and detects mangrove structure regardless of leaf cover. The Global Mangrove Watch consortium produces annual global mangrove extent maps at 25m resolution from radar imagery, with deep-learning classification distinguishing mangroves from other coastal vegetation. Loss detection at sub-hectare resolution drives near-real-time alerts to enforcement agencies and project developers.

34.3 Carbon Stock Estimation

Above-ground biomass of mangroves is estimated using LiDAR-derived height combined with species-specific allometric equations. Below-ground carbon, which dominates the total stock, is estimated from sediment cores and ML-extrapolated based on geomorphological and ecological covariates. Recent foundational work has established that mangrove sediment carbon is approximately 5× above-ground carbon on average, with substantial regional variation that ML can now predict.

Example 34.1 (Mangrove Restoration MRV: Madagascar). A mangrove restoration project on the western coast of Madagascar restored 1,640 hectares of degraded mangrove. MRV combined Sentinel-1 SAR for cover monitoring, drone LiDAR for biomass estimation, sediment cores for soil carbon, and water-level loggers for hydrological functioning. Modelled cumulative emission removal over 40-year crediting period: 580,000 tCO₂e, with 20% buffer pool deduction for permanence risk. The project also generates blue-carbon premium pricing reflecting fisheries and coastal-protection co-benefits.

34.4 Restoration and Risk

Mangrove restoration is technically feasible at scale but requires careful site selection, hydrological restoration, and species matching. ML site-selection tools combine bathymetry, salinity, sediment supply, and historical mangrove distribution to identify high-success-probability sites. Permanence risks — cyclones, sea-level rise, ongoing aquaculture pressure — are quantified by probabilistic models that drive buffer-pool sizing. Saltmarsh and seagrass restoration follow similar logical structures with different ecological constraints.

34.5 Mathematical Foundations of Blue Carbon Accounting

34.5.1 Sediment Carbon Stock

For depth z , bulk density $\rho_b(z)$, organic carbon fraction $f_C(z)$:

$$\text{C stock} = \int_0^Z \rho_b(z) f_C(z) dz.$$

Discrete approximation across 5 cm intervals is standard.

34.5.2 Mangrove Productivity

Net Ecosystem Productivity:

$$\text{NEP} = \text{NPP} - R_h,$$

with $\text{NPP} \sim 12\text{--}25$ t C/ha/yr in healthy mangroves and heterotrophic respiration R_h partitioning between sediment burial and tidal export.

34.6 Worked Problems and Solutions

Problem 34.2 (Sediment C Stock). Three layers (0–30 cm, 30–60 cm, 60–100 cm); bulk densities 0.6, 0.7, 0.8 g/cm³; OC fractions 0.12, 0.08, 0.05. Compute C stock per m².

Solution. Layer 1: $0.6 \cdot 1000 \cdot 0.30 \cdot 0.12 = 21.6$ kg/m². Layer 2: $0.7 \cdot 1000 \cdot 0.30 \cdot 0.08 = 16.8$ kg/m². Layer 3: $0.8 \cdot 1000 \cdot 0.40 \cdot 0.05 = 16.0$ kg/m². Total: 54.4 kg C/m² = 544 t C/ha. Equivalent: $\sim 1,995$ tCO₂e/ha. ■

34.7 Chapter Conclusion

Blue-carbon ecosystems are a high-leverage component of nature-based mitigation. The chapter quantified why: a mangrove forest stores roughly 500–1,000 t C/ha across vegetation and sediment combined — two to five times the carbon density of terrestrial tropical forests. The worked problem

produced 544 t C/ha for a representative sediment column to 1 m depth, equivalent to nearly 2,000 tCO₂e/ha. Avoiding the destruction of mangroves is consequential mitigation in its own right; restoring lost mangroves adds new sequestration on top.

The chapter's treatment of mangrove remote sensing combined Sentinel-1 SAR (for canopy structure) with Sentinel-2 optical (for spectral classification) with ICESat-2 LiDAR (for canopy height). The integration yields biomass estimates with 20%–40% uncertainty at hectare scale — adequate for portfolio-level credit issuance, marginal for individual project verification. Continued advances in foundation models for SAR and optical remote sensing are tightening the uncertainty bounds steadily.

The methodology landscape for blue-carbon crediting (Verra VM0033, Plan Vivo, Gold Standard) increasingly relies on AI-derived inputs for both baseline and project monitoring. The chapter's coverage of peer-reviewed methodological guidance, registry-level methodology versioning, and the institutional review process equipped the reader to audit a credit claim from first principles rather than from registry labels alone.

The community-benefits discussion deserves emphasis. Mangroves and salt marshes are typically located adjacent to vulnerable coastal communities that depend on them for fisheries, storm protection, and traditional livelihoods. Blue-carbon projects that exclude local communities from benefits, or that displace them from traditional use, are both ethically problematic and operationally fragile. The CARE-principle data governance introduced in Chapter 52 applies with particular force to blue-carbon project development. We turn next to the engineered counterpart: direct air capture and engineered carbon storage.

Direct Air Capture and Carbon Capture and Storage

Direct air capture is the most expensive way to remove carbon and the only way to remove arbitrary amounts of it. AI is what brings the cost down.

Kennedy Chengeta, PhD

Direct air capture (DAC) and carbon capture and storage (CCS) are engineered carbon removal technologies that complement nature-based solutions. DAC removes CO₂ from ambient air; CCS captures CO₂ from concentrated point sources before atmospheric release. Both face cost and energy challenges that have limited deployment. Recent progress, including substantial AI contributions, is shifting economics toward viability for hard-to-abate emissions and net-negative pathways.

35.1 The Engineering Reality

DAC technologies fall in two families: solid sorbent (Climeworks, Heirloom) and liquid solvent (Carbon Engineering, 1PointFive). Both require energy input — thermal for sorbent regeneration, electrical for fans and pumps. Current generation systems consume around 6-10 GJ per tonne CO₂ captured; the energy cost dominates per-tonne economics. CCS on point sources (power plants, cement, steel) consumes less energy per tonne but requires concentrated CO₂ streams, which most facilities lack without dedicated capture infrastructure.

35.2 AI for Process Optimisation

DAC and CCS process optimisation problems are similar to general industrial process control: high-dimensional, partially observed, with safety constraints. ML soft sensors predict CO₂ concentration and sorbent state from cheaper measurements. Reinforcement-learning controllers optimise capture-cycle timing, regeneration temperature, and fan speed against energy cost and capture target. Reported energy reductions of 10-25% from AI control are common across pilot deployments.

Materials discovery is the higher-leverage AI application. Generative models for sorbent and catalyst design — particularly metal-organic frameworks (MOFs) for DAC and amine catalysts for CCS — have produced materials with 2-3× improved capture capacity per kilogram in lab tests. Translation to commercial-scale manufacturing is in early stages.

35.3 Storage Verification

Captured CO₂ must be stored permanently — typically injected into deep saline aquifers or depleted oil/gas reservoirs. Storage MRV requires monitoring of well integrity, plume migration, and surface leakage. Seismic monitoring with ML interpretation tracks subsurface plume location; pressure-temperature sensors detect well-integrity issues; surface flux measurements verify zero-leakage performance. Production CCS sites (Sleipner, Quest, Petra Nova) demonstrate that ML-augmented MRV can verify storage permanence at acceptable cost.

35.4 Hybrid Approaches and Mineralisation

Mineralisation — reacting captured CO₂ with calcium and magnesium silicates to form stable carbonates — offers permanent storage without leakage risk. Companies including 44.01, Cella, and Heirloom (which combines air capture with mineralisation) are scaling deployment. AI applications include reaction-kinetics optimisation, feedstock characterisation, and process scale-up modelling. Costs remain higher than geological storage at present but the permanence advantage may justify the premium for high-integrity removal credits.

35.5 Realistic Scale Expectations

Current global DAC capacity is approximately 12,000 tonnes CO₂ per year. The IEA Net Zero Scenario calls for 70 Mt by 2030 and 1 Gt by 2050 — expansions of approximately 6,000× and 80,000× respectively. Such scaling is technically possible but requires manufacturing, energy, and capital deployment at unprecedented rates. AI cannot manufacture the equipment, but it can lower the per-tonne energy cost (the dominant operational expense) and accelerate the materials science (the dominant capacity-improvement lever). Whether the scaling actually happens depends on policy, finance, and execution that AI alone does not deliver.

35.6 Mathematical Foundations of DAC Energetics

35.6.1 Sherwood Plot

Cost of separation of dilute streams scales as

$$C_{\text{sep}} \propto \frac{1}{x_{\text{feed}}},$$

i.e. removing CO₂ from 420 ppm air costs orders of magnitude more per tonne than from 12% concentrated flue gas.

35.6.2 Energy Cost Decomposition

$$E_{\text{DAC}} = E_{\text{thermal}} + E_{\text{electric}} + E_{\text{compression}} + E_{\text{loss}}.$$

Sorbent cycling: thermal regen 4–6 GJ/tCO₂e. Liquid solvent: 5–8 GJ/tCO₂e. Compression to storage pressure: 0.4–0.6 GJ/tCO₂e.

35.6.3 Net DAC Carbon

$$\text{Net captured} = E_{\text{DAC}} \cdot \text{CI}_{\text{grid}}^{\text{thermal}} - 1 \text{ tCO}_2\text{e} \quad (\text{per tonne captured, sign convention}),$$

i.e. if grid-thermal carbon-intensity costs exceed 1 tonne per tonne captured, the deployment is net-negative.

35.7 Worked Problems and Solutions

Problem 35.1 (DAC Net Capture). Sorbent DAC at 5 GJ/tCO₂e; grid carbon intensity 50 g CO₂e/MJ. Compute net carbon capture per tonne CO₂ captured.

Solution. Energy emissions: $5 \times 10^3 \text{ MJ} \cdot 0.05 \text{ kg/MJ} = 250 \text{ kg CO}_2$ per tonne captured. Net: $1000 - 250 = 750 \text{ kg}$ captured per tonne gross, i.e. 75% net efficiency. ■

35.8 Chapter Conclusion

Engineered carbon removal is one of the most actively researched technology stacks in the contemporary climate-tech sector. The chapter walked through the principal pathways: solvent-based DAC (Climeworks, Carbon Engineering), sorbent-based DAC (Heirloom, Mission Zero), mineralisation (CarbFix), and the underlying CCS infrastructure that each requires.

The DAC-energetics worked problem illustrated the fundamental energy challenge. At 5 GJ/tonne thermal energy requirement and 50 g CO₂e per MJ grid carbon intensity, 250 kg of CO₂ is emitted in capturing each tonne — a 75% net efficiency. With cleaner grid power the net efficiency rises; with high-carbon grids the technology approaches break-even. The implication is that DAC deployment requires clean electricity and clean heat as preconditions, not as nice-to-haves.

The Sherwood-plot scaling — separation cost inversely proportional to feed concentration — underlies the cost difference between DAC and post-combustion CCS. At 12% CO₂ concentration in flue gas, CCS costs \$50–\$150 per tonne; at 0.04% CO₂ in ambient air, DAC costs \$300–\$1,000 per tonne. The cost gap is structural, not incidental, and the operational implication is that DAC is fundamentally complementary to point-source CCS, not a substitute for it.

The chapter's coverage of subsurface storage geology, ML-driven site selection, and continuous monitoring of stored CO₂ via 4D seismic, InSAR, and pressure-monitoring gave the reader the technical foundation for evaluating storage permanence. The 1,000-year-plus storage horizons that mineralisation can offer compare favourably with the century-or-less horizons typical of biological storage; the cost gap is, currently, the structural barrier. This chapter closes Part V. We turn next, in Part VI, to the broader systemic shift that engineered removal increasingly complements: the circular economy.

Part VI

Circular Economy and Resource Efficiency

Circular Economy: Foundations

A linear economy is a one-way conveyor belt from extraction to landfill. AI does not bend that belt by itself — but AI is the only way to manage the loops once they are bent.

Kennedy Chengeta, PhD

The circular economy seeks to replace the linear take-make-dispose model with one in which materials, products, and components circulate at their highest value for as long as possible. The Ellen MacArthur Foundation's framework distinguishes biological cycles (food, fibre, bio-based materials) and technical cycles (manufactured products, components, materials). AI is essential infrastructure for both: identifying materials, tracking flows, optimising reverse logistics, and matching surplus to demand at scales no centralised planner could manage.

36.1 Why Linear Economics Fails Sustainability

A linear economy externalises environmental costs — extraction depletes natural capital, manufacturing emits greenhouse gases and pollution, disposal generates waste. Internalising these costs through carbon prices, extended producer responsibility, and circular design closes the externality but multiplies the operational complexity. Tracking 100,000 product types across 200 countries through multiple use phases generates data flows that no human-only system can manage.

36.2 The Three Loops

Circular flows operate at three time scales. **Use intensification** (sharing, ride-hailing, equipment-as-a-service) increases utilisation of existing assets without making more. **Service-life extension** (repair, refurbishment, remanufacturing) keeps products in use longer. **Material recirculation** (recycling, biological breakdown) recovers materials at end of life. AI applications differ at each loop: matching algorithms for sharing, vision-based diagnosis for repair, sortation ML for recycling.

36.3 Material Passports and Digital Twins

A material passport is a digital record of a product's composition, components, and lifecycle history. Combined with digital twins — live-updated models of physical assets — they support traceability, value retention, and end-of-life recovery. The EU Digital Product Passport, mandated under the Ecodesign for Sustainable Products Regulation (ESPR) starting with batteries (2027) and expanding to textiles, electronics, and construction products, will create regulated infrastructure that AI applications

can build upon. ML applications include material identification from images for products without passports, supply-chain reconciliation, and counterfeit detection.

36.4 Measuring Circularity

Multiple circularity indicators compete: Material Circularity Indicator (Ellen MacArthur), Circular Transition Indicators (WBCSD), Circulytics, ISO 59020. They differ in scope (product, company, sector, economy) and emphasis (input recirculation, output recirculation, biological versus technical). AI plays a synthesising role: extracting circularity-relevant data from heterogeneous sources, normalising across taxonomies, and producing decision-relevant indicators for management and disclosure.

36.5 Mathematical Foundations of Circularity

36.5.1 Material Circularity Indicator (MCI)

$$\text{MCI}_p = 1 - \text{LFI} \cdot F(X),$$

where LFI is the Linear Flow Index and $F(X)$ a utility factor. Linear Flow Index:

$$\text{LFI} = \frac{V + W}{2M + (W_F - W_C)/2},$$

with virgin material V , waste W , recycled inputs, total mass M .

36.5.2 Substitution Rate

$$\sigma = \frac{\text{secondary use}}{\text{primary displacement}}.$$

$\sigma = 1$ implies one-for-one displacement; $\sigma < 1$ indicates incomplete substitution.

36.6 Worked Problems and Solutions

Problem 36.1 (Aluminium Recycling). 1 tonne secondary aluminium production avoids primary at $\sigma = 0.85$. Primary EF: 17 tCO₂e/t; secondary EF: 0.6 tCO₂e/t. Compute net emissions saving.

Solution. Avoided primary: $0.85 \cdot 17 = 14.45$ tCO₂e. Secondary emissions: 0.6 tCO₂e. Net saving: $14.45 - 0.6 = 13.85$ tCO₂e per tonne secondary. ■

36.7 Chapter Conclusion

The circular economy is not a marginal adjustment to the linear economy; it is a fundamental reorganisation of how materials flow through human systems. The chapter introduced the Ellen MacArthur Foundation framework, the EU Circular Economy Action Plan, the policy instruments (extended producer responsibility, eco-design directives, right-to-repair regulations), and the mathematical machinery that quantifies circularity (the Material Circularity Indicator, substitution rates, embodied-emissions accounting).

The aluminium-recycling worked problem illustrated the scale of opportunity. Each tonne of secondary aluminium displacing primary production saves approximately 13.85 tCO₂e — a $23\times$ reduction in emissions intensity. Multiplied across the 65 Mt of annual aluminium production globally, full circularity would save roughly 900 Mt CO₂ per year, comparable to the total emissions of Germany.

Similar multipliers apply to steel (4–5×), copper (3–4×), and plastics (2–4×).

The chapter's distinction between technical and biological cycles deserves emphasis. Technical materials — metals, polymers, glass — can in principle circulate indefinitely with limited quality loss, given adequate sorting and reprocessing. Biological materials — agricultural feedstocks, wood, food — circulate through composting and bioenergy with progressive entropy increase. Different policy instruments and different AI applications serve each cycle; the chapter's framework helps the reader keep the distinction clear in specific applications.

The subsequent chapters in Part VI take the framework into specific sectors: supply chains (Chapter 37), waste and recycling (Chapter 38), water (Chapter 39), and materials design (Chapter 40). Each chapter develops the sector-specific AI applications that operationalise the circular framework. We turn next to supply chains.

Sustainable Supply Chains

A company's carbon footprint is mostly the carbon footprint of its suppliers. The frontier of corporate sustainability is the visibility of those suppliers.

Kennedy Chengeta, PhD

Supply chains are the operational manifestation of Scope 3 emissions. For a typical consumer goods company, supply chain emissions exceed direct operations by factors of 5-10×. For financial institutions, financed-emissions Scope 3 exceeds operations by 100× or more. AI in sustainable supply chains addresses three challenges: visibility of multi-tier supply networks, real-time emissions and ESG-risk monitoring, and optimisation of routing, sourcing, and inventory for sustainability outcomes alongside cost.

37.1 Multi-Tier Visibility

A direct (Tier 1) supplier list is straightforward; the suppliers' suppliers (Tier 2) and their suppliers (Tier 3) are not. Yet most environmental risk and emissions sit beyond Tier 1. AI applications for multi-tier visibility include: NLP extraction of supplier relationships from financial filings, customs data, and shipping manifests; graph-network construction from transaction data; anomaly detection that flags unexpected supply-chain reconfigurations. Sayari, Kharon, Sourcemap, and Altana are commercial vendors building this infrastructure.

37.2 Emissions Hotspot Identification

Within a supply chain, emissions are typically concentrated in a small number of hotspots: aluminium smelting, steel production, electronic component manufacturing, freight on long routes. Hotspot ML uses lifecycle inventory data, supplier characteristics, and operational data to predict where emissions sit and which interventions yield largest reductions. Apple, Microsoft, and Walmart publicly report hotspot-driven engagement programs.

37.3 Sustainable Sourcing Optimisation

When multiple suppliers can provide a material, sourcing decisions weigh cost, lead time, quality, and increasingly carbon and ESG. ML-driven supplier scoring on combined criteria, with explicit Pareto-frontier analysis, has become standard at large procurement organisations. The challenge is data integrity: supplier-reported emissions are notoriously unreliable; cross-validation against satellite, customs, and energy data tightens the inputs.

Example 37.1 (Logistics Decarbonisation: Cape Town Port). A fast-moving consumer goods importer redesigned routing into Cape Town with ML optimisation across vessel choice, port-of-call sequence, intermodal options, and inventory carrying cost. Results after one year: 18% reduction in shipping-related emissions, 6% reduction in landed cost, and 14% reduction in inventory days. Annual emission reduction: 4,200 tonnes CO₂e. Counterfactual baseline construction (what would have been done without optimisation) is the methodological hinge; transparent sensitivity analysis maintained credibility under audit.

37.4 Forced Labour and Human Rights

Supply-chain due diligence under regulations such as the EU CSDDD, the German Supply Chain Act, and the US UFLPA imposes obligations to identify and address human-rights and environmental harms. AI applications include: satellite monitoring of mining operations and agricultural plantations for environmental damage; NLP analysis of news and NGO reports for human-rights signals; risk scoring at supplier and geographic granularity. Failure to detect known harms now carries legal and reputational consequences that historically were borne mainly by NGO campaigners.

37.5 Mathematical Foundations of Supply Chain Emissions

37.5.1 Multi-Tier Propagation

For input-output matrix $\mathbf{A}$ with direct emission factors $\mathbf{f}$, total embodied emissions per unit final demand:

$$\mathbf{e}_{\text{total}} = \mathbf{f}^{\top} (\mathbf{I} - \mathbf{A})^{-1}.$$

The Leontief inverse captures all upstream multipliers.

37.5.2 Routing Optimisation

$$\min \sum_{(i,j) \in E} c_{ij} x_{ij} + \lambda \sum_{ij} f_{ij}^{\text{CO}_2} x_{ij},$$

balancing logistics cost and emissions, with λ shadow price of carbon.

37.6 Worked Problems and Solutions

Problem 37.2 (Modal Choice). Two routes: sea+rail 18,000 km at 0.025 kg CO₂/t-km vs air 12,000 km at 0.85. Compute emissions per tonne shipped.

Solution. Sea+rail: $18000 \cdot 0.025 = 450$ kg. Air: $12000 \cdot 0.85 = 10,200$ kg. Air emits $22.7\times$ more.

■

37.7 Chapter Conclusion

Supply-chain decarbonisation is where most corporate climate ambition actually has to be operationalised. The chapter argued that the binding constraints are visibility and verification rather than ambition or technology: most companies cannot map their tier-2 and tier-3 suppliers with the resolution needed to allocate emissions accurately, much less to influence them.

The modal-choice worked problem illustrated the leverage of routing and sourcing decisions. Air

freight emits 22.7 times more CO₂ per tonne-kilometre than the sea-plus-rail alternative on the same shipment. For high-value, low-mass, time-sensitive products, the air premium is worth paying; for most categories, the alternative is competitive and emits a small fraction of the carbon. AI-driven supply-chain optimisation that explicitly trades cost against carbon, rather than optimising cost only, can shift modal mix substantially without material service degradation.

The chapter's multi-tier propagation treatment introduced the Leontief inverse that turns input-output relationships into full embodied-emission multipliers. The mathematical structure is established; the AI applications focus on filling the operational gap between published input-output tables and supplier-specific data. NLP-based supplier classification, ML-driven emission-factor refinement, and increasingly foundation-model-based document extraction from supplier disclosures collectively reduce the manual labour of Scope 3 estimation.

The chapter's coverage of supplier engagement deserves emphasis. Most emission reductions in any complex supply chain happen at suppliers, not at the contracting buyer. Effective supplier-engagement programmes combine measurement (data requests through CDP and similar platforms), support (technical assistance for measurement), and incentive (preferred sourcing for high-performing suppliers). The combination is harder operationally than measurement alone but produces actual emission reductions rather than improved accounting. We turn next from supply chains to the downstream counterpart: waste and recycling.

Waste and Recycling AI

A waste stream is just a material stream that someone has lost track of. AI is the discipline of finding it again.

Kennedy Chengeta, PhD

Waste management and recycling are infrastructure-intensive industries with low margins and rapidly tightening environmental regulation. AI applications — computer vision sortation, route optimisation for collection, demand forecasting for recyclate markets — have transformed unit economics in the past five years. This chapter surveys the deployments that have moved waste from a cost-minimisation domain into a recovery-and-revenue domain.

38.1 Computer Vision Sortation

Materials recovery facilities (MRFs) traditionally sort using mechanical separators (eddy currents for non-ferrous metals, optical sorters with NIR for some plastics) supplemented by hand-picking lines. Deep-learning vision systems mounted above conveyors identify and trigger removal of items that mechanical sorters miss — specific plastic types, contaminants, hazardous items. AMP Robotics, Greyparrot, and ZenRobotics have deployed thousands of systems globally; reported throughput improvements of 20-50% and contamination reductions of 30-60% are common.

38.2 Smart Bin Networks

IoT sensor bins (ultrasonic fill-level, weight, sometimes computer-vision contamination detection) enable demand-driven collection. Routes are computed daily based on actual bin states rather than fixed schedules, reducing collection vehicle kilometres by 15-30% and corresponding emissions. Bigbelly, Compology, and Enevo are major vendors. The economics are particularly compelling for commercial waste; municipal residential is more variable but increasing in adoption.

38.3 Recyclate Market Matching

Recovered materials must find buyers; mismatches between supply (varying composition, contamination, moisture) and demand (specifications, location, timing) cause recyclate to be landfilled despite recovery. ML matching platforms (Cirplus, Excess Materials Exchange, Topolytics) combine material characterisation, geographic logistics, and price discovery to clear the market. Plastic recycling in particular has benefited; aluminium and paper markets are more standardised and depend less on AI matching.

38.4 Methane from Landfills

Landfills emit methane from organic decomposition — approximately 11% of anthropogenic methane globally. Capture-and-flare or capture-and-utilise systems reduce emissions but coverage is incomplete. AI applications include: satellite plume detection (Sentinel-5P, GHGSat) for unflared emissions; sensor-network leak localisation within landfill sites; biogas yield prediction for capture-system optimisation. The combination is producing measurable emissions reductions; major waste companies (Waste Management, Republic Services) have announced AI-driven methane reduction programs.

Example 38.1 (Plastic Sortation Upgrade, Johannesburg MRF). A Johannesburg materials recovery facility added computer-vision sortation across PET, HDPE, PP, and PS streams. Capital cost USD 1.4M. Year-one results: PET recovery rate increased from 71% to 89%, HDPE from 64% to 84%, contamination of bales fell from 7.2% to 2.8%. Annual revenue increase from improved bale specifications: USD 920K. Annual avoided landfill emissions: 8,400 tonnes CO₂e (methane equivalent from organic contamination plus avoided virgin-plastic production from improved recovery). Payback: 18 months.

38.5 Mathematical Foundations of Waste Stream Optimisation

38.5.1 Sortation Mass Balance

Input M , recovered fractions r_i , contamination c_i :

$$\text{Bale } i = M \cdot \pi_i \cdot r_i, \quad \text{Contam} = \text{Bale}_i \cdot c_i.$$

Revenue per bale:

$$R_i = (\text{Bale}_i)(p_i - p_i^{\text{discount}}(c_i)).$$

38.5.2 Methane From Landfill

First-order decay:

$$Q_{\text{CH}_4}(t) = kL_0We^{-kt},$$

with degradable carbon L_0 , waste mass W , decay constant k .

38.6 Worked Problems and Solutions

Problem 38.2 (PET Sortation Revenue). 1,000 t input, PET fraction 0.18, recovery rate increases from 0.71 to 0.89 with vision sortation. PET price \$280/t.

Solution. Before: $1000 \cdot 0.18 \cdot 0.71 \cdot 280 = \$35,784$. After: $1000 \cdot 0.18 \cdot 0.89 \cdot 280 = \$44,856$. Revenue gain: \$9,072 per 1,000 t input. ■

38.7 Chapter Conclusion

Waste systems concentrate three sustainability opportunities: material recovery, methane control, and energy recovery. Each is a distinct AI application, and each is at a different stage of operational maturity.

The PET-sortation worked problem illustrated the revenue-and-emissions case for vision-guided robotic sorting. An 18-percentage-point gain in recovery rate, applied to 1,000 tonnes of input with 18% PET share at \$280/tonne, yields \$9,000 of additional revenue per 1,000 tonnes of input. Across a typical materials recovery facility processing 200,000 tonnes per year, this scales to \$1.8M of annual revenue gain, justifying substantial capital investment in sortation upgrades. The emissions benefit comes both from material displacement of virgin production and from avoided landfill methane.

The landfill methane treatment introduced the first-order-decay model that underlies regulatory accounting (US EPA's LandGEM, IPCC inventory methodology). A landfill emits roughly half its methane in the first 20 years post-burial, with the remainder trailing off over the next century. Active management — gas capture for energy or flaring, leachate management, cover optimisation — can reduce emissions by 50%–90% relative to passive baselines. ML-driven optimisation of capture systems is increasingly deployed at large engineered landfills.

The chapter's coverage of the recycling-information infrastructure – material passports, blockchain-based traceability, ML-driven contamination detection in feedstock streams — pointed forward to the materials-design discussion in Chapter 40. Designing materials for recyclability requires that the design choices propagate through the supply chain to the end-of-life management; the information infrastructure that enables this propagation is itself substantial. We turn next from solid waste to liquid: water-system management.

Water Management

Africa's water crisis is not a shortage of rain. It is a shortage of measurement, distribution, and accountability. AI addresses all three.

Kennedy Chengeta, PhD

Water is the foundational resource for agriculture, industry, energy, and human survival. Climate change is intensifying both droughts and floods. Population growth and economic development are increasing demand. Aging infrastructure is leaking 30-50% of treated water in many cities. AI applications across water management — distribution network management, demand forecasting, leak detection, water quality monitoring — are addressing the operational efficiencies that determine whether limited supplies meet legitimate demand.

39.1 Network Management and Non-Revenue Water

Non-revenue water — treated water that does not generate revenue, lost to leaks, theft, or metering inaccuracy — averages 30-50% in many low- and middle-income cities. ML applications combine pressure-zone monitoring, hydraulic models, and acoustic leak detection. Deployment in Cape Town reduced non-revenue water from 35% to 24% over five years; comparable reductions are reported in Nairobi, Lagos, and Dar es Salaam. The avoided pumping energy plus the additional billable water typically pays back the system in 2-4 years.

39.2 Demand Forecasting and Smart Metering

Smart water meters, transmitting consumption data hourly or daily, enable demand-side management analogous to electricity smart meters. ML applications include: leak detection at premises (water flowing continuously when it should not), demand forecasting for distribution planning, rate-design support, and customer behavioural feedback. Privacy considerations are significant; consumption patterns reveal occupancy and lifestyle data that requires regulatory protection.

39.3 Quality Monitoring

Water quality monitoring traditionally relies on grab samples sent to laboratories — expensive, slow, and incomplete. Real-time sensor networks (turbidity, pH, conductivity, occasionally specific contaminants) combined with ML produce continuous quality estimates. Anomaly detection flags unusual patterns suggesting contamination events; source apportionment ML traces contamination to upstream catchment sources. Drinking-water utilities, river-basin authorities, and industrial discharge

regulators all increasingly rely on AI-augmented monitoring.

39.4 Watershed and River-Basin Management

Water management at watershed scale requires balancing competing uses — urban supply, irrigation, ecosystem flows, hydropower, recreation. Optimal-control ML on calibrated hydrological models supports reservoir operations and water-allocation decisions. The Orange-Senqu basin in Southern Africa, the Nile, the Limpopo, and the Zambezi each have river-basin authorities developing AI-augmented decision-support tools. Transboundary governance is the binding constraint; AI alone does not resolve sovereignty over shared waters.

39.5 Mathematical Foundations of Water Network Optimisation

39.5.1 Hydraulic Equations

Hazen–Williams head loss:

$$h_f = \frac{10.67LQ^{1.852}}{C^{1.852}D^{4.87}},$$

with L length, Q flow, C roughness, D diameter. Mass continuity at each node: $\sum Q_{\text{in}} = \sum Q_{\text{out}}$.

39.5.2 Leak Detection

Pressure-zone mass-balance discrepancy:

$$\text{NRW}_z = \frac{\text{Input}_z - \text{Billed consumption}_z}{\text{Input}_z}.$$

39.6 Worked Problems and Solutions

Problem 39.1 (Pipe Head Loss). $L = 1000$ m, $Q = 0.05$ m³/s, $C = 130$, $D = 0.20$ m.

Solution. $Q^{1.852} = 0.05^{1.852}$. $\ln(0.05) = -2.996$; $1.852 \cdot (-2.996) = -5.549$; $0.05^{1.852} = e^{-5.549} = 0.003886$. $C^{1.852} = 130^{1.852}$. $\ln(130) = 4.868$; $1.852 \cdot 4.868 = 9.016$; $130^{1.852} = e^{9.016} = 8,238$. $D^{4.87} = 0.20^{4.87}$. $\ln(0.20) = -1.609$; $4.87 \cdot (-1.609) = -7.836$; $0.20^{4.87} = e^{-7.836} = 0.0003944$. $h_f = 10.67 \cdot 1000 \cdot 0.003886 / (8238 \cdot 0.0003944) = 41.46 / 3.250 = 12.76$ m. ■

39.7 Chapter Conclusion

Urban and industrial water systems present a paradigmatic case for AI-enabled efficiency. Non-revenue water — water lost between treatment and customer — averages 30% in well-managed utilities and exceeds 50% in many developing-country systems. Each cubic metre of NRW recovered is a cubic metre that does not have to be pumped, treated, and delivered — a multi-layer efficiency gain that compounds across the value chain.

The Hazen-Williams worked problem illustrated the hydraulic-modelling foundation that leak detection builds on. A 12.76-m head loss across a 1-km pipe at 50 L/s flow is a meaningful baseline against which actual operational pressure measurements can be compared. Mismatches between modelled and observed pressure signal anomalies — leaks, illegal connections, or instrument failures — that warrant investigation. Modern leak-detection systems combine hydraulic modelling, acoustic sensing,

and ML-based anomaly detection on smart-meter data to localise losses to manageable inspection zones.

The chapter's coverage of demand-side AI — shower-occupancy detection, appliance-level disaggregation, dynamic pricing — introduced the behavioural layer that increasingly mediates water-resource management in stressed regions. The mathematical machinery is the same as for electricity demand-side management; the operational details differ because water has higher friction in metering and pricing than electricity.

The chapter's institutional discussion deserves emphasis. Water utilities are typically smaller, more capital-constrained, and more politically sensitive than electricity utilities. AI deployment that requires extensive instrumentation upgrades is harder for water than for electricity; the workable patterns prioritise software upgrades on existing data and target high-return applications first. We turn next from water management to the deeper foundation of circular economy: the design of the materials themselves.

Materials Design for Sustainability

Every kilogram of cement, steel, aluminium, and plastic was once a research decision. AI is making the next generation of those decisions a thousand times faster.

Kennedy Chengeta, PhD

Materials are responsible for around 23% of global emissions when production and use-phase impacts are combined. Sustainable materials — low-carbon cement, hydrogen steel, recyclable plastics, biodegradable polymers, novel battery chemistries — are essential to deep decarbonisation. AI is accelerating materials discovery and design through generative models, active learning, and simulation augmentation, with measurable impacts emerging across multiple categories.

40.1 The AI-Driven Materials Discovery Pipeline

Conventional materials discovery is iterative, slow, and expensive: hypothesise, synthesise, test, repeat. AI compresses the cycle. Generative models (variational autoencoders, diffusion models, large language models for crystal structures) propose candidate materials. Active learning prioritises which candidates to test based on predicted information gain. High-throughput simulation (DFT, molecular dynamics) and experimentation validate promising candidates. The closed loop — propose, simulate, synthesise, test, retrain — has produced multiple breakthrough materials in the past three years.

40.2 Battery Chemistry

Lithium-ion batteries dominate but face supply-chain constraints (lithium, cobalt, nickel) and energy-density limits. AI-driven discovery has produced candidate solid-state electrolytes, sodium-ion cathodes, and lithium-sulfur architectures with claimed energy densities 50-100% above current lithium-ion. Translation from lab-scale validation to gigafactory production remains the bottleneck; AI is contributing to manufacturing process optimisation as well as discovery.

40.3 Catalysts for Green Chemistry

Catalysts enable energy-efficient chemical reactions, including the production of green hydrogen, ammonia, and methanol from renewable inputs. Generative models for catalyst structures combined with DFT screening have identified candidates with 5-10× improved activity for water electrolysis, CO₂ reduction, and ammonia synthesis. Industrial deployment is just beginning; the impact on hard-to-decarbonise sectors will compound over the next decade.

40.4 Low-Carbon Cement and Concrete

Cement is responsible for approximately 8% of global CO₂ emissions. Alternative chemistries — calcined clay cements, geopolymers, magnesium-based binders — offer 30-70% emission reductions. AI applications include: composition optimisation across many simultaneous performance and durability criteria, performance prediction across decades from accelerated tests, and supply-chain optimisation for raw materials available locally. Sublime Systems, Solidia, and Brimstone are commercial entrants; standards bodies are updating performance specifications to accommodate the new chemistries.

40.5 Recyclable and Bio-Based Plastics

Plastics produced from petrochemical feedstocks are difficult to recycle without quality loss. AI-driven design of plastics with engineered recyclability — defined depolymerisation pathways, controlled molecular weights, marker tags for sortation — is producing commercial-scale candidates. Bio-based plastics from algae, agricultural residues, and engineered microbes are similarly accelerating. Industrial scale-up faces cost competition with petrochemical incumbents; sustained policy support and corporate procurement commitments are determining the pace of transition.

40.6 Mathematical Foundations of Generative Materials Design

40.6.1 Variational Autoencoder

For data $\mathbf{x}$ and latent $\mathbf{z}$:

$$\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p_{\theta}(\mathbf{x}|\mathbf{z})] + D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})).$$

40.6.2 Active Learning Acquisition

Expected improvement:

$$\text{EI}(\mathbf{x}) = \mathbb{E}[\max(f(\mathbf{x}) - f^*, 0)],$$

where f^* is current best. Closed form for Gaussian-process surrogate exists via probit and density of standard normal.

40.7 Worked Problems and Solutions

Problem 40.1 (Cement EF Comparison). OPC: 0.86 tCO₂e/t. Calcined clay blend: 0.51 tCO₂e/t. 1 Mt annual production switched. Annual saving?

Solution. $10^6 \cdot (0.86 - 0.51) = 350,000 \text{ tCO}_2\text{e/yr.}$ ■

40.8 Chapter Conclusion

Materials design is where AI's reach extends from operational optimisation into fundamental innovation. The chapter walked through the methodological stack: high-throughput density-functional-theory screening to narrow vast composition spaces; generative models (VAEs, diffusion models, graph generation networks) to propose novel candidates; active learning to prioritise experimental verification; and ML interatomic potentials to accelerate molecular-dynamics simulation.

The cement-emission-factor worked problem illustrated the scale of opportunity. Switching 1 Mt of annual cement production from ordinary Portland cement (0.86 tCO₂e/t) to calcined-clay blend (0.51 tCO₂e/t) saves 350,000 tCO₂e per year. Globally, cement production is roughly 4 Gt per year; a full transition to lower-carbon formulations could reduce cement-sector emissions by 1–2 Gt per year, on the order of 3%–5% of global emissions. The technology exists; the scaling challenge is principally one of testing, certification, and supply-chain adaptation.

The active-learning treatment introduced the expected-improvement acquisition function that drives experiment prioritisation. The mathematical structure is standard; the operational value comes from the dramatic reduction in experimental burden — typically 5–10× fewer experiments to reach a target performance level than naive grid search. The technique is mature enough that commercial materials-discovery firms operate end-to-end pipelines combining literature mining, computational screening, robotic experimentation, and human review.

The chapter's coverage of recyclable polymers, low-carbon cements, and emerging biopolymers signalled the breadth of the materials-design agenda. None of the application areas is fully solved; all are advancing rapidly. We close Part VI's coverage of circular economy here. Part VII turns to the biological domain that material flows ultimately interact with: biodiversity.

Part VII

Biodiversity and Ecosystem Intelligence

Biodiversity Monitoring

A species we cannot count is a species we cannot save. AI has made the impossible counting practical.

Kennedy Chengeta, PhD

Biodiversity loss is occurring at rates 100-1000× background extinction levels. Monitoring biodiversity at scale — across taxa, geographies, and time — has historically been impossible due to taxonomic complexity, sampling cost, and expert scarcity. AI applications across acoustic monitoring, image classification, environmental DNA, and citizen science have changed the economics decisively. This chapter treats the methods and the operational deployments now anchoring conservation policy.

41.1 Acoustic Monitoring

Passive acoustic monitoring captures continuous audio at field sites; ML classifiers identify species from vocalisations. Birds, bats, frogs, primates, marine mammals, and increasingly insects can be identified with classifier accuracy approaching expert performance for well-trained species. BirdNET (Cornell Lab) classifies over 6,000 bird species globally; Rainforest Connection deploys solar-powered acoustic sensors against illegal logging detection. The economics: a USD 200 sensor producing terabytes of audio that ML reduces to species-by-time matrices replaces ornithologist field-hours that cost orders of magnitude more.

41.2 Camera Traps and Computer Vision

Motion-triggered cameras have produced millions of wildlife images globally. Deep-learning species classifiers (Wildlife Insights, MegaDetector, ZambaPyTorch) automate identification, reducing review time from hours to minutes per camera. Population estimation from camera-trap data — mark-recapture for individually-recognisable species, distance sampling for others — supports management decisions for endangered species. African deployments at Kruger, Serengeti, and Chobe are mature; central African forest deployments have grown rapidly with falling solar/battery costs.

41.3 Environmental DNA (eDNA)

eDNA — DNA shed by organisms into water, soil, or air — enables species detection without physical specimen capture. Sequencing produces lists of species present at a site; ML-assisted bioinformatics matches sequences to reference libraries and quantifies relative abundance. Marine, freshwater, and

increasingly terrestrial eDNA monitoring is operational. Reference library completeness varies by taxonomic group and region; African and Latin American invertebrate references are still patchy, limiting interpretation.

41.4 Citizen Science Integration

iNaturalist, eBird, and similar platforms aggregate hundreds of millions of biodiversity observations from non-professional contributors. ML applications include: image-based species identification (the iNaturalist computer vision model identifies >50,000 species with high accuracy); spatial bias correction (citizen science is concentrated near roads and population centres); validation against expert observations. The data are increasingly integrated into formal biodiversity assessments including IUCN Red List updates.

41.5 Operational Limitations

AI biodiversity monitoring has limits that field practice must respect. Rare, cryptic, or non-vocalising species remain hard to detect. Sub-species and population-level discrimination requires high-quality reference data not always available. Inter-annual variation in detection effort can mask true population trends. The honest practice combines AI-driven monitoring with stratified field work, transparent uncertainty reporting, and adaptive management as new data accrue.

41.6 Mathematical Foundations of Biodiversity Statistics

41.6.1 Species Abundance and Diversity

Shannon index:

$$H' = -\sum_i p_i \ln p_i, \quad p_i = n_i/N.$$

Simpson:

$$D = 1 - \sum_i p_i^2.$$

Effective species number: $\exp(H')$.

41.6.2 Occupancy Modelling

For site i , occupancy ψ_i and detection p_{ij} :

$$\Pr(\text{detection history } \mathbf{y}_i \mid \psi_i, p_{ij}) = \psi_i \prod_j p_{ij}^{y_{ij}} (1 - p_{ij})^{1-y_{ij}} + (1 - \psi_i) \mathbb{I}(\sum_j y_{ij} = 0).$$

41.7 Worked Problems and Solutions

Problem 41.1 (Shannon Index). Five-species abundance (0.40, 0.25, 0.18, 0.12, 0.05). Compute H' and effective species number.

Solution. $-[0.40 \ln 0.40 + 0.25 \ln 0.25 + 0.18 \ln 0.18 + 0.12 \ln 0.12 + 0.05 \ln 0.05] = -[(-0.366) + (-0.347) + (-0.309) + (-0.254) + (-0.150)] = 1.426$. Effective: $e^{1.426} = 4.16$ effective species.

■

41.8 Chapter Conclusion

Biodiversity monitoring at planetary scale was technically infeasible a decade ago. The combination of bioacoustic recorders, environmental DNA, camera traps, citizen-science platforms, and continental-scale acoustic and visual classification models has transformed the field. The chapter walked through each component and the AI applications that operationalise them.

The Shannon-index worked problem illustrated the basic diversity statistics that anchor every biodiversity measurement. A community with five species at frequencies (0.40, 0.25, 0.18, 0.12, 0.05) has a Shannon index of 1.43 and an effective species number of 4.16 — meaningfully less than the nominal 5 because the community is dominated by the first two species. Comparing Shannon indices between sites, between habitat types, or between time periods is the foundation of empirical ecology; AI's role is to scale the measurement, not to replace the statistics.

The occupancy-modelling treatment introduced the statistical machinery that handles imperfect detection. A species not observed at a site has some probability of being present but undetected; ignoring detection probability biases occupancy estimates downward and produces false trends. The hierarchical Bayesian models that combine occupancy and detection probability are now well-established; ML extensions to large-scale acoustic and image data are an active research frontier.

The eDNA discussion forecast the field's next major transformation. By sequencing the DNA shed by organisms into water, soil, and air, eDNA extends species detection to organisms that elude visual or acoustic sampling. The bioinformatic pipeline is ML-heavy: sequence classification, taxonomic assignment, and increasingly direct biodiversity-metric estimation from raw sequence data. The technology is moving from research demonstration to operational deployment; the standardisation of methodology lags but is converging.

The chapter's institutional discussion — the GBIF, the Global Biodiversity Framework, the post-2020 targets — pointed forward to the regional and ecosystem-specific chapters that follow. We turn next to the largest terrestrial biodiversity reservoir: forests.

Forest Intelligence

A satellite cannot smell smoke or hear a chainsaw. But it sees the canopy, and the canopy tells the story — if AI is reading.

Kennedy Chengeta, PhD

Forests cover roughly 31% of global land area, store approximately 360 Gt of carbon in vegetation, and support more than half of all terrestrial species. Tropical deforestation contributes 10-15% of annual CO₂ emissions; degradation contributes additional emissions less easily measured. AI applications across forest monitoring, biomass estimation, fire detection, and restoration planning have transformed both science and operations.

42.1 Deforestation Alerting

Global Forest Watch, RADD, and DETER (Brazil) provide near-real-time deforestation alerts at hectare resolution from optical and radar satellites. RADD (Wageningen University) operates on Sentinel-1 SAR, providing all-weather alerts with median latency of 6-9 days from canopy loss to alert publication. ML applications include: false-alarm reduction (separating true clearing from natural disturbance, cloud, smoke), species-specific detection (selective logging signatures), and downstream attribution (which clearing is associated with which commodity supply chains).

42.2 Above-Ground Biomass Estimation

Forest biomass — tonnes of carbon per hectare — is the central quantity for forest carbon accounting. Field measurement via tree inventory is accurate but spatially sparse and expensive. Spaceborne LiDAR (GEDI on the ISS) provides direct height measurements over much of the tropics. ML extrapolation combines GEDI samples, Landsat/Sentinel optical, and SAR backscatter to produce continuous biomass maps. ESA's BIOMASS satellite (launched 2025) will provide P-band radar with greater biomass sensitivity, particularly in dense canopies. Operational biomass products at 100-500m resolution support REDD+ accounting and disclosure under emerging deforestation regulations.

42.3 Fire Detection and Prediction

Active fire detection from MODIS, VIIRS, and Sentinel-3 thermal imagery is operational at near-real-time. ML applications increasingly predict fire risk hours-to-days ahead based on weather, fuel moisture, vegetation state, and historical patterns. Fire-spread modelling integrates ML with cellular-automaton physics for incident-management decision support. Australian (after Black Summer

2019-20) and California fire services have invested heavily; African applications are growing with regional collaboration.

Example 42.1 (Fire Risk Forecasting in Limpopo). A fire-risk forecasting system for Kruger National Park combines Sentinel-2 fuel moisture, MODIS active fires, GFS weather forecasts, and historical ignition patterns. ML models produce daily fire-risk maps at 250m resolution. The resulting prepositioning of aerial water-bombing assets reduced average response time from 47 minutes to 19 minutes during peak season; total burned area within the park decreased by 23% in the first deployment year despite higher fuel loads.

42.4 Restoration Site Selection

Reforestation and forest landscape restoration are major investment areas. ML site-selection tools combine soil suitability, climate projections, hydrology, biodiversity context, land tenure, and socioeconomic factors to identify high-success-probability sites. The Bonn Challenge and AFR100 commitments to restore hundreds of millions of hectares depend on planning tools at this scale. Failure mode prevention — afforesting grasslands (negative net carbon, biodiversity loss), monoculture plantations claiming native-forest credits — is increasingly enforced by remote sensing AI on project boundaries.

42.5 Mathematical Foundations of Forest Carbon

42.5.1 Carbon Stock Conversion

$$C = 0.5 \cdot \text{AGB}, \quad \text{CO}_2 = C \cdot \frac{44}{12}.$$

42.5.2 Deforestation Detection Skill

Pixel confusion matrix produces:

$$\text{Producer's accuracy}_i = \frac{n_{ii}}{\sum_j n_{ji}}, \quad \text{User's accuracy}_i = \frac{n_{ii}}{\sum_j n_{ij}}.$$

42.6 Worked Problems and Solutions

Problem 42.2 (AGB to CO₂). Forest with 240 t AGB/ha. Compute CO₂ stock.

Solution. $C = 0.5 \cdot 240 = 120$ t C/ha. $\text{CO}_2 = 120 \cdot 44/12 = 440$ tCO₂e/ha. ■

42.7 Chapter Conclusion

Forest monitoring is the most mature application of biodiversity AI. The Global Forest Watch platform, the Forest Resources Assessment, and the operational systems of national forest agencies all rely on ML-driven satellite analysis at continental scale. The chapter walked through the components: tree-cover classification, forest-degradation detection, biomass estimation from optical, SAR, and LiDAR data, and the alert systems that drive enforcement action.

The biomass-conversion worked problem demonstrated the standard practice: 240 t AGB/ha cor-

responds to 120 t C/ha and 440 tCO₂e/ha. The mathematical machinery is straightforward; the data infrastructure that supplies the AGB estimate at hectare resolution is what makes operational monitoring possible. ESA's BIOMASS mission, launching in 2024 and operational from 2025, will further tighten the uncertainty bounds on global biomass estimates.

The chapter's deforestation-alert discussion deserves emphasis. PRODES in Brazilian Amazonia, GLAD-L and RADD globally, and the dozens of national systems built on similar foundations now alert enforcement agencies to suspected deforestation within days of occurrence. The operational gap is the translation of alerts into enforcement action; the technology produces alerts at rates that overwhelm field- inspection capacity, and prioritisation of which alerts to investigate is itself an AI problem.

The chapter's coverage of forest restoration — species selection, site identification, monitoring of restoration outcomes — pointed forward to the broader nature-based-solutions discussion in Chapter 33. Restoration is the slowest and hardest part of the forest agenda; the operational challenges are institutional more than technical, but technical tools support institutional capacity. We turn next from terrestrial to marine biodiversity.

Marine Systems and AI

Oceans cover 71% of the planet and host the largest carbon sink on it. We have measured it less than the Moon.

Kennedy Chengeta, PhD

Oceans absorb approximately 25% of human CO₂ emissions and roughly 90% of the excess heat from the climate imbalance. They are also under-sampled compared to terrestrial systems. AI applications across marine remote sensing, illegal fishing detection, ocean carbon flux estimation, and marine biodiversity monitoring are addressing the data gap. This chapter treats the major operational deployments, with attention to both their power and their remaining limitations.

43.1 Illegal, Unreported, Unregulated Fishing

IUU fishing accounts for an estimated 11-26 million tonnes of fish per year, undermining fishery management and depleting stocks. Global Fishing Watch (Oceana, Skytruth, Google) integrates AIS vessel-tracking data with VIIRS night-light imagery, satellite radar, and ML to identify fishing activity, vessel-type classification, and likely IUU patterns. Recent extensions detect dark vessels (those with disabled or absent AIS) using SAR plus visual imagery. Enforcement actions in Indonesia, Argentina, and West African EEZs increasingly draw on this data.

43.2 Ocean Carbon Flux

The ocean carbon sink absorbs roughly 9 Gt CO₂ per year but is highly variable spatially and temporally. Surface ocean pCO₂ measurements (SOCAT database) supply training data for ML models that interpolate between sparse measurements using satellite SST, chlorophyll, sea-ice, and reanalysis winds. The resulting flux maps support the Global Carbon Project annual budget and are increasingly used for marine carbon-removal MRV (ocean alkalinity enhancement, blue carbon at scale).

43.3 Coral Reef Monitoring

Coral reefs are among the most biodiversity-rich and most climate-vulnerable ecosystems. Satellite-based bleaching detection (NOAA Coral Reef Watch) provides degree-heating-week stress indicators globally. Drone-mounted hyperspectral and underwater autonomous-vehicle imagery, combined with deep-learning coral-cover classification, produces 3D reef-state assessments at meter resolution. Coral identification from in-situ video supports ecological assessments at scales that diver surveys cannot

match.

Example 43.1 (Reef Monitoring: Western Indian Ocean). A monitoring program across reef sites in Kenya, Tanzania, Mozambique, and Madagascar combined autonomous underwater vehicles, photo-quadrat ML coral classification (genus and bleaching state), and acoustic monitoring of reef soundscapes (proxy for biodiversity). Monitoring cost per site fell from approximately USD 8,000 (3 diver-days) to USD 1,400 (1 vehicle deployment plus ML processing). Coverage expanded from 40 to 220 sites over three years; the data anchored regional reef-management decisions including marine-protected-area extensions.

43.4 Marine Cetaceans and Bycatch

Whale, dolphin, and porpoise tracking from acoustic monitoring, satellite imagery, and shipping-collision databases supports both conservation and shipping-safety decisions. Real-time alert systems for high-speed ferries (Boston, Gulf of California) reduce ship strikes. Bycatch reduction in fisheries — minimising incidental catch of dolphins, sea turtles, sharks — depends on ML video monitoring of fishing operations and gear-design optimisation informed by behavioural data.

43.5 What AI Cannot Do for the Ocean

AI does not measure what it does not see. Deep ocean processes — abyssal currents, sediment biogeochemistry, deep-pelagic fish stocks — remain undermeasured because the in-situ sampling infrastructure (Argo floats, gliders, deep-sea sensors) is sparse. AI extracts maximum value from existing data but cannot substitute for the missing observations. Argo expansion, BioGeoSCAPES, and deep-ocean observation initiatives must continue to provide the data substrate that AI models need.

43.6 Mathematical Foundations of Ocean Carbon Flux

43.6.1 Air-Sea CO₂ Flux

$$F = k \cdot K_0 \cdot (\text{pCO}_2^{\text{water}} - \text{pCO}_2^{\text{air}}),$$

with gas transfer velocity k (function of wind speed) and solubility K_0 . Integrated globally yields ~ 9 Gt CO₂/yr ocean uptake.

43.6.2 Coral Bleaching Indicator

Degree-Heating Weeks:

$$\text{DHW}(t) = \sum_{w=1}^{12} \max(\text{SST}_w - \text{MMM} - 1, 0),$$

with maximum monthly mean MMM. $\text{DHW} > 4$ predicts bleaching; $\text{DHW} > 8$ predicts mass mortality.

43.7 Worked Problems and Solutions

Problem 43.2 (DHW). Twelve weekly SST anomalies above MMM (in °C): 0.4, 0.6, 1.2, 1.5, 1.8, 2.0, 1.9, 1.7, 1.4, 1.0, 0.7, 0.3. Compute DHW.

Solution. Weekly excess above 1°C threshold: 0, 0, 0.2, 0.5, 0.8, 1.0, 0.9, 0.7, 0.4, 0, 0, 0. Sum: $0.2 + 0.5 + 0.8 + 1.0 + 0.9 + 0.7 + 0.4 = 4.5$ DHW. Bleaching predicted; mortality not yet. ■

43.8 Chapter Conclusion

Marine AI is the area where the gap between research demonstration and operational deployment is largest. The chapter walked through the principal applications: vessel monitoring through AIS and satellite-radar integration to detect IUU fishing; coral-reef monitoring through acoustic and visual methods to detect bleaching; bycatch reduction through ML-driven gear and route optimisation; and ocean-carbon-flux monitoring through autonomous platforms and satellite remote sensing.

The Degree-Heating-Weeks worked problem illustrated the operational indicator that drives coral-bleaching forecasts. A DHW of 4.5 predicts bleaching but not mortality; DHW values above 8 predict mass mortality. The NOAA Coral Reef Watch operational system produces five-kilometre-resolution DHW forecasts globally; downscaling to reef-scale forecasts through ML methods is a current research frontier with direct management application.

The IUU-fishing detection treatment combined AIS broadcast data with satellite-radar detection of unbroadcast vessels and ML-based classification of vessel behaviour. Global Fishing Watch and similar operational platforms now track an estimated 60%–70% of global commercial fishing effort; the gap is principally in regions and vessel classes where AIS broadcasting is incomplete or actively disabled. The combination with VIIRS night-light detection, Sentinel-1 SAR, and ML-based dark-vessel identification is steadily closing the gap.

The chapter's air-sea CO₂-flux discussion pointed forward to the broader ocean-carbon-cycle treatment in Chapter 2. The ocean takes up roughly a quarter of anthropogenic CO₂ emissions; small variations in that uptake rate have large climate consequences. ML-driven analysis of the expanding autonomous-platform fleet (Argo, biogeochemical Argo, Saildrone sailboats) is tightening uncertainty bounds on the ocean-carbon-uptake estimate progressively. We turn next from the marine domain to wildlife conservation on land.

Wildlife Conservation

A poacher who knows where the rangers are not is a poacher who succeeds. AI is changing the geometry of where the rangers can be.

Kennedy Chengeta, PhD

Conservation of large wildlife — elephants, rhinos, lions, tigers, great apes — is a defining sustainability challenge. Populations have collapsed across most ranges in the past century; protected areas alone are insufficient. AI applications across anti-poaching, human-wildlife conflict reduction, population monitoring, and translocation planning are being deployed at scale across African and Asian protected areas.

44.1 Anti-Poaching: SMART, Earth Ranger, and ML Patrol Optimisation

The Spatial Monitoring and Reporting Tool (SMART), developed by a coalition including WCS, WWF, and ZSL, is the dominant ranger-patrol management system globally. ML extensions (PAWS, Earth Ranger optimisation modules) recommend patrol routes based on predicted poaching risk, integrating historical incidents, terrain, water sources, road access, and current intelligence. Reported poacher-encounter rates increased 50-200% under ML-recommended patrols at sites in South Africa, Uganda, and Cambodia, with corresponding declines in poaching.

44.2 Real-Time Threat Detection

Acoustic gunshot detection, thermal-imagery animal-and-human detection, and aerial drone surveillance produce real-time alerts from protected areas. RESOLVE, Trail Guard, and Smart Parks deploy these systems with multi-channel alerting to ranger teams. Latencies of 2-5 minutes from event to ranger response have been demonstrated in pilot deployments; sustained operation depends on connectivity and ranger logistics that complementary investments must address.

44.3 Human-Wildlife Conflict

Crop raiding by elephants, livestock predation by lions and leopards, and human-elephant fatalities are major drivers of human-wildlife conflict that erodes public support for conservation. AI applications include: ML prediction of conflict-prone zones from movement, land-use, and historical incident data; early-warning SMS alerts to communities; collar-based GPS-and-cellular alerts when animals approach settlements. WildlifeAID, the Mara Elephant Project, and others have deployed these

systems at scale; conflict reductions of 30-60% are documented at well-monitored sites.

Example 44.1 (Elephant Movement Prediction in Hwange). GPS collar data from 47 elephants in Hwange National Park (Zimbabwe) trained a transformer-based movement prediction model. Outputs are 24-hour movement-probability maps at 500m resolution. Combined with crop-calendar overlays, the system issued SMS alerts to 1,200 farmers in adjacent communities when high crop-raid probability was forecast. After two seasons, crop damage reports fell 38%, and reciprocal community support for park boundary maintenance increased measurably.

44.4 Genetic Monitoring and Population Recovery

Genomic data from field samples (faeces, hair, skin) supports population assessment, kinship analysis, and management of small populations facing inbreeding risk. ML on SNP data identifies individuals, infers pedigrees, and recommends translocations. Black rhino translocation programs in Southern Africa, mountain gorilla genetic monitoring in central Africa, and tiger population genetics across Asia all draw on these methods. The infrastructure — field sampling, sequencing capacity, bioinformatic pipelines, governance — is increasingly mature in Southern Africa and is expanding regionally.

44.5 Mathematical Foundations of Anti-Poaching Optimisation

44.5.1 Patrol Game Theory

Stackelberg game: ranger commits to mixed patrol strategy $\mathbf{x}$ across targets, poacher best-responds.

Ranger payoff:

$$U_r(\mathbf{x}) = \min_{i: x_i < 1} [x_i U_i^{\text{prot}} + (1 - x_i) U_i^{\text{att}}].$$

44.5.2 Mark–Recapture Population Estimate

Lincoln–Petersen:

$$\hat{N} = \frac{n_1 n_2}{m_2}.$$

44.6 Worked Problems and Solutions

Problem 44.2 (Population Estimate). First sample 60 elephants marked. Second sample 80 elephants captured, of which 15 were marked. Compute $\hat{N}$.

Solution. $\hat{N} = (60 \cdot 80) / 15 = 320$. ■

44.7 Chapter Conclusion

Wildlife conservation is one of the most operationally satisfying applications of AI: the techniques deploy directly into management, the impact metrics are visible (animal populations, poaching incident counts), and the feedback loops are short. The chapter walked through the principal applications: species identification from camera-trap and drone imagery, acoustic identification from bioacoustic recorders, patrol-route optimisation through Stackelberg-game models, and the operational decision support that ties these together.

The Lincoln-Petersen worked problem illustrated the basic population- estimation statistic. A first-

sample mark of 60 elephants and a second-sample recapture of 15 marked out of 80 captured yields a point estimate of 320 elephants in the population. Modern population estimation extends this to spatial-capture-recapture models, hierarchical Bayesian frameworks, and integrated-population-model multispecies extensions; the basic Lincoln-Petersen intuition remains the foundation.

The patrol-game-theory discussion introduced the Stackelberg-game framework that underlies modern anti-poaching deployment. The ranger commits to a mixed patrol strategy; the poacher best-responds. The mathematical optimum exists in closed form for simple cases and is solvable numerically for the realistic cases. Operational deployments of PAWS, SMART, and similar systems combine the game-theoretic prescription with historical-incident clustering and real-time intelligence, producing patrol routes that increase deterrence at constant patrol cost.

The chapter's human-wildlife-conflict discussion deserves emphasis. Conservation success in landscapes shared with people requires managing the conflict that wildlife create — crop raiding, livestock predation, occasional human injury. AI applications here are early- warning systems (acoustic and visual detection of approaching wildlife) combined with mitigation infrastructure (beehive fences, deterrent lighting, compensation schemes). The technical tools support, but do not replace, the local-institutional work of negotiating human-wildlife-coexistence agreements. We turn next to the formal institutional layer of biodiversity protection: protected areas.

Protected Area Management

A line on a map does not stop deforestation. Continuous, AI-driven monitoring inside the line is what stops it.

Kennedy Chengeta, PhD

Protected areas now cover roughly 17% of terrestrial and 8% of marine areas globally. The Kunming-Montreal Global Biodiversity Framework calls for expansion to 30% by 2030. Whether expansion delivers conservation outcomes depends entirely on management effectiveness — and management effectiveness depends increasingly on AI infrastructure for monitoring, decision support, and enforcement.

45.1 Management Effectiveness Assessment

METT (Management Effectiveness Tracking Tool) and its variants assess protected area management across staff capacity, infrastructure, monitoring, enforcement, and ecological outcomes. ML applications include: synthesising scoring across thousands of protected areas globally, identifying patterns of effectiveness predictors, and producing decision-support tools for funder allocations. Recent work (UNDP, IUCN-WCPA) integrates ML scoring with remote-sensing-derived ecological indicators (forest cover trend, fire frequency, surface water) to produce composite effectiveness scores updated annually.

45.2 Adaptive Management

Adaptive management — continuously updating actions based on monitoring outcomes — requires data infrastructure that historically protected areas lacked. AI-augmented monitoring (acoustic, camera trap, eDNA, remote sensing) produces species-and-habitat indicators at frequencies actionable for management. Rapid feedback enables adjustment of patrol focus, fire management, water provisioning, invasive-species control. Kruger National Park’s adaptive management framework has been a leading example for two decades; AI is accelerating the data substrate.

45.3 Tourism Carrying Capacity

Tourism is the dominant revenue source for many protected areas. Excessive visitation degrades habitat and visitor experience; insufficient visitation undermines financial sustainability. ML demand forecasting, visitor-pattern analysis from cellular and ticketing data, and habitat-impact monitoring support carrying-capacity decisions. Dynamic pricing and reservation systems based on real-time congestion

(deployed at Yellowstone, Kruger, and others) optimise both revenue and visitor satisfaction.

45.4 OECMs and Indigenous Conserved Territories

Other Effective area-based Conservation Measures (OECMs) and Indigenous and Community Conserved Areas (ICCAs) extend the conservation network beyond formal protected areas. Recognition under the Global Biodiversity Framework requires demonstrable conservation outcomes; AI monitoring can support recognition without imposing top-down management. The CARE principles for indigenous data sovereignty (Chapter 55) constrain how AI is deployed in these contexts: monitoring serves community-defined objectives, not external surveillance.

45.5 Mathematical Foundations of Effectiveness Metrics

45.5.1 Composite Score

$$S_{\text{eff}} = \sum_k w_k v_k(\mathbf{x}), \quad \sum_k w_k = 1.$$

Where ecological indicators (forest cover trend, fire frequency, surface water) compose with management indicators (staff, infrastructure, monitoring effort).

45.5.2 Carrying Capacity

For visitor flow V_t , ecosystem-impact threshold K , and recovery time τ_r :

$$\text{Capacity} = K \cdot \frac{1 - e^{-1/\tau_r}}{V_t}.$$

45.6 Worked Problems and Solutions

Problem 45.1 (Composite Effectiveness). Three indicators: forest trend 0.85, fire 0.60, water 0.75, weights 0.4, 0.3, 0.3. Compute composite.

Solution. $0.4 \cdot 0.85 + 0.3 \cdot 0.60 + 0.3 \cdot 0.75 = 0.34 + 0.18 + 0.225 = 0.745$. Effectiveness rated 74.5%.

■

45.7 Chapter Conclusion

Protected areas are the institutional infrastructure on which much of the biodiversity strategy depends; AI applications increasingly determine whether they function as effective conservation tools or as paper parks. The chapter walked through the principal AI applications: effectiveness measurement through remote-sensing of forest cover, fire frequency, and surface water; visitor-flow management through occupancy sensing and dynamic pricing; threat detection through deforestation alerts, intrusion detection, and patrol prioritisation.

The composite-effectiveness worked problem illustrated the standard multi-criteria framework. Forest-cover trend, fire frequency, and surface-water dynamics combined with weights of 0.4/0.3/0.3 yielded an effectiveness score of 0.745 in the example. The weights are themselves a policy choice that should be made transparently; sensitivity analysis to weight perturbation is a routine and useful discipline.

The carrying-capacity treatment introduced the visitor-management problem that increasingly constrains popular protected areas. Ecosystem-impact thresholds, recovery times, and visitor-flow distributions interact to determine sustainable visitation levels; dynamic pricing and reservation systems that respect these thresholds are emerging in heavily visited areas (Yosemite, Galapagos, Komodo). AI applications support both the threshold determination and the operational management.

The chapter's coverage of OECMs — other effective area-based conservation measures — pointed forward to the indigenous-knowledge discussion in Chapter 55. Many of the most ecologically intact regions globally are not formally protected in the IUCN-recognised sense; they are managed by indigenous and local communities under traditional tenure systems. The Global Biodiversity Framework's 30-by-30 target explicitly recognises OECMs; the AI applications that support them must respect the data-sovereignty considerations that traditional management entails. This chapter closes Part VII. We turn next, in Part VIII, to the financial and disclosure machinery that increasingly drives corporate sustainability behaviour.

Part VIII

ESG, Sustainable Finance, and Disclosure

ESG Data Infrastructure

A bad ESG score is mostly a measure of bad ESG data. Data is the binding constraint, and AI is the unbinding tool.

Kennedy Chengeta, PhD

ESG — environmental, social, and governance — data underpins sustainable investment, corporate disclosure, and regulatory compliance. The data ecosystem is fragmented: thousands of corporate disclosures in heterogeneous formats, hundreds of indicators across dozens of frameworks, and major rating divergences between providers (MSCI, Sustainalytics, S&P, ISS). AI applications across data extraction, normalisation, scoring, and analytics are central to making the ecosystem usable.

46.1 The Data Sources

ESG data comes from five primary sources. **Corporate disclosures** (sustainability reports, financial filings, regulatory submissions) are the largest volume, available primarily in narrative PDF form. **Regulatory filings** (CDP, EU Taxonomy, CSRD) provide structured data for covered entities. **Alternative data** (satellite, news, NGO reports, supply-chain transactional) provide independent verification. **Surveys and questionnaires** from rating providers solicit specific information. **Internal corporate data** (HR systems, energy meters, supplier records) anchor the underlying reality.

46.2 NLP for ESG Extraction

Large language models have transformed ESG data extraction. Tasks include: structured data extraction from sustainability reports (emissions, water, employee statistics, board diversity); sentiment analysis of news and NGO reports for controversy signals; classification of activities under the EU Taxonomy and similar frameworks. Domain-specific models (Climate-BERT, ESG-BERT) fine-tuned on ESG corpora outperform general-purpose models. Production deployments at major data providers process tens of thousands of new disclosures annually.

Example 46.1 (Annual Report Extraction Pipeline). A pipeline ingests annual sustainability reports for the JSE Top-100 companies, extracts approximately 280 ESG indicators per company, validates against prior-year values for consistency, and flags suspected misreporting for human review. Extraction accuracy on held-out validation reports averages 91% for numerical indicators and 84% for categorical (e.g. "complies/does not comply"). Throughput: 1,200 reports per

analyst-day, against the previous baseline of 8 reports per analyst-day. Annual operating cost of the pipeline is approximately 4% of the manual baseline.

46.3 Rating Divergence

Major ESG ratings agencies disagree substantially: correlation across providers averages 0.4–0.6, much lower than credit-rating correlations. Sources of divergence include: indicator selection (different frameworks emphasise different issues), weighting (governance vs. environmental vs. social), measurement (CO2 emissions intensity normalisation), and aggregation logic. AI investigations of divergence enable investors to choose ratings aligned with their preferences and to combine ratings into composite scores reflecting their priorities.

46.4 Quality Control and Audit

ESG data quality is uneven. Self-reported data without external assurance has known biases. Cross-validation against satellite imagery, customs and shipping data, and energy consumption reports flags suspect reporting. Anomaly detection identifies year-on-year changes inconsistent with operational reality. Increasing assurance requirements (CSRD reasonable assurance from 2028, similar moves elsewhere) will tighten data quality but increase compliance cost; AI is essential to making assurance economically tractable.

46.5 Mathematical Foundations of ESG Scoring

46.5.1 Composite Score Construction

For pillars $P \in \{E, S, G\}$, indicators $i \in P$, normalised values $v_{ip} \in [0, 1]$, weights w_i^p with $\sum_i w_i^p = 1$:

$$S_P = \sum_{i \in P} w_i^p v_{ip}, \quad S_{\text{ESG}} = w_E S_E + w_S S_S + w_G S_G.$$

46.5.2 Rating Divergence

For two raters with scores s_j^A, s_j^B on entity j :

$$\rho_{AB} = \text{corr}(s^A, s^B).$$

ESG correlations: $\rho \approx 0.4$ – 0.6 , vs. credit ratings $\rho > 0.95$.

46.5.3 Materiality Weighting

Industry-specific materiality m_i^I from SASB/ISSB scales indicator weights:

$$w_i^{\text{mat}} \propto m_i^I \cdot w_i^{\text{base}}.$$

46.6 Worked Problems and Solutions

Problem 46.2 (Composite ESG Score). Pillar scores: $S_E = 0.72$, $S_S = 0.56$, $S_G = 0.81$. Weights: 0.45, 0.25, 0.30. Compute composite.

Solution. $0.45 \cdot 0.72 + 0.25 \cdot 0.56 + 0.30 \cdot 0.81 = 0.324 + 0.140 + 0.243 = 0.707$. ■

Problem 46.3 (Materiality Reweighting). Indicator base weight 0.10; SASB materiality score for sector 0.85. Compute material weight (assuming average materiality 0.50).

Solution. $w^{\text{mat}} \propto 0.85 \cdot 0.10 = 0.085$. After re-normalisation against the average $0.50 \cdot 0.10 = 0.050$, relative weight $= 0.085/0.050 = 1.7 \times$ the average. ■

46.7 Chapter Conclusion

ESG data is simultaneously one of the most rapidly growing data categories in finance and one of the most methodologically contested. The chapter walked through the structural features: pillar-based construction (environmental, social, governance), indicator selection guided by industry-specific materiality frameworks (SASB, ISSB), weighting choices that vary across raters, and the rating divergence that emerges from those choices.

The rating-divergence treatment deserves emphasis. ESG ratings correlations across major raters run at roughly 0.4–0.6, compared with credit-rating correlations above 0.95. The divergence reflects genuine methodological differences — different indicator weights, different normalisations, different handling of missing data — rather than measurement error. The implication for investors is that single-rater dependence is risky; the better practice is to triangulate across raters, examine the underlying indicator structures, and form independent judgements on material issues.

The composite-score worked problem illustrated the standard aggregation framework. Environmental, social, and governance pillar scores of (0.72, 0.56, 0.81), weighted (0.45, 0.25, 0.30), yield a composite of 0.707. The weights are themselves a policy choice; the materiality-adjustment worked problem demonstrated how industry-specific materiality can shift effective weights substantially.

The chapter's AI-application coverage focused on the operational plumbing: NLP for sustainability-report parsing, ML-driven controversy detection from news feeds, computer vision for verification against satellite imagery, and large-language-model-based summarisation of voluminous disclosures. The combination is reducing the data-extraction cost of ESG analysis by roughly an order of magnitude; investing in the analysis itself — judgement on what the data means — is what differentiates leading from following ESG practice. We turn next to the financial instruments that ESG data informs.

Green Finance

Green finance does not happen because investors are virtuous. It happens because investors believe the data, and AI is what makes the data believable.

Kennedy Chengeta, PhD

Green finance — capital allocation that supports environmental objectives — has grown into a multi-trillion-dollar market across green bonds, sustainable loans, transition finance, and impact investing. AI applications support deal-flow identification, eligibility verification, performance monitoring, and reporting. This chapter treats the major instruments and the AI infrastructure that anchors their integrity.

47.1 Green Bonds and Sustainable Loans

Green bonds (governed by the ICMA Green Bond Principles) and sustainability-linked bonds (governed by the SLB Principles) are the dominant fixed-income instruments. Green bond proceeds finance specific eligible projects; sustainability-linked instruments tie coupons to KPI achievement (emissions intensity, renewable share, etc). AI applications include: project-eligibility verification against taxonomy criteria (EU Taxonomy, Climate Bonds Initiative, country-specific taxonomies), KPI monitoring across thousands of underlying projects, and second-party-opinion automation. Issuance volumes have risen rapidly; integrity concerns parallel those in voluntary carbon markets and motivate similar AI infrastructure.

47.2 Climate Investment Funds

Climate-focused funds — private equity, infrastructure, mezzanine debt — finance specific decarbonisation projects. Diligence and monitoring are data-intensive: technology risk, policy risk, off-take risk, country risk. AI applications include: pipeline screening against fund mandates, due-diligence document processing, and ongoing portfolio monitoring against KPIs. The blended-finance segment, where concessional capital de-risks commercial investment in emerging markets, is particularly data-intensive because of weak local data infrastructure.

47.3 Microfinance and Financial Inclusion

Smallholder farmers, micro-enterprises, and informal-sector workers represent the largest gap in financial inclusion. Climate-vulnerable populations need credit, savings, and insurance to manage risks that conventional financial institutions cannot price. AI applications include: alternative credit scoring

from mobile-money data, satellite-derived agricultural risk assessment, and portfolio monitoring at scale. Apollo Agriculture, M-Kopa, and Pula are leading African examples; cross-pollination with climate-AI capabilities is accelerating.

47.4 Risk and the Cost of Capital

Climate-vulnerable countries face higher cost of capital than equivalent climate-resilient countries — a self-reinforcing problem because the higher cost makes adaptation investment harder. AI-driven climate risk pricing, when integrated transparently into sovereign and corporate credit assessments, can both improve risk pricing accuracy and reveal risk-reduction interventions that improve borrowing terms. The IMF, MDBs, and major credit rating agencies are integrating ML climate risk modules into their assessments.

47.5 Mathematical Foundations of Green Finance Pricing

47.5.1 Green Bond Yield Differential (Greenium)

$$\text{Greenium} = y_{\text{conventional}} - y_{\text{green}}.$$

Empirically 1–8 bps in mature markets.

47.5.2 Sustainability-Linked Bond Coupon Step-Up

$$c_{t+1} = c_t + \Delta c \cdot \mathbb{1}(\text{KPI not met}),$$

typically $\Delta c = 25$ bps.

47.5.3 Blended Finance Catalysis

Catalytic ratio:

$$\text{CR} = \frac{\text{Private capital mobilised}}{\text{Concessional capital deployed}}.$$

Targets are 2–4× in well-structured deals.

47.6 Worked Problems and Solutions

Problem 47.1 (SLB Cost Impact). Bond face \$100M, coupon 5.5%; KPI miss triggers 25 bps step-up. Annual extra cost?

Solution. $0.0025 \cdot 100,000,000 = \$250,000/\text{yr}$. ■

Problem 47.2 (Catalytic Ratio). \$50M concessional drew \$180M commercial. Compute CR.

Solution. $180/50 = 3.6\times$. Above target. ■

47.7 Chapter Conclusion

Sustainable finance instruments have moved from niche to mainstream in under a decade. Green bonds, sustainability-linked bonds, transition bonds, and the broader labelled-finance category now account for roughly 5%–10% of new global fixed-income issuance, with substantial pipeline growth. AI applications support each part of the ecosystem.

The SLB-step-up worked problem illustrated the structural feature that distinguishes sustainability-

linked from green bonds. A green bond ring-fences proceeds for eligible projects; an SLB places the issuer's entire balance sheet behind a sustainability KPI, with coupon step-up triggered if the KPI is missed. The \$250,000 annual cost from a 25-basis-point step-up on a \$100M bond is enough to focus management attention; it is not, by itself, transformative. The structural value of the SLB lies in the public commitment to a specific, measurable target.

The catalytic-ratio worked problem introduced the blended-finance metric that drives multilateral and concessional capital deployment. A $3.6\times$ catalytic ratio — \$180M of private capital mobilised for \$50M of concessional — represents efficient use of public funds. Catalytic-ratio targets vary by mechanism, instrument, and geography; the worked-example ratio is above the $2\text{--}3\times$ target typical for well-structured blended deals.

The chapter's coverage of greenium evidence, AI-driven taxonomy- eligibility assessment, and the emerging integration with the EU Taxonomy, the UK Green Taxonomy, and equivalent jurisdictional frameworks gave the reader the operational toolkit for navigating the contemporary sustainable-finance landscape. The taxonomic infrastructure is still maturing; the direction of travel is settled. We turn next to the disclosure infrastructure that grounds all sustainable-finance instruments: climate-related disclosure.

Climate Disclosure and the ISSB Standards

A disclosure framework is only as strong as the verification behind it. AI is becoming the verification of choice because the alternative is human-only audit at impossible scale.

Kennedy Chengeta, PhD

Climate disclosure regulation has moved from voluntary to mandatory across major economies. The IFRS Sustainability Standards (S1, S2) under the ISSB became operational in 2024. The EU CSRD and ESRS apply from FY2024 reporting. The US SEC climate disclosure rule and parallel state regulations expand US coverage. South Africa's JSE listing rules require TCFD-aligned disclosure for major issuers. Implementing compliant disclosure at firm scale demands AI infrastructure no manual process can match.

48.1 The ISSB Framework

IFRS S1 (general sustainability) and S2 (climate) provide a global baseline for sustainability disclosure under the IFRS Foundation, with structure derived from TCFD. Required disclosures include: governance, strategy (including transition plans), risk management, and metrics and targets including Scope 1, 2, and material Scope 3 emissions. Industry-specific guidance derived from SASB applies. Adoption has been rapid: as of 2025, IFRS S1/S2 has been adopted or proposed by jurisdictions covering a substantial share of global market capitalisation.

48.2 CSRD and the European Sustainability Reporting Standards

CSRD applies to roughly 50,000 EU and non-EU companies meeting size thresholds, with phased implementation from FY2024 to FY2028. ESRS specify approximately 1,000 disclosure datapoints across 12 standards covering environment, social, and governance topics. Double materiality — assessing both impact on the company and impact of the company — distinguishes CSRD from US-style frameworks. AI applications are essential at the data-gathering scale CSRD demands; without AI, compliance cost would consume material portions of profitability for affected entities.

48.3 Transition Plans

Transition plans — documented strategies for moving towards net-zero objectives — are increasingly mandated. The TPT (Transition Plan Taskforce) framework, IFRS S2 guidance, and EU regulation specify content. AI applications include: scenario modelling against NGFS and IEA pathways,

capital expenditure modelling under transition scenarios, and continuous monitoring of plan execution against milestones. Investor-side AI applications include credibility assessment of stated plans against operational data — whether the capex actually happens, whether emissions actually fall on the trajectory.

48.4 AI in Disclosure Production

Producing compliant disclosure for a global enterprise involves data gathering across hundreds of legal entities, normalisation against 1,000+ datapoints, validation against external benchmarks, and narrative generation that satisfies both technical and accessibility requirements. AI applications at every stage: data extraction from operational systems, NLP-driven cross-reference against framework requirements, anomaly detection in compiled data, automated narrative generation from quantitative templates, and audit-trail production. Major sustainability software vendors (Workiva, Sphera, Sweep, Watershed) embed these capabilities; in-house platforms at large enterprises combine them with domain-specific AI for unique processes.

48.5 Mathematical Foundations of Climate Disclosure

48.5.1 Implied Temperature Rise

For portfolio with weighted intensity I_{port} and reference trajectories I_T for warming T :

$$\text{ITR} = \arg \min_T |I_{\text{port}} - I_T|.$$

48.5.2 Transition Plan Credibility

Trajectory deviation from stated path $\hat{E}_t$:

$$\Delta_t = E_t^{\text{actual}} - \hat{E}_t, \quad \text{Credibility score} = 1 - \frac{1}{T} \sum_t \frac{|\Delta_t|}{\hat{E}_t}.$$

48.6 Worked Problems and Solutions

Problem 48.1 (ITR). Portfolio intensity 240 g CO₂/USD revenue. Reference: 1.5C target 180; 2.0C 240; 2.5C 320; 3.0C 400. Compute ITR.

Solution. Closest match is 240 $\Rightarrow$ ITR = 2.0C. ■

48.7 Chapter Conclusion

Climate disclosure is the regulatory infrastructure on which most of the rest of sustainable finance ultimately rests. The chapter walked through the principal frameworks — TCFD/ISSB IFRS S2, the EU CSRD with its European Sustainability Reporting Standards, the SEC climate-disclosure rule, and the parallel jurisdictional regimes emerging in the UK, Australia, Brazil, Japan, and elsewhere.

The Implied-Temperature-Rise worked problem illustrated the methodological core of forward-looking disclosure. A portfolio intensity of 240 g CO₂/USD revenue, compared against reference trajectories of (180, 240, 320, 400) for (1.5C, 2.0C, 2.5C, 3.0C), yields an ITR of 2.0C. The methodology is

established but contested: different ITR providers produce materially different scores for the same portfolio depending on benchmark choices, sector treatment, and Scope 3 inclusion. Transparency on methodology is the operational discipline that distinguishes credible from cosmetic disclosure.

The transition-plan-credibility treatment introduced the trajectory- deviation framework that increasingly distinguishes leading from laggard corporate practice. A transition plan that announces 2050 targets without 2025, 2030, and 2035 milestones is unverifiable; a plan that includes such milestones can be tracked against actual performance. The mathematical machinery for credibility scoring is straightforward; the harder work is institutional — ensuring that the milestones reflect genuine strategic commitment rather than rhetorical positioning.

The chapter's AI-application coverage focused on the data-extraction and verification plumbing that disclosure increasingly requires. NLP- based extraction from sustainability reports, computer-vision-based verification against satellite imagery, ML-driven anomaly detection in disclosed time series, and the emerging integration of large- language-model-based assistance into disclosure preparation collectively reduce the manual labour of compliance by roughly an order of magnitude. The judgement work — what to disclose, how to characterise risk, how to position the firm — remains unambiguously human. We turn next to the supply-chain extension of disclosure: due-diligence regulation.

Supply Chain Due Diligence

A diligence obligation that ends at Tier 1 is a diligence obligation that ends where most of the harm begins.

Kennedy Chengeta, PhD

Due diligence regulations now require companies to identify, prevent, and mitigate human-rights and environmental harms in their supply chains. The EU CSDDD, the German Supply Chain Act (LkSG), the French Duty of Vigilance, the UK Modern Slavery Act, the US UFLPA, and parallel national laws together create overlapping obligations. AI applications in due diligence span risk assessment, supplier screening, satellite-based environmental monitoring, and grievance-mechanism analytics.

49.1 The Diligence Obligation

Modern due diligence obligations require ongoing risk identification (not one-time supplier surveys), action to prevent and mitigate identified risks, grievance mechanisms accessible to affected stakeholders, transparent reporting, and remediation when harms occur. The expansion from Tier 1 to upstream tiers and from environmental to human-rights scope has increased operational complexity by orders of magnitude. AI is essential because manual due diligence at this scope is impossible.

49.2 Geographic and Sector Risk Mapping

Risk maps overlay geographic risk indicators (governance, conflict, child labour prevalence, environmental sensitivity) with sectoral risk indicators (forced labour in cotton, palm oil deforestation, cobalt mining child labour). ML scoring across hundreds of indicators produces supplier-level risk assessments that direct diligence resources to highest-risk relationships. The Walk Free Foundation, Verisk Maplecroft, and Sayari publish risk indices that AI integrates with company-specific supplier data.

49.3 Forced Labour and the UFLPA

The US Uyghur Forced Labor Prevention Act presumes that goods produced wholly or in part in Xinjiang are made with forced labour and prohibits import unless companies rebut the presumption with clear and convincing evidence. Compliance requires deep supply-chain traceability: cotton bales traced to fields, polysilicon batches traced to foundries, lithium batteries traced to cells. AI applications include: customs-data ML for shipment-pattern analysis, isotopic and DNA traceability of agricultural

commodities, and corporate-relationship graph analysis for hidden ownership. Enforcement actions have produced multi-billion-dollar import detentions; compliant alternative sourcing has been a major procurement pivot.

49.4 Environmental Due Diligence and the EUDR

The EU Deforestation Regulation (EUDR), applicable from late 2025, prohibits import of cattle, cocoa, coffee, palm oil, rubber, soy, and wood (and derived products) sourced from land deforested after 31 December 2020. Compliance requires geolocation of every production plot to that date plus deforestation-free verification. ML deforestation alerting (Global Forest Watch, MapBiomas, RADD) underpins verification at scale. African coffee and cocoa supply chains, smallholder-dominant and historically unmapped, face particular implementation challenges; AI plus mobile-phone geolocation by farmer cooperatives is the dominant pathway.

Example 49.1 (Cocoa Traceability in Ghana). A Ghana cooperative serving 14,000 smallholders mapped every member farm using farmer-collected GPS points and satellite imagery cross-validation. Each cocoa bag is tagged at collection point and traced through fermentation, drying, and export. ML deforestation analysis confirms zero plot-level deforestation post-2020. Compliance cost: USD 2.40 per tonne of cocoa traced — absorbable into the EUDR-driven price premium for verified deforestation-free cocoa. Without AI, manual verification at this granularity would have been economically infeasible.

49.5 Mathematical Foundations of Supply Chain Risk Scoring

49.5.1 Composite Supplier Risk

$$R_{\text{supplier}} = \alpha_g R_{\text{geographic}} + \alpha_s R_{\text{sector}} + \alpha_h R_{\text{historical}}.$$

49.5.2 EUDR Plot Compliance

For plot p with cover state $s_p \in \{\text{forest}, \text{non-forest}\}$ at reference date t_0 and current t_1 :

$$\text{Compliant}(p) = \mathbb{K}(s_p(t_1) = \text{forest}) \vee \mathbb{K}(s_p(t_0) = \text{non-forest}).$$

49.6 Worked Problems and Solutions

Problem 49.2 (Geographic Risk Score). Forced labour prevalence 4.2/1,000; weight 0.6. Conflict 6.8/10; weight 0.4. Compute composite (normalise to [0,1]).

Solution. Normalise: $4.2/10 = 0.42$, $6.8/10 = 0.68$. $0.6 \cdot 0.42 + 0.4 \cdot 0.68 = 0.252 + 0.272 = 0.524$.

■

49.7 Chapter Conclusion

Supply-chain due-diligence regulation is the policy instrument through which environmental and human-rights expectations are being made operationally binding on global supply chains. The chapter walked through the principal regimes: EUDR for deforestation, UFLPA for forced labour, CSDDD for

general sustainability due diligence, and the parallel national regimes that supplement them.

The geographic-risk-score worked problem illustrated the operational prioritisation that due-diligence regulation requires. Few companies can perform exhaustive due diligence on every supplier in every geography; the workable practice is risk-based, with deeper investigation triggered by elevated geographic, sector, or behavioural risk indicators. The mathematical machinery — composite scoring, multi-attribute risk decomposition — is straightforward; the operational challenge is calibrating thresholds, validating methodology, and defending decisions against regulatory and stakeholder scrutiny.

The EUDR-plot-compliance treatment introduced the specific operational machinery of the deforestation regulation. Plot-level geolocation combined with satellite-derived forest-cover history at the December 2020 reference date determines compliance; non-compliant plots disqualify their products from EU market access. The technical infrastructure — geolocation collection, satellite-based forest-cover analysis, traceability platforms — is being built rapidly under regulatory deadline pressure.

The chapter's coverage of AI-enabled investigation workflows — NLP-based vendor-screening across sanctions and adverse-media databases, computer-vision-based facility verification, network analysis of corporate ownership structures, and increasingly foundation-model-based document review — gave the reader the operational toolkit. The combination reduces investigation cost by an order of magnitude; the structural insight that due-diligence regulation is not principally a technical problem but an institutional-design problem remains valid. We turn next to the social dimension that all of this work intersects with: just transition.

Just Transition

A transition that decarbonises the economy while leaving workers and communities behind is not a transition — it is a relocation of the cost.

Kennedy Chengeta, PhD

Just transition is the principle that decarbonisation must be implemented in ways that protect workers, communities, and disadvantaged groups from the dislocation it causes. Coal-region transitions in the US, Germany, Poland, and South Africa; the closure of internal-combustion-engine manufacturing across automotive supply chains; the realignment of energy-export-dependent economies — all face just-transition challenges. AI applications support workforce transition planning, regional economic modelling, and equitable allocation of transition resources.

50.1 Workforce Modelling

Decarbonisation creates jobs (renewable installation, building retrofits, EV manufacturing, public transit) and destroys jobs (coal mining, oil refining, ICE manufacturing). The geographic distribution rarely matches: jobs created in growth regions may not absorb workers displaced in legacy regions. ML applications model job flows by occupation, region, and skill, identifying mismatches that targeted retraining can address. The IEA, IRENA, and ILO publish projections; country-specific applications support national just-transition plans.

50.2 Regional Economic Transition

A coal region with three thousand mining jobs and a power plant facing retirement is a complex economic system: direct employment, indirect employment in services and construction, induced employment from local spending, fiscal effects on schools and infrastructure, real-estate values, tax bases. Computable general equilibrium models, agent-based simulation, and ML scenario analysis support transition planning. Komati (South Africa) is a contemporary case study; the Just Energy Transition Partnership funding mechanisms direct concessional capital based partly on these modelling outputs.

50.3 Energy Poverty and Equity

Energy transition can either reduce or worsen energy poverty depending on design. Carbon pricing without rebates is regressive; targeted rebates (Canada's federal carbon-tax rebate, California's climate dividend) preserve fairness. Building-efficiency programs benefit upper-income owners

disproportionately unless designed with explicit equity targeting. AI applications include: equity-impact modelling of policy proposals, targeting low-income households for energy-efficiency support, and identifying communities likely to be left behind without active intervention.

50.4 African Just Transition

African just-transition challenges are distinctive. Most African countries have low historical emissions but face significant transition impacts because exports are concentrated in fossil fuels (Nigeria, Angola, Ghana) or fossil-intensive minerals (South Africa coal, copper-belt countries). The Just Energy Transition Partnerships — South Africa (USD 8.5B announced), Indonesia, Vietnam, Senegal — direct concessional capital toward managed transitions. Implementation challenges — governance, labour-market institutions, regional political economy — are real; AI is necessary infrastructure but not sufficient policy.

50.5 Mathematical Foundations of Just-Transition Modelling

50.5.1 Worker Reallocation

Bipartite matching minimising distance:

$$\min_{\mathbf{x}} \sum_{ij} d_{ij} x_{ij} \quad \text{s.t.} \quad \sum_j x_{ij} = 1 \quad \forall i, \quad \sum_i x_{ij} = c_j \quad \forall j.$$

50.5.2 Regional Multiplier

Direct + indirect + induced employment:

$$\Delta L_{\text{total}} = \Delta L_{\text{direct}} \cdot k_{\text{reg}},$$

with regional multiplier $k_{\text{reg}} \approx 1.5\text{--}3$ for manufacturing, $\sim 2\text{--}4$ for energy.

50.6 Worked Problems and Solutions

Problem 50.1 (Job Multiplier). 3,000 direct coal-mining job losses; multiplier 2.4. Total regional employment impact?

Solution. $3000 \cdot 2.4 = 7,200$ total job-equivalents. ■

50.7 Chapter Conclusion

Just transition is the social-policy framework that distinguishes managed from chaotic decarbonisation. The chapter walked through the analytical components: identifying affected workers and communities, modelling regional economic impacts, designing replacement-employment programmes, and tracking outcomes over the multi-year transition horizon.

The job-multiplier worked problem illustrated the regional-economic scale of the transition. Three thousand direct coal-mining job losses, combined with a 2.4 multiplier, produce 7,200 total job-equivalents of regional employment impact. The multiplier captures the indirect and induced effects — suppliers of mining equipment, local services dependent on miner spending, government revenues that

fund regional public services — that determine whether the transition is manageable or catastrophic for the region.

The bipartite-matching treatment for worker reallocation introduced the optimisation framework that increasingly supports active-labour-market policy. The mathematical structure is the assignment problem; the practical work is in the distance metric — what does it mean for a laid-off coal miner to be “close” to an opening as a renewable- energy technician, or a building-retrofit installer, or a hydrogen- plant operator? The metrics combine occupational-skills overlap, geographic distance, wage compatibility, and training-requirement proximity; calibrating them is itself a substantial body of work.

The chapter’s coverage of the South African Just Energy Transition Partnership, the German coal-region transitions, and the parallel programmes in Indonesia, Vietnam, and Senegal pointed forward to Part X. The patterns differ across jurisdictions — the scale of fiscal support, the strength of organised labour, the diversification options of regional economies all matter — but the underlying analytical framework generalises. This chapter closes Part VIII. Part IX turns to the broader governance and ethics layer that all sustainability AI must navigate.

Part IX

Governance, Ethics, and Regulation

AI Ethics in the Climate Context

A climate model that improves average accuracy while increasing failure on the most vulnerable communities is a more dangerous model than the one it replaced.

Kennedy Chengeta, PhD

AI ethics and climate ethics intersect in ways that neither field alone fully addresses. AI systems trained on data from richer countries fail in poorer ones. Optimisation against aggregate efficiency can worsen distributional outcomes. Algorithmic decisions about who receives climate finance, drought aid, or insurance pay-outs reinforce historical inequities. This chapter treats the ethical questions specific to deploying AI for sustainability, with attention to the governance frameworks that have emerged in response.

51.1 The Geographic Bias Problem

Most ML training data are produced in regions with strong data infrastructure: North America, Europe, parts of East Asia. Models trained on this data systematically underperform on other regions. Climate models miss tropical convection. Crop models miss African agroecological zones. ESG models miss informal-sector activities that dominate emerging-economy economies. The remedy is local data collection, model adaptation, and validation — effort that is unevenly funded and politically contested.

51.2 Distributional Impacts

AI applications change who bears costs and who receives benefits. Smart-grid optimisation may shift load-shedding to lower-income suburbs. Precision-irrigation gains on commercial farms may release water that flows to other commercial farms rather than to ecosystems or smallholders. Climate-risk pricing in insurance may exclude vulnerable populations who most need cover. Distributional analysis must accompany every deployment; equity-aware loss functions, fairness constraints, and policy overlays redistribute outcomes that purely efficiency-optimal AI would concentrate.

51.3 Consent and Surveillance Risk

Detailed monitoring infrastructure — satellite, sensor, smart-meter, mobile-data — has clear sustainability benefits and clear surveillance risks. Continuous deforestation monitoring also tracks indigenous communities. Smart-meter data reveals occupancy patterns. Smartphone-based agronomic

advisory generates farmer behavioural data. The same systems that enable beneficial monitoring enable harmful surveillance. Governance frameworks — consent regimes, data minimisation, purpose limitation, oversight bodies — determine whether the trade-off is acceptable.

51.4 Who Decides What Sustainability Means

Sustainability metrics encode values: which species matter, which costs are externalities, what time horizon is appropriate, whose welfare counts equally. AI systems instantiate these choices, often without explicit articulation. The governance question — who decides what sustainability means and therefore what the AI optimises — is political, not technical. Frameworks that formally include affected communities, indigenous knowledge holders, and future generations in design and oversight are increasingly required by responsible-AI standards (UNESCO Recommendation on AI Ethics, the OECD AI Principles, sectoral standards).

51.5 Mathematical Foundations of Algorithmic Fairness

51.5.1 Demographic Parity

For protected attribute A and prediction $\hat{Y}$:

$$\Pr(\hat{Y} = 1 \mid A = a) = \Pr(\hat{Y} = 1 \mid A = b).$$

51.5.2 Equalised Odds

$$\Pr(\hat{Y} = 1 \mid Y = y, A = a) = \Pr(\hat{Y} = 1 \mid Y = y, A = b), \quad y \in \{0, 1\}.$$

51.5.3 Group Calibration

$$\Pr(Y = 1 \mid \hat{Y} = p, A = a) = p \quad \forall a.$$

The impossibility result: only one of demographic parity, equalised odds, and calibration can hold simultaneously (Chouldechova 2017, Kleinberg et al. 2017) when base rates differ.

51.6 Worked Problems and Solutions

Problem 51.1 (Disparity Test). Approval rates: group A 0.72, group B 0.48. Compute disparate-impact ratio.

Solution. $0.48/0.72 = 0.667$. Below the 0.80 four-fifths rule, indicating potential disparate impact.

■

51.7 Chapter Conclusion

AI ethics in the sustainability context is more demanding than AI ethics in conventional applications, for three reasons identified at the chapter's opening: physical rather than informational stakes, global rather than national affected populations, and inter- generational rather than annual time horizons. The chapter walked through the principal ethical considerations: distributional fairness, algorithmic transparency, accountability for adverse outcomes, and the participation of affected communities in design and deployment decisions.

The disparate-impact worked problem illustrated the four-fifths-rule calculation that anchors much of US discrimination law. An approval- rate ratio of 0.667 between groups A and B falls below the 0.80 threshold and creates legal exposure under disparate-impact doctrine. The mathematical machinery is straightforward; the harder work is in the threshold setting, the protected-class definition, and the remediation pathway when disparity is detected.

The fairness-impossibility result introduced by Chouldechova and Kleinberg et al. deserves emphasis. The three desirable properties — demographic parity, equalised odds, and calibration — cannot all be satisfied simultaneously when base rates differ across groups. This is a mathematical fact, not a research challenge to be solved. The operational implication is that fairness in any specific application requires explicit choice among the irreconcilable properties; pretending that all can be achieved is at best naive and at worst deceptive.

The chapter's coverage of climate-justice frameworks, the Aarhus Convention on environmental information access, the IPCC's treatment of equity and ambition, and the emerging principles for AI deployment in vulnerable communities pointed forward to the remaining chapters in Part IX. The principles are increasingly well- established; the operational implementation lags. We turn next to the data-governance layer that operationalises many of the principles in practice.

Data Governance for Sustainability

Data sovereignty is not a constraint on sustainability work — it is a precondition for it.

Kennedy Chengeta, PhD

Data governance — who holds, controls, and benefits from data — is increasingly central to sustainability AI deployments. Data sovereignty regulations, indigenous data principles, personal data protection, and emerging data-marketplace frameworks shape what is possible. This chapter treats the major frameworks and the operational practices they require.

52.1 National Data Sovereignty

Many jurisdictions require that data about their territory and citizens be stored, processed, and accessed under domestic legal frameworks. South Africa's POPIA, Kenya's Data Protection Act, the African Union Malabo Convention, India's DPDP Act, and parallel laws across Latin America and Southeast Asia formalise this. Operational implications: cloud regions, contractual safeguards, data-localisation architectures. Many sustainability AI projects now design for data residency from the outset rather than retrofitting compliance after deployment.

52.2 Indigenous Data Sovereignty and CARE Principles

The CARE principles — Collective benefit, Authority to control, Responsibility, Ethics — supplement the FAIR principles for data concerning indigenous communities and territories. Operationalising CARE means: free prior informed consent for data collection, community-defined access controls, community participation in research design, and benefit sharing from outcomes. Biodiversity monitoring, traditional knowledge documentation, and locations of culturally significant sites are governed by relevant communities rather than open by default. Implementation requires institutional infrastructure — community data trusts, indigenous data alliances — that is being built in Aotearoa, Canada, and increasingly Africa.

52.3 Personal Data Protection

Sustainability AI generates substantial personal data: smart-meter consumption, smartphone agroeconomic advisory, ride-sharing trip data, building-occupancy patterns. GDPR, POPIA, and parallel laws set requirements for lawful basis, data minimisation, purpose limitation, retention limits, and individual rights. AI applications must accommodate these requirements through anonymisation

(with care — naive anonymisation leaks identity), federated learning that keeps data on device, and differential privacy for aggregated analytics.

52.4 Open Data and the Public Goods Problem

Climate data, satellite imagery, and biodiversity records have characteristics of public goods: non-rival, with spillover benefits across applications. Open data policies (Copernicus open licence, NASA open data, Argo CC0 terms) have produced enormous downstream value. Conflicts arise when commercial value emerges (high-resolution imagery, proprietary species records) and when open data could enable harm (rare-species locations to poachers, infrastructure data to attackers). Governance frameworks balance access and protection through licensing, controlled access, and tiered disclosure.

52.5 Mathematical Foundations of Privacy-Preserving Analytics

52.5.1 ϵ -Differential Privacy

A mechanism M is ϵ -DP if for all neighbouring datasets $\mathcal{D}, \mathcal{D}'$ and outputs S :

$$\Pr(M(\mathcal{D}) \in S) \leq e^\epsilon \Pr(M(\mathcal{D}') \in S).$$

Laplace mechanism: $M(x) = f(x) + \text{Lap}(\Delta f / \epsilon)$.

52.5.2 Federated Averaging

For K clients with local datasets $\mathcal{D}_k$:

$$\theta_{t+1} = \sum_{k=1}^K \frac{|\mathcal{D}_k|}{|\mathcal{D}|} \theta_k^{(t)}.$$

Privacy strengthened with secure aggregation and DP-noise addition.

52.6 Worked Problems and Solutions

Problem 52.1 (Laplace Noise). Sensitivity $\Delta f = 1$, $\epsilon = 0.5$. Compute Laplace noise scale parameter.

Solution. $b = \Delta f / \epsilon = 1 / 0.5 = 2$. Standard deviation $= \sqrt{2} \cdot 2 = 2.83$. ■

52.7 Chapter Conclusion

Data governance is where ethical principles become operationally binding constraints on system design. The chapter walked through the principal frameworks: the FAIR principles (findable, accessible, interoperable, reusable) that govern scientific data infrastructure; the CARE principles (collective benefit, authority to control, responsibility, ethics) that govern indigenous data sovereignty; the GDPR and equivalent regional data-protection regimes; and the emerging privacy-preserving analytics stack.

The differential-privacy worked problem illustrated the mathematical machinery of formal privacy guarantees. A Laplace mechanism with sensitivity 1 and $\epsilon = 0.5$ requires noise scale $b = 2$ (standard deviation 2.83). The privacy-utility trade-off is explicit: smaller ϵ produces stronger privacy but more noise; larger ϵ admits more information leakage but supports more accurate analytics. Practical

deployments typically operate at ϵ between 0.1 and 10 depending on data sensitivity and accuracy requirements.

The federated-learning treatment introduced the distributed training paradigm that increasingly supports sensitive data analytics across institutional and jurisdictional boundaries. Hospital networks for medical data, cross-utility coalitions for grid data, and emerging agricultural cooperatives for farm data are deploying federated training to extract aggregate insights without centralising raw data. The technical infrastructure is mature; the governance infrastructure that determines who participates, who validates, and who benefits remains in active development.

The chapter's coverage of indigenous data sovereignty deserves emphasis. Many biodiversity and traditional-knowledge datasets are generated in or about indigenous communities; default open-data practices that ignore community ownership and consent are increasingly recognised as ethically and operationally inadequate. The CARE principles supplement rather than replace FAIR; the combination is the contemporary best practice. We turn next to the most reflexive of the governance questions: the carbon footprint of AI itself.

The Carbon Footprint of AI

An AI system that optimises sustainability while consuming more energy than it saves is a polluting machine with a marketing problem.

Kennedy Chengeta, PhD

AI has a carbon footprint of its own. Training large models, serving inference at scale, and the embodied carbon of hardware all contribute. As AI workloads have grown rapidly, data centre electricity demand has grown with them. Honest sustainability AI must account for the costs of the AI itself, not just the benefits of its outputs. This chapter treats the measurement, the technological mitigation, and the strategic choices that make AI net-positive for sustainability.

53.1 Training and Inference Costs

Training a large language model can emit hundreds of tonnes CO₂ depending on compute requirements, energy mix, and hardware efficiency. Strubell et al. (2019) catalysed attention to the issue; Patterson et al. (2021) refined the methodology emphasising that location, hardware generation, and software efficiency dominate the variability. Inference at scale, repeated billions of times, can exceed training emissions for popular deployed models.

Definition 53.1 (AI Carbon Accounting). For a deployed AI system, total emissions over deployment lifetime τ are

$$E_{\text{AI}} = E_{\text{embodied}} + \int_0^{\tau} \text{PUE}(t) \cdot P(t) \cdot \text{CI}(t) dt$$

where E_{embodied} is hardware embodied carbon, PUE is data centre power usage effectiveness, P is IT power, CI is grid carbon intensity.

53.2 Mitigation: Hardware, Software, and Location

Hardware efficiency: each generation of TPU, GPU, and inference accelerator improves performance per watt. Software efficiency: model distillation, quantisation, sparsity, mixture-of-experts, and inference caching all reduce per-query compute by factors of 5-100× for common workloads. Location: data centre carbon intensity varies by 10-50× between fossil-dominated and renewable-dominated grids; geographically routing workloads to low-carbon regions and times produces large reductions without algorithm changes. Major cloud providers report grid-aware scheduling delivering 5-15% emission reductions in operational deployments.

53.3 Model Choice and Right-Sizing

A 100-billion-parameter model is not always better than a 1-billion-parameter model for the application. Right-sizing — choosing the smallest model that delivers required quality — has both cost and emission benefits. Foundation-model fine-tuning produces smaller, task-specific models that match generalist-model performance on the specific task at fractions of the inference cost. Retrieval-augmented generation reduces required model size for knowledge-intensive tasks by sourcing knowledge from documents at inference time.

53.4 The Net-Positive Test Revisited

Chapter 1 introduced the net-positive concept. Operationalising it requires honest accounting: count inference at the actual deployment scale, count embodied carbon prorated over realistic lifetimes, count the carbon of model retraining cycles. Most operational sustainability AI deployments have net-positive ratios in the tens to thousands; some research and pilot deployments do not yet break even. Transparent reporting — disclosing the energy and carbon costs of the AI alongside the benefits claimed — is essential to credibility and to avoiding the embarrassment of belated revelations that climate-AI was net-negative.

53.5 Industry Standards Emerging

The AI Energy Score, Carbontracker, ML CO₂ Impact, and similar projects standardise reporting of training and inference emissions. The Green Software Foundation's Software Carbon Intensity (SCI) standard provides a generic framework. Integration into enterprise AI procurement is patchy but growing. Regulatory pressure (the EU AI Act includes some sustainability disclosure provisions; CSRD covers Scope 2 from data centres) is accelerating adoption.

53.6 Mathematical Foundations of AI Energy and Carbon

53.6.1 Operational Emissions

$$E_{\text{op}}(\tau) = \int_0^\tau \text{PUE}(t) \cdot P_{\text{IT}}(t) \cdot \text{CI}(t) dt.$$

53.6.2 Embodied Hardware Carbon

$$E_{\text{embodied}} = \frac{\text{Mfg footprint}}{\text{Useful life}} \cdot \tau_{\text{deploy}}.$$

GPU manufacture: $\sim 150\text{--}250$ kg CO₂ per accelerator; amortised over 3–5 years.

53.6.3 Inference Cost

Per-query energy: $E_q = T_q \cdot P_{\text{avg}}$, with query latency T_q (s) and average power P_{avg} (W).

53.7 Worked Problems and Solutions

Problem 53.2 (Training Emissions). Training run: 1,000 GPU-hours at average 350 W per GPU; PUE 1.15; grid CI 0.45 kg CO₂/kWh.

Solution. Energy: $1000 \cdot 0.350 \cdot 1.15 = 402.5$ kWh. Emissions: $402.5 \cdot 0.45 = 181.1$ kg CO₂ = 0.181 tCO₂e. ■

Problem 53.3 (Inference at Scale). Model serves 50M queries/day at 0.2 J each; 365-day operation; CI 0.30 kg/kWh.

Solution. Annual energy: $50 \times 10^6 \cdot 0.2 \cdot 365 \text{ J} = 3.65 \times 10^9 \text{ J} = 1,014 \text{ kWh}$. Emissions: $1,014 \cdot 0.30 = 304 \text{ kg CO}_2/\text{yr}$. ■

53.8 Chapter Conclusion

The carbon footprint of AI is an internal sustainability problem of the sustainability-AI field. The chapter walked through the components: operational energy consumption (training and inference), embodied hardware carbon (manufacture and disposal of accelerators), data- centre overhead (cooling, power conversion, building infrastructure), and the lifecycle accounting that integrates them.

The training-emissions worked problem illustrated the standard calculation. A 1,000-GPU-hour training run at 350 W average power, PUE 1.15, and 0.45 kg CO₂/kWh grid carbon intensity produces 0.181 tCO₂e of operational emissions. Larger models scale roughly linearly with GPU-hours; the 1.4-million-GPU-hour training of GPT-4 class models implies operational emissions of order 250 tCO₂e per training run before counting embodied hardware. The inference arithmetic is comparable per query: a fraction of a gram of CO₂ per query, multiplied by billions of queries per day, becomes operationally material.

The chapter's mitigation strategies — regional siting of training runs at clean-grid locations, time-shifting to clean-power hours, model distillation to reduce inference cost, sparse activation in mixture-of-experts architectures, hardware-specific kernel optimisation, and increasingly carbon-aware scheduling at data-centre fleet level — collectively can reduce training emissions by 5–10× without performance loss. The operational practice is maturing rapidly but is not yet universal.

The chapter's net-positive recap returned to the framework of Chapter 1. An AI system whose lifecycle emissions are not exceeded by the lifecycle reductions it enables is not net- positive, regardless of the language used to describe it. The discipline of honest accounting is the precondition for the discipline of net-positive deployment. We turn next from the carbon footprint to the broader regulatory landscape that increasingly governs AI deployment.

AI and Sustainability Regulation

*A regulation that does not anticipate AI is a regulation that AI will route around.
A regulation that mandates AI is a regulation that may concentrate power.*
Kennedy Chengeta, PhD

Regulation of AI for sustainability is emerging across overlapping frameworks: the EU AI Act, sector-specific carbon regulations, ESG disclosure rules, deforestation due diligence, and emerging Article 6 mechanisms under the Paris Agreement. This chapter surveys the regulatory landscape and the implications for AI design and deployment.

54.1 The EU AI Act

The EU AI Act, applicable from 2025 with phased implementation, regulates AI systems based on risk classification. Prohibited practices include certain manipulative and surveillance applications; high-risk systems include those used in critical infrastructure, education, employment, and essential services. Limited-risk systems (chatbots) face transparency obligations. Sustainability applications can fall into multiple categories: AI used in critical-infrastructure operation (high-risk), AI in employment for green-skill assessment (high-risk), generative AI for ESG reporting (subject to GP-AI obligations). Compliance requires risk management, data governance, technical documentation, and human oversight.

54.2 Sector-Specific Regulation

Carbon regulations (ETS, CBAM, EUDR) impose data and verification requirements that AI in infrastructure must meet. Financial regulations (CSRD, ISSB adoption, CFIUS-style review of AI investments) impose disclosure and risk-management requirements. Energy regulations (FERC and equivalent bodies) increasingly govern AI in grid operations with stability and reliability standards. The fragmented landscape requires regulatory mapping at deployment scale; large enterprises maintain dedicated regulatory-AI coordination functions.

54.3 International Mechanisms

The Paris Agreement Article 6 governs international transfer of mitigation outcomes. Implementation requires extensive MRV that AI underpins. The Global Methane Pledge, the Forest and Climate Leaders' Partnership, and similar arrangements rely on satellite-AI verification. Export control regimes (US BIS controls on advanced AI chips and models) intersect with sustainability deployments by

constraining hardware and model availability in certain jurisdictions.

54.4 Africa-Specific Regulatory Development

African regulatory frameworks for AI sustainability are at varied stages. South Africa's POPIA and the proposed AI Code of Practice are the most developed. Kenya's ICT framework includes draft AI provisions. The African Union's Continental AI Strategy (2024) provides high-level guidance. Sustainability-specific frameworks — the African Carbon Markets Initiative, regional climate-disclosure roadmaps under the African Capacity Building Foundation — are emerging. AI deployments must navigate this evolving landscape with regulatory engagement that anticipates rather than reacts to changes.

54.5 Mathematical Foundations of Regulatory Risk

54.5.1 Compliance Cost Function

$$C_{\text{compliance}} = c_{\text{fixed}} + \sum_r v_r \cdot c_r,$$

with v_r the regulatory volume metric (covered tonnes, products, etc.) and c_r unit cost.

54.5.2 Penalty Expected Value

$$\mathbb{E}[\text{Penalty}] = \text{Pr}(\text{violation}) \cdot \text{Sanction}.$$

54.6 Worked Problems and Solutions

Problem 54.1 (CSRD Compliance). 1,000 datapoints at \$1,200 per point; fixed costs \$80k. Total?

Solution. $80,000 + 1,000 \cdot 1,200 = \1.28M . ■

54.7 Chapter Conclusion

AI regulation is one of the most rapidly evolving aspects of the technology environment that sustainability practitioners must navigate. The chapter walked through the principal regimes: the EU AI Act with its risk-tier-based regulation; the US Executive Order and emerging agency rulemakings; the UK pro-innovation regulatory framework; the Chinese deep-synthesis and generative-AI regulations; and the proliferating national regimes in Singapore, Japan, Brazil, India, and others.

The CSRD-compliance worked problem illustrated the operational scale. A thousand datapoints at \$1,200 per point combined with \$80,000 of fixed costs produces \$1.28M of annual compliance cost for a moderately complex disclosure. CSRD applies to all large EU companies and non-EU companies with substantial EU operations; the fixed costs of compliance support a competitive advantage for sophisticated incumbents and create barriers to entry for smaller firms. The same pattern recurs across other regulatory regimes.

The chapter's risk-tier discussion deserves emphasis. The EU AI Act distinguishes among prohibited AI (social scoring, real-time biometric identification in specific contexts), high-risk AI (critical-infrastructure operation, education, employment, essential services), limited-risk AI (chatbots, deep-fakes), and minimal-risk AI (the rest). Sustainability applications cluster mostly in the limited-risk and minimal-risk tiers; specific applications — grid operation, agricultural advisory in subsidised

contexts, automated ESG rating with material market consequence — can fall into the high-risk tier and incur substantially heavier compliance obligations.

The chapter's regulatory-trajectory section pointed forward to the likely developments of 2026–2030: sectoral regulation of AI in finance, healthcare, and infrastructure; emerging regulation of foundation models; growing coordination across jurisdictions on fundamental principles; and persistent divergence on operational details. Practitioners should expect compliance complexity to increase, not decrease, over the planning horizon. We turn next to the cultural-knowledge layer that regulation often under-addresses: indigenous knowledge integration.

Indigenous Knowledge Systems

A model that learns sustainability from indigenous knowledge holders is a model that learns from the people who have practised it longest.

Kennedy Chengeta, PhD

Indigenous and traditional knowledge systems — developed over millennia of human-environment co-evolution — contain substantial wisdom about sustainable resource management, climate adaptation, and biodiversity conservation. AI applications can either erase, exploit, or partner with these systems. The choice is structural and the consequences are durable. This chapter treats the principled integration of indigenous knowledge with AI sustainability work.

55.1 What Indigenous Knowledge Includes

Indigenous knowledge spans practical agricultural and ecological knowledge (crop varieties, soil management, weather indicators, fire management), governance and resource-allocation institutions, spiritual and cosmological frameworks, and language carrying culture-specific environmental categories that scientific languages may lack. Examples: African pastoralist drought-coping strategies, Pacific Islander navigation and fisheries, Amazonian agroforestry systems, Australian Aboriginal fire-management practices. Each is place-specific and resists reduction to generalised models.

55.2 Risks of AI Engagement

AI engagement with indigenous knowledge has produced harms historically: extractive research that benefits external researchers, monetisation of knowledge without community benefit, oversimplification that erodes cultural specificity, and surveillance disguised as monitoring. The CARE principles (Chapter 52) and frameworks like the Local Contexts TK and BC Notices respond to these patterns by reasserting community authority over knowledge.

55.3 Models of Partnership

Successful partnerships share several features: community-defined research questions; community ownership of data; community-controlled benefit sharing; long-term relationships that survive funding cycles; interdisciplinary teams including indigenous scholars and elders; and willingness to abandon projects that communities determine should not proceed. Examples include the Indigenous Climate Hub (Canada), the Pacific Climate Change Portal, and emerging African initiatives at IIED, ILRI, and university partnerships. AI applications within partnerships serve community priorities.

55.4 Indigenous Knowledge in Climate Adaptation

Climate adaptation in many regions builds on indigenous practices: drought-resilient varieties, water-harvesting techniques, biodiversity-supporting agriculture. Documenting and amplifying these practices through AI tools (decision-support systems, language-localised advisory, intergenerational knowledge transfer applications) can be net-positive when community-led. Top-down deployment of AI tools that displace indigenous practices often fails because the local knowledge embedded in traditional practice exceeds what generic AI provides.

55.5 Languages and AI

Indigenous languages are repositories of environmental knowledge encoded in terminology, narrative, and grammatical structure that translation cannot fully convey. Linguistic AI work — speech recognition, machine translation, text-to-speech — on under-resourced languages enables digital infrastructure to serve indigenous-language communities. Mozilla Common Voice, Masakhane (African NLP), the Indigenous Languages Technology project, and similar initiatives are building the foundations. Sustainability AI deployments serving indigenous communities must operate in their languages or risk replicating colonial information patterns.

55.6 Mathematical Foundations of Knowledge Integration

55.6.1 Bayesian Knowledge Combination

For evidence E from indigenous knowledge with prior $\pi(H)$ and scientific likelihood $L(E | H)$:

$$\pi(H | E) \propto L(E | H) \cdot \pi(H).$$

The prior encodes traditional knowledge; the likelihood incorporates modern observation.

55.6.2 Weighted Ensemble Forecast

$$\hat{f}_{\text{combined}} = w_T \hat{f}_{\text{traditional}} + w_M \hat{f}_{\text{modern}}.$$

Skill-weighted updating where each system's track record informs w_T, w_M over time.

55.7 Worked Problems and Solutions

Problem 55.1 (Weighted Forecast). Traditional rainfall forecast 380 mm; modern model 420 mm; track-record skill weights $w_T = 0.4$, $w_M = 0.6$.

Solution. $0.4 \cdot 380 + 0.6 \cdot 420 = 152 + 252 = 404$ mm. ■

55.8 Chapter Conclusion

Indigenous-knowledge integration is the area where sustainability AI most directly confronts the limits of conventional research-and- deployment practice. The chapter argued, and the worked example confirmed, that indigenous and traditional knowledge can be quantitatively combined with scientific observation when both are treated as legitimate sources of evidence. The Bayesian framework for knowledge combination, the weighted-ensemble forecast formulation, and the skill-based updating of weights together provide the technical machinery.

The weighted-forecast worked problem illustrated the operational practice. Traditional and modern rainfall forecasts of 380 mm and 420 mm, weighted by track-record-derived skill (0.4 traditional, 0.6 modern), yield a combined forecast of 404 mm. The combined forecast is not merely a numerical average; it is a statement that traditional forecasts carry information that scientific models can lack — typically about microclimate, seasonality, and ecological indicators that scientific models do not observe at the relevant resolution.

The chapter's coverage of free, prior, and informed consent (FPIC), the CARE data-sovereignty principles, and the operational practice of research partnership rather than research extraction gave the reader the institutional toolkit for ethically grounded indigenous-knowledge integration. The technical work is, perhaps surprisingly, the easier part; the harder and more important work is the institutional relationship-building that creates the conditions for legitimate collaboration.

The chapter's case-study examples — traditional fire management in Australia, herbal knowledge in southern African pharmacology, indigenous-led conservation in the Amazon, traditional seasonal forecasting in Sahel agriculture — pointed forward to Part X. African contexts present distinctive opportunities for indigenous-knowledge integration; the continent's traditional ecological knowledge bases remain under-systematised, and the AI tools that could change this are increasingly accessible. This chapter closes Part IX. Part X turns explicitly to the African context.

Part X

African Case Studies and the Sustainable Future

African Food Systems

Africa will feed itself or it will not feed itself. The question is whether the AI infrastructure is built in time.

Kennedy Chengeta, PhD

Africa's food systems face a uniquely concentrated set of challenges: a growing population, climate change accelerating in tropical and arid zones, soils degraded by decades of nutrient mining, fragmented value chains, and acute capital constraints. The continent is also the largest reservoir of undeveloped agricultural potential globally and host to the world's most rapidly growing AI agriculture deployments. This chapter synthesises the application of sustainability AI to African food systems with attention to deployments that have moved from pilot to scale.

56.1 Smallholder Production Systems

Approximately 70% of African food is produced by smallholder farmers on plots typically below 2 hectares. Average maize yields across Sub-Saharan Africa are roughly 2 tonnes per hectare, against achievable yields of 5-7 tonnes with adequate inputs and management. Closing the gap requires AI advisory delivered at smallholder scale, financial products that fund input adoption, and market access that rewards quality. Apollo Agriculture (Kenya), One Acre Fund, Hello Tractor (Nigeria), Farmerline (Ghana), and dozens of smaller actors have demonstrated unit economics that work; scaling within and across countries is the binding challenge.

56.2 Climate Adaptation in African Agriculture

Climate change is shifting growing seasons, increasing temperature stress, intensifying both droughts and floods, and altering pest and disease dynamics across African agroecological zones. AI applications support adaptation through: variety recommendations updated against shifting climate regimes (using ML on multi-decade weather and yield data), planting-window advisory based on seasonal forecasts (Chapter 18), pest and disease early warning (the FAO-CABI fall armyworm system, locust-watch analogues), and parametric insurance triggered by remote-sensing-verified shocks (ARC, ACRE Africa, Pula).

56.3 Soil Restoration at Continental Scale

African soils have lost an estimated 30-50% of their pre-cultivation organic carbon. Restoration through cover cropping, agroforestry, and integrated soil-fertility management is technically feasible

and economically attractive, but requires extension at scales that human-only systems cannot deliver. The AfSIS (African Soil Information Service), the African Forest Landscape Restoration Initiative (AFR100), and the Comprehensive Africa Agriculture Development Programme (CAADP) provide framing; AI infrastructure for soil mapping, practice recommendation, and outcome monitoring is being built across these initiatives.

Example 56.1 (Smallholder Soil Restoration: Eastern Cape). A project across 4,200 smallholder farms in the Eastern Cape of South Africa combined SoilGrids and AfSIS soil data, satellite-monitored cover crop adoption, and ML modelling of soil-organic-carbon trajectories under various practices. Year-three results: average increase in topsoil SOC of 0.31 t C/ha, against a paired-control increase of 0.04 t C/ha. Total project SOC increase: approximately 1,330 tonnes carbon, or 4,880 tCO₂e over three years. Verra-registered VM0042 credits issued to participating farmers cover the input costs of cover cropping; net farm income increased materially through both yield gains and credit revenue.

56.4 Value Chains and Market Access

Smallholder yield gains do not translate to household income gains without functional value chains. Storage losses, transport bottlenecks, intermediation costs, and quality discounts erode farm-gate prices. AI applications include: storage-condition monitoring (low-cost IoT in cooperative warehouses), market-price forecasting (Esoko, Tigeranga deployments), demand-aggregation platforms matching cooperatives to processors, and quality-grading via computer vision. The combined effect on farm-gate income substantially exceeds production-side AI gains in many contexts.

56.5 What is Working and What is Not

Honest assessment requires distinguishing what is working from what is not. Working: input-financing-bundled advisory for cash crops with strong off-take partners; satellite-driven parametric insurance at sovereign and commercial-farmer scale; pest early-warning where outbreak dynamics align with weekly satellite revisit. Not yet working at scale: precision irrigation in dispersed smallholder rainfed systems; soil-organic-carbon credits at smallholder economics; AI-driven extension competing with peer-to-peer learning. The honest assessment improves the next round of deployment design.

56.6 Mathematical Foundations of Yield Gap Analysis

56.6.1 Yield Gap Decomposition

$$Y_p - Y_a = \underbrace{(Y_p - Y_w)}_{\text{water-limited gap}} + \underbrace{(Y_w - Y_n)}_{\text{nutrient gap}} + \underbrace{(Y_n - Y_a)}_{\text{management gap}}.$$

Y_p potential, Y_w water-limited, Y_n nutrient-limited, Y_a actual.

56.6.2 Aggregation Across Smallholders

For region with N farms:

$$Y_{\text{regional}} = \frac{\sum_i A_i Y_i}{\sum_i A_i}.$$

56.7 Worked Problems and Solutions

Problem 56.2 (Yield Gap). $Y_p = 8.0$, $Y_w = 6.5$, $Y_n = 5.2$, $Y_a = 2.1$ t/ha. Decompose the gap.

Solution. Water gap: 1.5; nutrient gap: 1.3; management gap: 3.1. Total 5.9 t/ha. Management is the dominant constraint at 53%; nutrient 22%, water 25%. ■

56.8 Chapter Conclusion

African food systems are simultaneously the largest single deployment opportunity for agricultural AI and the most operationally challenging context for it. The chapter walked through the structural features — predominantly smallholder production, low fertiliser use, rain-fed agriculture vulnerable to climate variability, post-harvest losses exceeding 30% in many value chains — and the AI applications that address each.

The yield-gap-decomposition worked problem illustrated the diagnostic machinery. A potential yield of 8.0 t/ha against actual of 2.1 t/ha decomposes into 1.5 t/ha water-limited gap, 1.3 t/ha nutrient-limited gap, and 3.1 t/ha management gap. The management gap is the largest, which means that improved agronomy — timing of planting, weeding, input application, harvest — has the highest near-term return. Nutrient and water interventions matter but face higher operational and financial barriers in the smallholder context. The diagnostic informs the intervention sequencing.

The chapter's coverage of one-stop-shop digital advisory platforms — Hello Tractor, FarmDrive, Apollo Agriculture, OneAcre Fund's digital arm — pointed forward to the broader institutional discussion. The technical machinery is largely solved; the operational challenge is delivery at scale, particularly the integration with finance, insurance, and input supply that the smallholder context requires. Bundled services consistently outperform unbundled advisory; the unbundled model has been tried repeatedly and consistently fails to achieve scale.

The chapter's specific case studies — maize in Eastern Africa, sorghum in Sahelian West Africa, cassava in the Congo basin, coffee in highland East Africa — gave the reader a tour of the dominant African food systems. Each has distinct agronomy, distinct value-chain structure, and distinct AI deployment patterns; the common framework applies, but the operational details differ. We turn next from food to the other dominant material flow on the continent: energy.

African Energy Transition

A grid that load-sheds in Johannesburg and a solar home system that powers a clinic in Kenya are both part of the same energy transition. AI manages both.

Kennedy Chengeta, PhD

Africa's energy transition is distinctive. Approximately 600 million Africans lack electricity access; expanding access is a humanitarian and developmental imperative. Existing grids are stressed — South Africa load-sheds routinely; many West African systems have inadequate generation; East African grids depend heavily on hydropower vulnerable to drought. Renewable resources are abundant: solar irradiance among the highest globally, wind in the Horn and Southern coasts, hydro in central and East Africa. AI is essential to integrating distributed renewables, managing weak grids, and serving off-grid populations efficiently.

57.1 Mini-Grids and Off-Grid Solar

Mini-grids and off-grid solar home systems serve populations beyond economic reach of grid extension. PAYGo solar (M-Kopa, BBOXX, Sun King, Easy Solar) has connected tens of millions of African households since 2013. AI applications include: customer-default prediction from payment history and usage patterns, system-failure prediction from telematics, dispatch optimisation for hybrid solar-battery-diesel mini-grids, and demand-forecasting for productive-use loads. Unit economics improvements from AI have been a key driver of geographic expansion.

57.2 Grid Stability in Constrained Systems

Eskom (South Africa), KenGen (Kenya), VRA (Ghana), and similar national utilities face grid stability challenges that mature European or US grids do not. Generation capacity is constrained, transmission infrastructure is aged, demand exceeds reliable supply during peak. AI applications include: load forecasting against constrained supply for load-shedding scheduling that minimises economic damage, predictive maintenance that reduces forced-outage rates, voltage and frequency control under high renewable penetration, and consumer demand-response programs that shift load. The South African Just Energy Transition Investment Plan includes substantial AI infrastructure investment alongside generation and transmission projects.

57.3 Productive Use of Energy

Electricity access alone does not transform livelihoods — access combined with productive use does. AI-assisted matching of energy supply to productive uses (irrigation pumps, food processing, refrigeration, light manufacturing) is increasing the developmental impact of mini-grid and SHS investments. Pioneering work by Energy 4 Impact, Power for All, and IFC programs combines AI customer segmentation with appliance financing and skills development.

Example 57.1 (Productive-Use Pump Sizing in Rwanda). A mini-grid operator in Rwanda used ML demand-forecasting on 18 months of consumption data to design productive-use programs offering financed solar irrigation pumps to farmers within 1km of mini-grid lines. Pump sizing matched to farm area, crop water needs, and grid generation profile. Year-one outcomes: 312 pumps deployed, average household revenue increase from irrigated horticulture USD 380 per year, mini-grid daytime utilisation rose from 32% to 58% (improving operator economics), local food prices in dry season fell 14%. Combined CO₂ avoidance from displaced diesel pumping plus expanded local food supply: approximately 220 tonnes per year.

57.4 Continental Renewable Resource Mapping

High-quality renewable resource data are foundational to investment. The IFC-Africa Renewable Energy Atlas, IRENA Global Atlas, and World Bank ESMAP Solar/Wind Atlases provide continental coverage. AI applications refine these into project-development tools: site-specific yield estimation, infrastructure-aware routing, and grid-impact assessment. The Africa Power Pool roadmaps — SAPP, EAPP, WAPP — depend on this data for cross-border interconnection planning.

57.5 Just Energy Transition Partnerships

JETPs — the South Africa USD 8.5B partnership announced 2021, plus Indonesia, Vietnam, Senegal — channel concessional capital toward managed transitions. South African implementation has faced challenges including governance, project pipeline, and political contestation. AI applications support implementation through: pipeline-project diligence, monitoring of transition outcomes against plans, and just-transition impact tracking on coal-region communities. The honest assessment is that AI is necessary infrastructure but not sufficient policy; institutional capacity, political will, and financial structuring are determining ultimate success.

57.6 Mathematical Foundations of Mini-Grid Economics

57.6.1 Tariff Calculation

$$T = \frac{C_{\text{capital,annualised}} + C_{\text{O\&M}} + C_{\text{fuel}}}{E_{\text{annual}} \cdot \eta_{\text{collection}}}.$$

57.6.2 Productive-Use Demand Multiplier

$$E_{\text{post}} = E_{\text{pre}} \cdot (1 + \alpha \rho_{\text{PU}}),$$

with ρ_{PU} productive-use uptake rate and α per-user demand multiplier (typically 2–6 for irrigation pumps, 3–8 for processing equipment).

57.7 Worked Problems and Solutions

Problem 57.2 (Solar Tariff). Annualised capital \$25k; O&M \$5k; fuel 0; energy 60 MWh/yr; collection 85%.

Solution. $T = (25,000 + 5,000)/(60,000 \cdot 0.85) = 30,000/51,000 = \$0.588/\text{kWh}$. ■

57.8 Chapter Conclusion

African energy transition is a different problem from European or North American energy transition. The chapter walked through the distinctive features: large energy-access deficit alongside substantial existing infrastructure; abundant renewable resources alongside grid weaknesses that limit their integration; rapidly growing demand alongside fiscal constraints on capital deployment; and a development-and-climate agenda that emphasises co-benefits more strongly than mature-economy agendas typically do.

The mini-grid-tariff worked problem illustrated the economics. A \$0.588/kWh tariff for the example mini-grid is high relative to grid tariffs in developed countries (\$0.10–\$0.30/kWh typical) but low relative to the diesel-generator alternative (often \$0.60–\$1.00/kWh inclusive of fuel and maintenance) that mini-grids replace. The economic case closes for productive use — irrigation pumps, grain mills, cold-chain refrigeration — where the cost is recovered through productivity gains; residential consumption is more challenging without subsidy or cross-subsidisation.

The productive-use-multiplier discussion pointed forward to the deployment patterns that distinguish successful from struggling mini-grid operators. PowerGen Renewable Energy in East Africa, Husk Power in West Africa and India, and the various pay-as-you-go solar-home-system operators have established the operational patterns: anchor productive-use loads first, build to scale around them, integrate financial services to support customer purchase, and use AI-driven demand forecasting and predictive maintenance to keep operational costs low.

The chapter's coverage of grid-scale AI in South Africa — load-shedding optimisation, demand-response programmes, renewable-energy forecasting for the Just Energy Transition Partnership — showed how the same technical machinery operates at the other end of the gradient. The unifying theme is that African energy AI must respect the institutional and infrastructural context far more carefully than mature-economy energy AI typically does. We turn next to the continent's emerging role in global carbon markets.

African Carbon Markets

Africa holds nearly a fifth of global biological carbon and receives roughly two percent of global carbon finance. The arithmetic alone is the case for African carbon markets.

Kennedy Chengeta, PhD

African carbon markets have grown rapidly but remain underdeveloped relative to the continent's carbon resources. The Congo Basin alone stores approximately 70 Gt carbon. Mangroves around the western Indian Ocean and the Gulf of Guinea store substantial blue carbon. Soil-carbon restoration potential across African drylands is large. AI infrastructure for measurement, verification, and benefit sharing is critical to whether African carbon markets serve sustainability outcomes or replicate extractive patterns of historical commodity markets.

58.1 The African Carbon Markets Initiative

The ACMI, launched at COP27 with stretch targets of 300M+ credits per year by 2030 and 1.5B by 2050, is the continental coordinating framework. Member countries (Kenya, Senegal, Egypt, Nigeria, South Africa, others) commit to enabling regulatory frameworks, project pipelines, and revenue distribution mechanisms. AI infrastructure — MRV, registry, marketplace — is identified as critical enabling infrastructure; multiple investment vehicles are funding its development.

58.2 Article 6 and Sovereign Mechanisms

Paris Agreement Article 6 enables international transfer of mitigation outcomes (ITMOs). African countries are positioned as ITMO sellers given the carbon-resource base. Implementation requires bilateral agreements, MRV infrastructure, and benefit-sharing frameworks. Switzerland's agreements with Ghana, Senegal, and others are early examples. Singapore and South Korea are following. AI MRV underpins the credibility of the underlying credits; corresponding-adjustment accounting requires registry interoperability that AI-managed data infrastructure enables.

58.3 Community Benefit Sharing

Carbon markets without community benefit sharing replicate extractive patterns: external developers capture value while host communities bear restrictions on traditional land use. Equitable carbon markets require: free prior informed consent processes, formalised benefit-sharing agreements, community participation in MRV, and grievance mechanisms. The Climate, Community, and Biodiversity

Standards (CCB) require demonstrable community benefits. AI applications support equitable models through transparent revenue tracking, community-controlled MRV data, and monitoring of agreed benefit flows.

Example 58.1 (Mangrove Carbon: Mtwara, Tanzania). A mangrove restoration and avoided-deforestation project on the Mtwara coast of Tanzania spans 14,000 hectares of mangrove ecosystem. Community institutions (six village councils) hold project rights; an external implementer provides technical and financial services with revenue split 60-30-10 (community-implementer-government). MRV combines Sentinel-1 SAR for cover monitoring, drone LiDAR for biomass, and sediment sampling for soil carbon. ML modelling of cumulative removals produces credits issued under Verra VM0033. Year-five revenue: USD 2.4M total, of which USD 1.4M to communities — approximately USD 240 per household per year, materially significant in local income terms. Continued enforcement of community-led mangrove protection is the operational success metric, not just credit volume.

58.4 Soil Carbon and Smallholder Agriculture

Smallholder soil-carbon credits face economic challenges: per-hectare credit volumes are small (1-3 tCO₂e per hectare per year), MRV cost per hectare is high, and aggregation across thousands of farmers is administratively complex. Models that work — aggregator-led, embedded MRV, bundled with input financing — are emerging in Kenya, Ghana, and Zambia. AI is the enabler of unit economics: at sufficiently low MRV cost per farm and sufficient automation of aggregation, smallholder soil-carbon becomes economically viable.

58.5 Risks and Failure Modes

African carbon markets face the same integrity risks as voluntary markets globally (Chapter 31) plus continent-specific risks: weak land-tenure systems creating disputed project rights, political-risk exposures affecting project longevity, capacity constraints in regulators, and the historical extractive pattern of commodity markets that carbon could replicate. The constructive response is institutional: building capable regulators (the Kenya Climate Change Directorate, the Egyptian Carbon Market Regulator), strong civil-society oversight, and genuine community institutions. AI infrastructure helps but does not substitute.

58.6 Mathematical Foundations of African Carbon Markets

58.6.1 ITMO Corresponding Adjustment

For host country H transferring volume V to acquiring country A :

$$\text{NDC}'_H = \text{NDC}_H + V, \quad \text{NDC}'_A = \text{NDC}_A - V.$$

Both adjustments apply to honest accounting.

58.6.2 Benefit-Sharing Allocation

For revenue R split among community c , implementer i , government g :

$$R_c = \alpha_c R, R_i = \alpha_i R, R_g = \alpha_g R, \quad \alpha_c + \alpha_i + \alpha_g = 1.$$

Equitable splits typically have $\alpha_c \geq 0.5$ for community-led projects.

58.7 Worked Problems and Solutions

Problem 58.2 (Benefit Share). Project annual revenue \$2.4M. Splits 60-30-10 community-implementer-government. Per-household: 6,000 households.

Solution. Community: $0.6 \cdot 2.4M = \$1.44M$. Per household: $1,440,000/6,000 = \$240/\text{yr}$. ■

58.8 Chapter Conclusion

African carbon markets are at an inflection point. The chapter walked through the structural features: substantial natural-capital endowment, emerging policy frameworks (the African Carbon Markets Initiative, the Africa Climate Summit Nairobi Declaration, the increasing role of carbon revenue in NDCs), and the persistent capacity gaps that constrain project quality. AI applications support each part of the ecosystem.

The benefit-sharing worked problem illustrated the operational economics. A 60-30-10 community-implementer-government split of \$2.4M of annual project revenue produces \$240 per household per year across 6,000 households — material at the household level, transformative at the village scale. The 60% community share is at the high end of contemporary practice; many projects historically operated with much smaller community shares, and the rebalancing toward community-majority allocation is one of the defining trends of the past five years.

The ITMO-corresponding-adjustment treatment introduced the Article 6 mechanism that increasingly governs cross-border carbon transfers. The mathematical structure is straightforward; the operational difficulty is in the bilateral negotiation that establishes which mitigation outcomes are transferable and which are needed to meet the host country's own NDC. African countries are at varying stages of this negotiation; the AI applications that support both the technical MRV and the negotiation analytics are still maturing.

The chapter's coverage of African-led project developers, the emerging continental MRV infrastructure, and the role of pan-African institutions (the African Carbon Markets Initiative, the African Development Bank, the African Union Commission) pointed forward to the broader institutional discussion in Chapter 59. African carbon markets will succeed or fail based on institutional capacity as much as on technical infrastructure; building both is the work of the coming decade. We turn next to the capacity discussion explicitly.

Capacity Building and Skills

AI deployed by external consultants is at best a temporary improvement. AI deployed by the people who live where the impact is made is the only durable improvement.

Kennedy Chengeta, PhD

Capacity building — the development of skills, institutions, and infrastructure that enable countries and communities to deploy and govern AI for sustainability — determines whether AI applications produce durable outcomes or temporary improvements. African data science talent has grown substantially in the past decade but remains far below scale for the continental needs. This chapter treats the major capacity-building initiatives, the skills required, and the institutional arrangements that translate skills into impact.

59.1 The Skills Stack

Sustainability AI requires multidisciplinary skills. Foundational: programming, statistics, mathematics. ML: classical methods, deep learning, MLOps. Domain: climate science, agriculture, energy systems, ESG frameworks. Operational: software engineering, cloud infrastructure, data engineering. Communication: scientific writing, policy translation, community engagement. The stack is broad; few individuals master all of it. Effective teams combine specialists; effective organisations combine teams; effective ecosystems combine organisations.

59.2 African AI Education and Research

African AI education has expanded substantially. AIMS (African Institute for Mathematical Sciences) campuses across the continent. AI for Africa, the African Master's in Machine Intelligence, the Carnegie Mellon Africa programme. Data Science Africa workshops. Indaba and DSI conferences. Industry training: Andela, Decagon, Moringa School, Genesys Tech Hub. The pipeline is improving; the binding constraint has shifted from training capacity to retention and deployment opportunity — many trained data scientists move to international employers due to local opportunity gaps.

59.3 Open Source and Locally-Developed Models

Foundation models for African languages, African crops, African climates, and African ecosystems are increasingly developed locally rather than imported. Masakhane (NLP for African languages), AfricaNLP, and the African National Vegetation projects are examples. Local development matters

not only for performance but for governance: models trained on local data, by local teams, governed by local institutions are more likely to serve local priorities.

59.4 Institutional Capacity

Skills without institutions do not produce sustained impact. Government statistical offices, regulators, universities, civil-society organisations, and private-sector companies all need data-and-AI capacity. The African Data Initiative, the African Centre for Statistics, and emerging climate-data institutes are building the institutional substrate. Donor and international funding has historically privileged short-term project delivery over institutional capacity; the balance is shifting but slowly.

59.5 Public-Sector AI Capacity

Effective regulation, equitable resource allocation, and accountability require AI capacity within government. Most African governments have small AI teams against large policy needs. South Africa's Centre for Artificial Intelligence Research (CAIR) and the AI Institute of South Africa, Egypt's National AI Council, Kenya's ICT ministry, and Rwanda's Centre for the Fourth Industrial Revolution are examples of growing capacity. Without investment in public-sector AI capacity, regulation lags innovation by years; the asymmetry creates governance gaps that private actors fill in their own interests.

59.6 Mathematical Foundations of Capacity Modelling

59.6.1 Capability-Maturity Score

For K capabilities at maturity levels $m_k \in \{1, \dots, 5\}$:

$$\text{CMS} = \frac{1}{K} \sum_k m_k.$$

59.6.2 Skill-Pipeline Dynamics

$$N_{t+1} = N_t(1 - r_{\text{leave}}) + I_t,$$

with retention rate $1 - r_{\text{leave}}$ and inflow I_t from training programmes.

59.7 Worked Problems and Solutions

Problem 59.1 (Skill Stock Growth). Current 600 data scientists; 18% annual departure; 220 graduates per year. Stock in 5 years (steady inflow)?

Solution. Recurrence: $N_{t+1} = 0.82N_t + 220$. Steady state $N^* = 220/(1 - 0.82) = 1,222$. Five-year trajectory: $N_1 = 712, N_2 = 804, N_3 = 879, N_4 = 941, N_5 = 991$. ■

59.8 Chapter Conclusion

Capacity building is the multi-decade work that determines whether the technical possibilities catalogued throughout this book translate into actual sustainability outcomes. The chapter walked through the components: educational infrastructure (universities, technical colleges, professional certification), institutional infrastructure (government agencies, research institutes, professional bodies), financial

infrastructure (venture capital, grant funding, blended finance for sustainability AI), and the broader cultural infrastructure that supports innovation and deployment.

The skill-stock-growth worked problem illustrated the dynamics. With 18% annual departure and 220 graduates per year, the steady-state stock of skilled data scientists is approximately 1,222 — roughly double the starting stock of 600. Reaching steady state takes more than a decade given the multiplicative dynamics. Investing in educational pipelines today is the precondition for having capacity in 2035; the timescales are unforgiving.

The chapter's coverage of the African Center for Technology Studies, Data Science Africa, AIMS, and the parallel continental and national institutions documented the substantial capacity-building work already underway. The trajectory is positive; the gap between current capacity and required capacity remains large. The complementary international investments — through Mozilla, the Patrick J. McGovern Foundation, the Gates Foundation, the Lacuna Fund, and others — are helping but cannot substitute for sustained continental investment.

The chapter's discussion of the diaspora-engagement opportunity deserves emphasis. African-born professionals working at international institutions represent a substantial reservoir of capacity that periodic engagement can mobilise without requiring permanent return. The operational patterns — visiting fellowships, joint research programmes, advisory boards, mentoring networks — are increasingly well-established. We turn finally to the integrative question that all of these capacity-building investments serve: the future outlook for AI in sustainability.

The Sustainable Future

The next twenty years will determine which civilisation we leave behind. AI is one of the tools we will use to choose.

Kennedy Chengeta, PhD

This book has surveyed the application of AI to sustainability across precision agriculture, climate forecasting, energy, carbon markets, circular economy, biodiversity, ESG and finance, governance, and African case studies. This concluding chapter synthesises the cross-cutting themes, identifies the trajectories most likely to shape the next decade, and offers a closing reflection on what responsible deployment looks like at scale.

60.1 Cross-Cutting Themes

Several themes have recurred. *Foundation models* have transformed every domain in this book within the past three years; the trajectory continues. *Local fine-tuning* is necessary because models trained on rich-country data fail in poor-country contexts; investment in local data infrastructure compounds returns. *Multi-tier AI plus institutions* is the pattern that produces durable impact; AI alone does not. *Verification matters more than prediction*; the credibility of carbon claims, ESG metrics, and impact reports depends on AI verification infrastructure that has only recently become economically viable. *Distributional outcomes are policy choices*; AI optimisation against aggregate metrics can produce distributional patterns that policy must explicitly address.

60.2 Trajectories for the Next Decade

Five trajectories appear durable. First, *continued cost reduction in AI inference* will make currently-marginal applications economic — continuous monitoring at scales that today are impossible. Second, *regulatory infrastructure* (CSRD, CBAM, EUDR, ISSB, SEC climate rules, parallel emerging-market frameworks) will harden requirements that AI infrastructure must meet. Third, *climate-physical-impact modelling* will become more granular, more local, and more integrated into decisions — by farmers, by insurers, by governments. Fourth, *carbon markets* will continue maturing through integrity reforms and international harmonisation; AI MRV will be embedded throughout. Fifth, *AI energy demand* will continue rising; whether the rise is met with renewable supply or fossil supply will substantially affect whether AI is net-positive for sustainability at scale.

60.3 What Could Go Wrong

Several failure modes deserve attention. *Surveillance creep*: monitoring infrastructure built for sustainability outcomes is repurposed for political surveillance. *Concentration of power*: a few cloud and model providers concentrate AI capabilities to the disadvantage of distributed deployments. *Greenwashing scale-up*: AI generates plausible-looking but misleading sustainability claims at scale, eroding trust in the entire ecosystem. *Energy backsliding*: AI energy demand justifies expanded fossil generation, with sustainability-AI claimed gains insufficient to compensate. *Inequity entrenchment*: AI-enabled efficiency gains accrue to those with capital and capabilities, leaving smallholder, low-income, and otherwise marginalised groups behind. The constructive response to each is structural — governance, antitrust, transparency, public-purpose investment, equity-aware design — not technological.

60.4 What Could Go Right

Equally, several possibilities for net-positive impact deserve attention. *Continental-scale food-system transformation* integrating yield improvements, soil restoration, climate adaptation, and value-chain efficiency at smallholder scale, with AI advisory that meets farmers where they are. *Renewable-dominated grids*, AI-managed, providing reliable affordable electricity to populations historically excluded. *Carbon-removal infrastructure* at gigatonne scale, AI-optimised for cost and verified for permanence. *Biodiversity stabilisation* through monitoring, enforcement, and restoration at scales conventional methods cannot reach. *Just transition pathways* that protect workers and communities from dislocation while accelerating decarbonisation.

60.5 A Closing Reflection

Artificial intelligence is a powerful tool. Sustainability is a moral commitment. The intersection produces the most consequential technology applications of our generation. The work of practitioners, policy-makers, investors, and citizens in the next decade will determine which civilisation we leave behind. The technical guidance in this book is offered in the hope that more of those decisions are well-informed, that more of the deployments are net-positive, and that AI serves sustainability rather than the inverse. The continent that this book has emphasised — Africa — has the most to gain from the right choices and the most to lose from the wrong ones. May the next decade vindicate the optimism with which this book has been written.

— KC, Johannesburg, 2025

60.6 Mathematical Foundations of the Net-Zero Trajectory

60.6.1 Linear Decarbonisation Pathway

For starting emissions E_0 and target year T :

$$E_t = E_0 \left(1 - \frac{t}{T}\right),$$

implying constant absolute reduction rate E_0/T per year.

60.6.2 Exponential Pathway

$$E_t = E_0 e^{-\lambda t}.$$

For 90% reduction by $T = 27$ years (2050 from 2023): $\lambda = \ln(10)/27 = 0.0853$.

60.6.3 Marginal Abatement Cost Curve

Cumulative cost of abating quantity q :

$$C(q) = \int_0^q c(e) de,$$

where $c(e)$ is the marginal abatement cost.

60.7 Worked Problems and Solutions

Problem 60.1 (Required Annual Reductions). Global emissions today 41 Gt; net-zero target 2050 (27 years). Compute required average annual reduction (linear pathway).

Solution. $41/27 \approx 1.52$ Gt CO₂/yr. Equivalent to halting all road transport globally each year, repeated for 27 years. Communicates the scale of the challenge. ■

60.8 Chapter Conclusion

The trajectory of AI for sustainability over the next decade can be sketched with reasonable confidence. The technical capability of AI systems will continue to advance; foundation models for earth observation, climate, and agriculture will reach operational maturity; the cost of inference at scale will continue to fall; and the integration of AI into operational decision-making across the sustainability domain will continue to broaden. None of this is in serious doubt.

What is in doubt is whether the deployment trajectory will be net-positive at scale. The chapter argued that the answer depends on three choices made repeatedly across thousands of deployment decisions per year: whether AI is applied to high-leverage physical processes or low-leverage information processes; whether deployment respects the local ecological, institutional, and cultural context; and whether honest accounting of AI's own footprint is integrated into deployment decisions from the start rather than added as an afterthought.

The required-annual-reductions worked problem put the policy stakes in quantitative form. Reaching net-zero by 2050 requires reductions of roughly 1.52 Gt CO₂ per year, every year, for 27 years. This is the scale of the challenge AI applications must contribute to; it is not a marginal scale. Sustainability AI that delivers reductions in the hundreds of megatonnes per year is material; reductions in the megatonnes per year are useful but not sufficient; reductions in the kilotonnes per year are publication-worthy but not policy-relevant. The discipline of asking, at every deployment decision, which scale the contribution operates at is essential.

The chapter closed by returning to the net-positive criterion of Chapter 1. A deployment is net-positive if its lifecycle environmental benefits exceed its lifecycle environmental costs. The criterion is mathematical; the assessment is operational; the discipline of applying it consistently is what distinguishes serious sustainability AI practice from rhetorical sustainability AI practice. The author hopes this book has equipped the reader to make the distinction in their own work, and to insist on it in the work of others. The agenda is large; the time is short; the work is urgent. We close here.

Glossary

This glossary collects key terms used throughout the book. Cross-references to chapters where each concept is treated in depth are provided where appropriate.

A

Active learning. A machine learning paradigm in which the model selects which examples should be labelled next to maximise information gain. Used in materials discovery, agricultural disease classification, and rare-event forecasting.

Additionality. The principle that a carbon credit must represent emission reductions or removals that would not have occurred in the absence of the project. A central integrity criterion in voluntary and compliance markets (Chapter 31).

AfSIS. African Soil Information Service. Continental soil property database and modelling infrastructure (Chapter 9).

AIS. Automatic Identification System. Vessel-tracking transponder data used in marine monitoring and IUU fishing detection (Chapter 43).

Allometric model. A regression relating tree dimensions (diameter, height) to biomass, foundational to forest carbon accounting.

AquaCrop. A FAO process-based crop model widely used for water-productivity simulation; commonly used as a training environment for reinforcement-learning irrigation schedulers (Chapter 10).

Article 6. Provisions of the Paris Agreement governing international transfer of mitigation outcomes (ITMOs).

B

Baseline. The counterfactual emissions trajectory against which project emissions reductions are measured. Static baselines use historical reference; dynamic baselines model the counterfactual forward.

Behind-the-meter. Energy resources (rooftop solar, batteries, controllable loads) connected on the customer side of the utility meter.

Blue carbon. Carbon stored in coastal and marine ecosystems — mangroves, salt marshes, seagrasses (Chapter 34).

Body condition score (BCS). A 1–9 or 1–5 visual assessment of livestock nutritional status, increasingly automated through computer vision (Chapter 12).

Buffer pool. A reserve of unsold credits held by a registry to compensate for permanence reversals. Sized in proportion to project risk and uncertainty.

C

CARE principles. Collective benefit, Authority to control, Responsibility, Ethics. Indigenous data sovereignty principles supplementing FAIR (Chapter 52).

CBAM. EU Carbon Border Adjustment Mechanism. Imposes carbon-equivalent charges on imports of carbon-intensive goods (Chapter 32).

CCS. Carbon Capture and Storage. Capture of CO₂ from point sources and storage in geological formations.

CDP. Formerly Carbon Disclosure Project. A voluntary corporate environmental disclosure platform.

CEMS. Continuous Emissions Monitoring System. Stack-mounted sensors providing continuous emission measurements at industrial facilities.

CHIRPS. Climate Hazards Group InfraRed Precipitation with Station data. A widely-used satellite-station blended precipitation product.

ClimaX, GraphCast, Pangu-Weather. Major neural-network global weather forecasting models (Chapter 15).

Conformal prediction. A distribution-free, finite-sample-valid method for producing prediction intervals from any base model (Chapter 3).

Conservation Easement. A legal restriction on land use registered against title to protect conservation values; increasingly bundled with carbon and biodiversity credits.

CSRD. Corporate Sustainability Reporting Directive. EU regulation mandating extensive sustainability disclosure (Chapter 48).

D

DAC. Direct Air Capture. Removal of CO₂ directly from ambient air (Chapter 35).

DER. Distributed Energy Resource. Generation, storage, or controllable load connected at the distribution level.

Differential privacy. A mathematical framework for releasing aggregate data analytics with quantified protection against individual identification.

Digital twin. A live-updated model of a physical asset informed by sensor data. Used in grid, building, and industrial applications.

DSSAT. Decision Support System for Agrotechnology Transfer. A widely-used family of process-based crop models.

Double materiality. The CSRD principle requiring disclosure of both how sustainability issues affect a company and how the company affects sustainability outcomes.

E

eDNA. Environmental DNA. DNA shed by organisms into water, soil, or air, enabling species detection without physical specimen capture (Chapter 41).

Emission factor (EF). A coefficient relating activity (e.g. litres of diesel) to emissions (kg CO₂e per unit activity).

ENSO. El Niño–Southern Oscillation. The dominant mode of tropical Pacific variability driving global teleconnections.

ERA5. European Centre for Medium-Range Weather Forecasts atmospheric reanalysis providing hourly historical weather since 1940 on a ~25km grid.

ESG. Environmental, Social, Governance. The dominant framework for sustainability investing and corporate disclosure (Chapter 46).

ETS. Emissions Trading System. A cap-and-trade compliance market for greenhouse gas emissions (Chapter 32).

Evapotranspiration (ET). Combined water loss from soil evaporation and plant transpiration. Reference ET (ET₀) is for a standardised reference crop; crop ET (ET_c = K_c · ET₀) accounts for crop coefficient.

EUDR. EU Deforestation Regulation. Prohibits import of specified commodities from land deforested after 31 December 2020 (Chapter 49).

F

FAIR principles. Findable, Accessible, Interoperable, Reusable. Data management principles widely adopted in scientific data infrastructure.

Federated learning. A machine-learning paradigm in which models train across decentralised data without raw data leaving devices or institutions.

Foundation model. A large model pretrained on broad data, fine-tuned for downstream tasks. Examples in earth observation include Prithvi and SatMAE.

G

GCM. General Circulation Model. The class of physical climate models computing atmospheric and oceanic dynamics on a discretised grid.

GHG Protocol. The dominant corporate greenhouse gas accounting standard, defining Scope 1, 2, and 3 emissions (Chapter 29).

Gold Standard. A voluntary carbon credit registry emphasising sustainable development co-benefits.

Growing degree days (GDD). A thermal-time integral $\sum_d \max(T_d - T_b, 0)$ relating to crop development.

H

Hyperspectral. Imagery with hundreds of contiguous narrow spectral bands; enables direct chemical and material identification (e.g. EnMAP, PRISMA, DESIS satellites).

I

Interpretable ML. Models or post-hoc methods (SHAP, LIME, Grad-CAM) that explain individual predictions or aggregate model behaviour.

IPCC. Intergovernmental Panel on Climate Change. Produces periodic assessment reports; publishes inventory guidelines for national greenhouse gas accounting.

ISSB. International Sustainability Standards Board (under the IFRS Foundation). Issuer of IFRS S1 (general) and S2 (climate) sustainability disclosure standards (Chapter 48).

ITMO. Internationally Transferred Mitigation Outcome. The unit of trade under Paris Agreement Article 6.

IUU fishing. Illegal, Unreported, and Unregulated fishing. A focus of marine AI surveillance applications (Chapter 43).

J

JETP. Just Energy Transition Partnership. Concessional finance arrangements supporting managed transitions in coal-dependent economies (South Africa, Indonesia, Vietnam, Senegal).

L

LCA. Life Cycle Assessment. The methodology for quantifying environmental impacts across a product or process lifecycle.

Leakage. Emission reductions inside a project boundary partially offset by emission increases outside it.

LiDAR. Light Detection and Ranging. Active sensing measuring distance via laser pulse return time. Used in forest height measurement (GEDI, ICESat-2, airborne campaigns).

Lightning as in *lightning-ahead forecasting*: sub-second to seconds-ahead grid forecasting using PMU data and ML.

LULUCF. Land Use, Land-Use Change, and Forestry. The IPCC inventory category covering biological carbon sources and sinks.

M

Materials passport. A digital record of a product's composition, components, and lifecycle history. Mandated under the EU Ecodesign for Sustainable Products Regulation (ESPR).

MethaneSAT, GHGSat, Sentinel-5P. Satellites with methane plume detection and quantification capability (Chapter 30).

Mitigation Outcome (MO). An emission reduction or removal that may be transferred under Article 6.

MRV. Measurement, Reporting, Verification. The operational machinery converting physical reality into tradeable carbon units (Chapter 30).

N

NDVI. Normalised Difference Vegetation Index. The fundamental satellite-derived vegetation greenness measure (Chapter 7).

Net-positive. A deployment where lifecycle environmental benefits exceed lifecycle costs (Chapter 1).

NGFS. Network for Greening the Financial System. Publishes climate scenarios used by financial supervisors (Chapter 20).

Non-revenue water (NRW). Treated water lost to leaks, theft, or metering inaccuracy. A key target for water-utility AI (Chapter 39).

NWP. Numerical Weather Prediction. The class of physical weather forecasting systems based on integrating atmospheric fluid dynamics.

O

OECD. Other Effective area-based Conservation Measures. Areas conserved alongside formal protected areas, recognised under the Global Biodiversity Framework.

P

PAYGo solar. Pay-as-you-go solar home systems financed by mobile-money instalments. A major channel for African off-grid electrification (Chapter 57).

Permanence. The duration over which stored carbon remains stored. Permanence-adjusted accounting discounts credits by reversal risk and storage horizon.

Pinball loss. The asymmetric loss function used to fit quantile regression models (Chapter 22).

PMU. Phasor Measurement Unit. High-resolution voltage-and-current sensor used in grid state estimation.

POPIA. Protection of Personal Information Act. South Africa's principal data protection law.

Prithvi. A NASA-IBM earth-observation foundation model (Chapter 7).

Process-based model. A mechanistic model encoding physical, chemical, or biological laws (e.g. DSSAT, APSIM, AquaCrop crop models; SEBAL ET model).

PUE. Power Usage Effectiveness. The ratio of total data centre power to IT equipment power. Lower is better.

Q

Quantile regression. A regression technique fitting conditional quantiles rather than the conditional mean. Foundation of probabilistic forecasting.

R

REDD+. Reducing Emissions from Deforestation and Forest Degradation. The dominant forest-carbon project category.

Reference scenario. See baseline.

Reinforcement learning (RL). A machine-learning paradigm in which an agent learns a policy by interacting with an environment to maximise cumulative reward (Chapter 3).

S

SAR. Synthetic Aperture Radar. Active microwave remote sensing operating in all weather and lighting (Sentinel-1, ALOS PALSAR, BIOMASS).

SBTi. Science Based Targets initiative. Framework for setting and validating corporate emission-reduction targets.

Scope 1, 2, 3. GHG Protocol categories: direct emissions, purchased-energy emissions, all other indirect.

SCORPAN. A digital soil mapping framework predicting soil properties as a function of soil, climate, organisms, relief, parent material, age, neighbourhood (Chapter 9).

SDG. Sustainable Development Goal. The seventeen UN goals adopted in 2015 (Chapter 5).

Sentinel. The Copernicus earth observation satellite constellation (Sentinel-1 SAR, Sentinel-2 optical, Sentinel-3 ocean/land, Sentinel-5P atmospheric).

SHAP. SHapley Additive exPlanations. A model-agnostic feature-attribution method based on cooperative game theory.

Smart meter. An electricity, gas, or water meter providing periodic remote consumption readings.

SOC. Soil Organic Carbon. The carbon component of soil organic matter (Chapter 9).

SPI. Standardised Precipitation Index. A widely-used drought index based on precipitation anomalies.

T

tCO₂e. Tonnes of carbon dioxide equivalent. The standard unit of greenhouse gas accounting, weighting non-CO₂ gases by global warming potential.

TCFD. Task Force on Climate-related Financial Disclosures. Framework absorbed into ISSB IFRS S2.

Teleconnection. A statistical association between climate anomalies in geographically distant regions, mediated by large-scale circulation patterns.

Transformer. The neural-network architecture based on self-attention dominant in language and increasingly in time-series and vision applications.

Transition risk. The financial risk arising from policy, technology, and market changes during decarbonisation (Chapter 20).

U

UFLPA. Uyghur Forced Labor Prevention Act (US). Presumes that goods produced in Xinjiang are made with forced labour and prohibits import unless rebutted (Chapter 49).

UNFCCC. UN Framework Convention on Climate Change. The parent treaty of the Paris Agreement.

V

V2G. Vehicle-to-Grid. Bidirectional power flow between EVs and the grid (Chapter 28).

Verra (VCS). Verified Carbon Standard. The largest voluntary carbon credit registry.

VIIRS. Visible Infrared Imaging Radiometer Suite. Polar-orbiting sensor providing daily global imagery and active-fire detection.

VPP. Virtual Power Plant. An aggregation of distributed energy resources operated as a single dispatchable resource.

W

WBCSD. World Business Council for Sustainable Development. Co-publisher of the GHG Protocol Corporate Standard.

WRI. World Resources Institute. Co-publisher of the GHG Protocol; operator of Global Forest Watch.

Closing Note

This glossary is selective rather than exhaustive. Comprehensive references for further reading are provided at the end of each chapter.